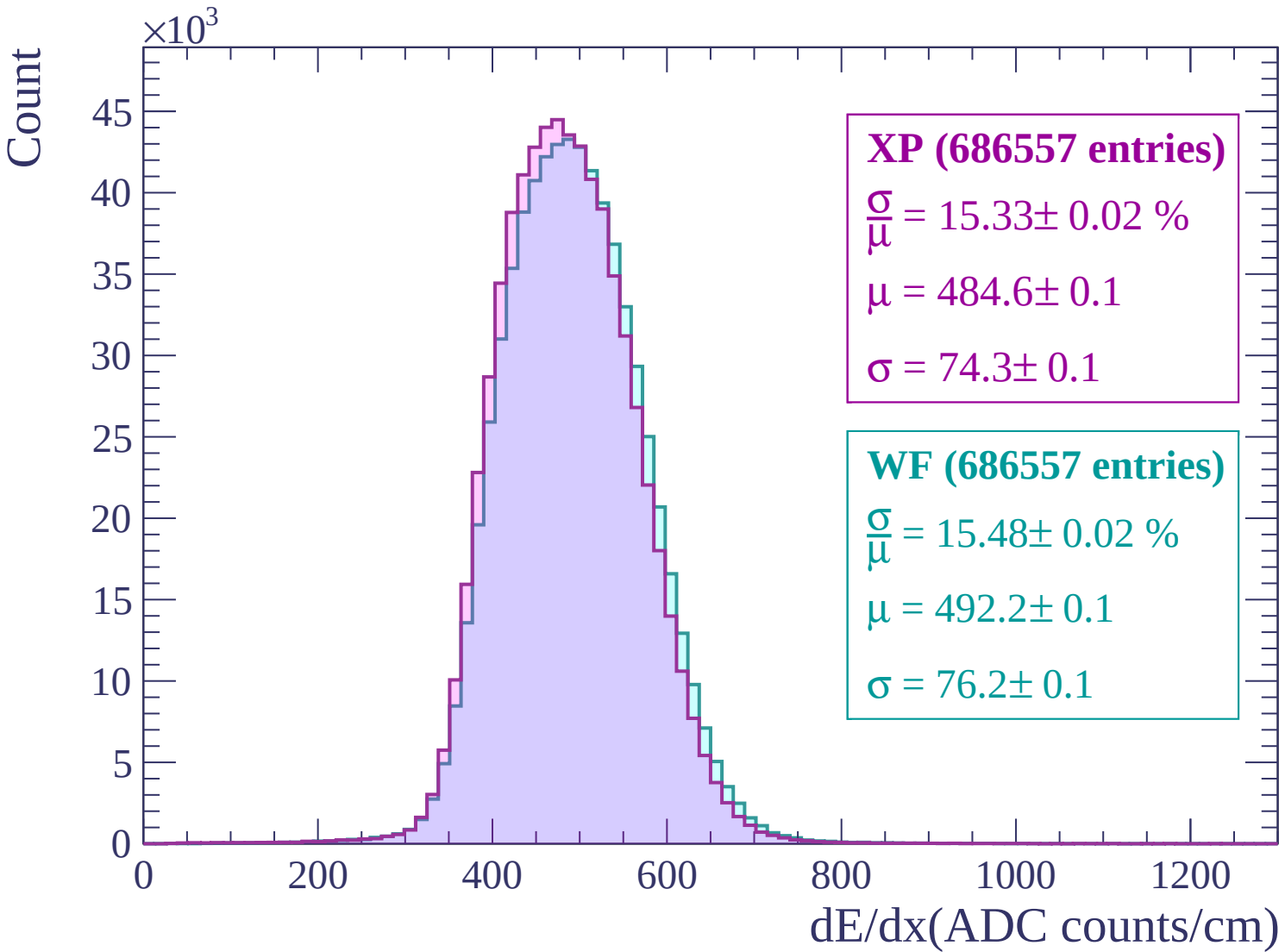
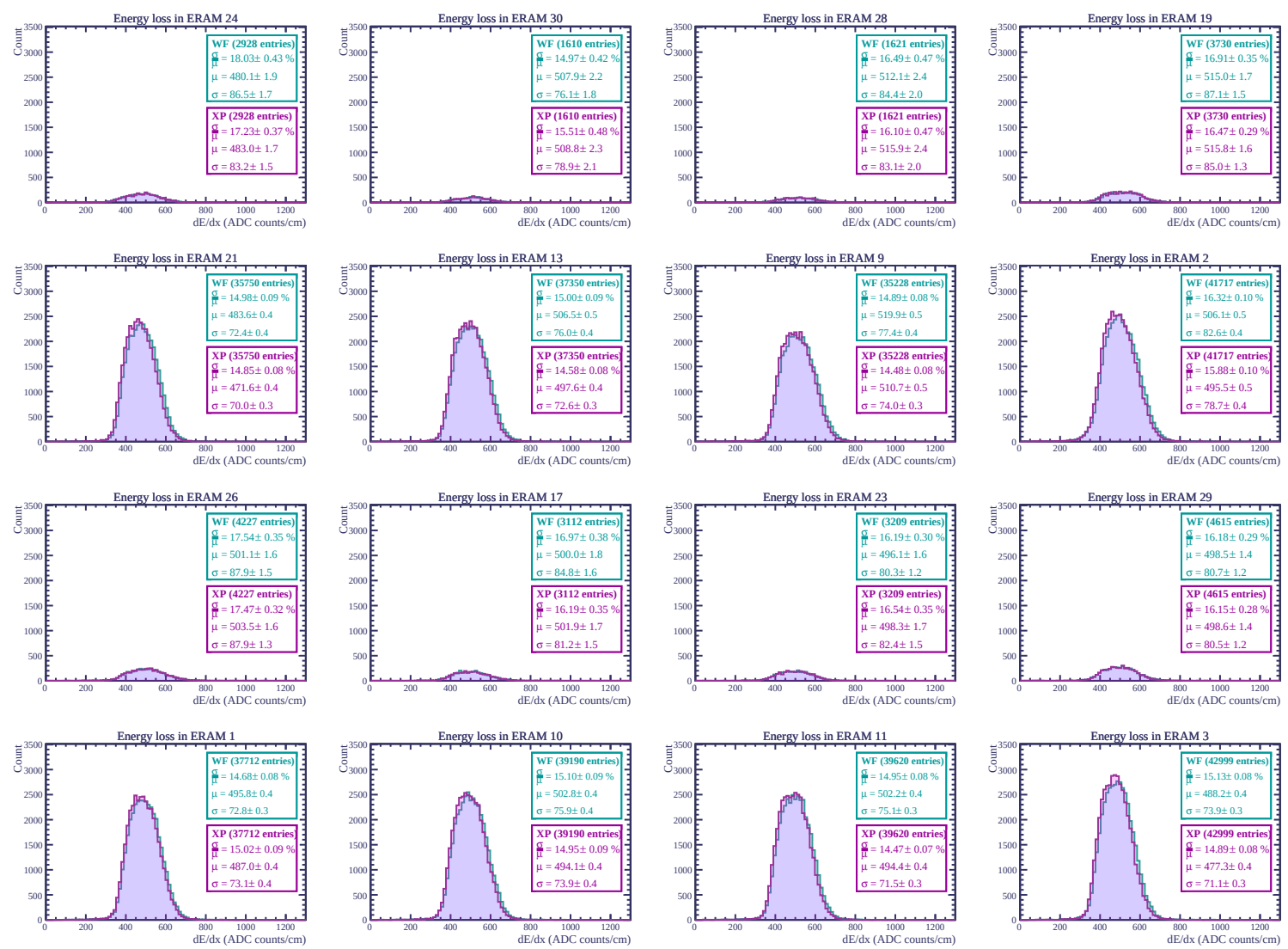
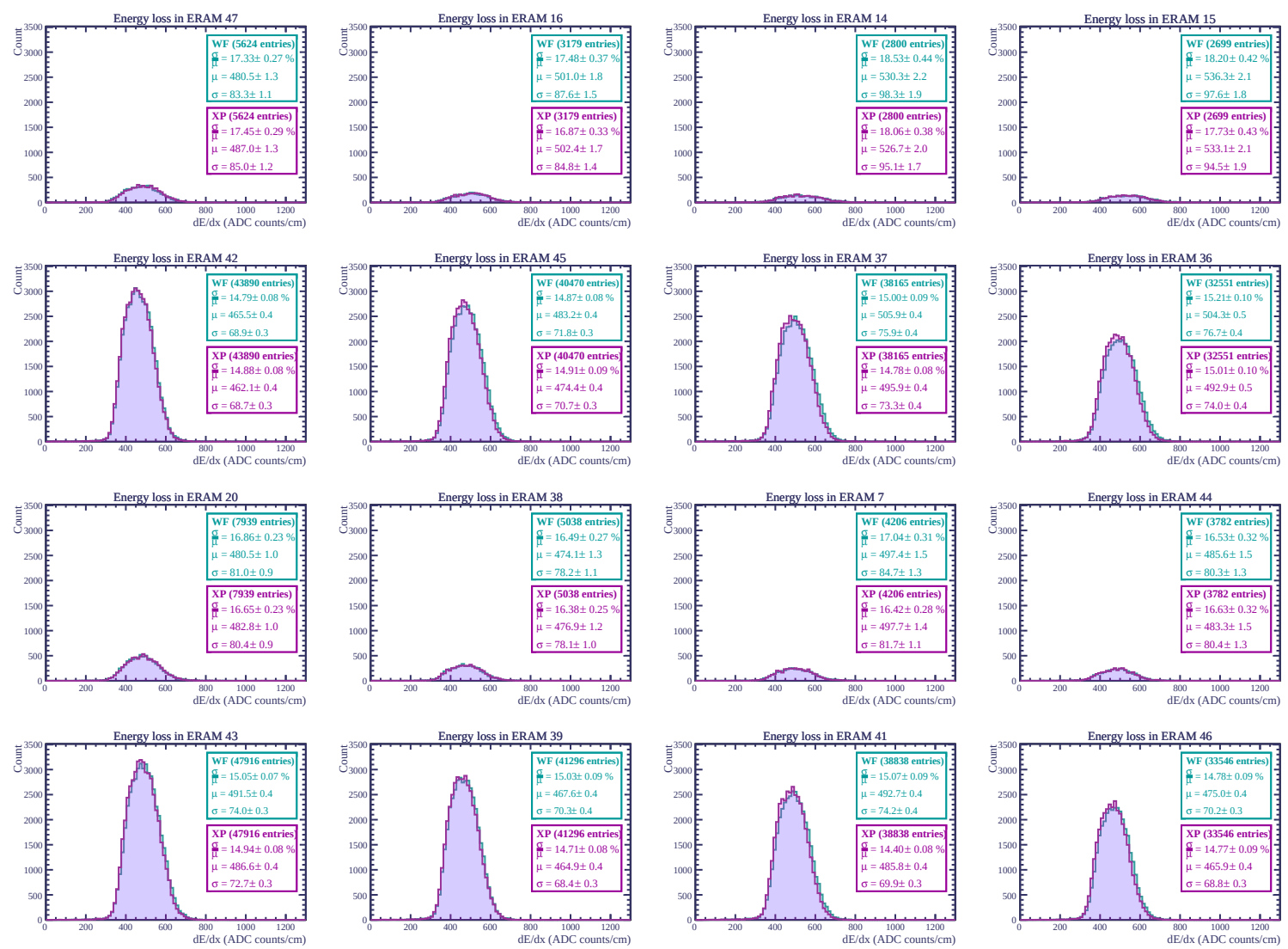
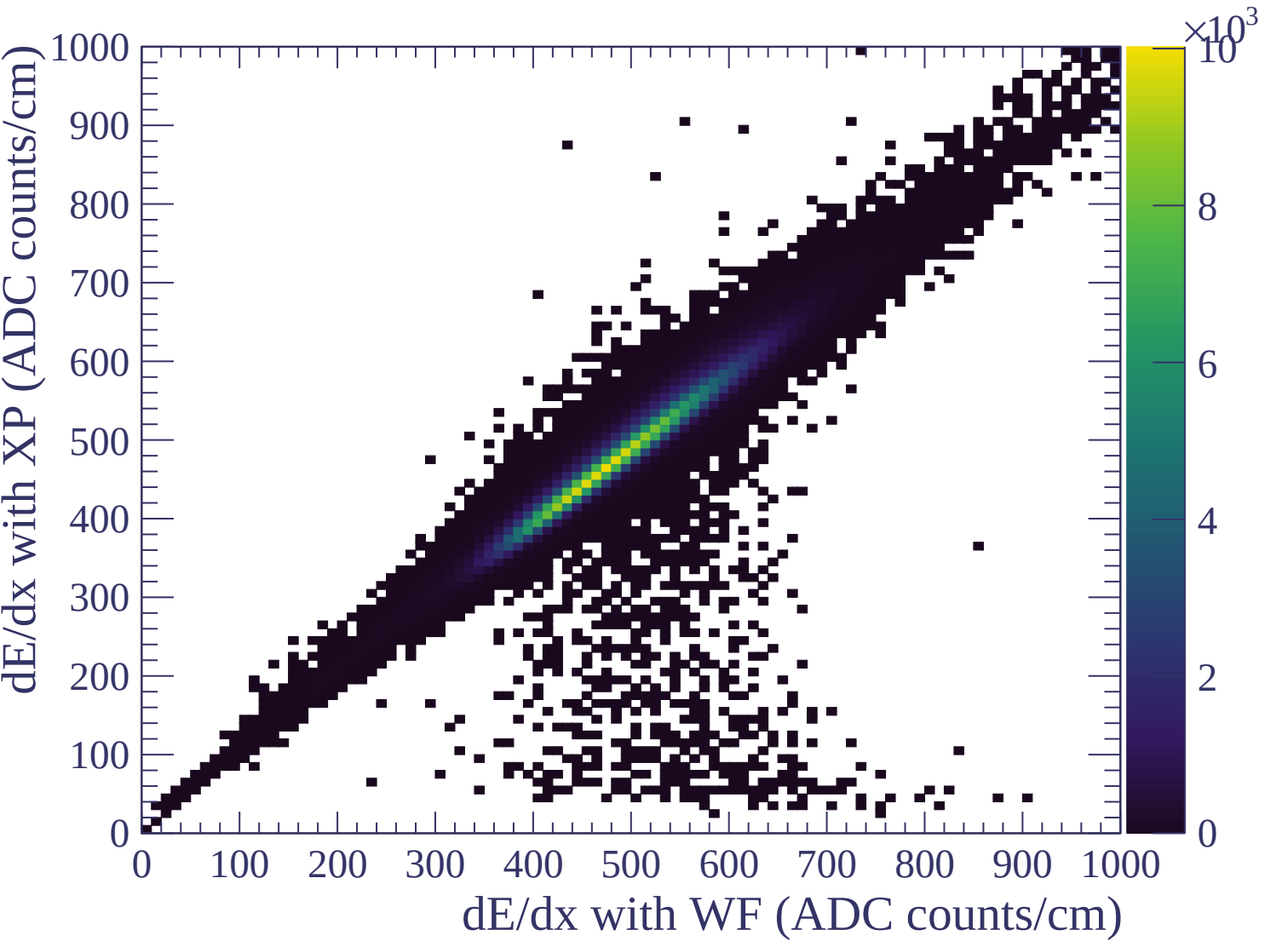
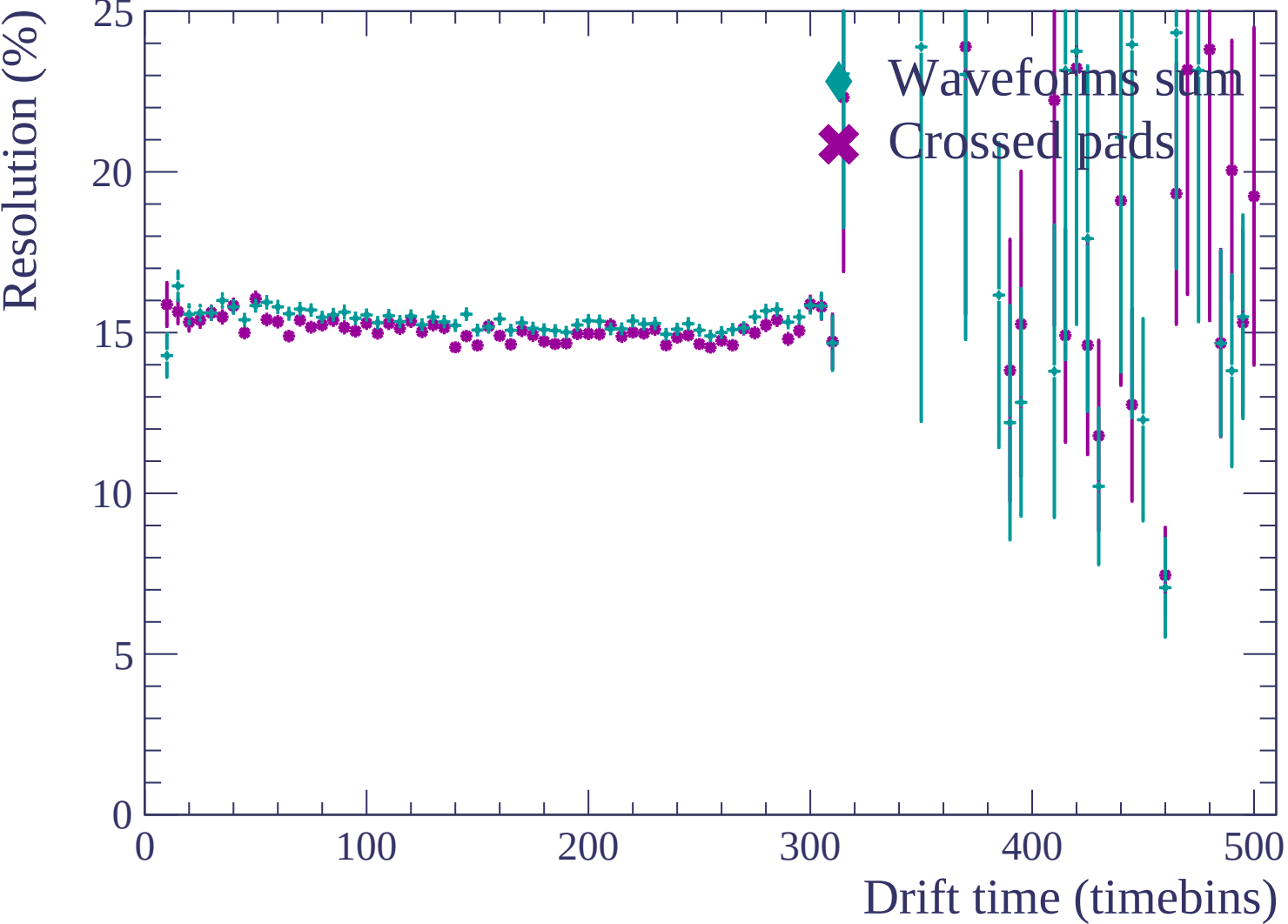


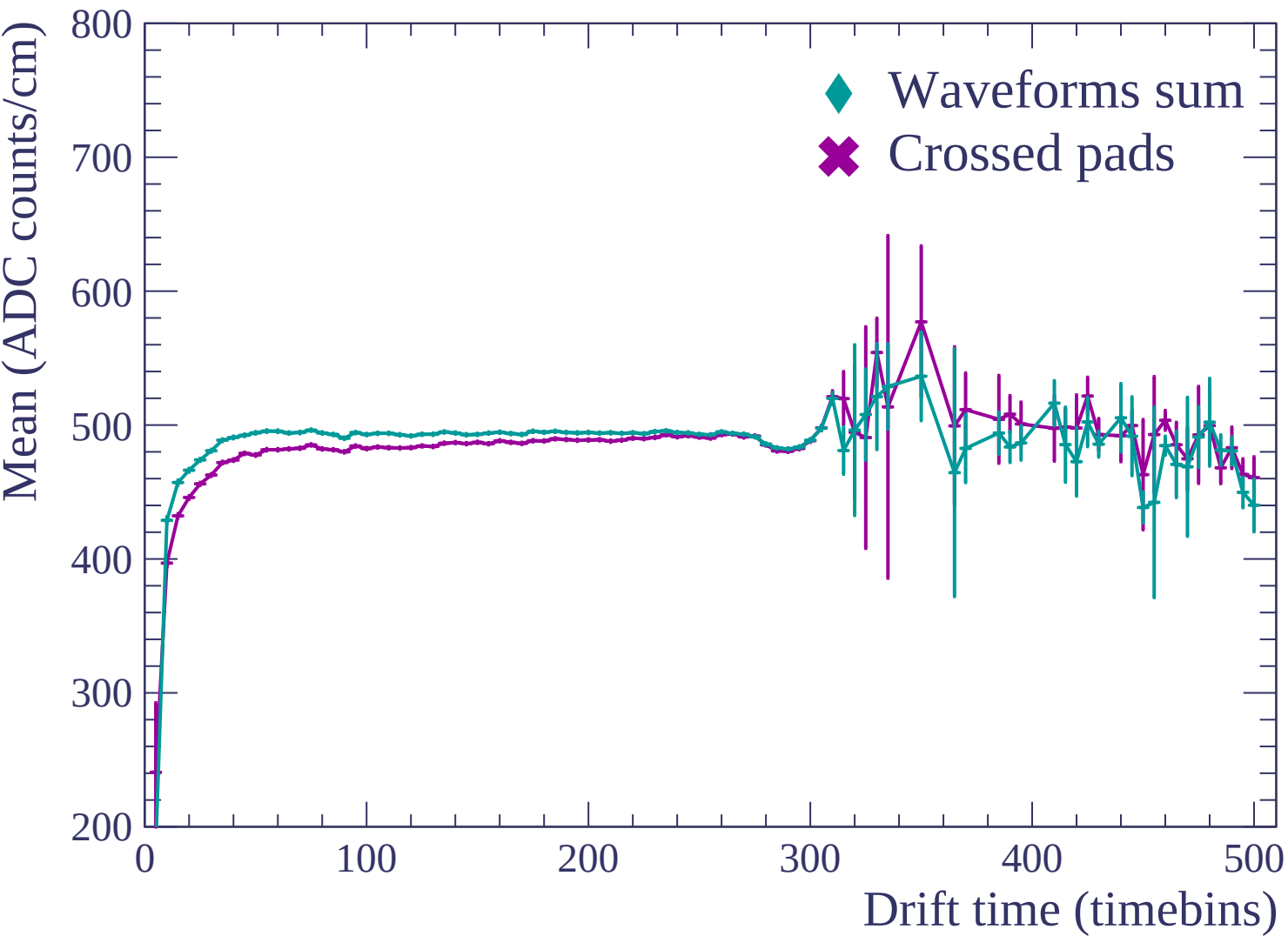


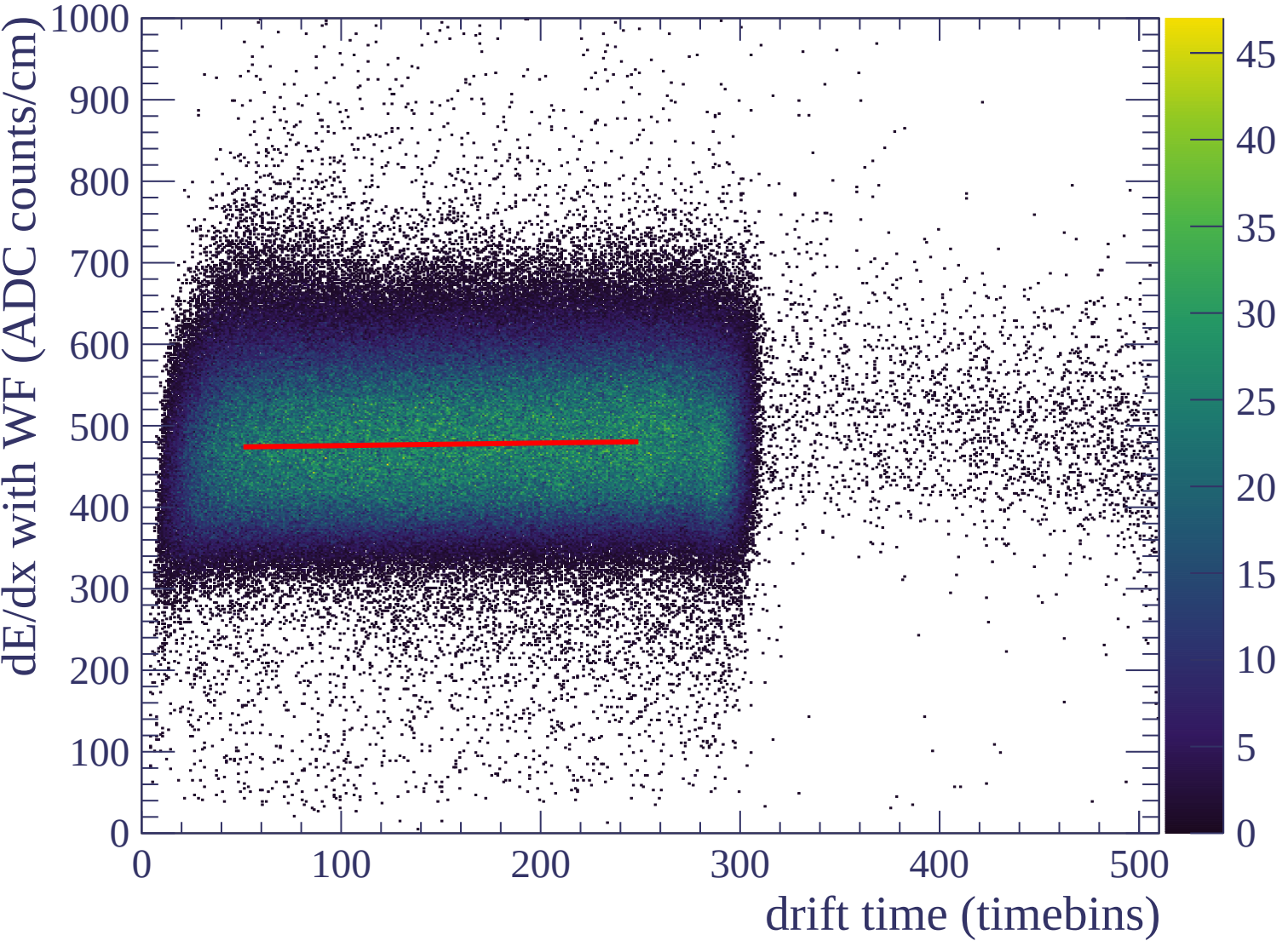


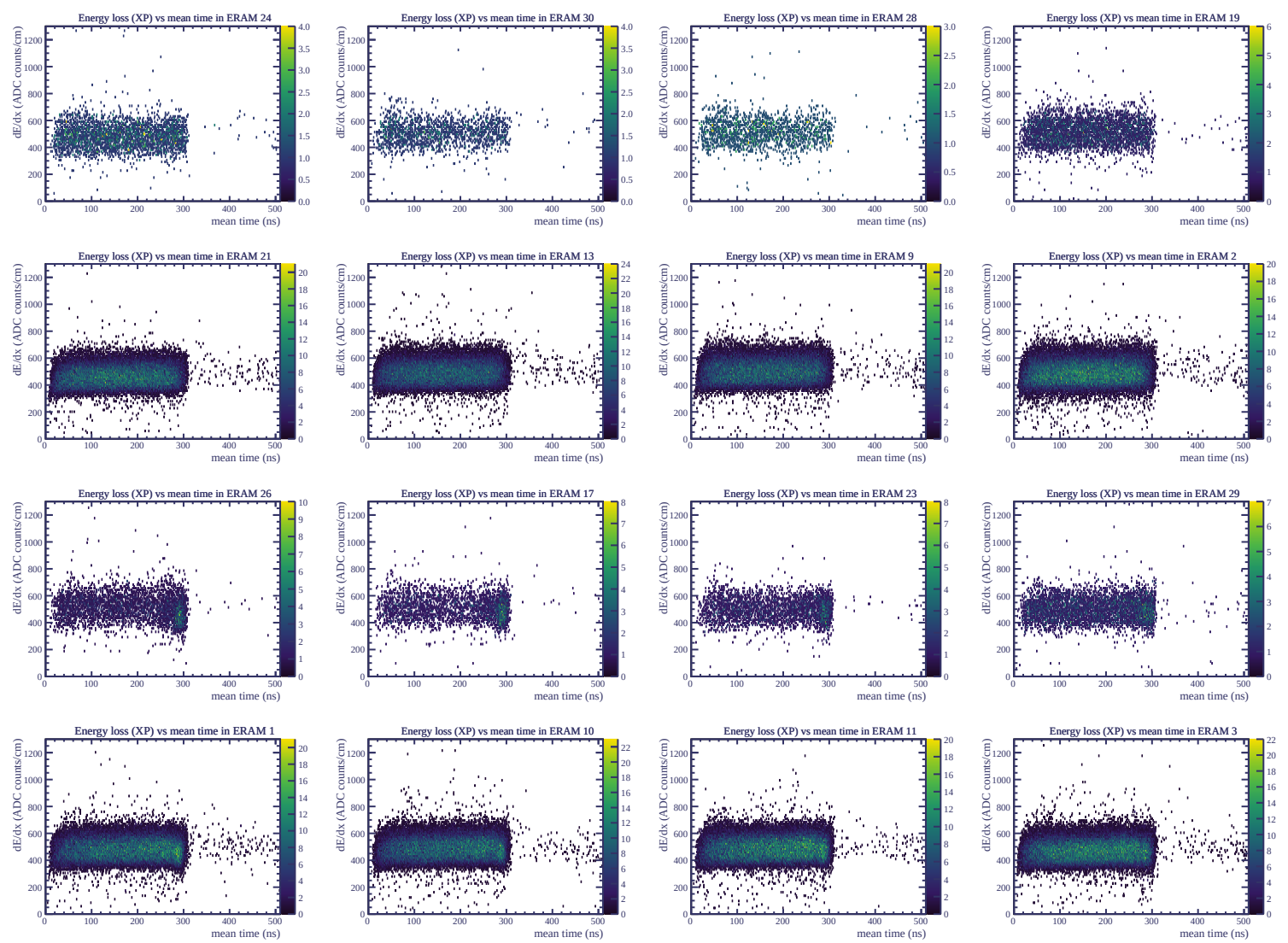


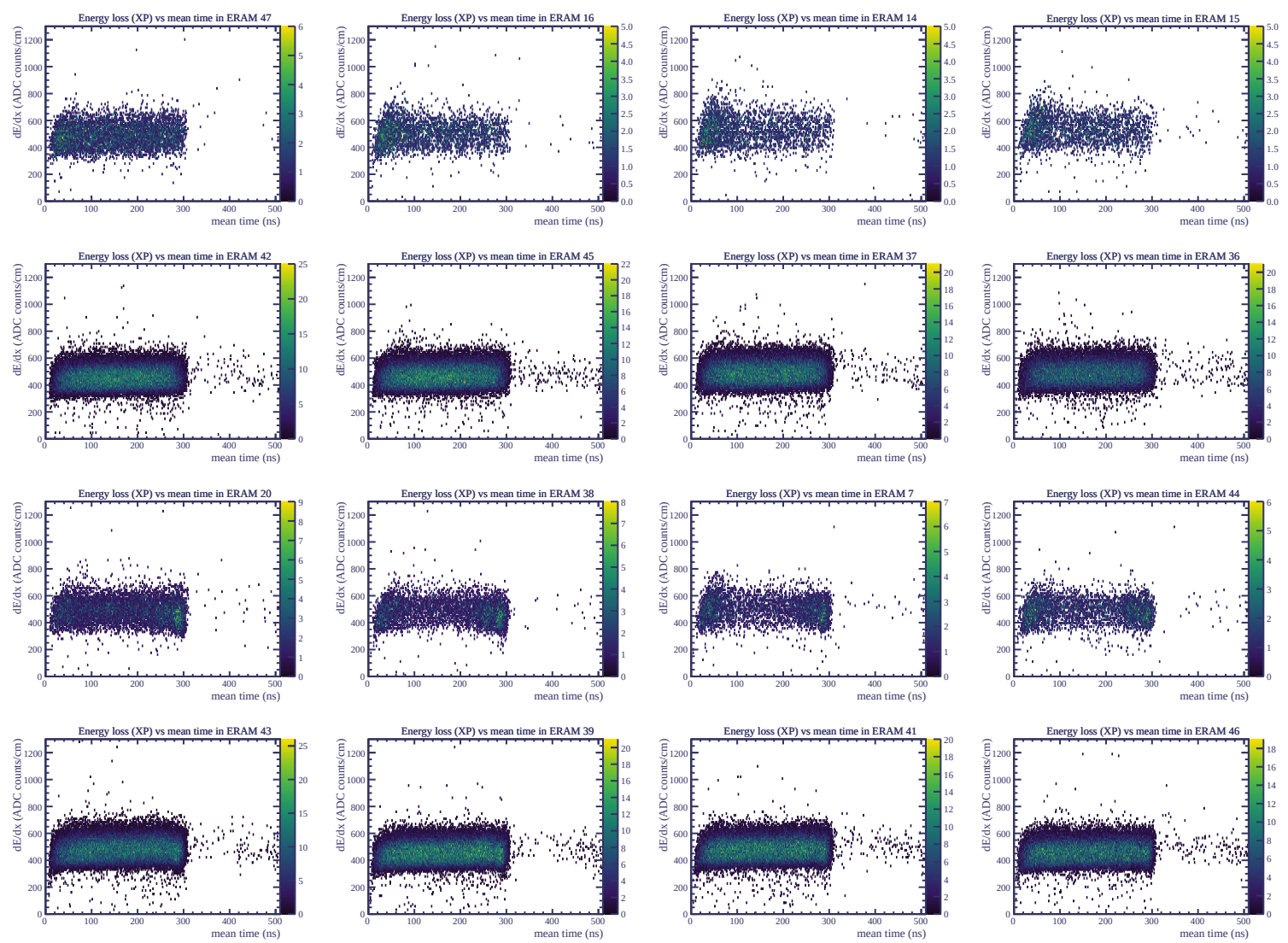


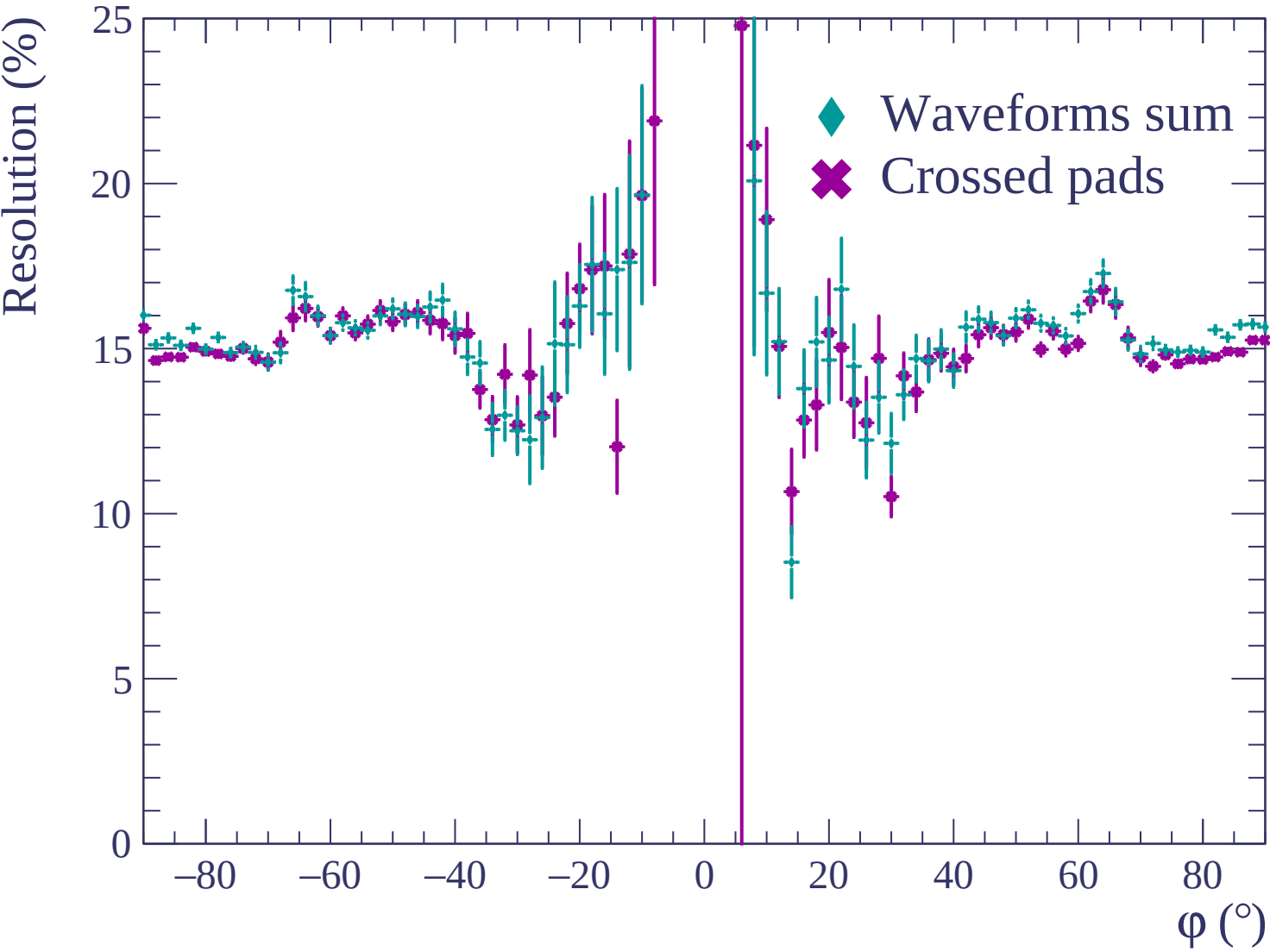


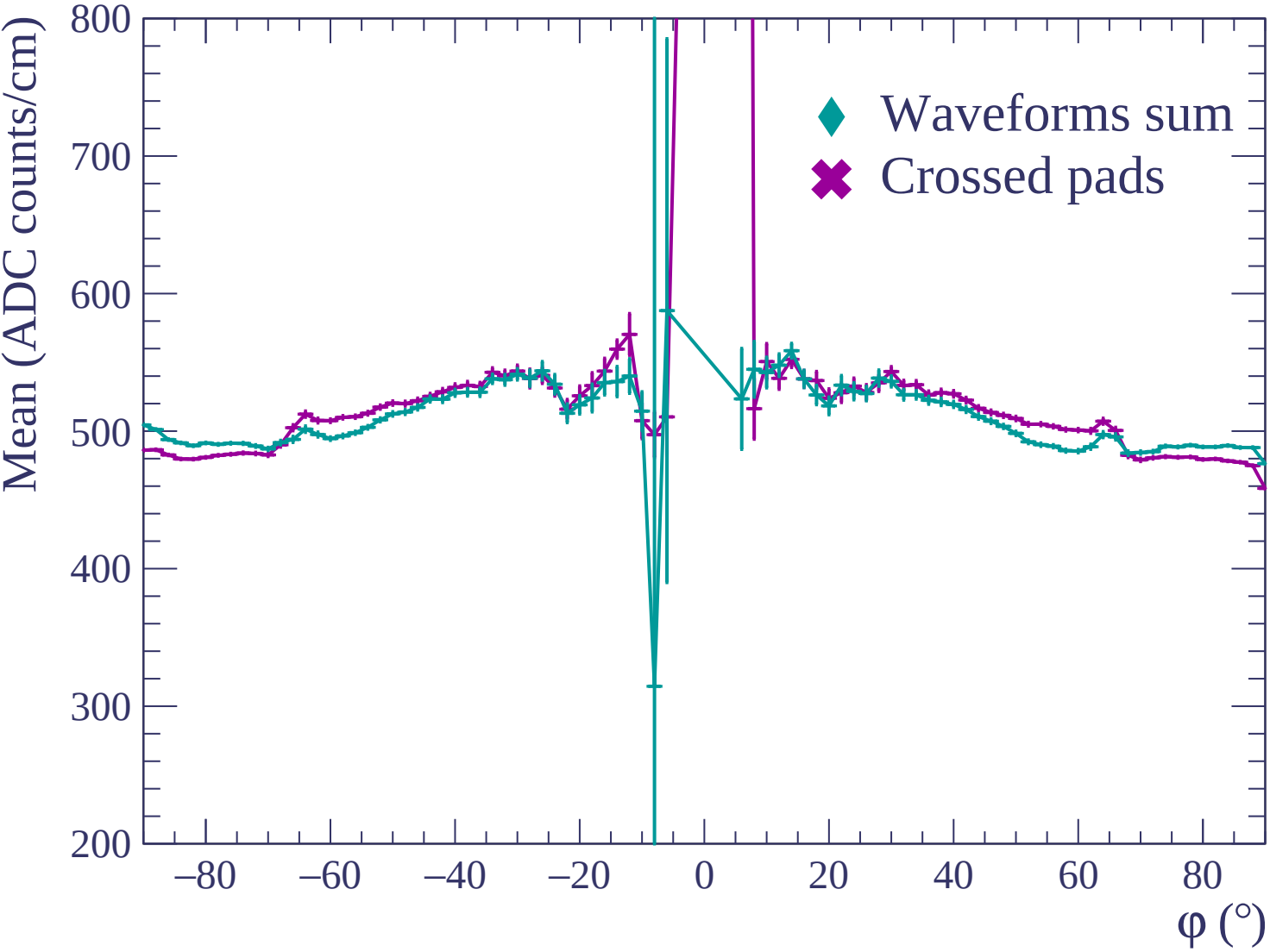


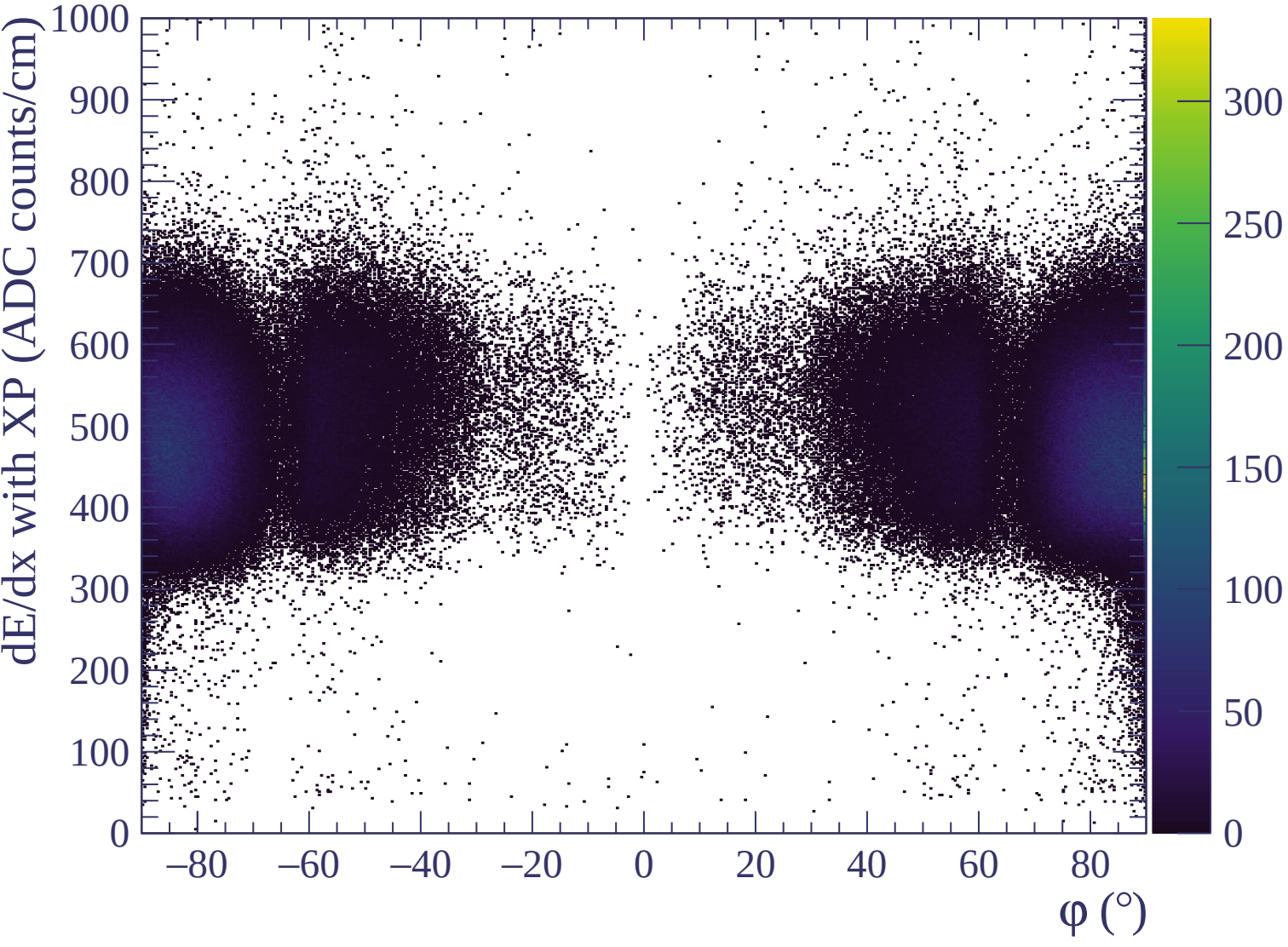


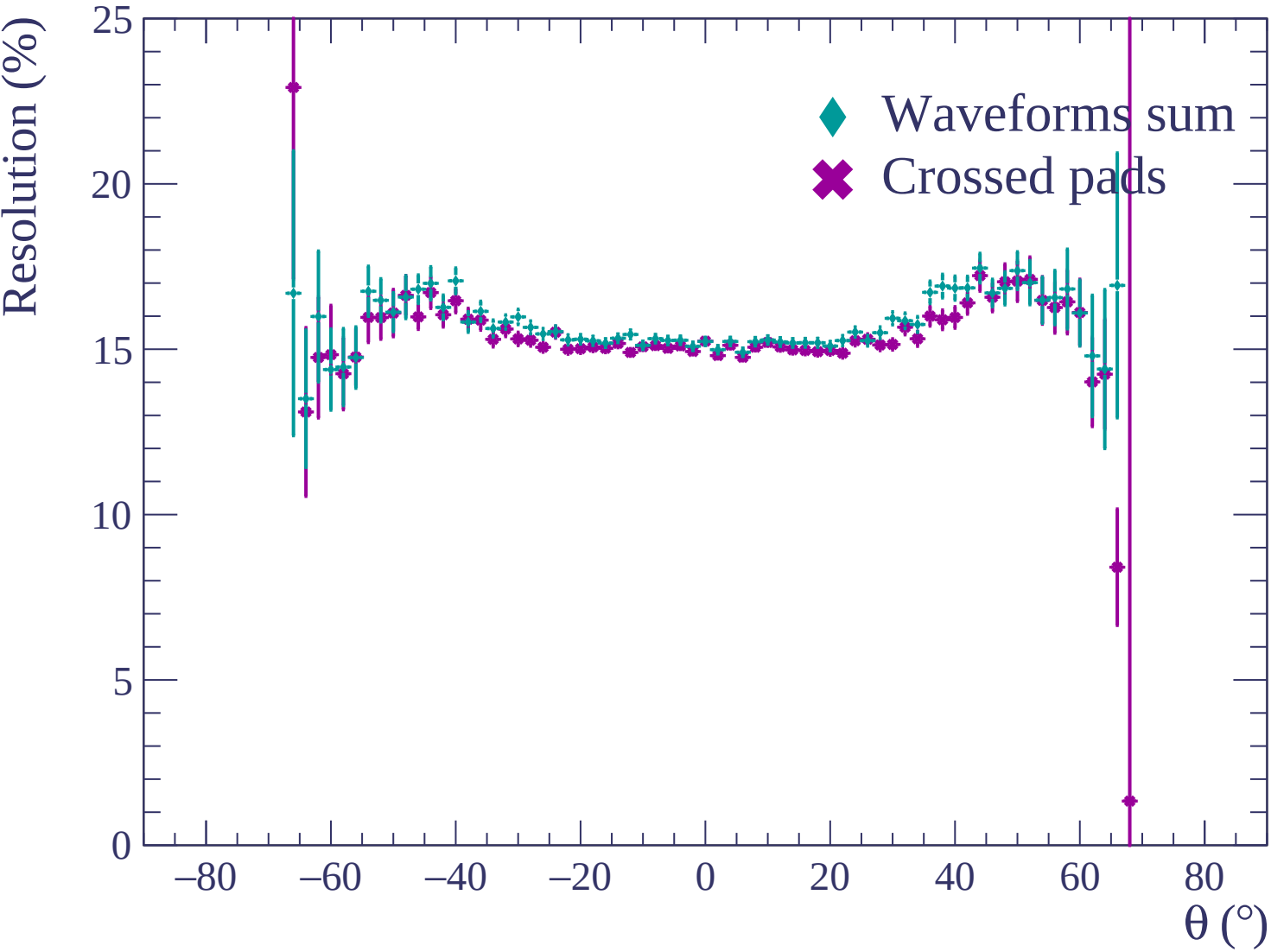


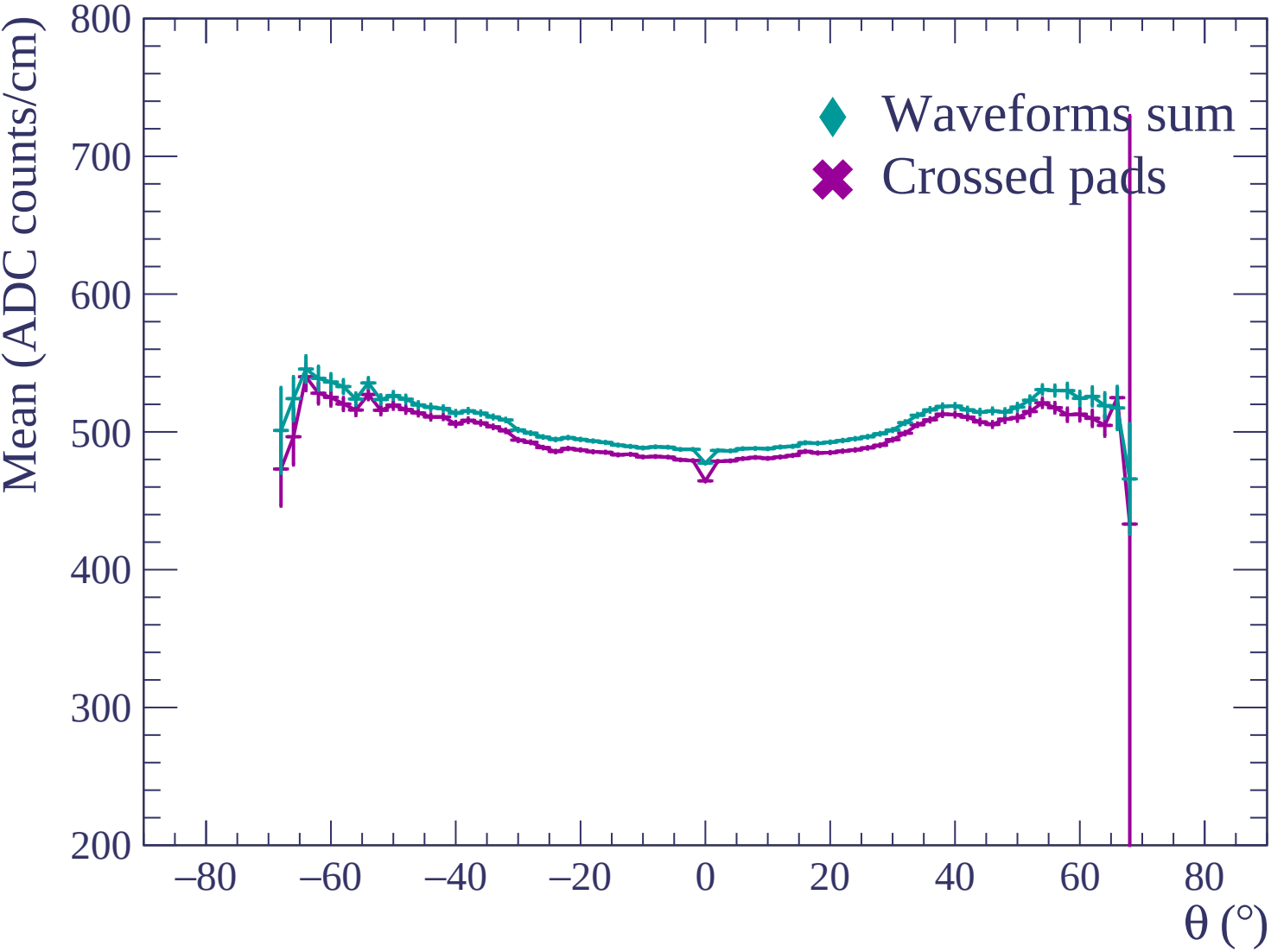


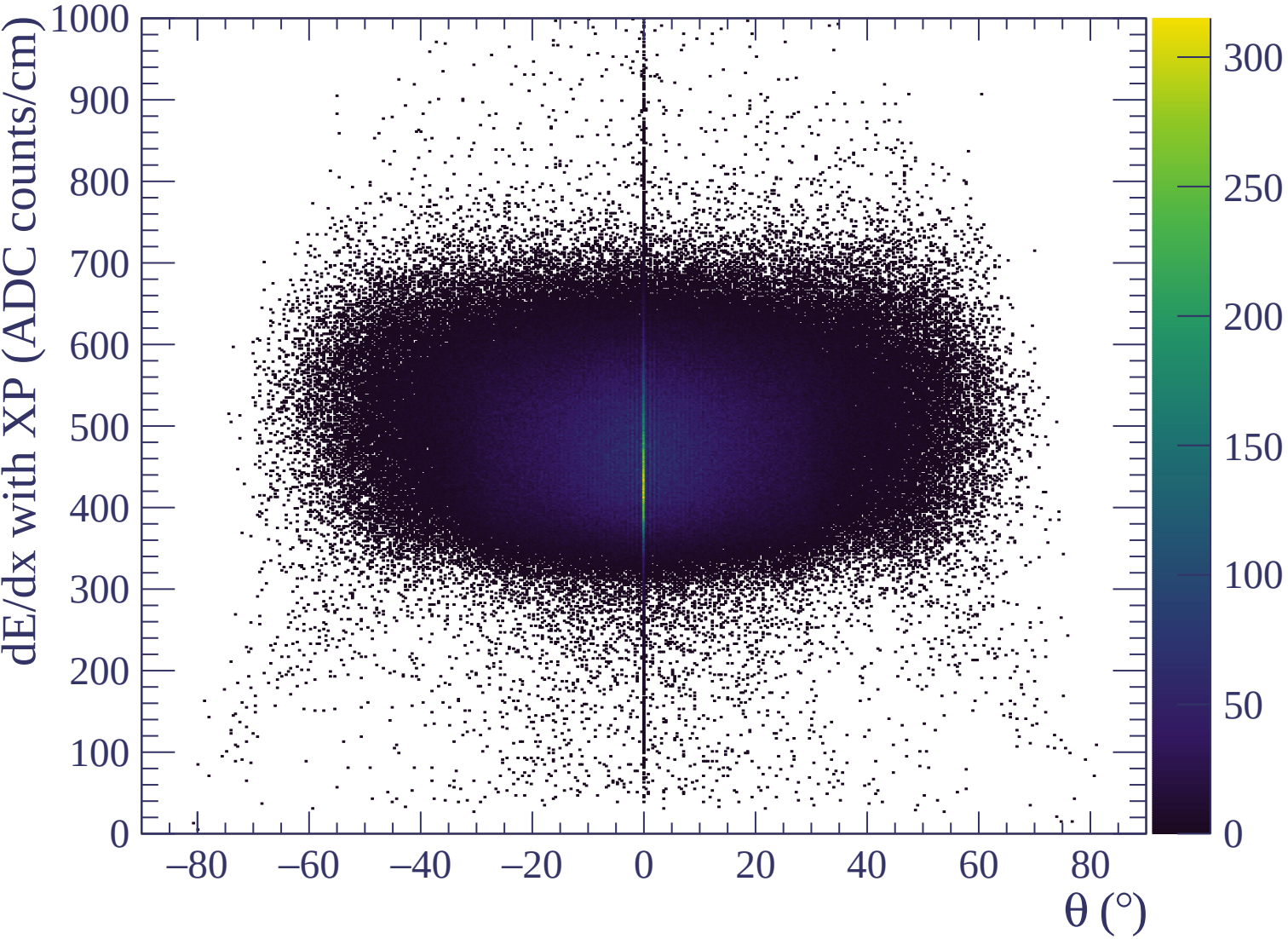


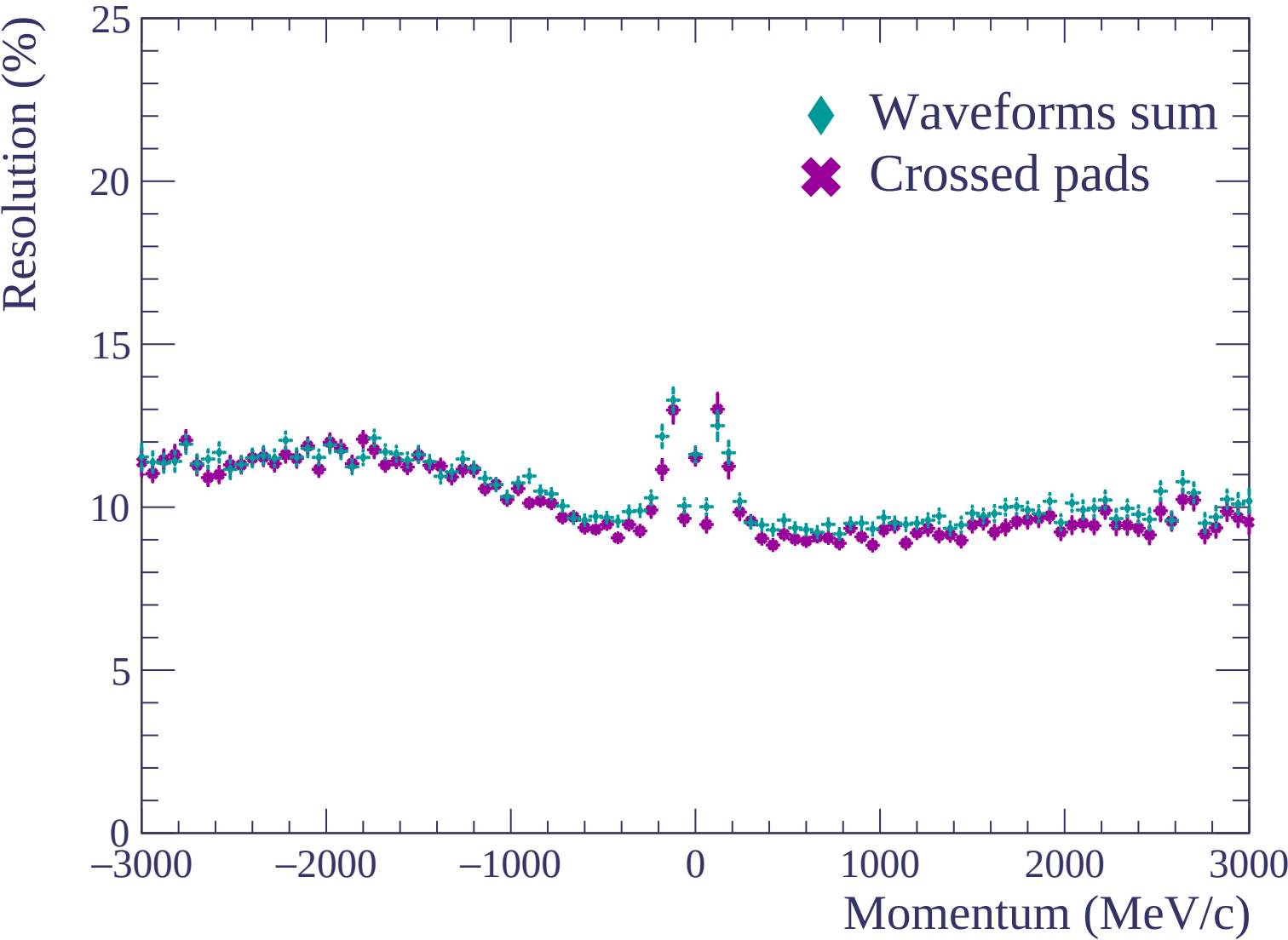


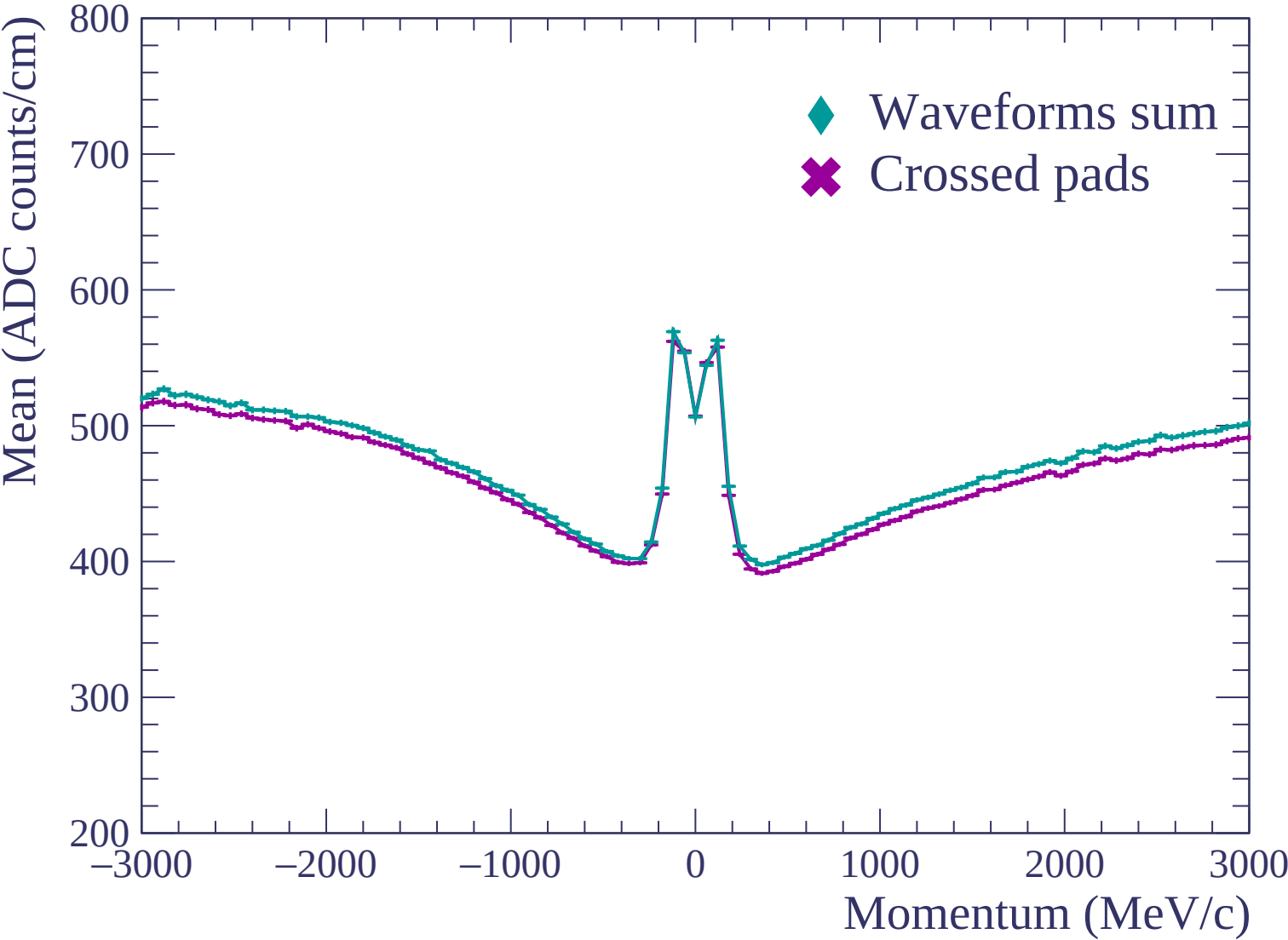


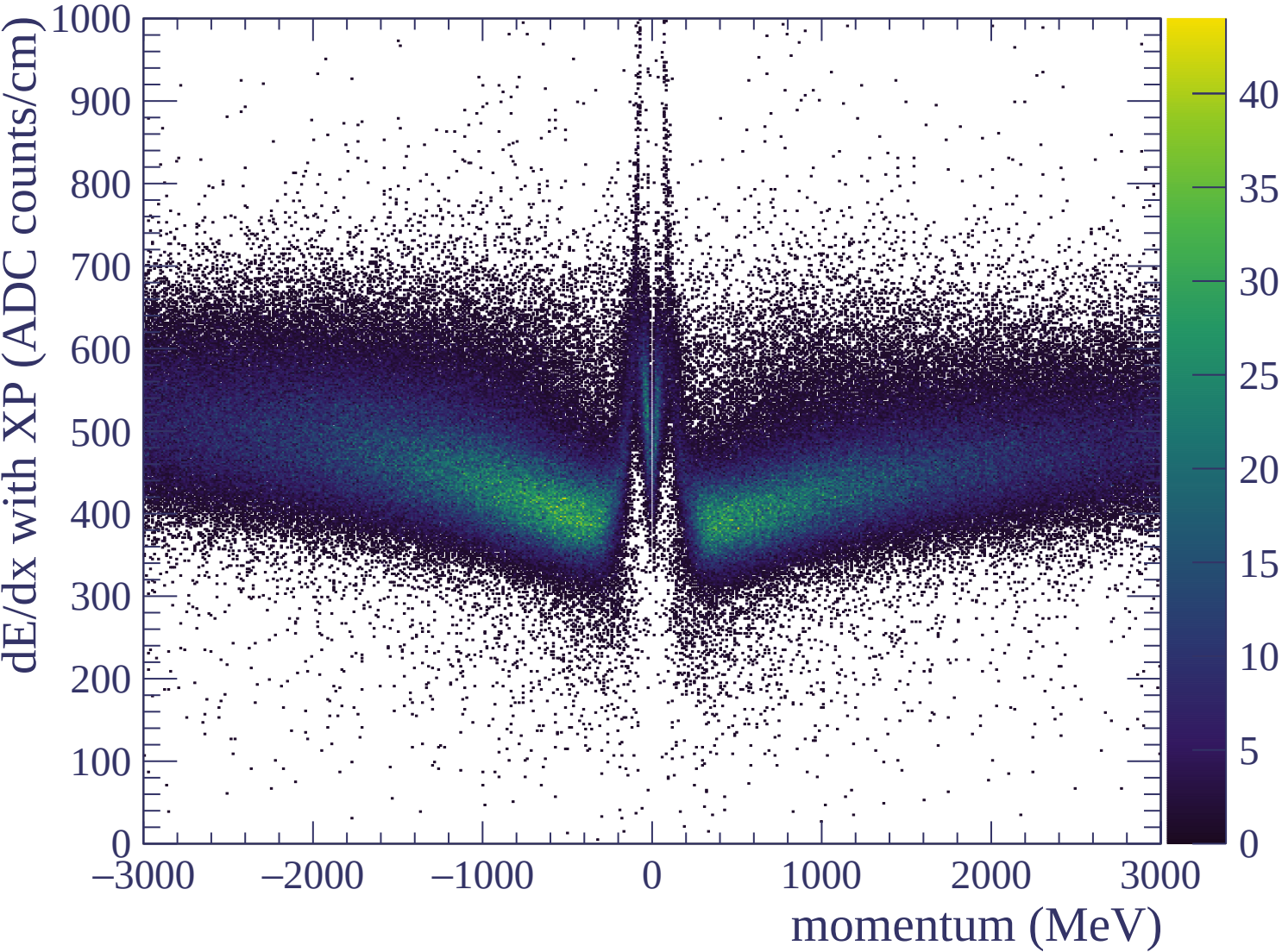


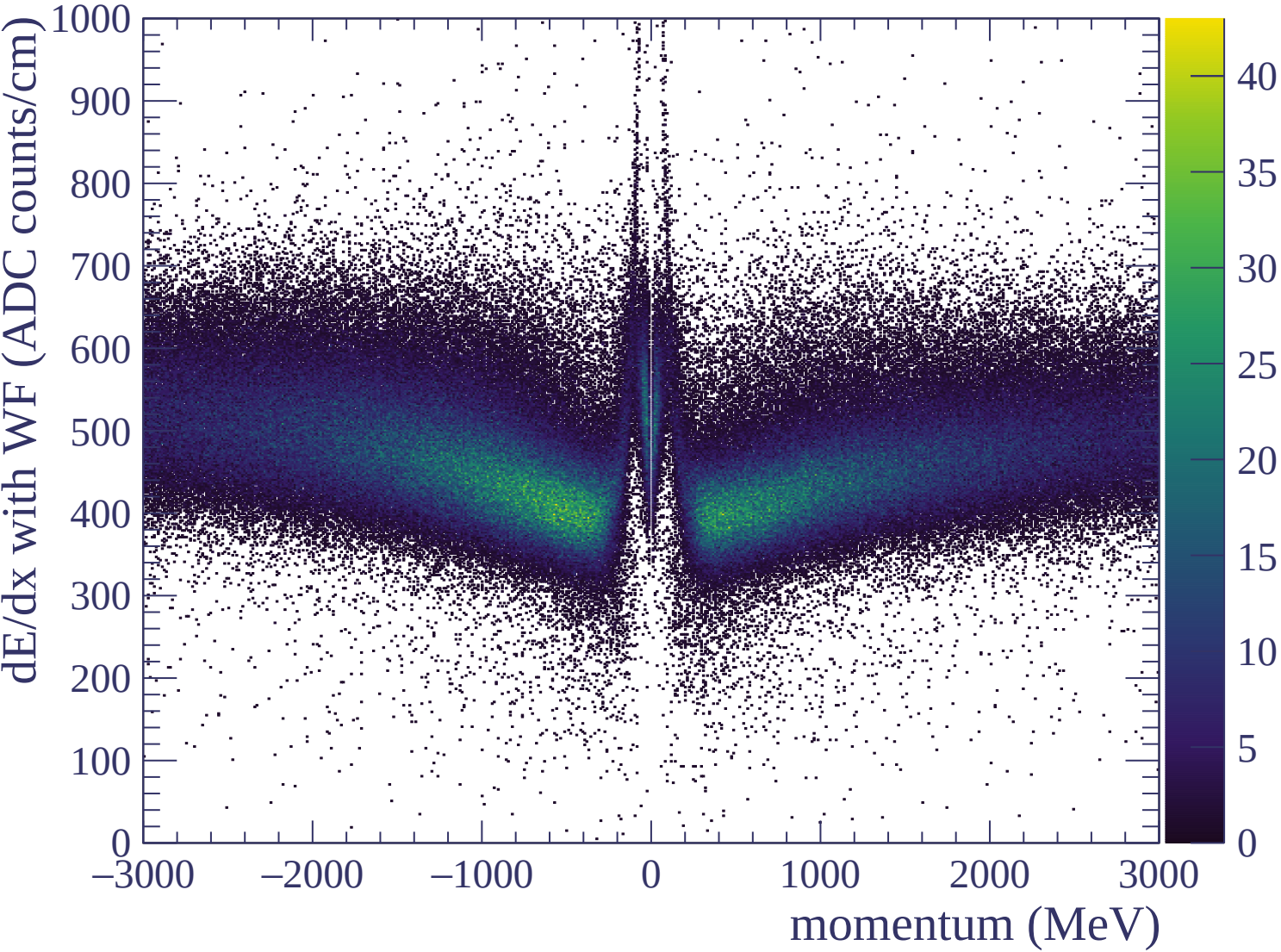


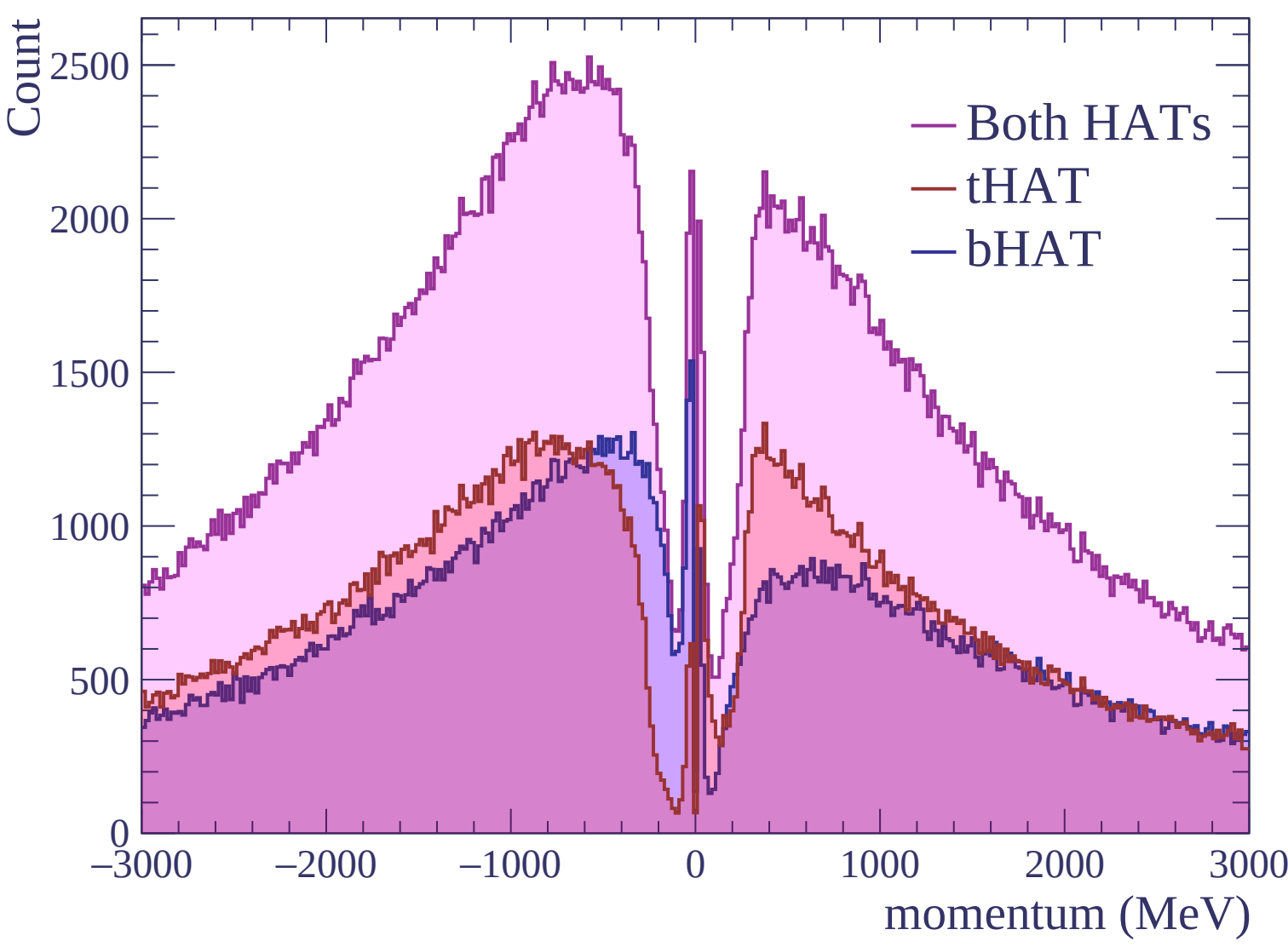


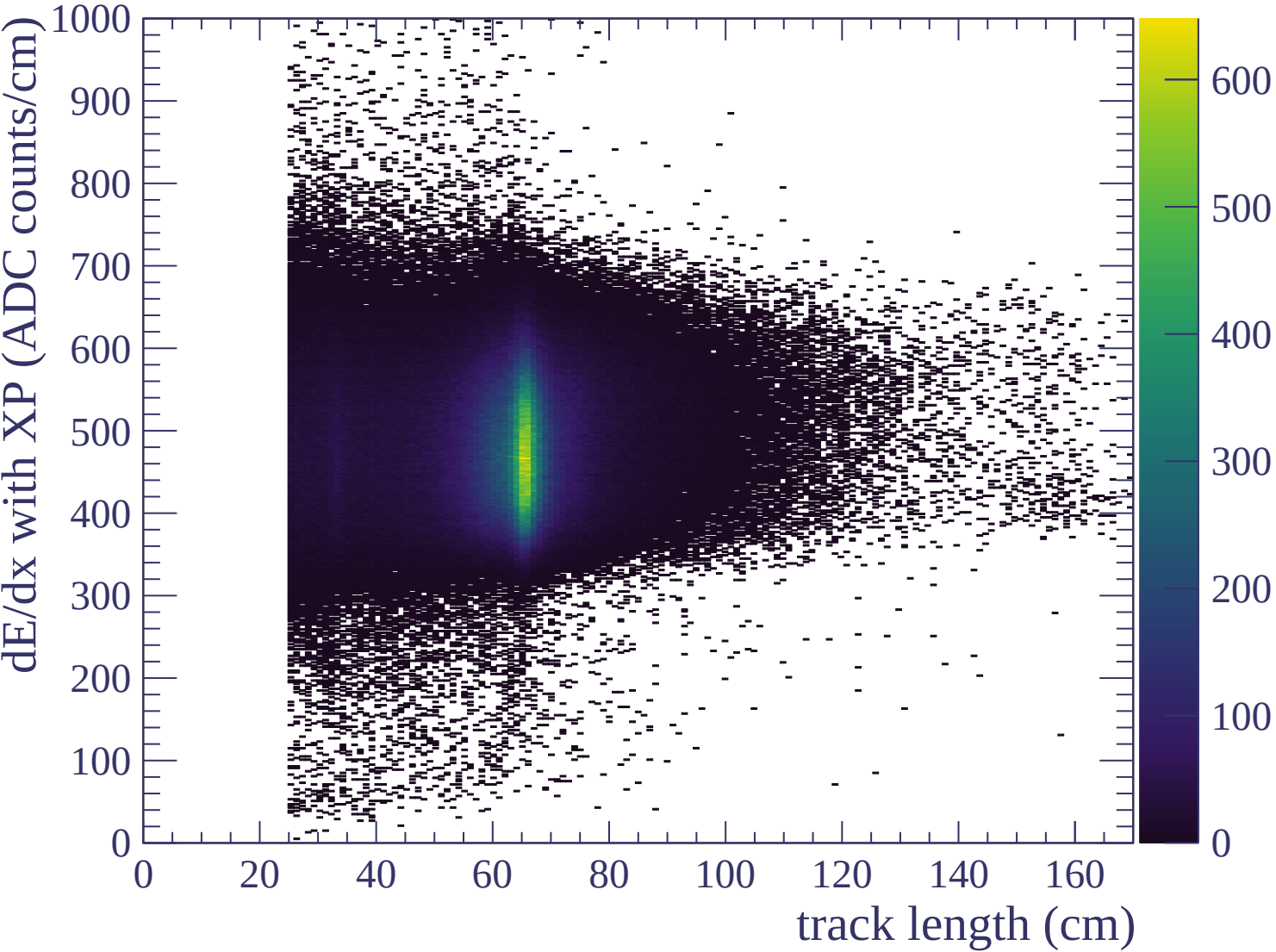


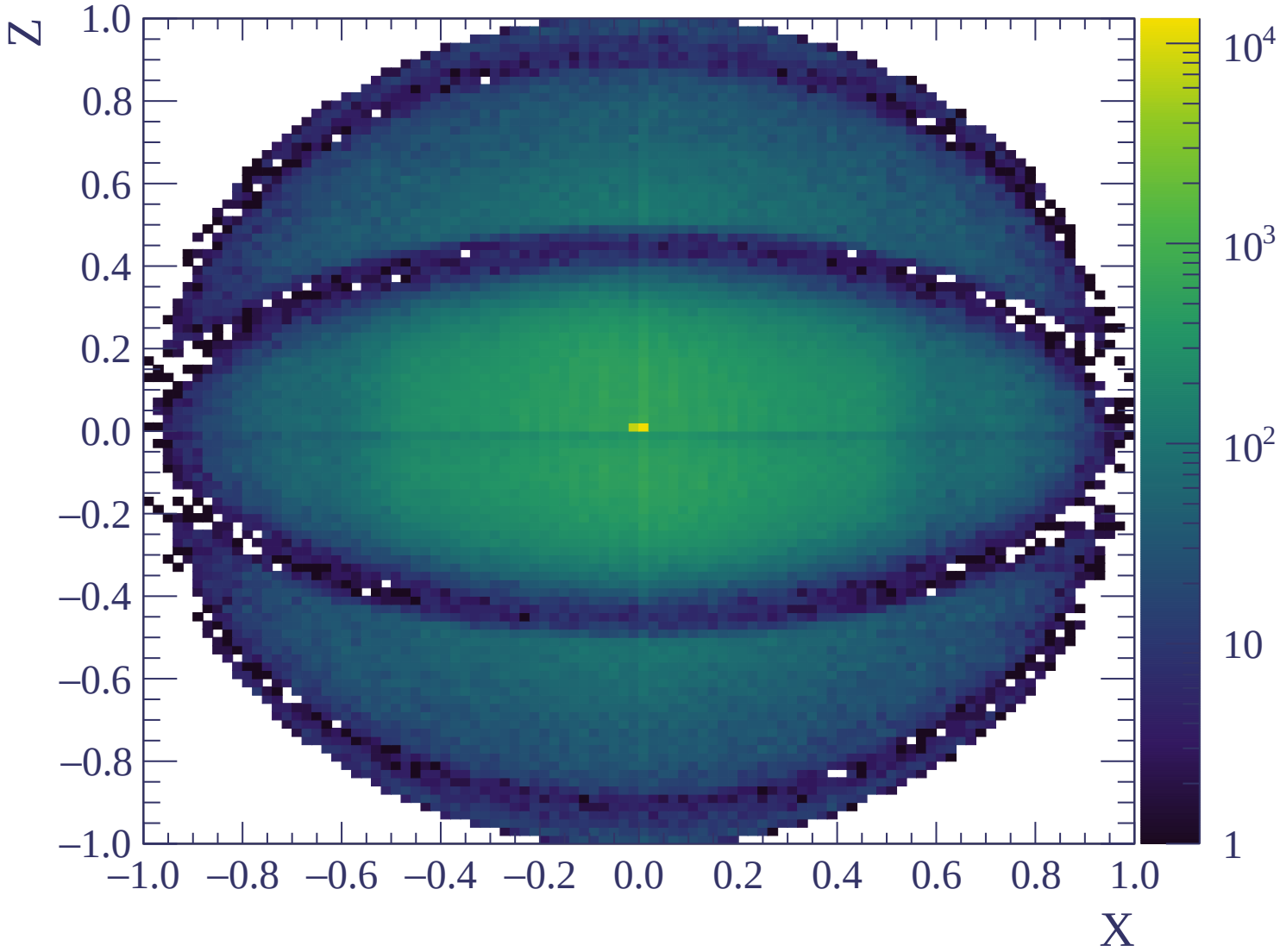


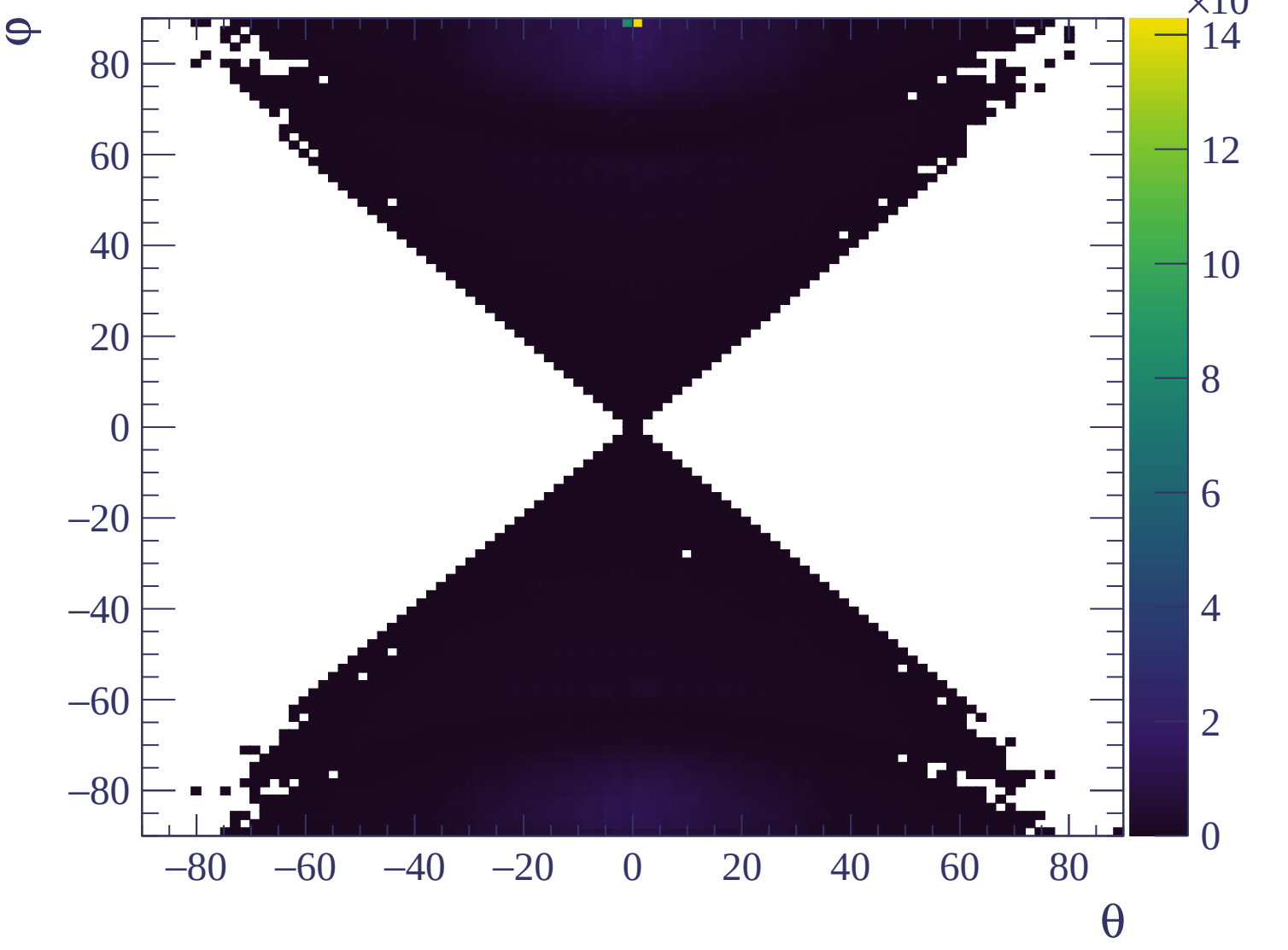


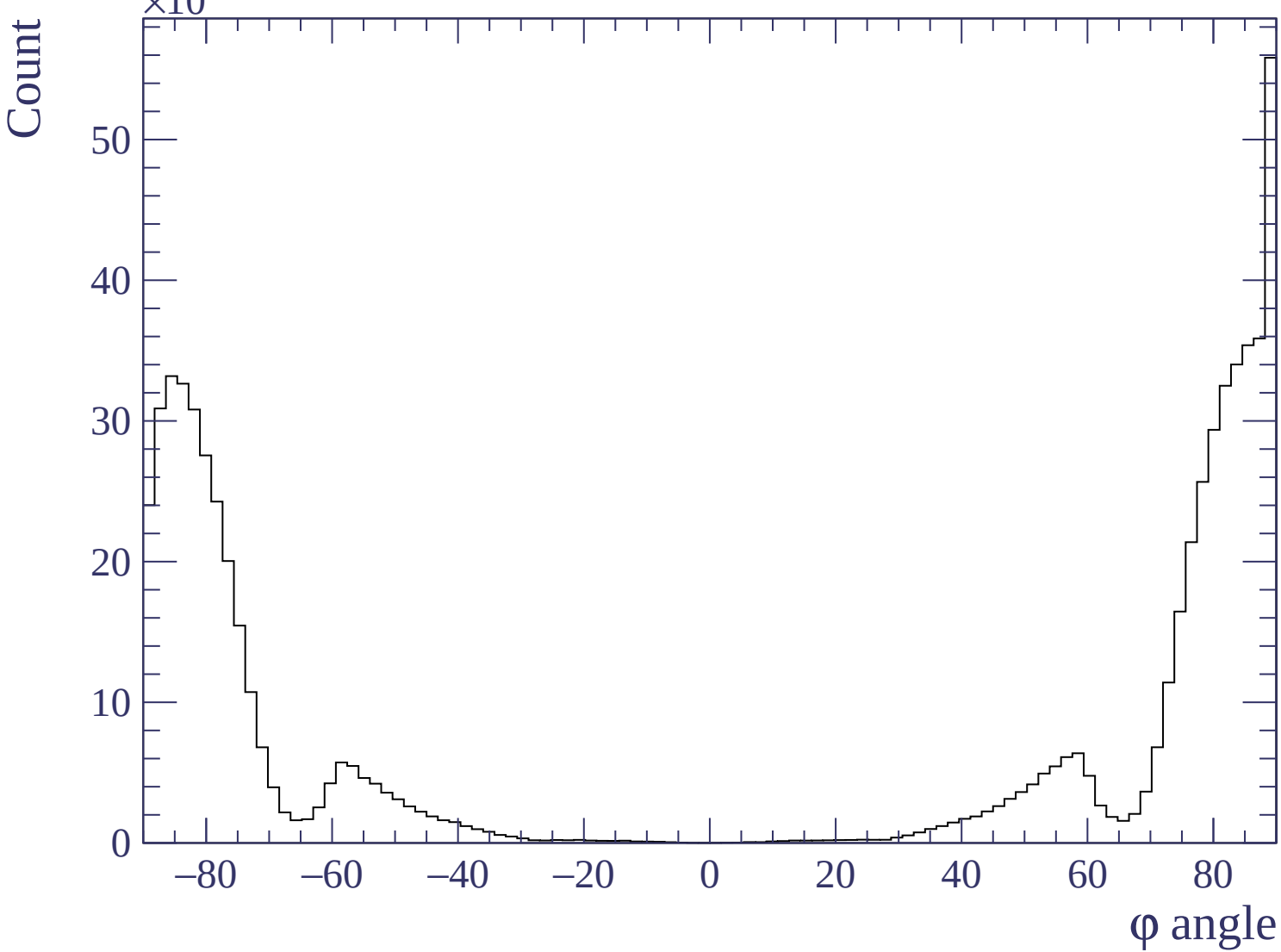


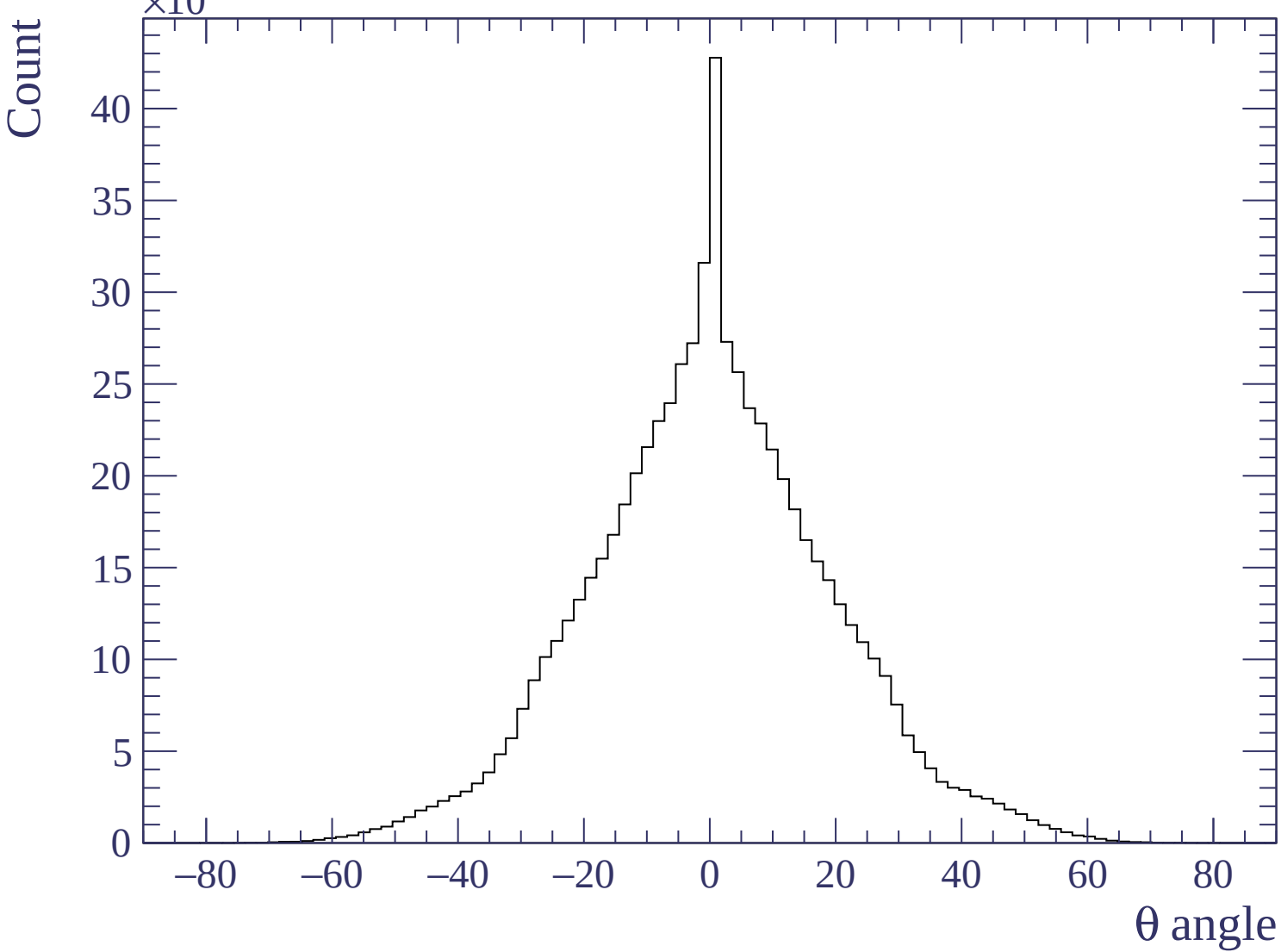


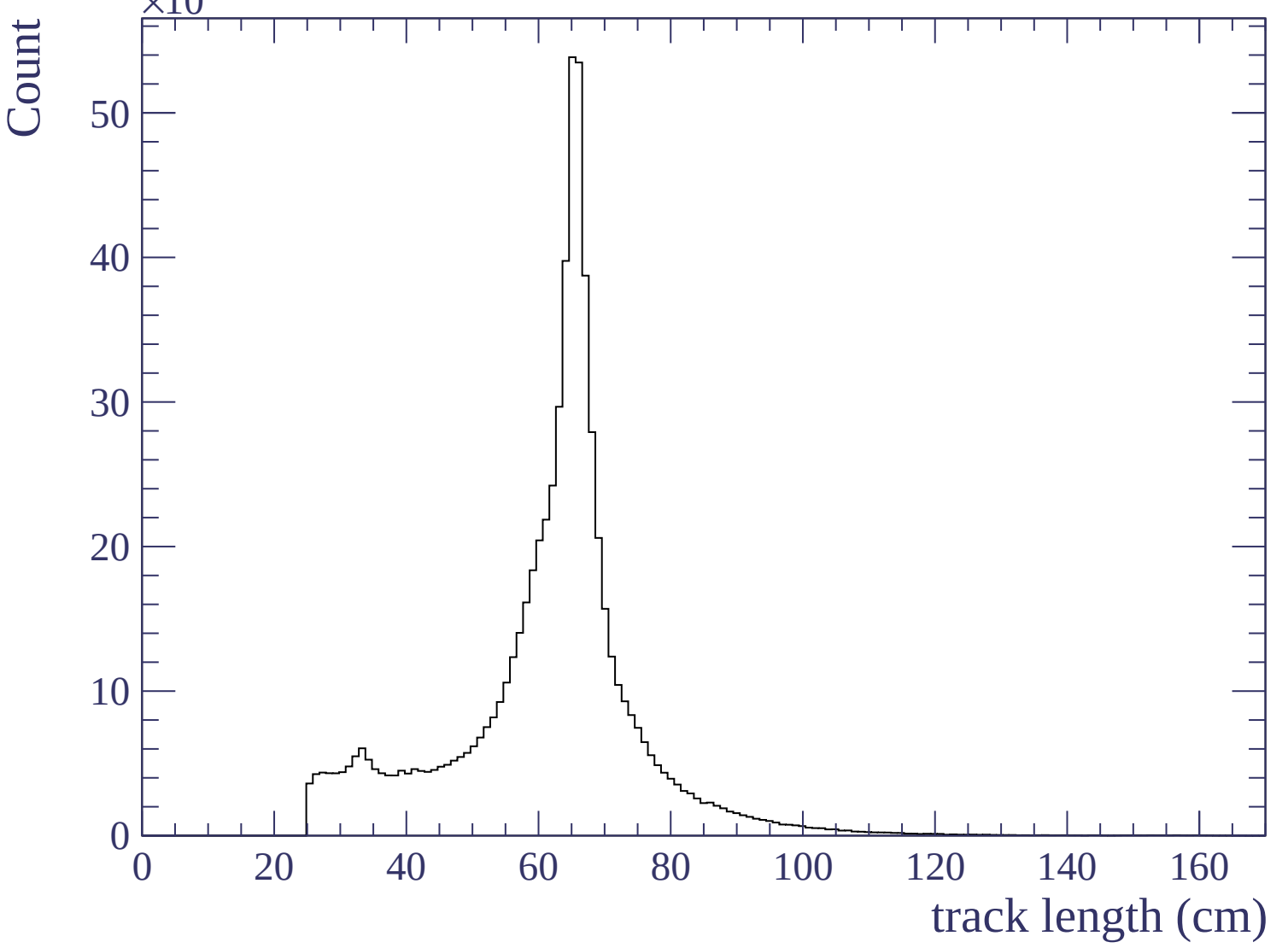


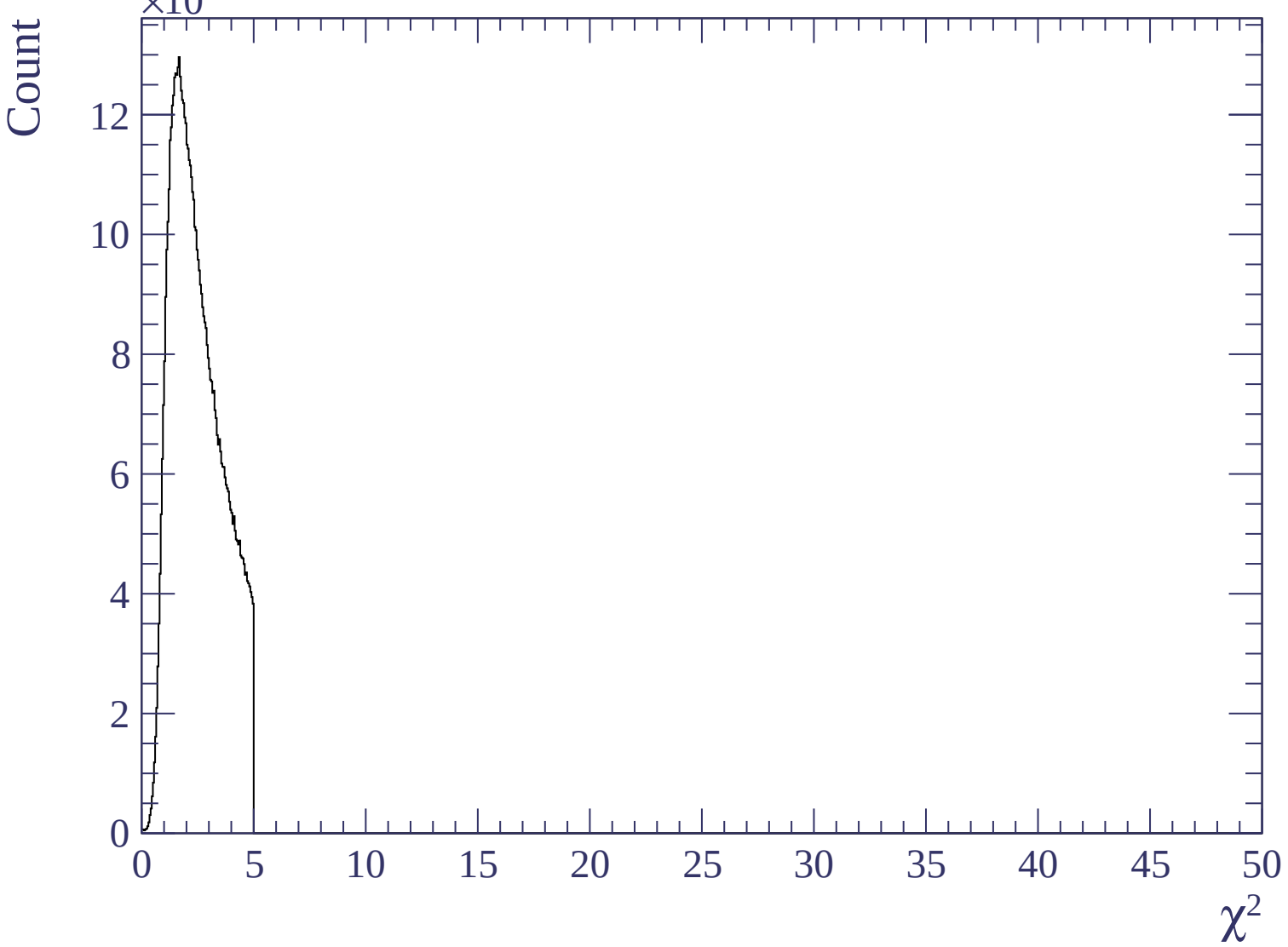


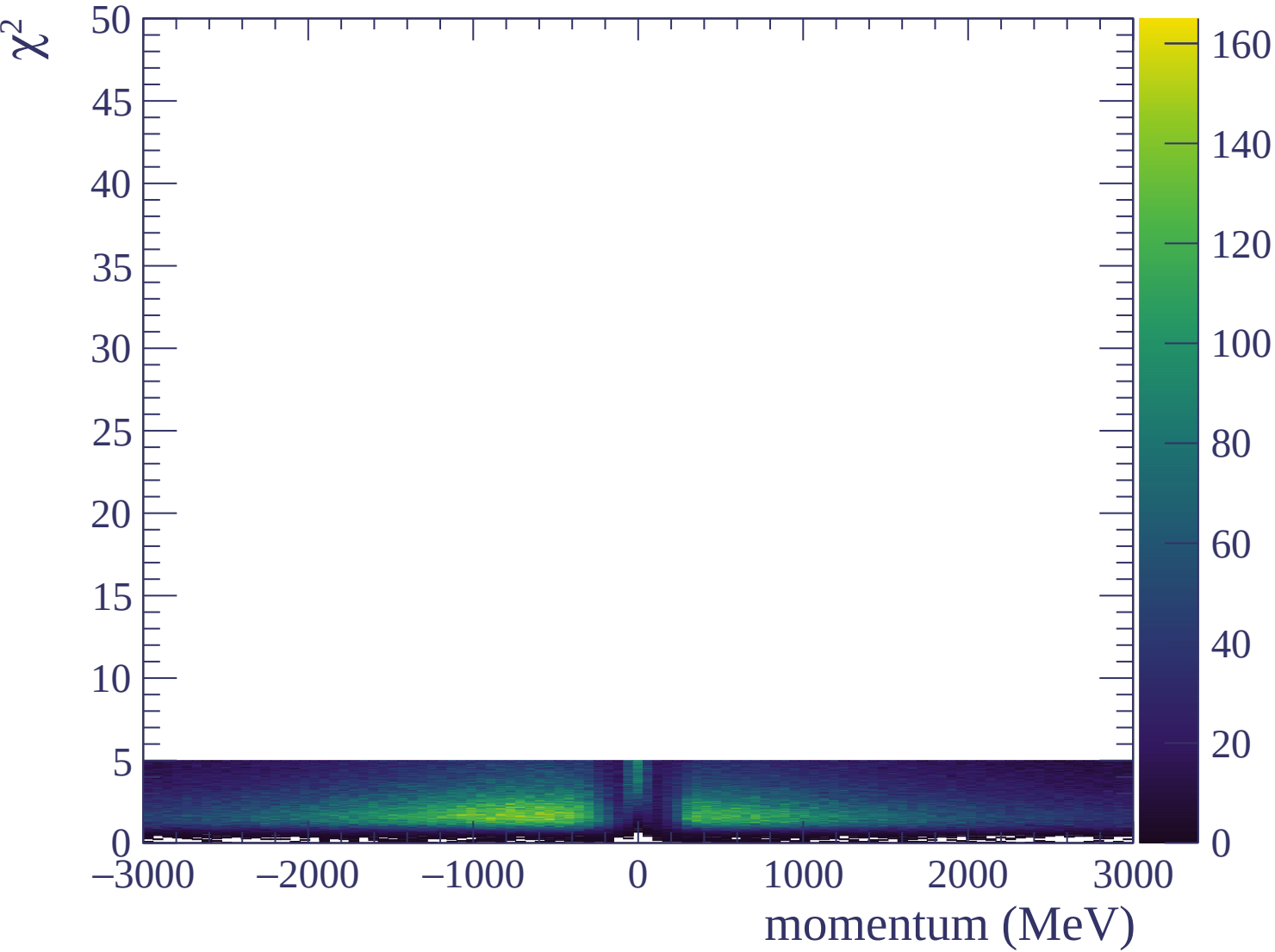


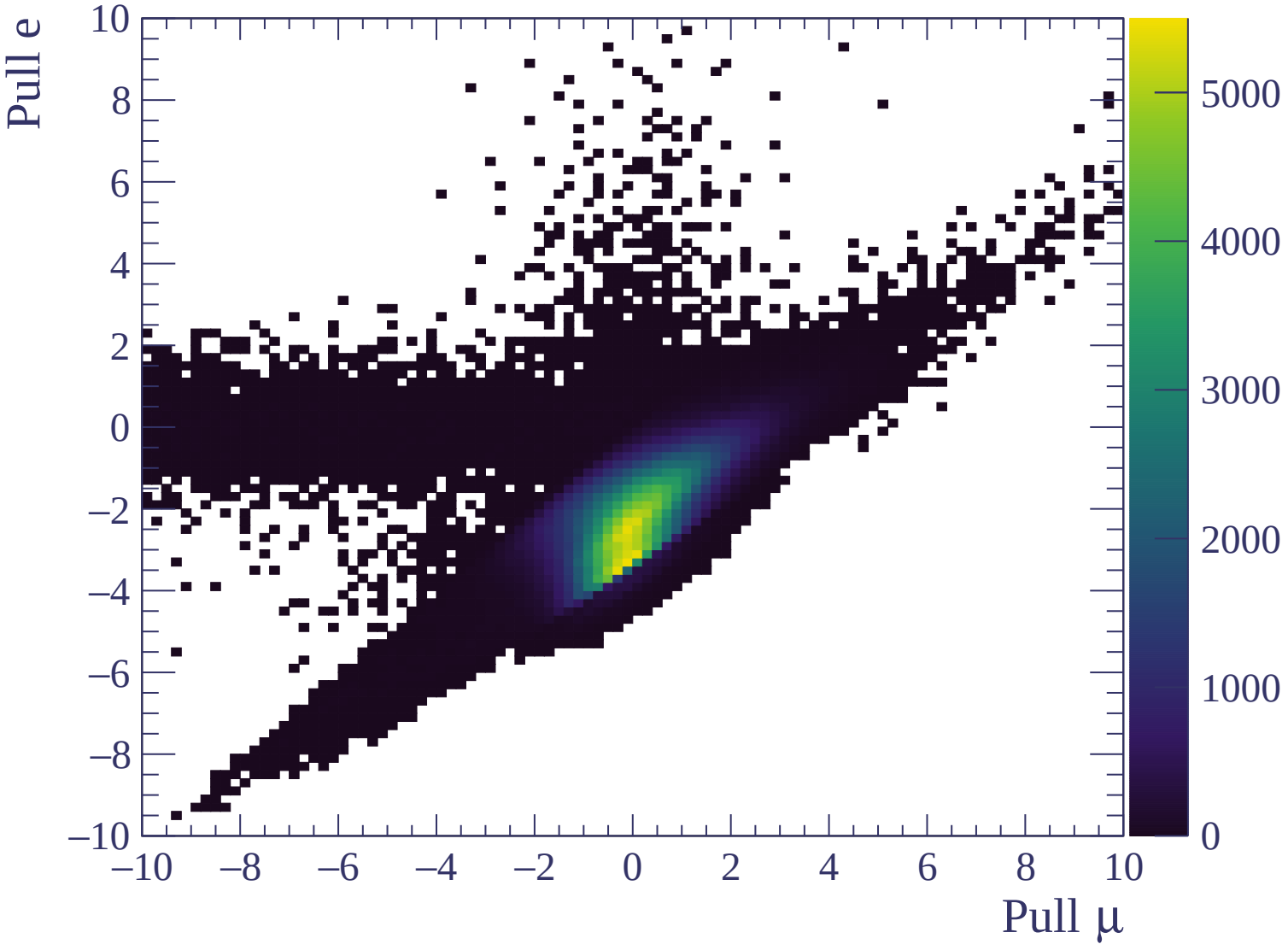


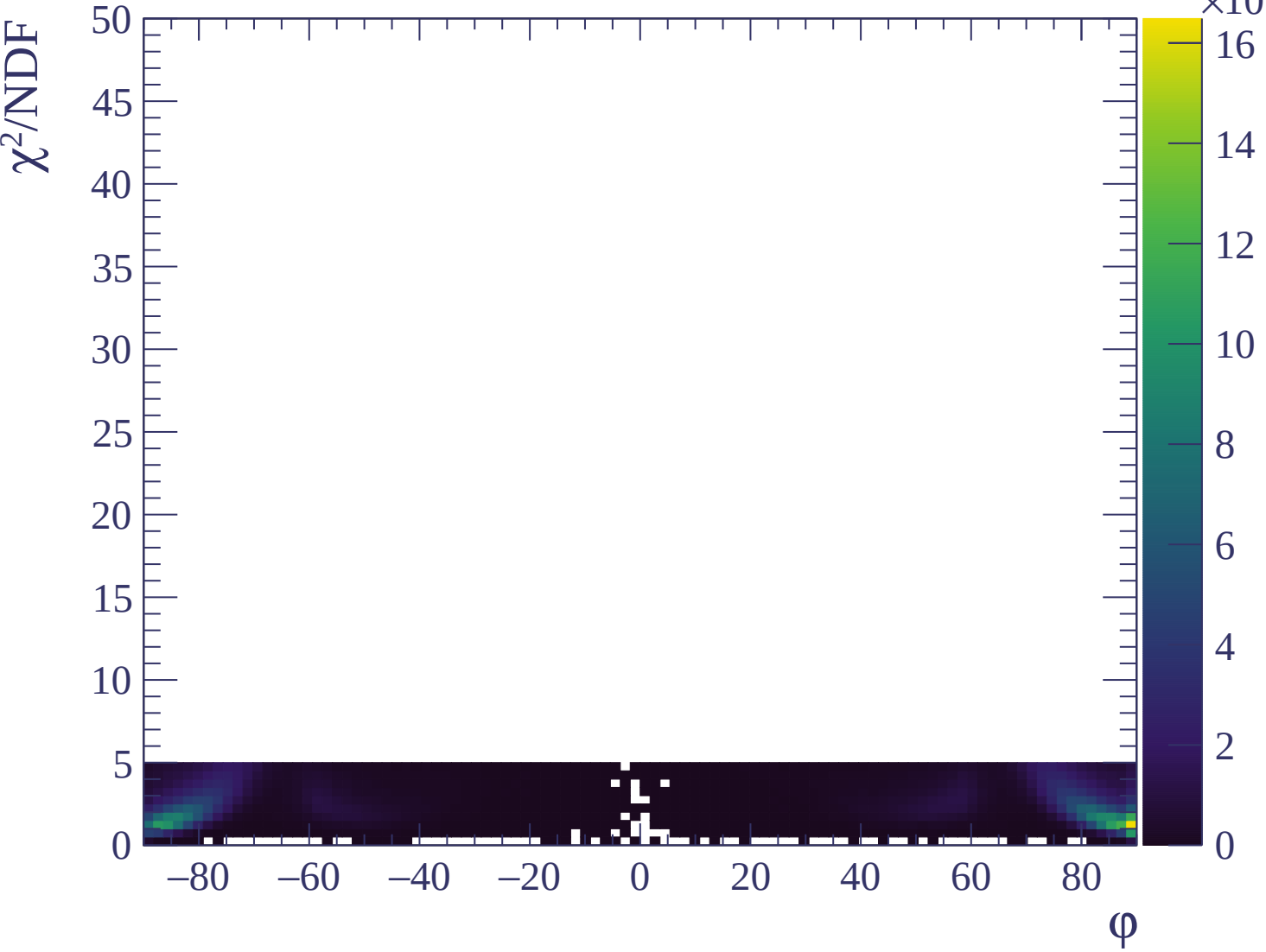


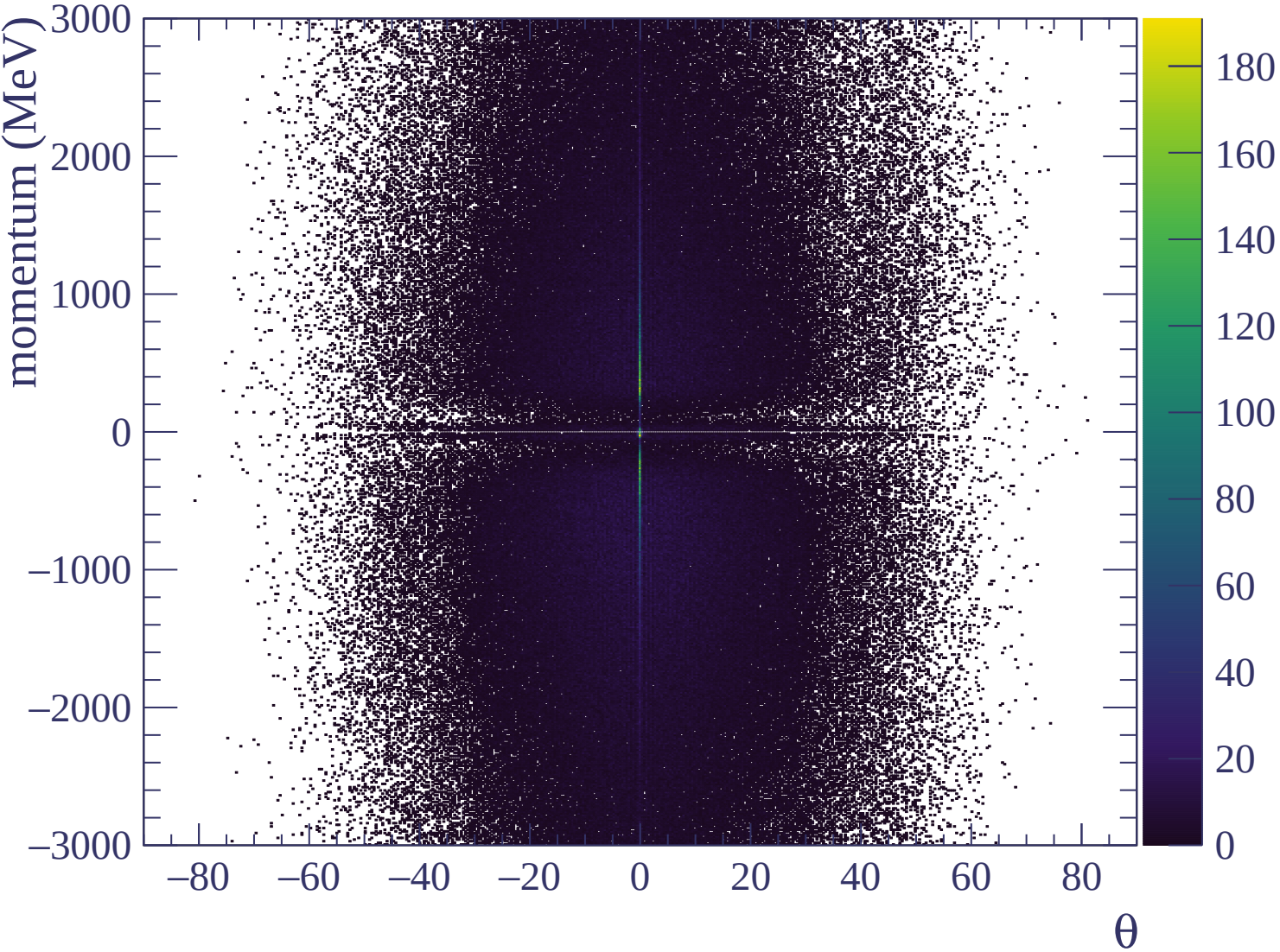


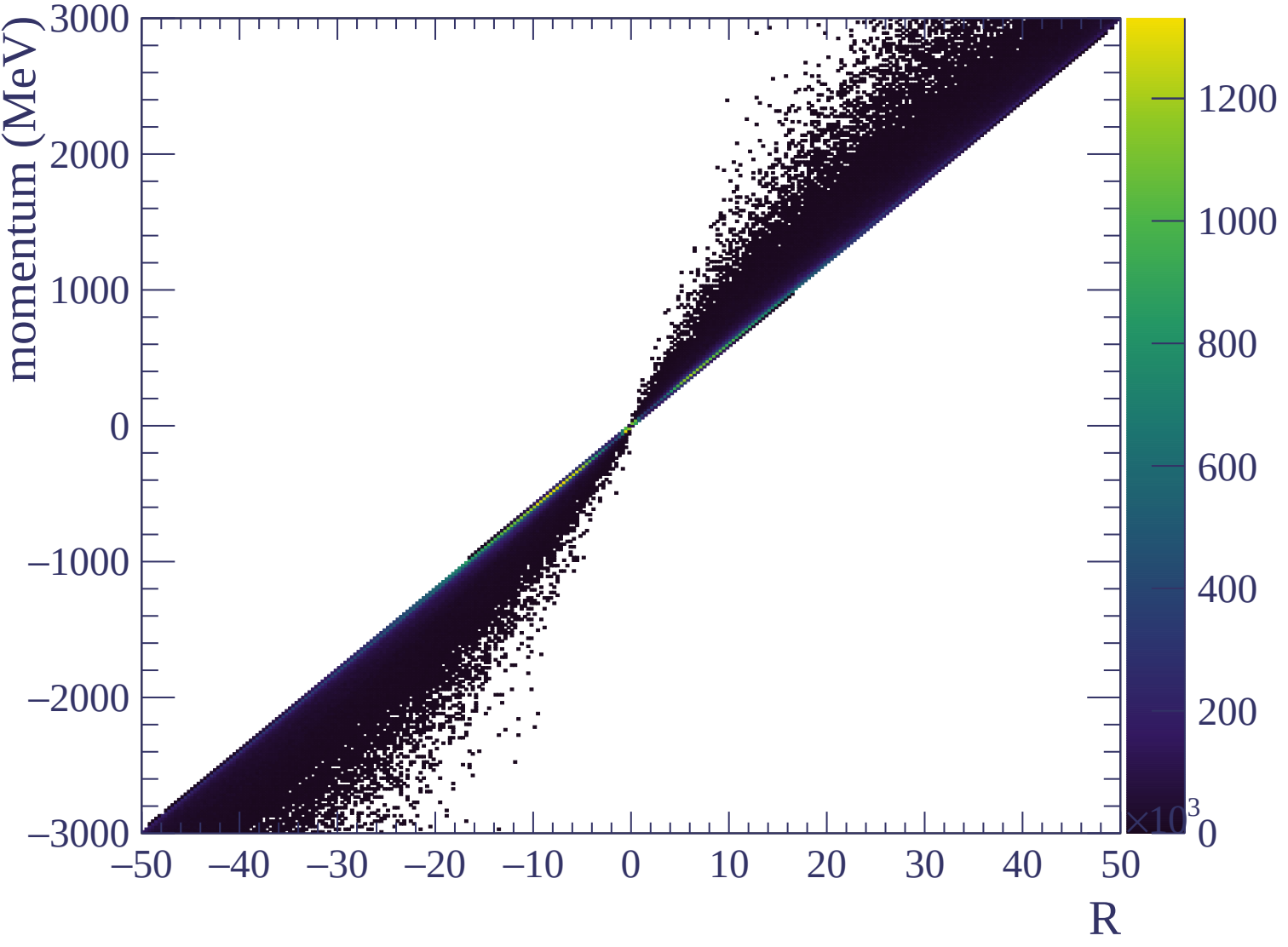


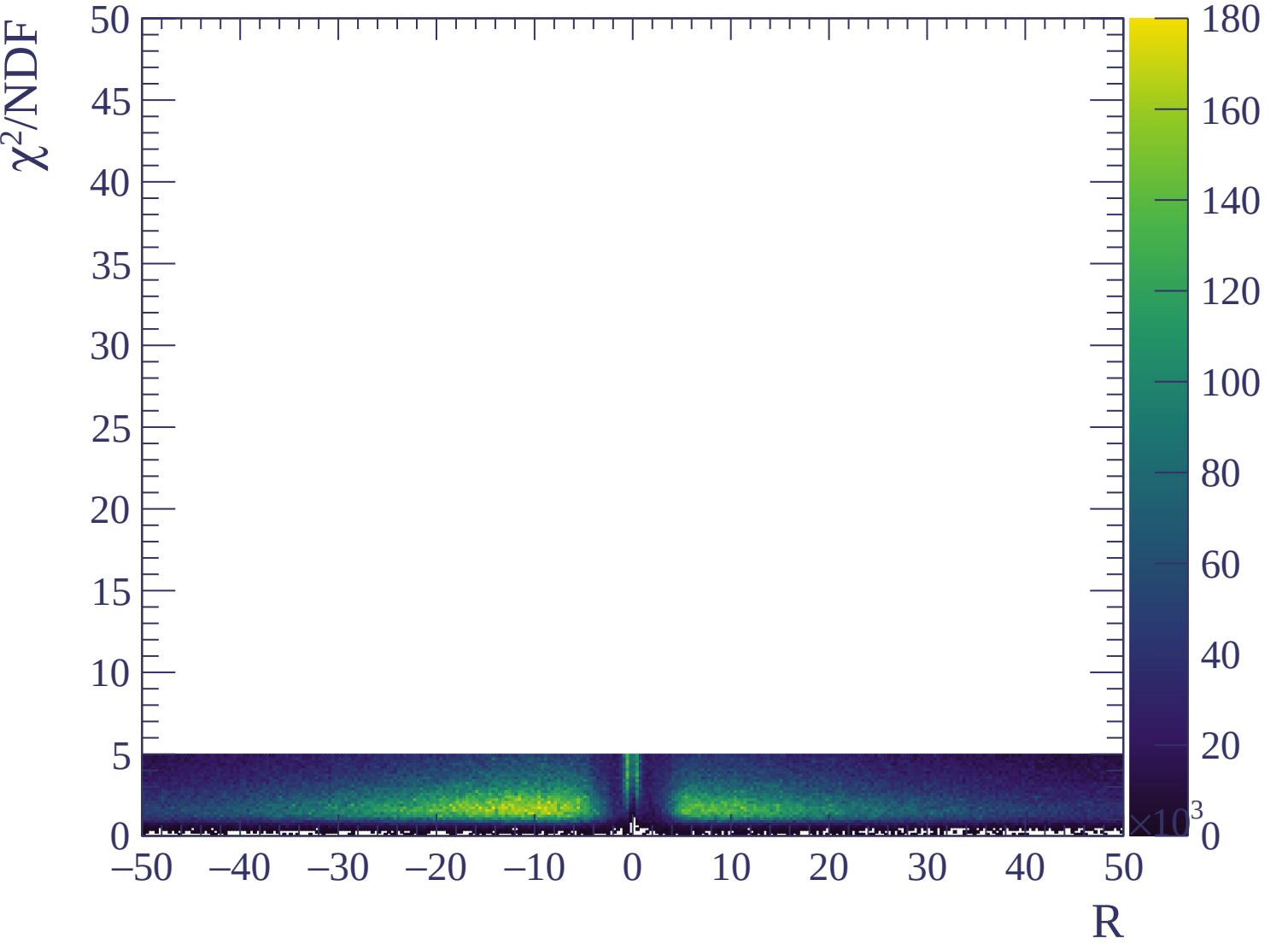


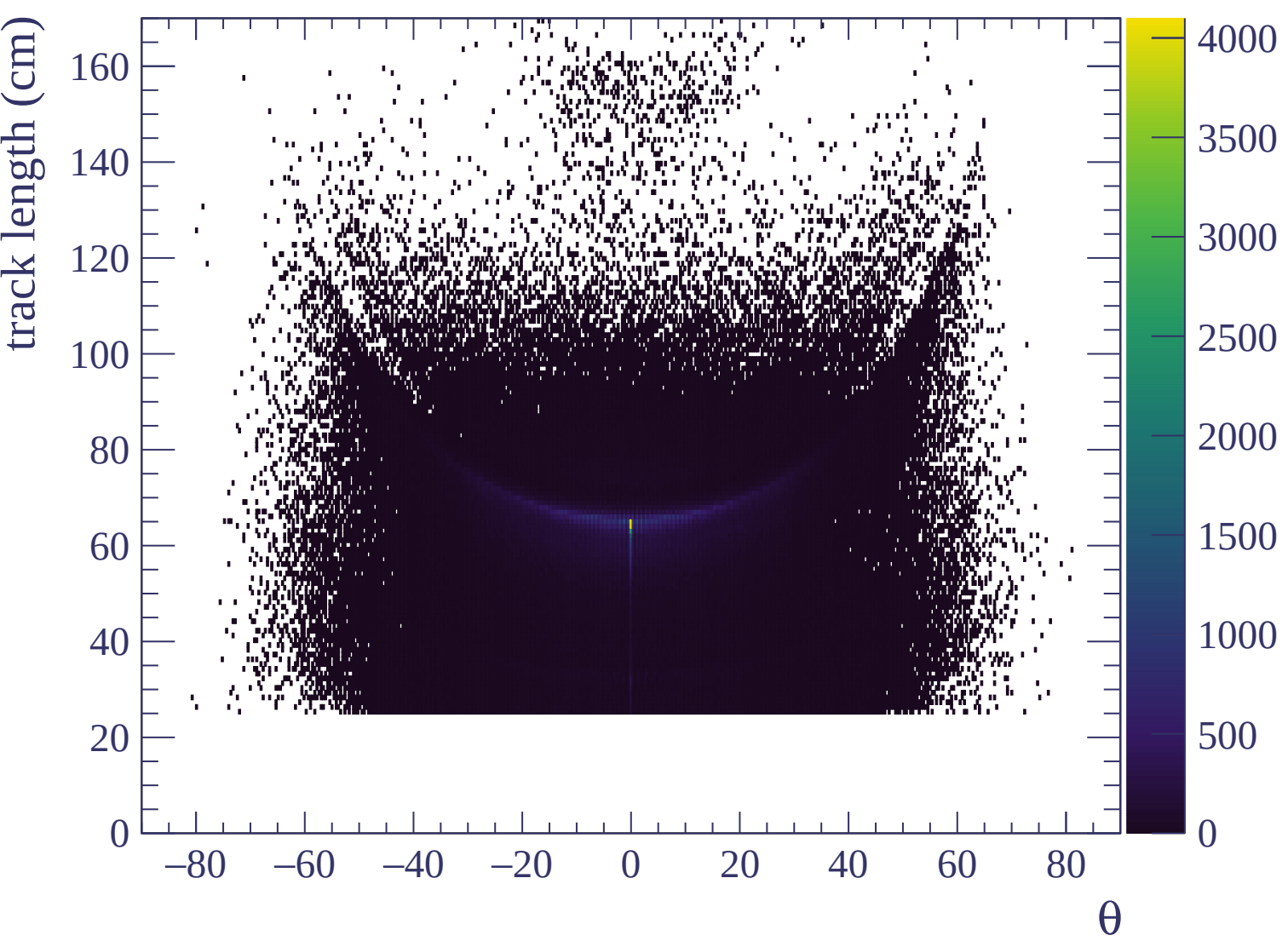


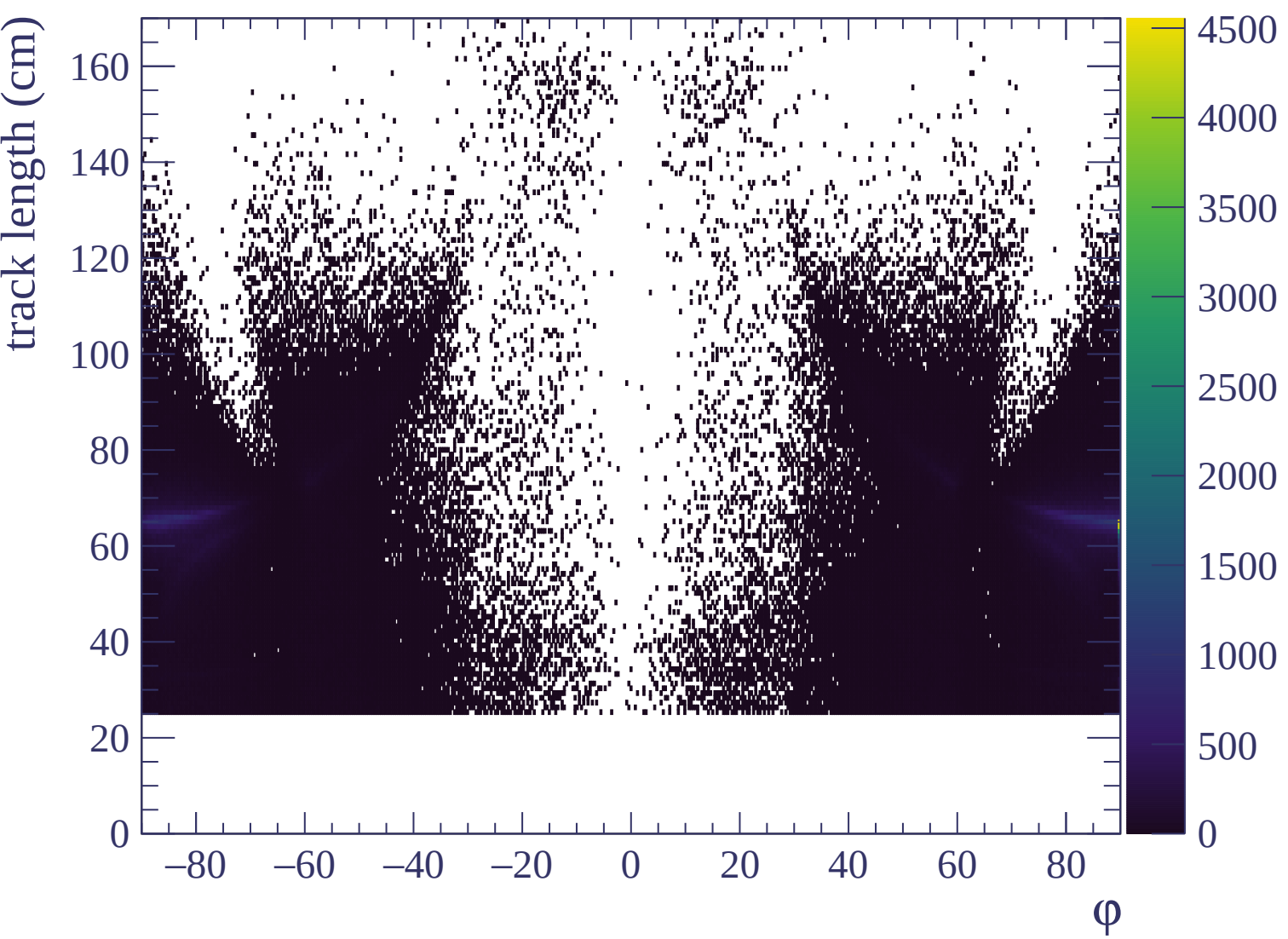






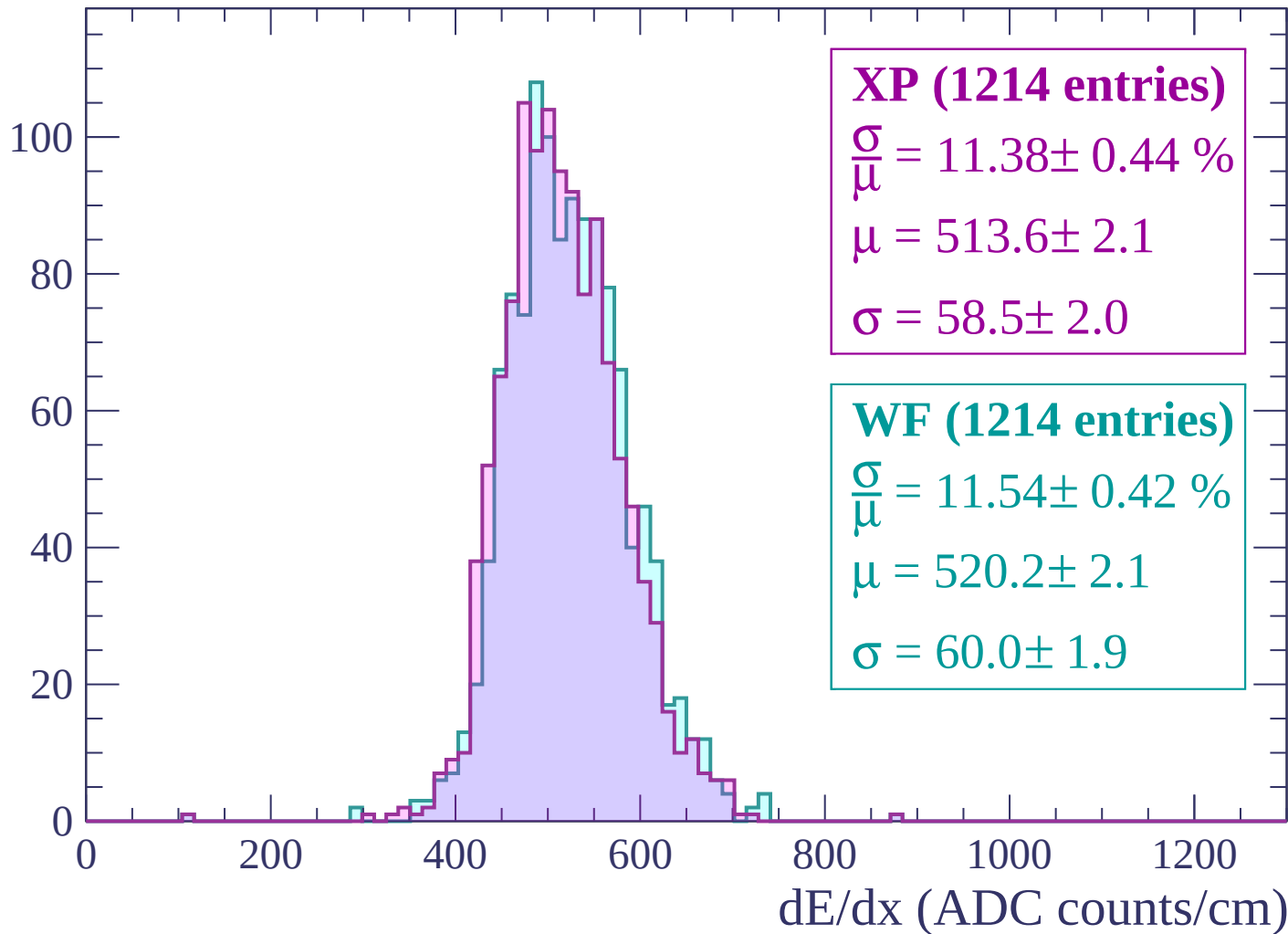






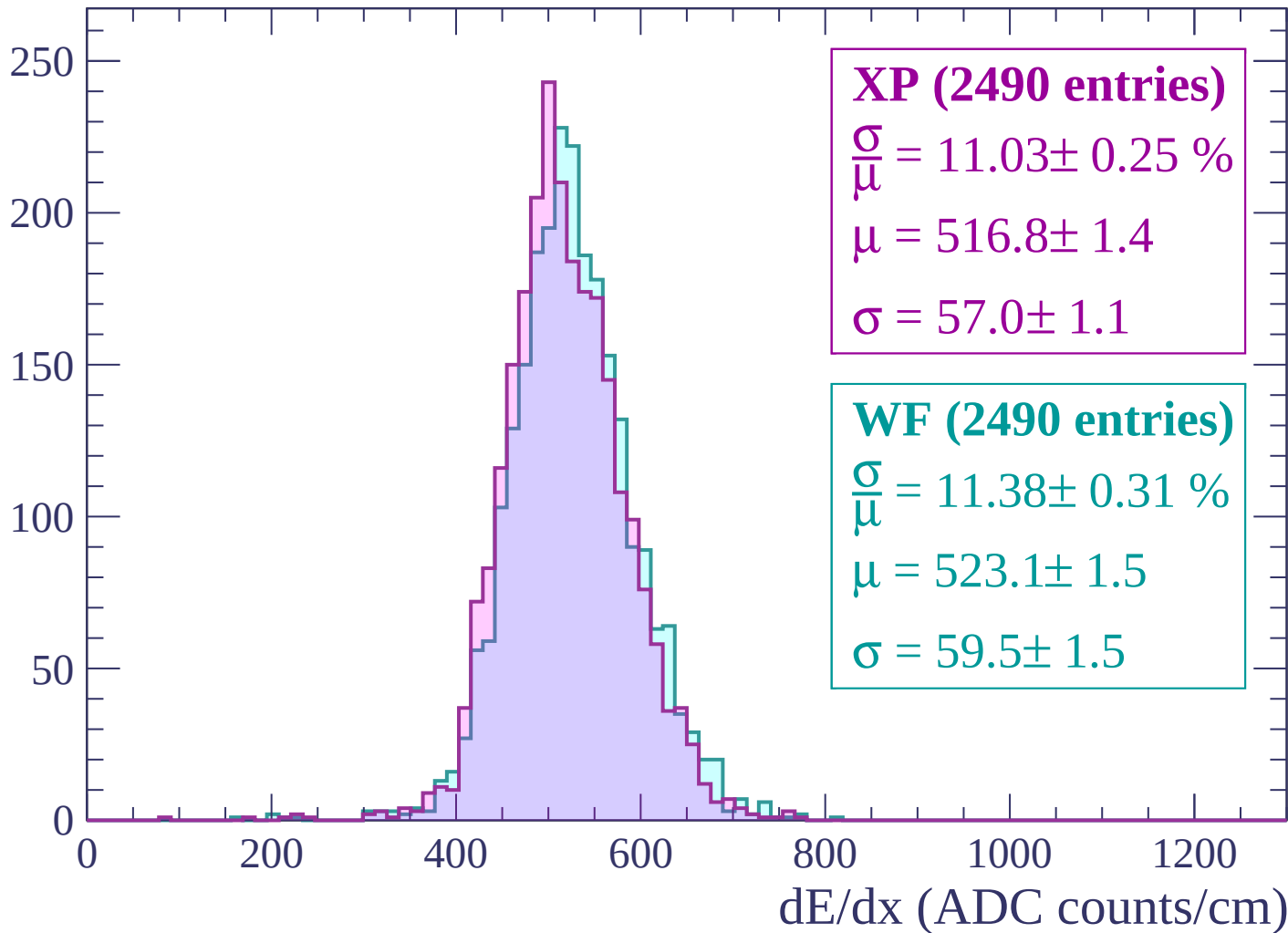
Energy loss | $-3000 < p < -2940$

Count



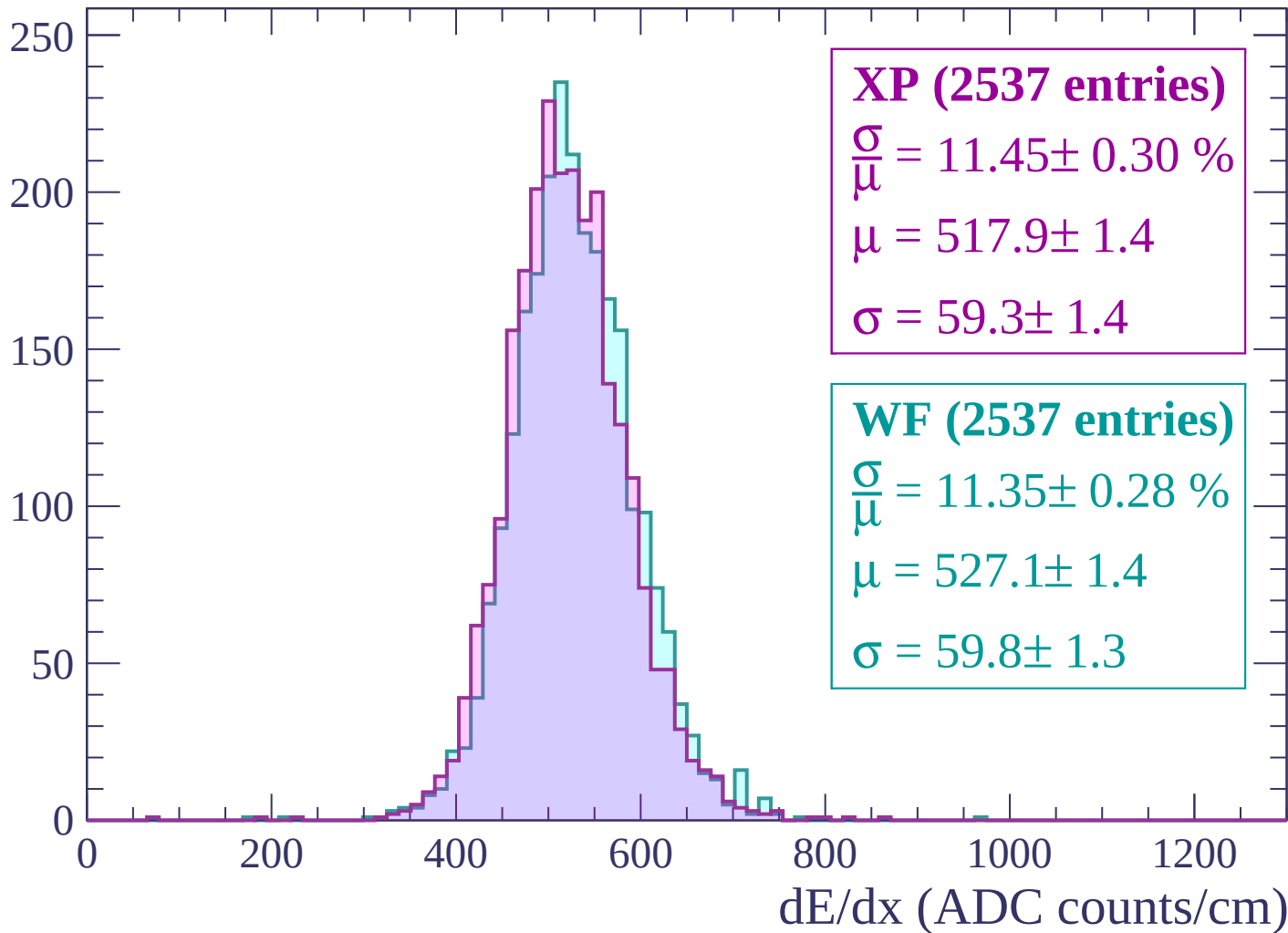
Energy loss | $-2940 < p < -2880$

Count



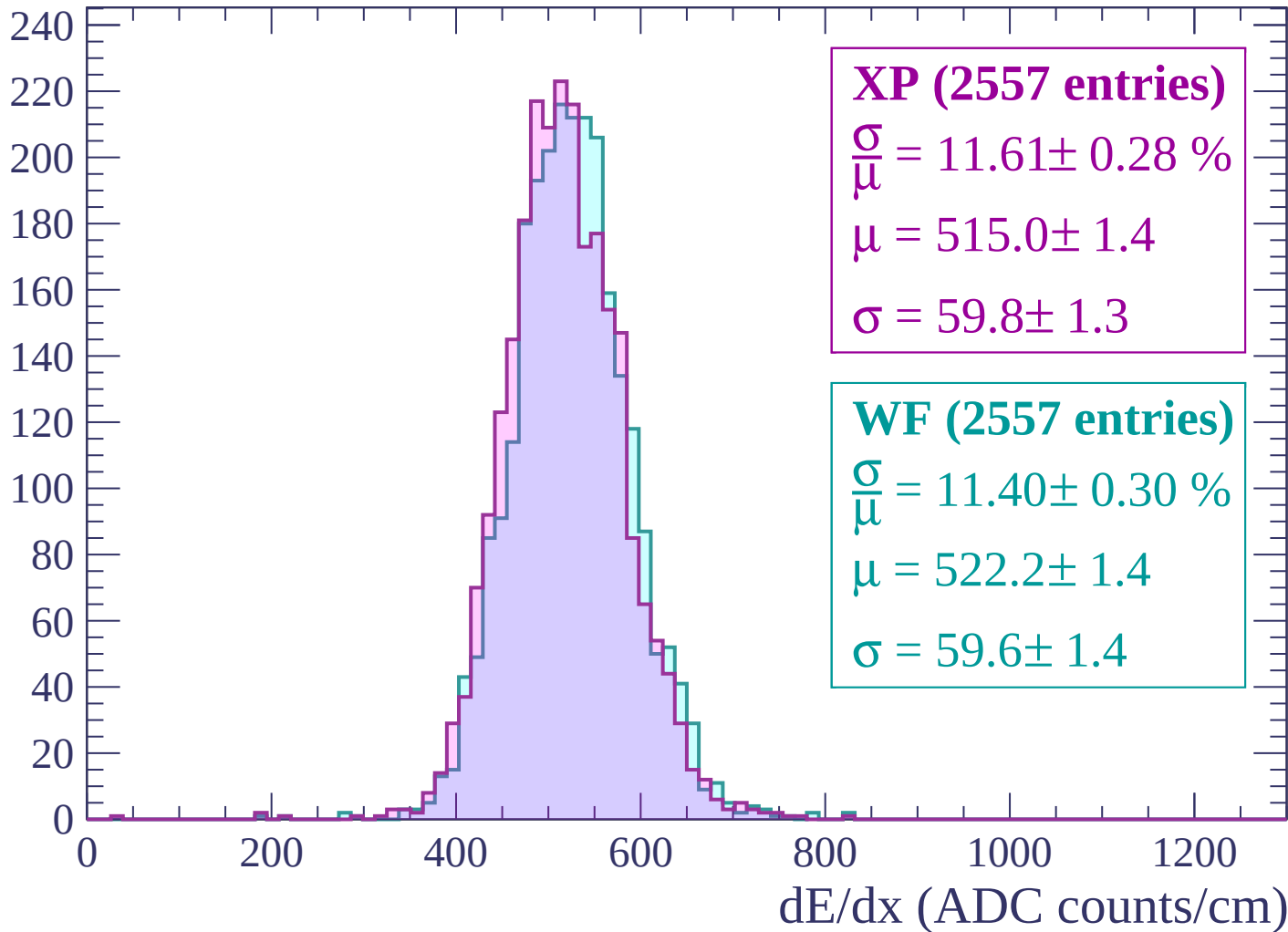
Energy loss | $-2880 < p < -2820$

Count



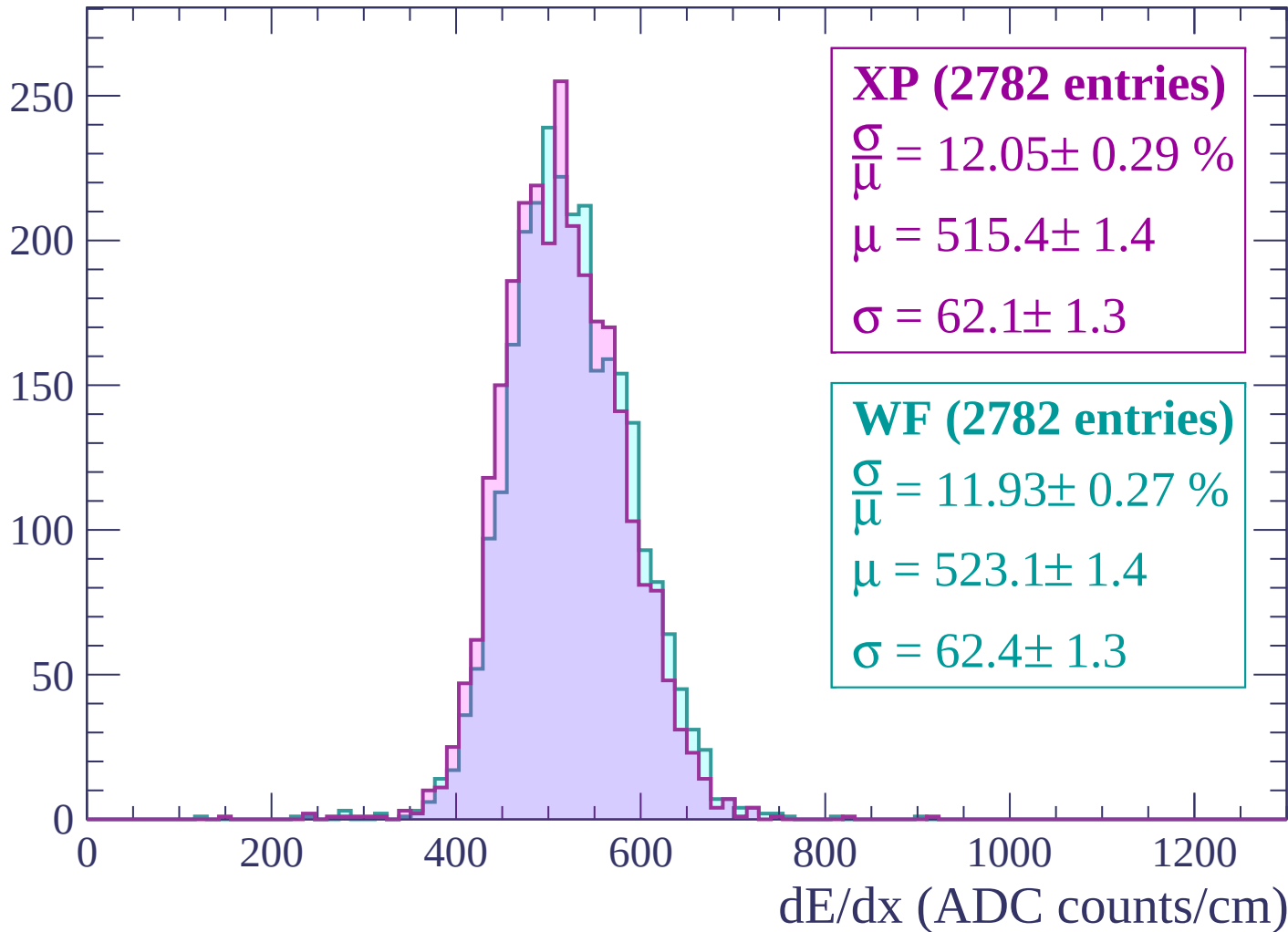
Energy loss | $-2820 < p < -2760$

Count



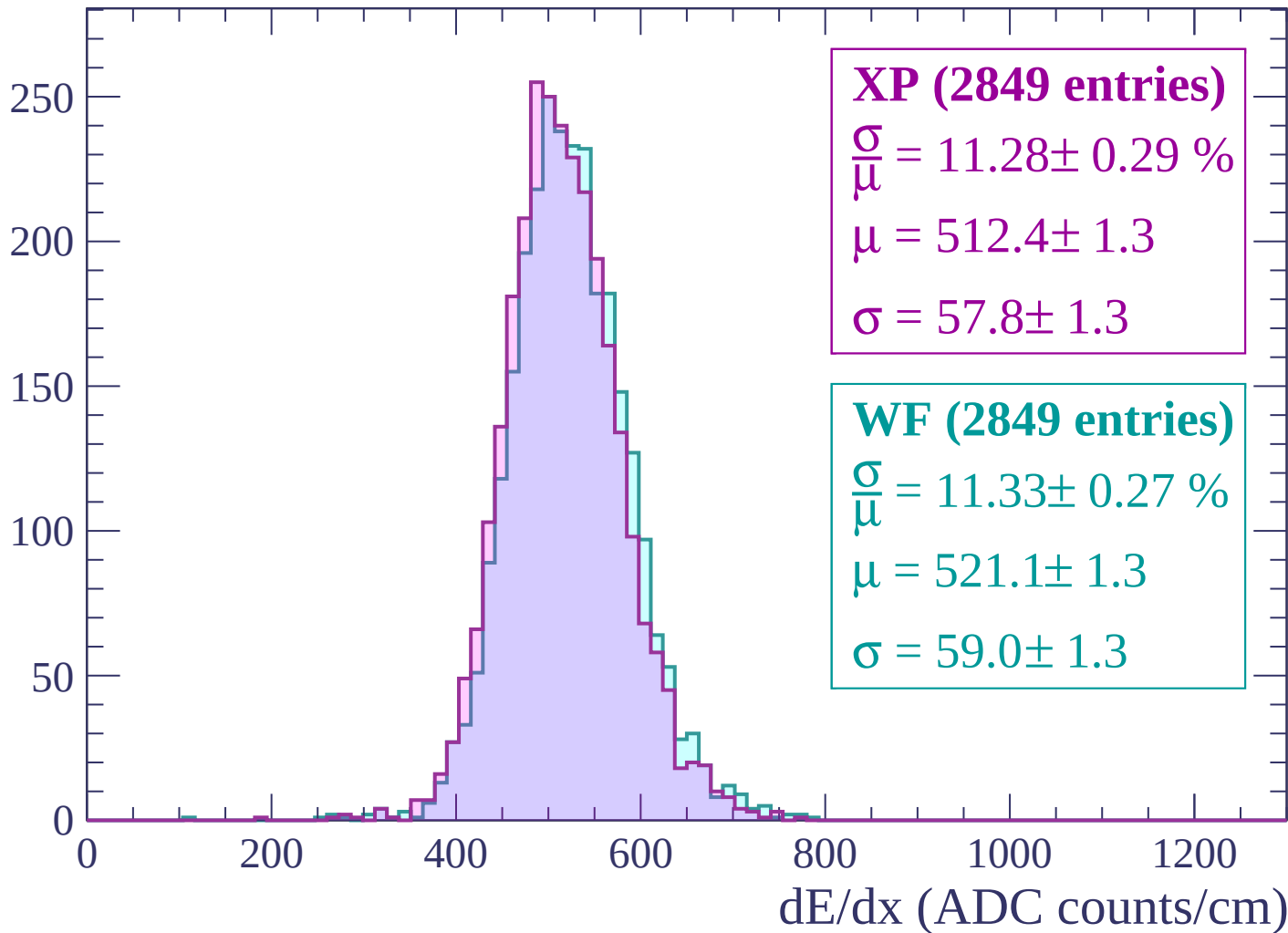
Energy loss | -2760 < p < -2700

Count



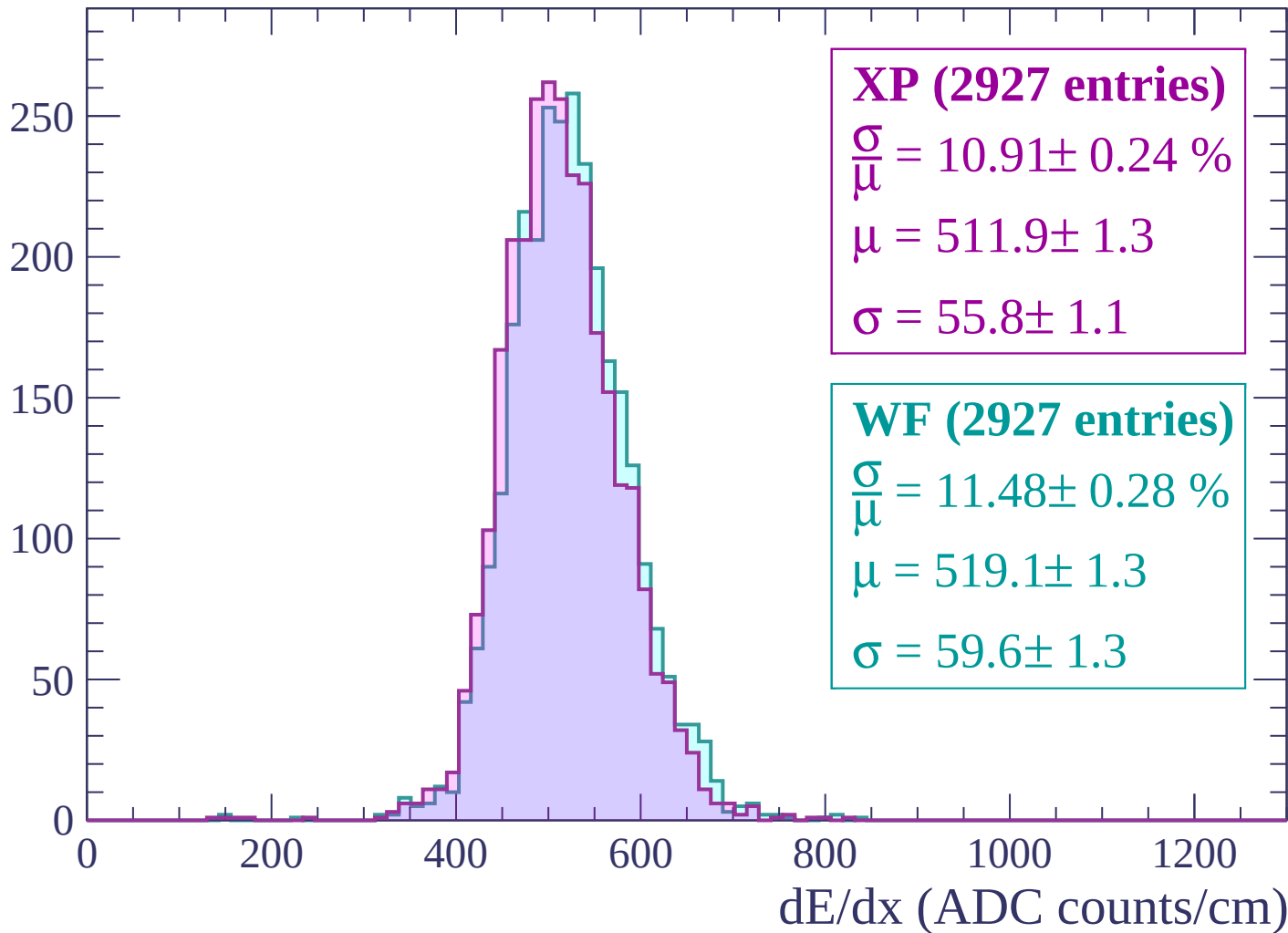
Energy loss | -2700 < p < -2640

Count



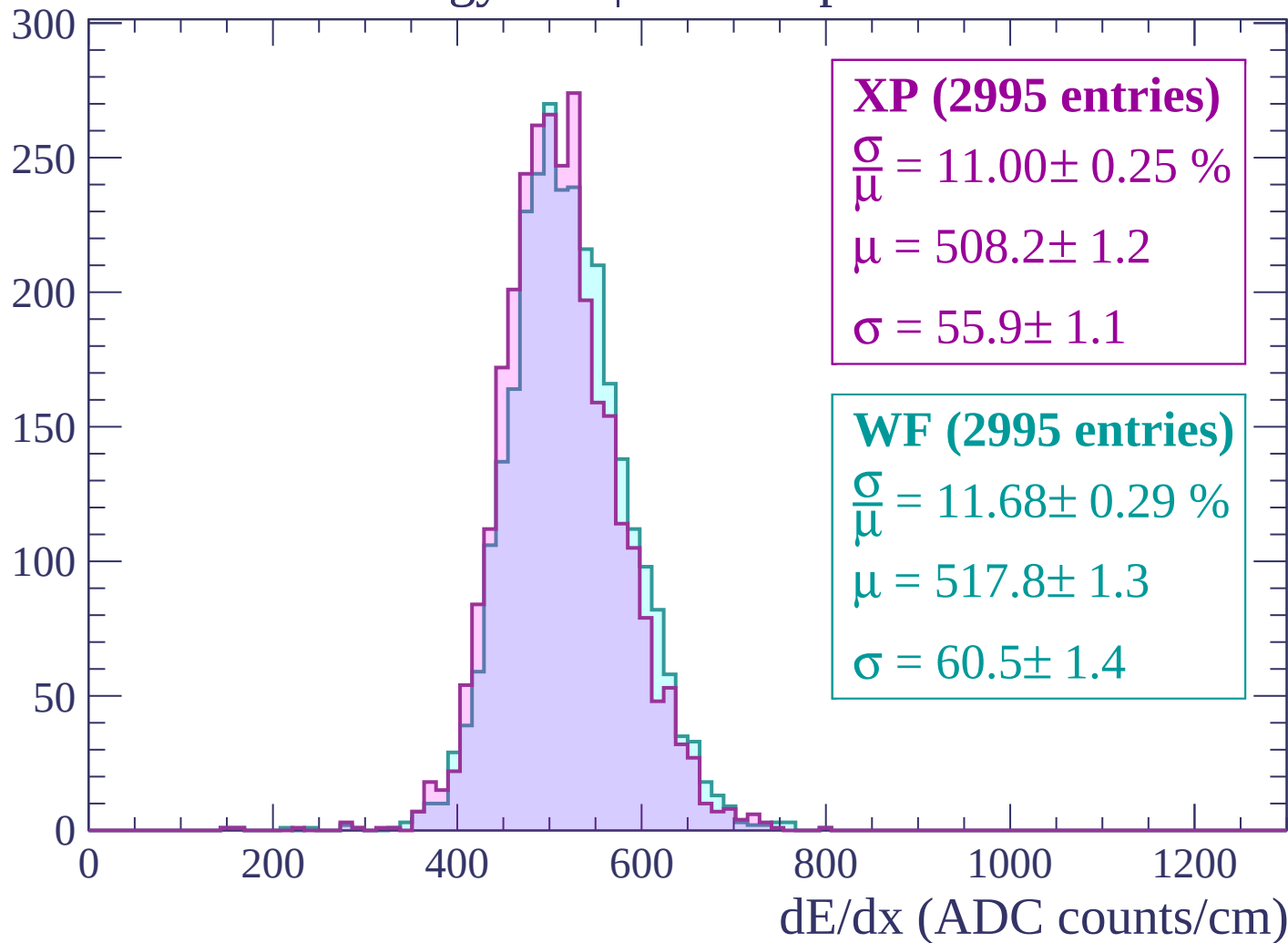
Energy loss | $-2640 < p < -2580$

Count



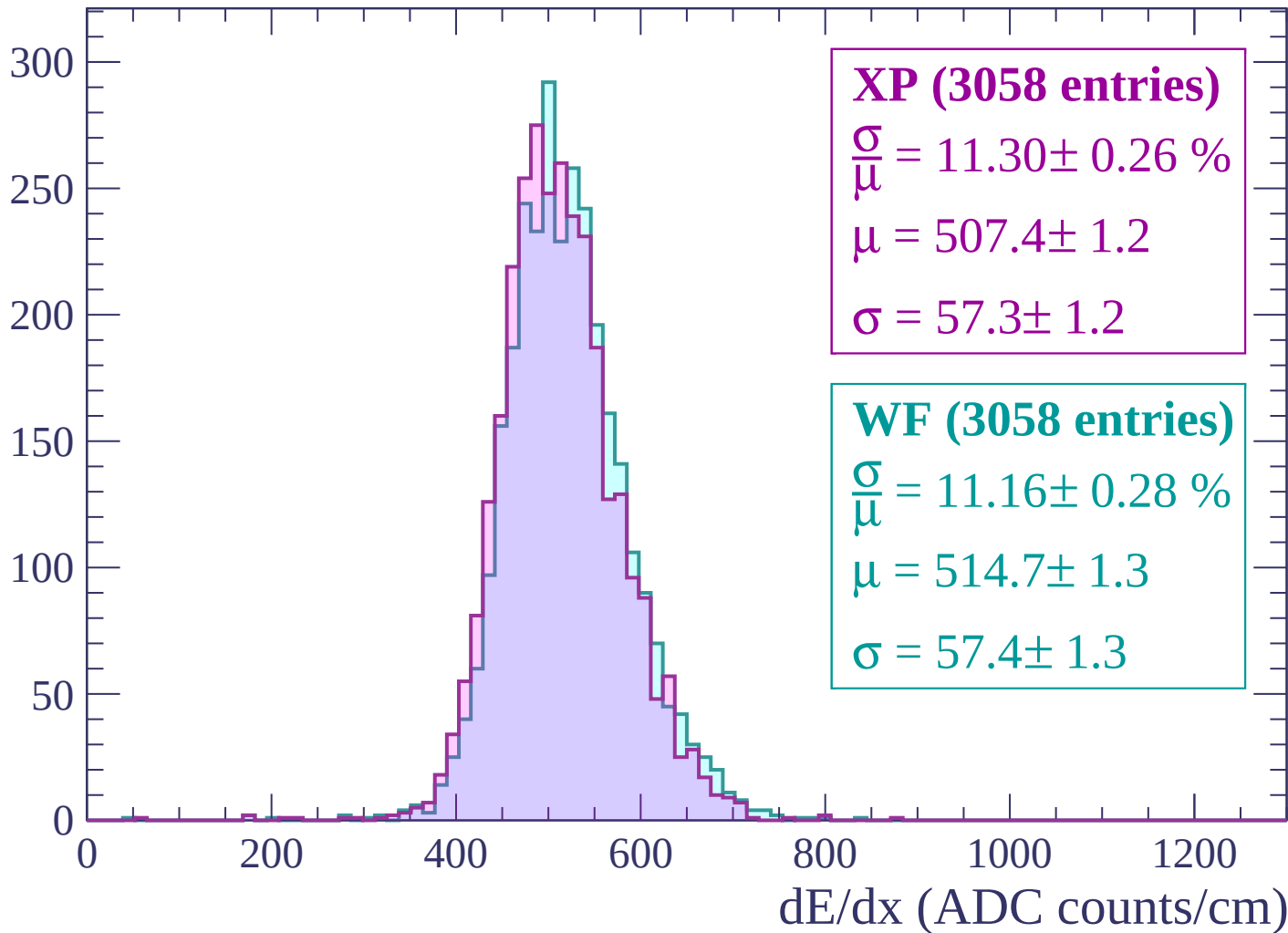
Energy loss | $-2580 < p < -2520$

Count



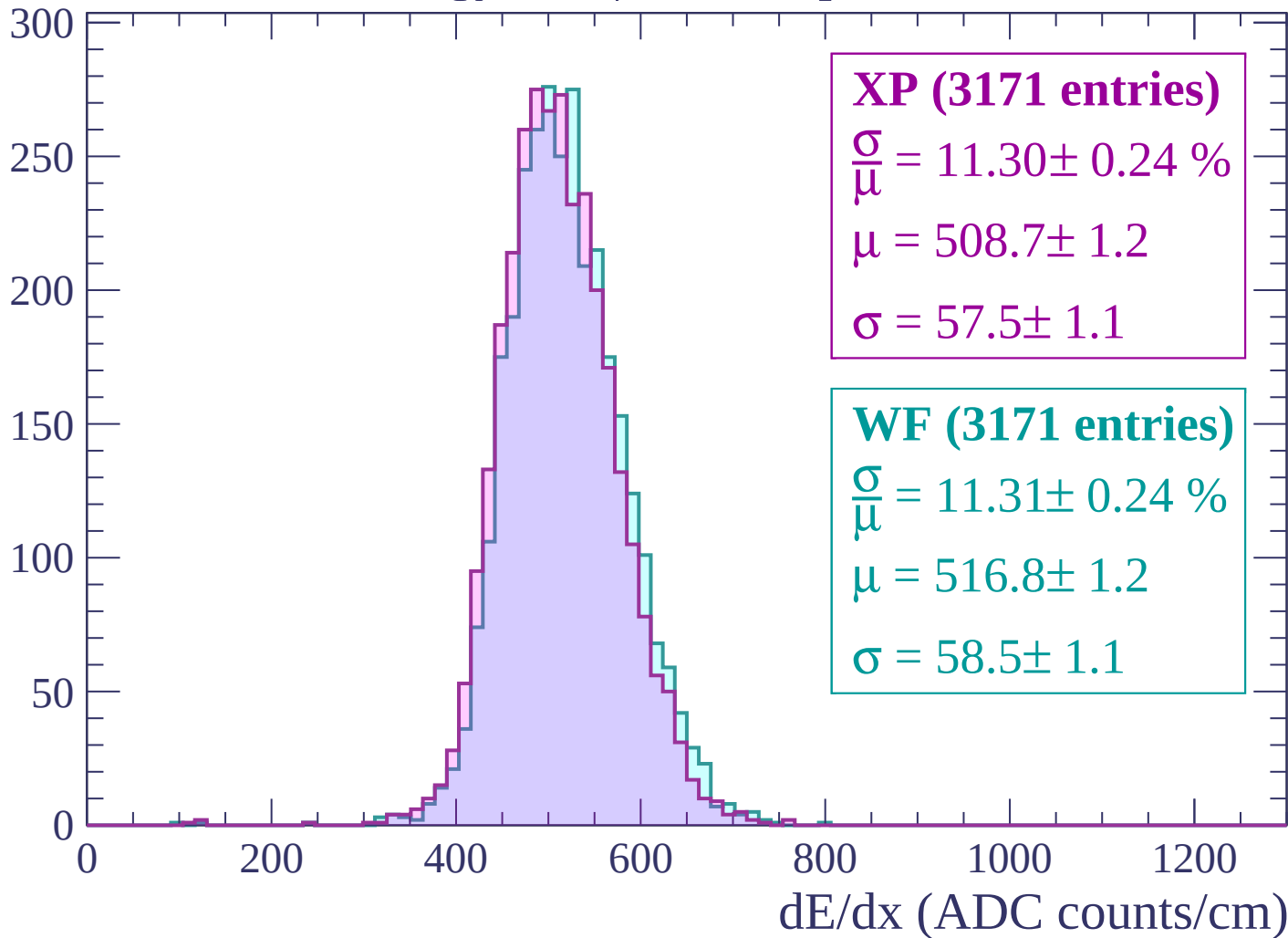
Energy loss | -2520 < p < -2460

Count



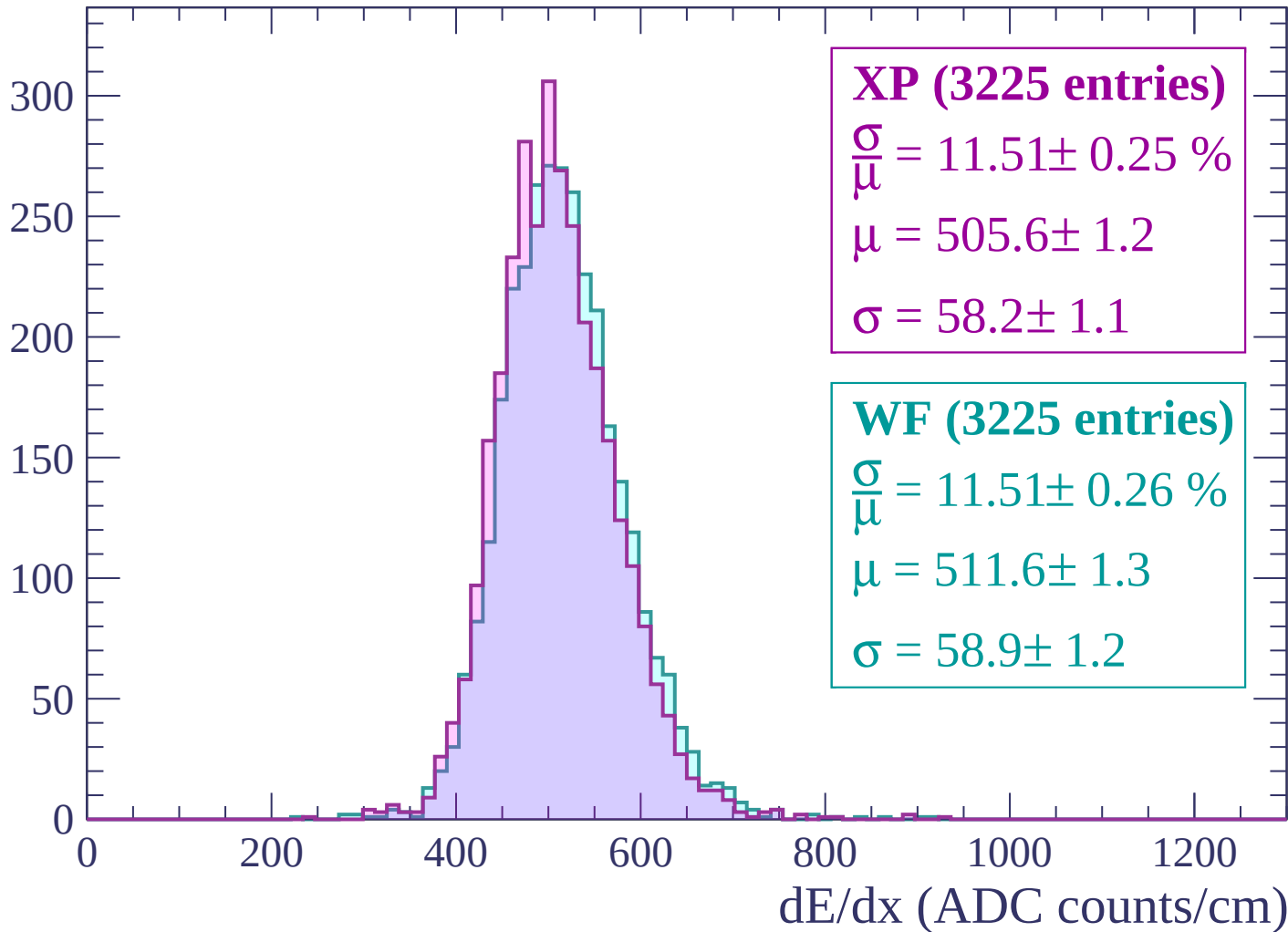
Energy loss | $-2460 < p < -2400$

Count



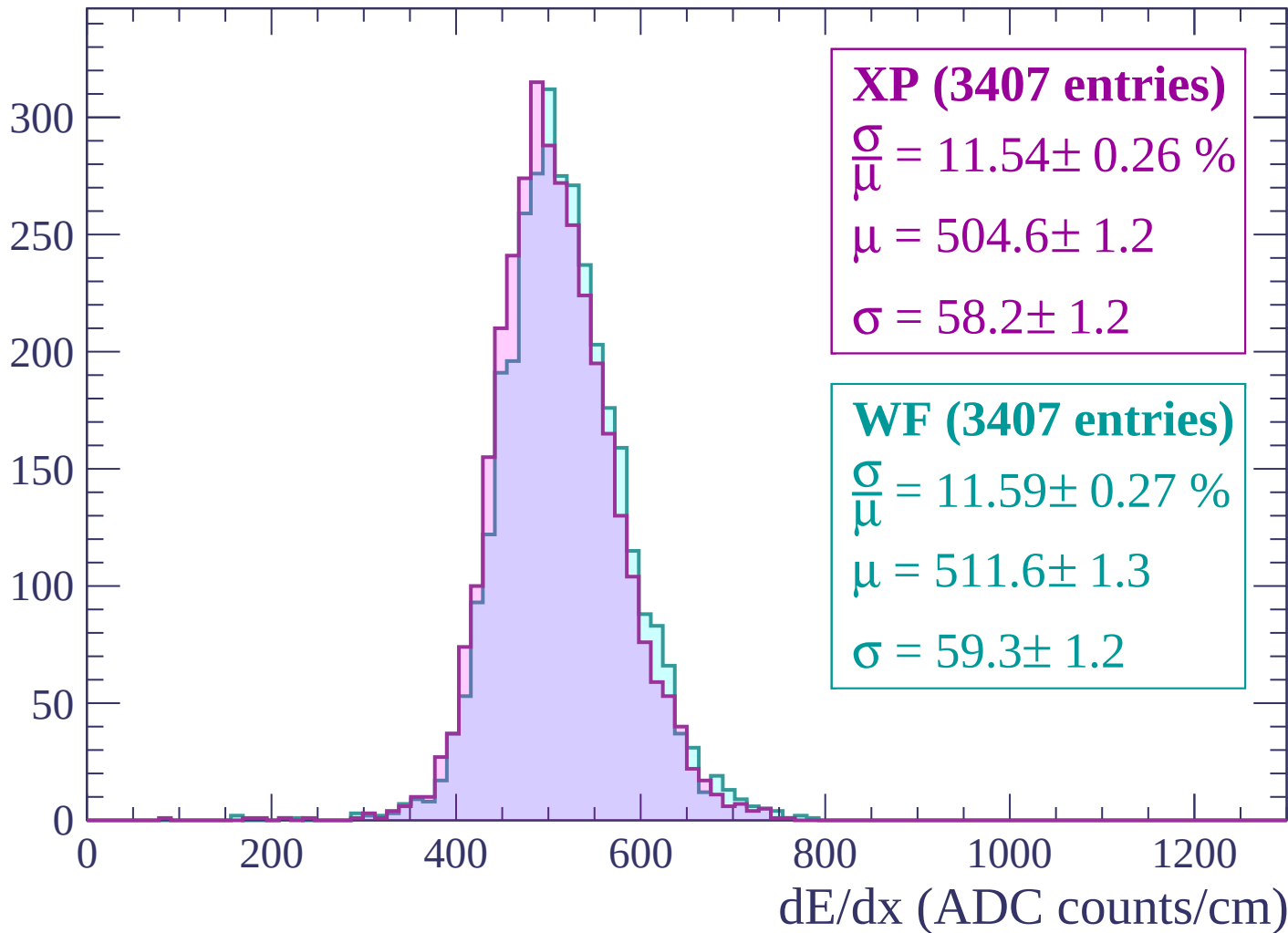
Energy loss | $-2400 < p < -2340$

Count



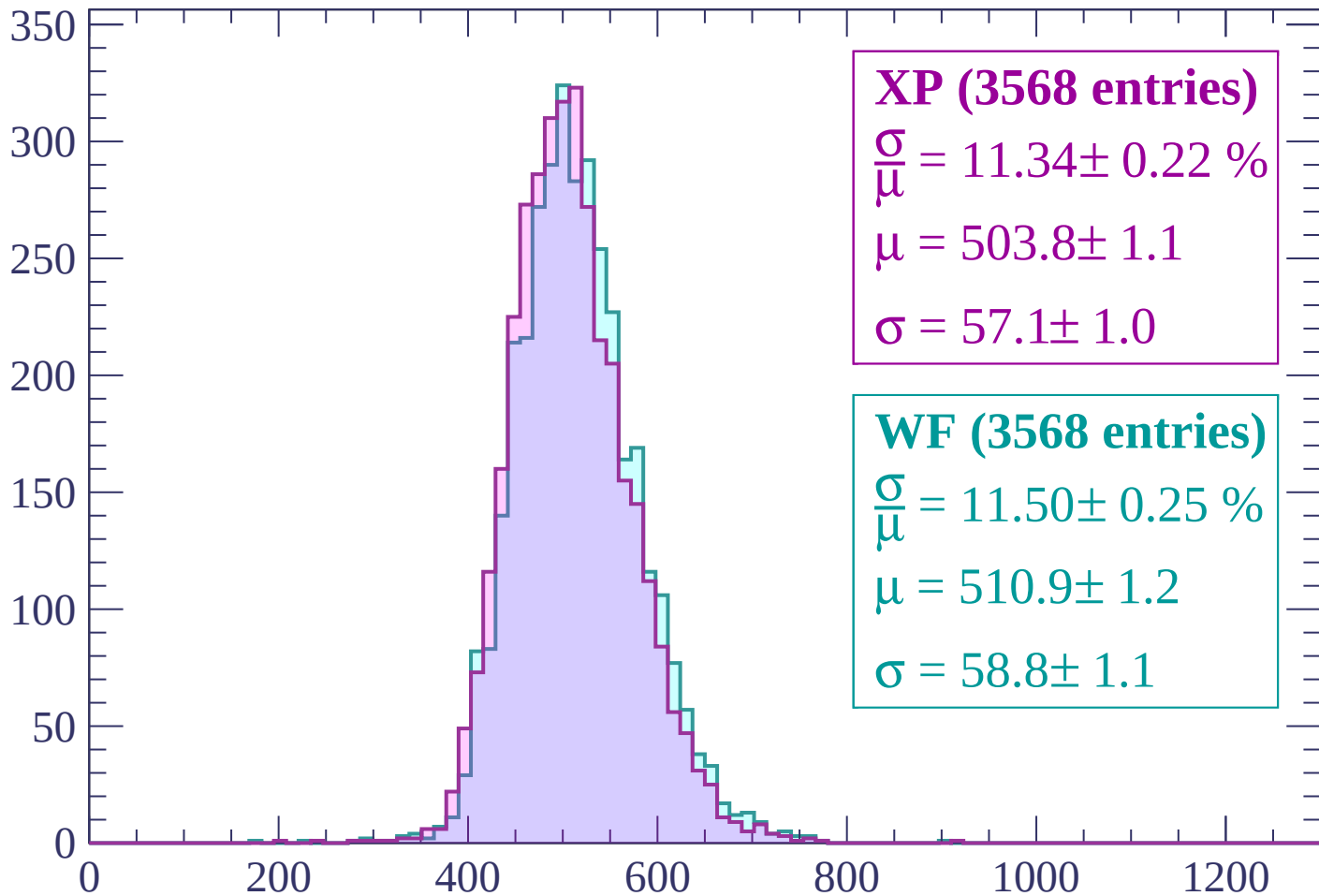
Energy loss | $-2340 < p < -2280$

Count



Energy loss | $-2280 < p < -2220$

Count



XP (3568 entries)

$$\frac{\sigma}{\mu} = 11.34 \pm 0.22 \%$$

$$\mu = 503.8 \pm 1.1$$

$$\sigma = 57.1 \pm 1.0$$

WF (3568 entries)

$$\frac{\sigma}{\mu} = 11.50 \pm 0.25 \%$$

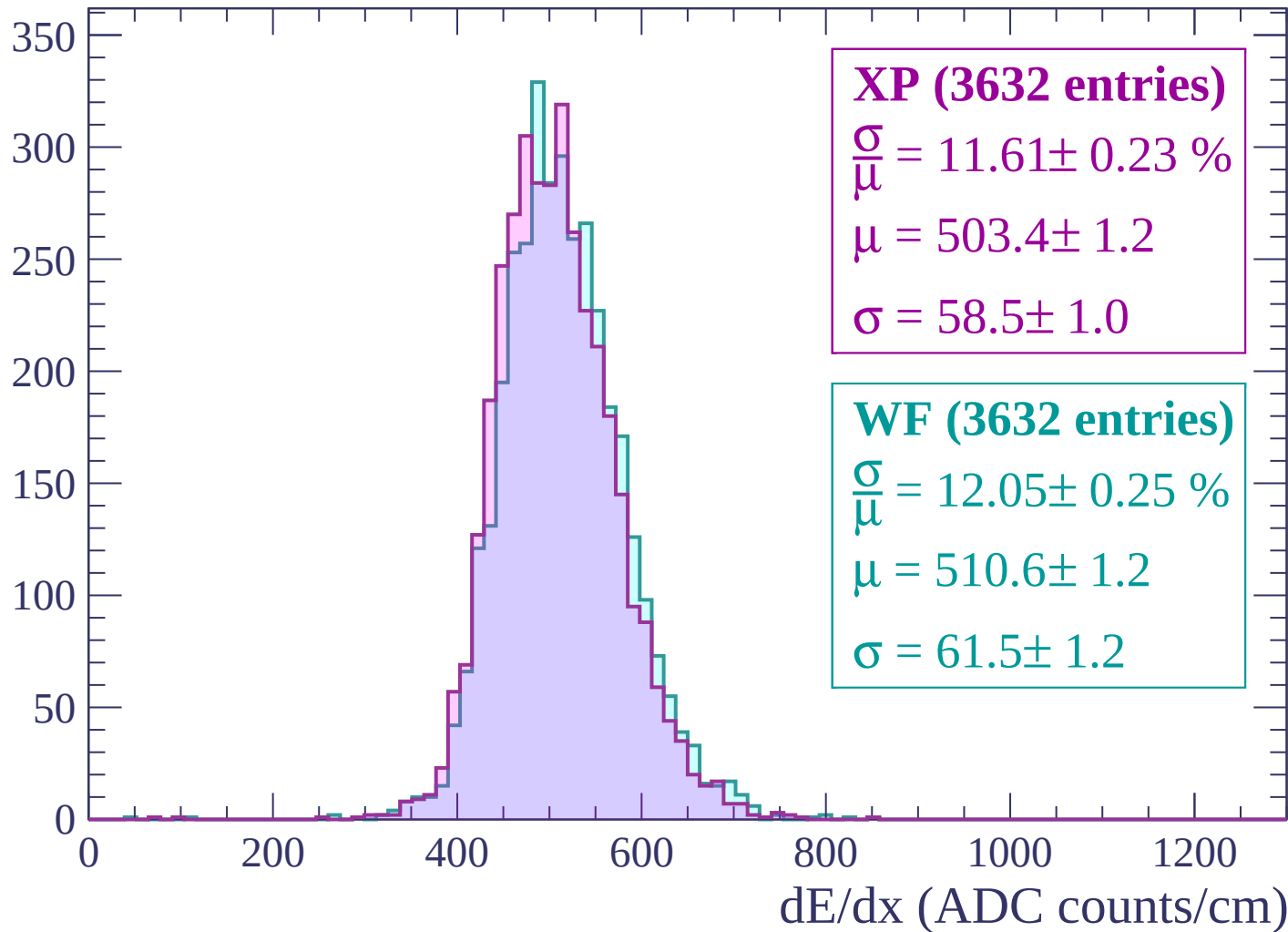
$$\mu = 510.9 \pm 1.2$$

$$\sigma = 58.8 \pm 1.1$$

dE/dx (ADC counts/cm)

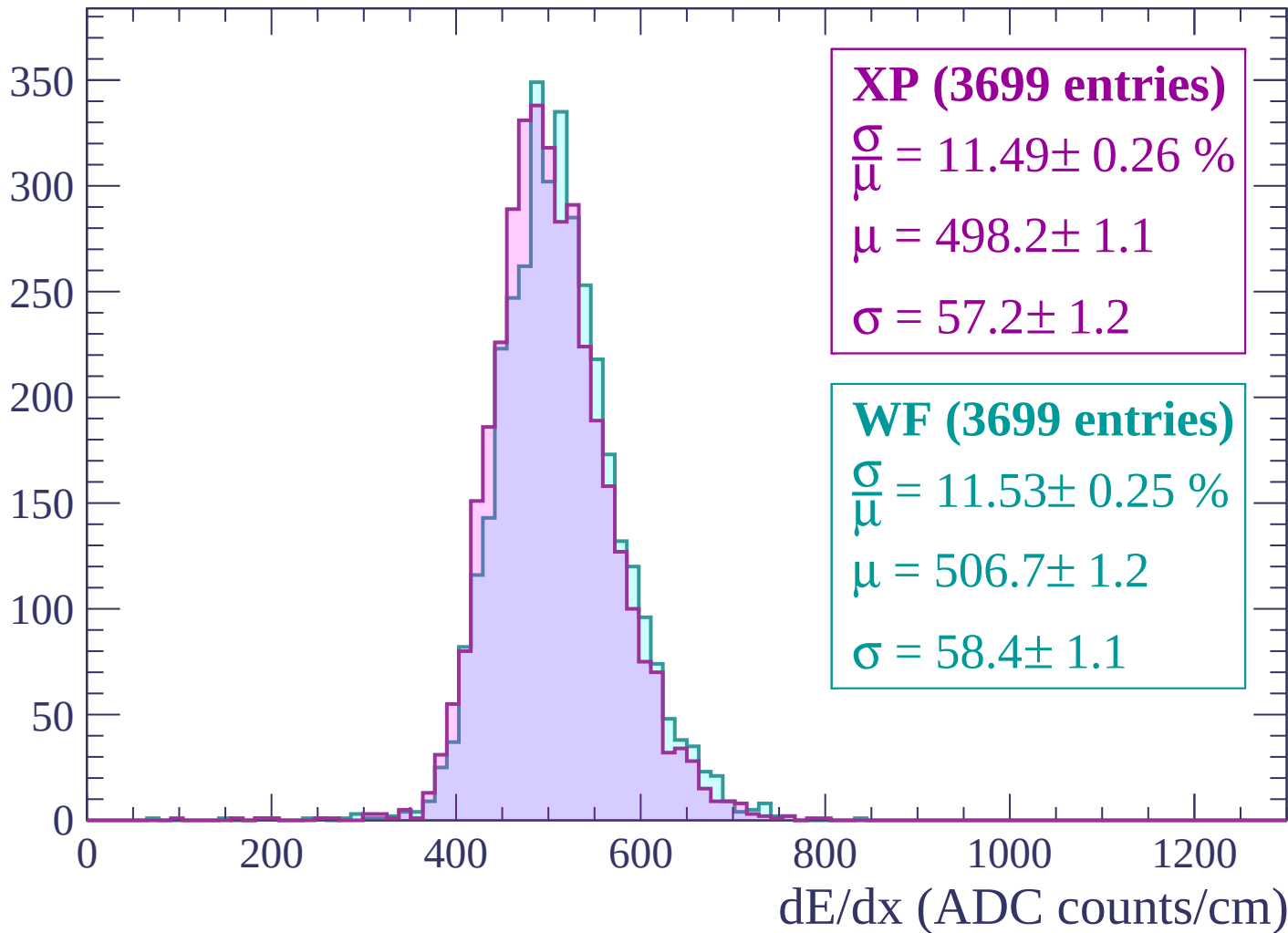
Energy loss | -2220 < p < -2160

Count



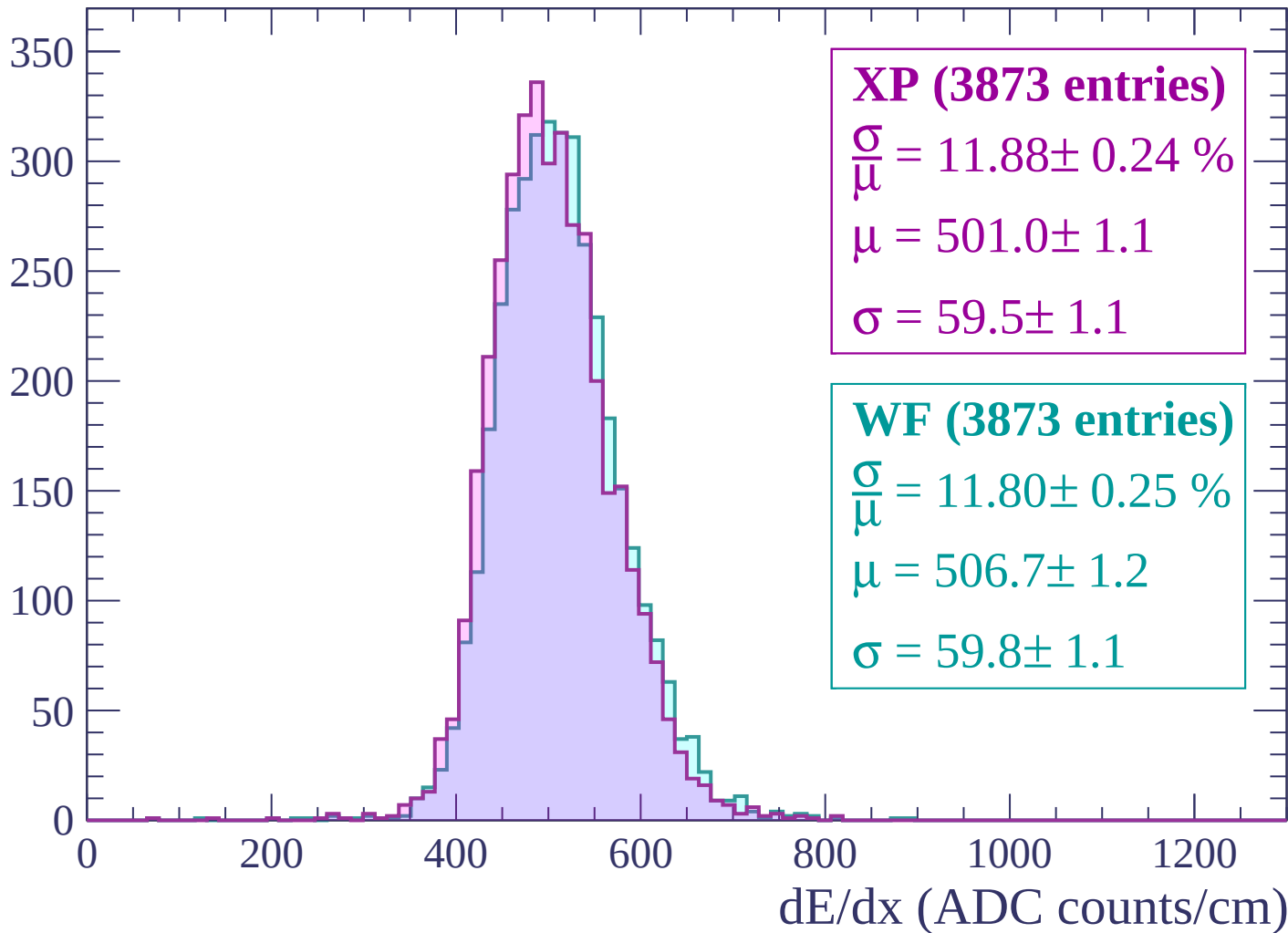
Energy loss | -2160 < p < -2100

Count



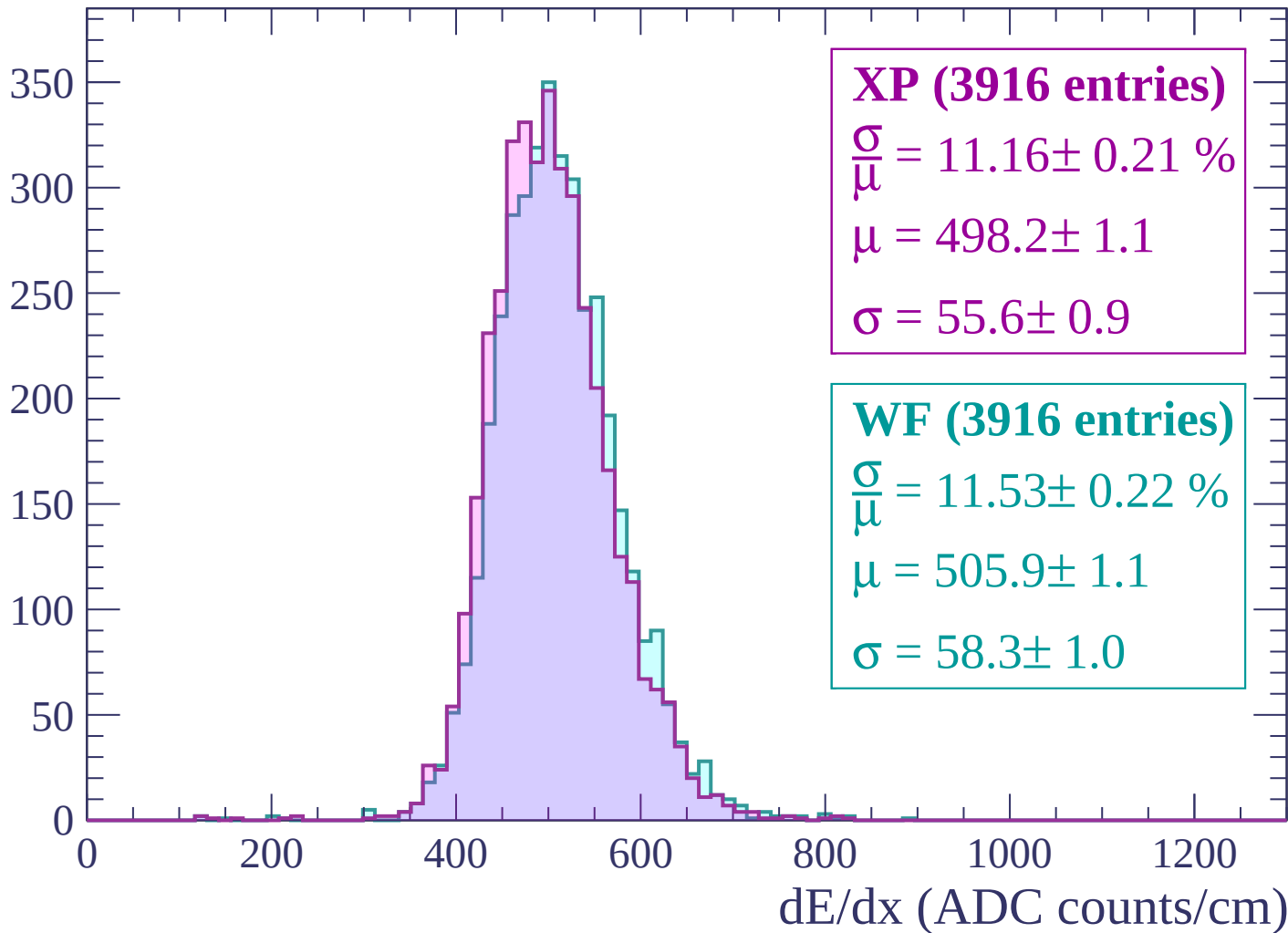
Energy loss | -2100 < p < -2040

Count



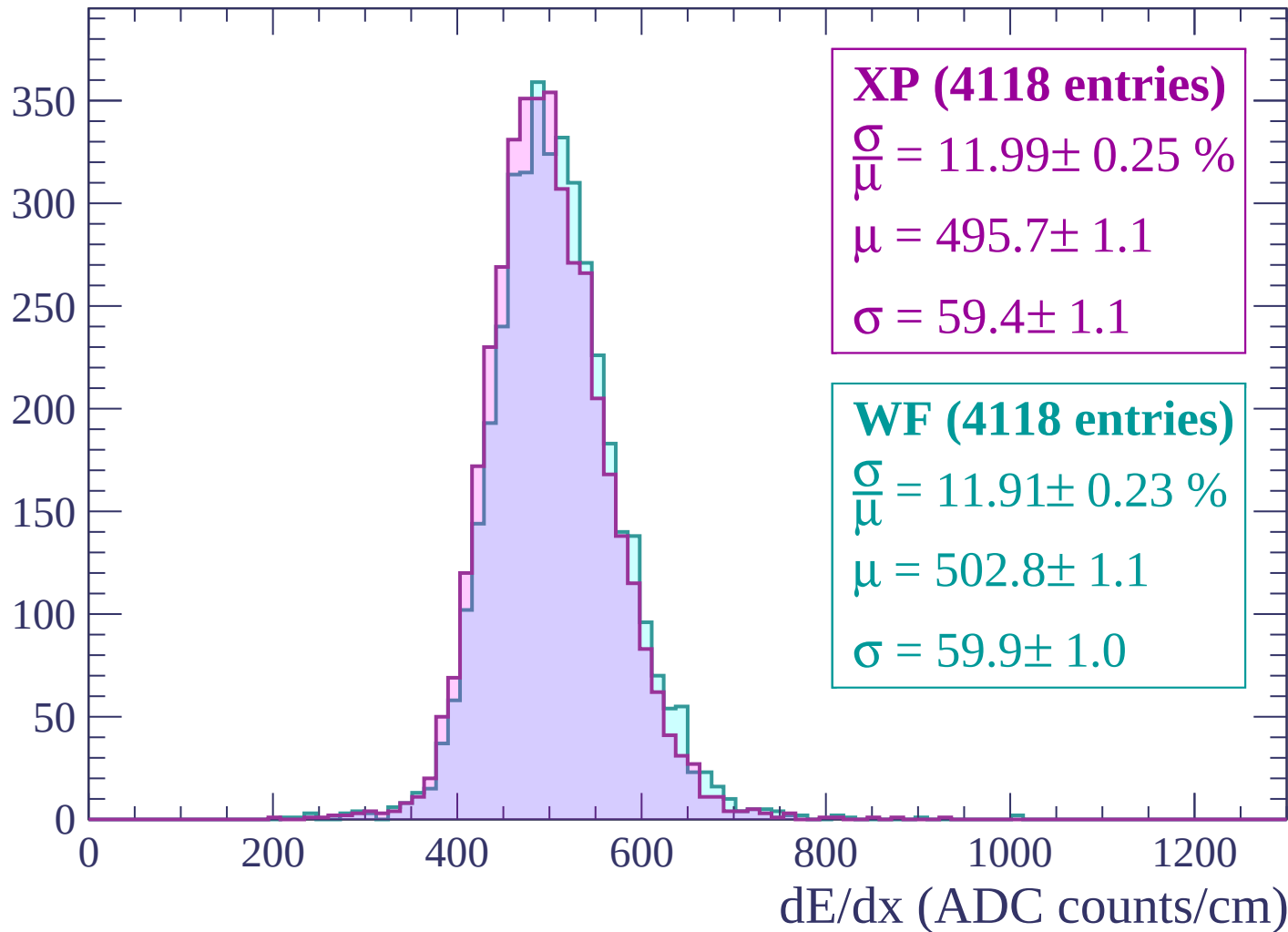
Energy loss | -2040 < p < -1980

Count



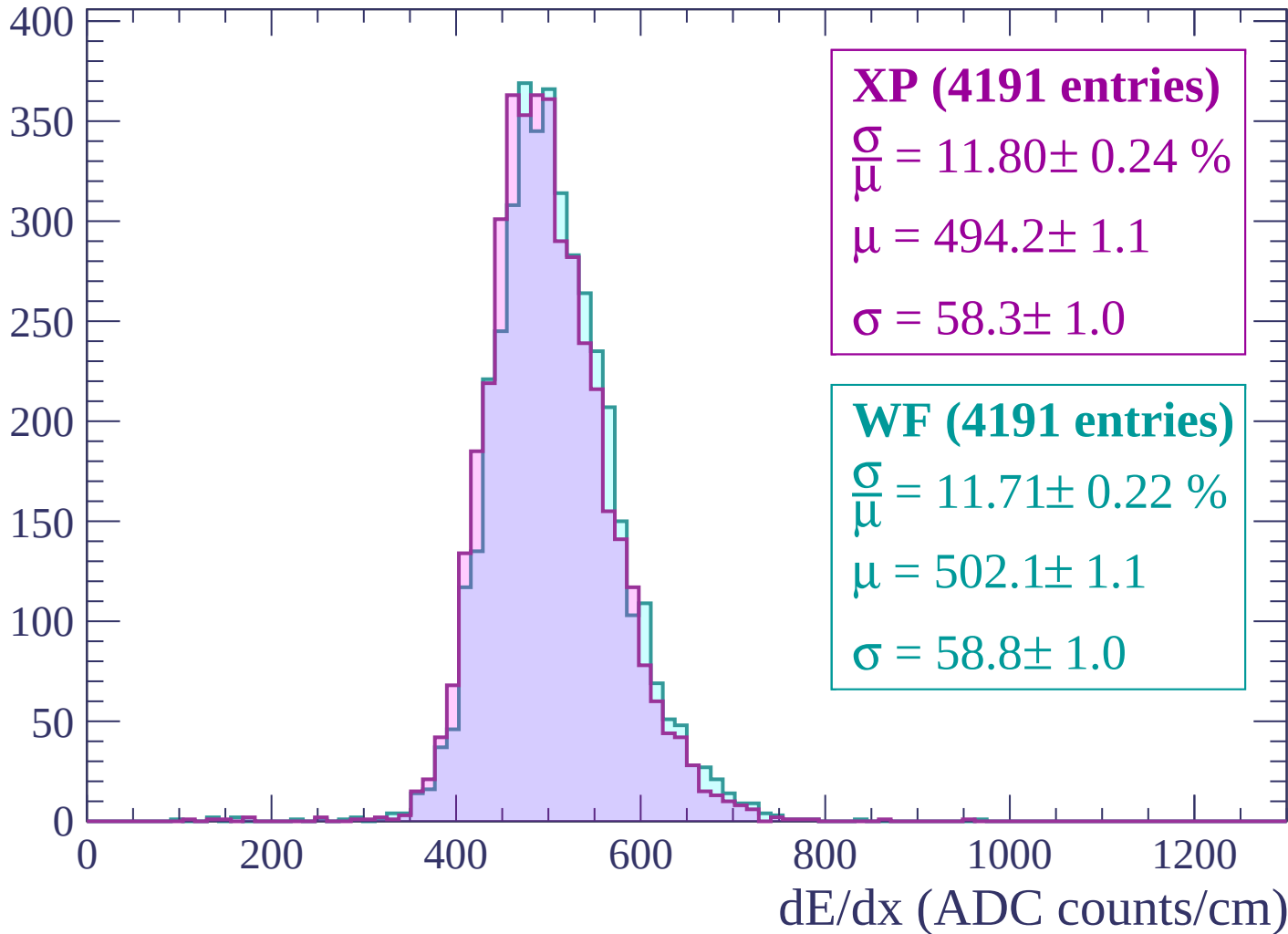
Energy loss | -1980 < p < -1920

Count



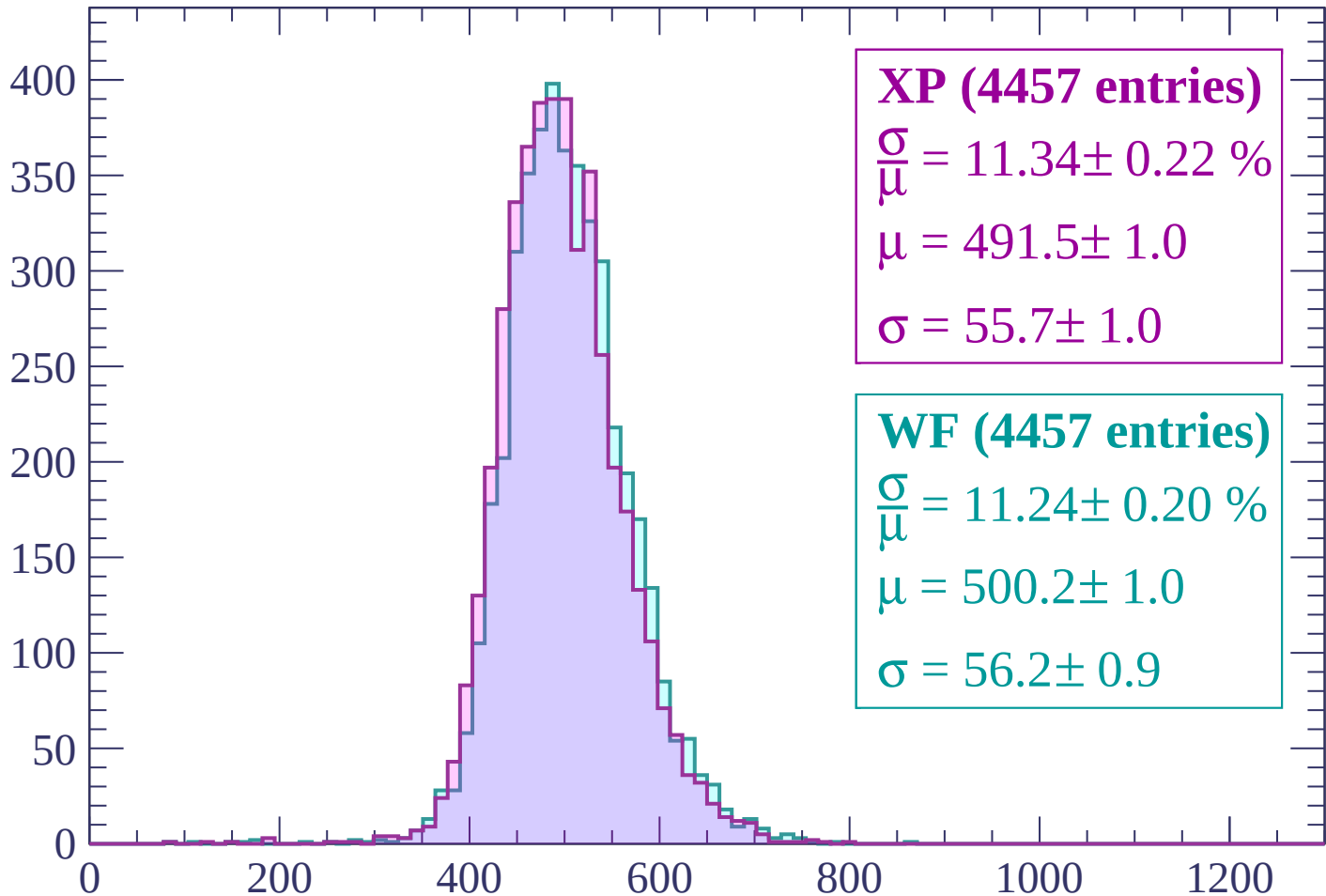
Energy loss | -1920 < p < -1860

Count



Energy loss | -1860 < p < -1800

Count



XP (4457 entries)

$$\frac{\sigma}{\mu} = 11.34 \pm 0.22 \%$$

$$\mu = 491.5 \pm 1.0$$

$$\sigma = 55.7 \pm 1.0$$

WF (4457 entries)

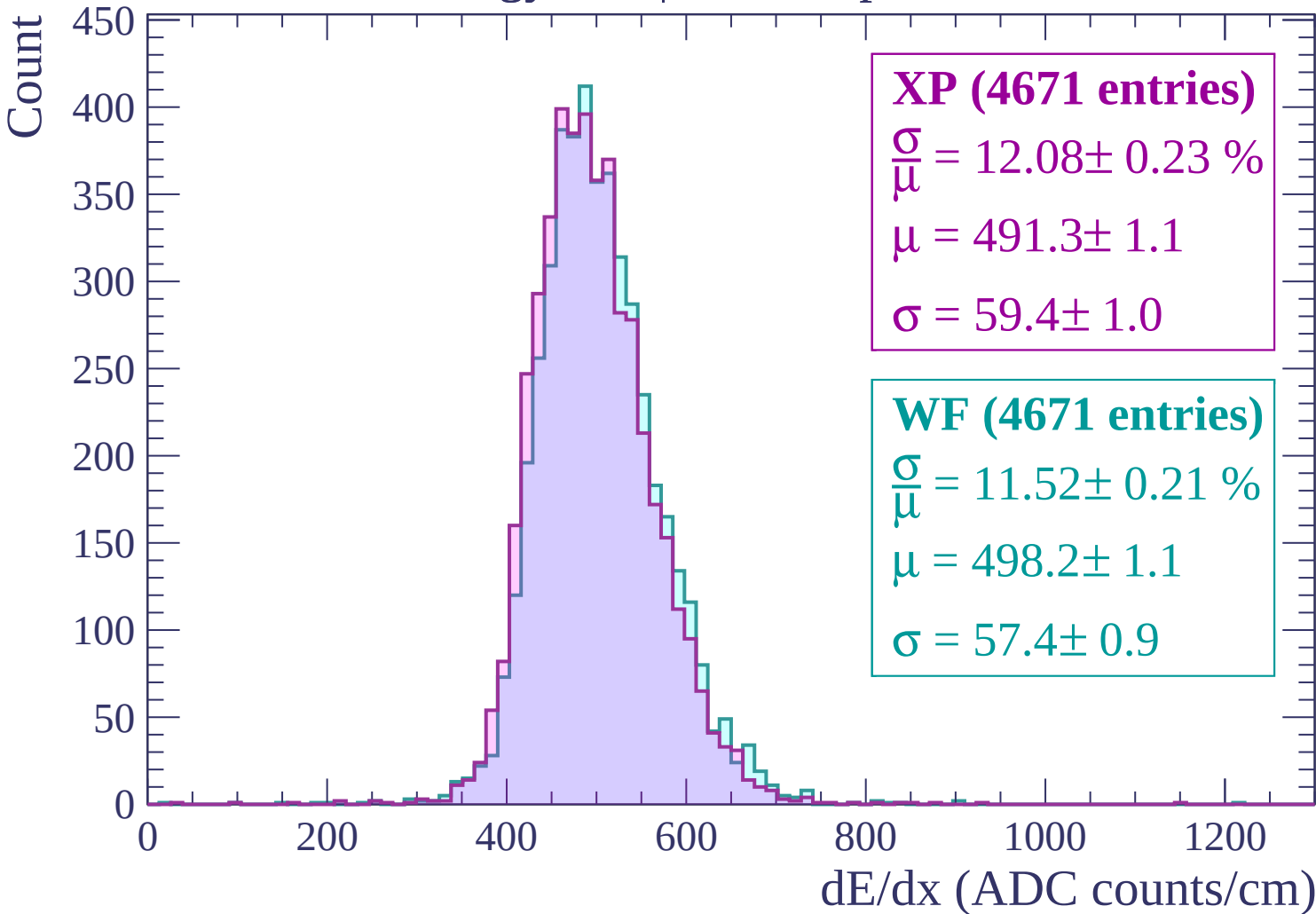
$$\frac{\sigma}{\mu} = 11.24 \pm 0.20 \%$$

$$\mu = 500.2 \pm 1.0$$

$$\sigma = 56.2 \pm 0.9$$

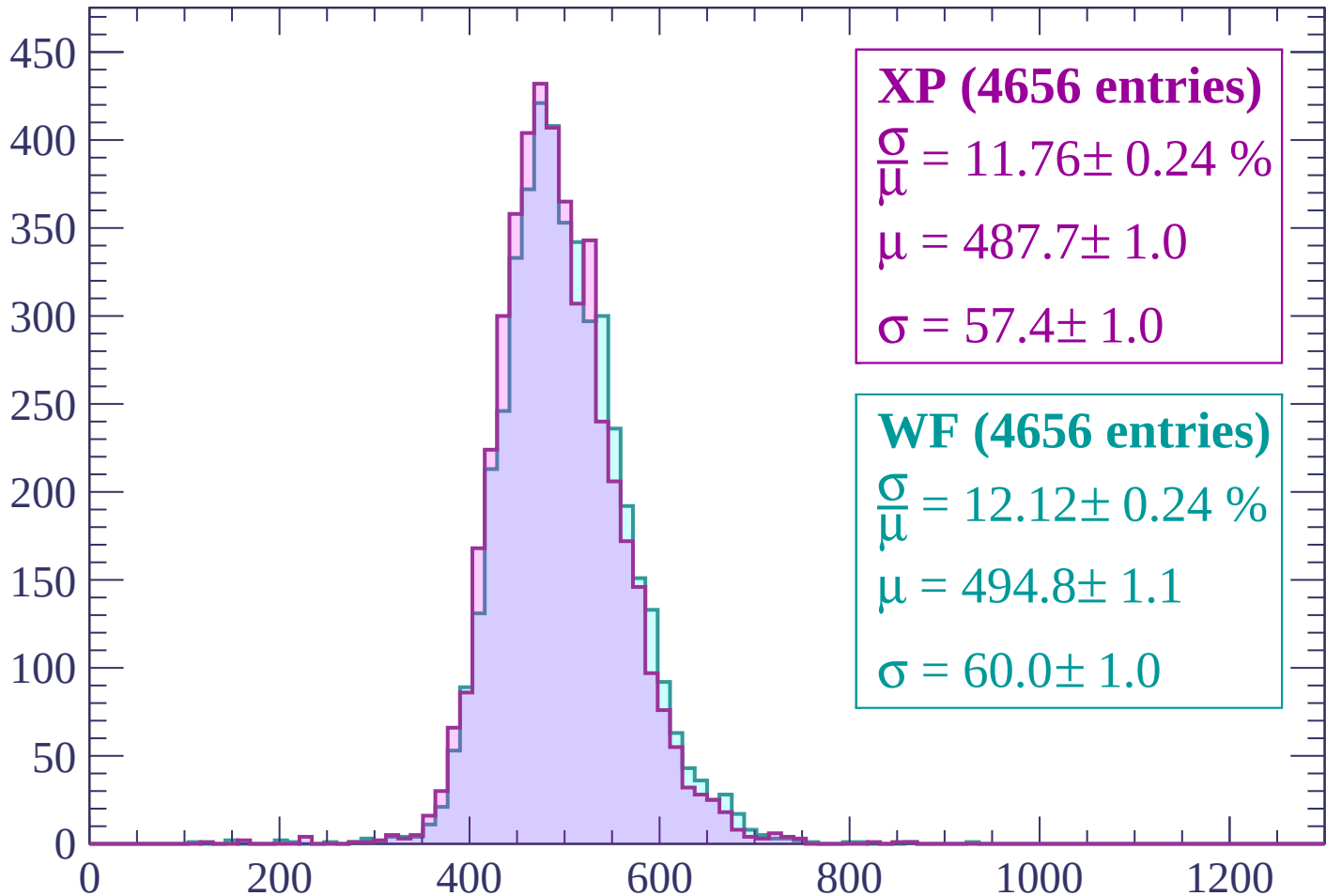
dE/dx (ADC counts/cm)

Energy loss | -1800 < p < -1740



Energy loss | $-1740 < p < -1680$

Count



XP (4656 entries)

$$\frac{\sigma}{\mu} = 11.76 \pm 0.24 \%$$

$$\mu = 487.7 \pm 1.0$$

$$\sigma = 57.4 \pm 1.0$$

WF (4656 entries)

$$\frac{\sigma}{\mu} = 12.12 \pm 0.24 \%$$

$$\mu = 494.8 \pm 1.1$$

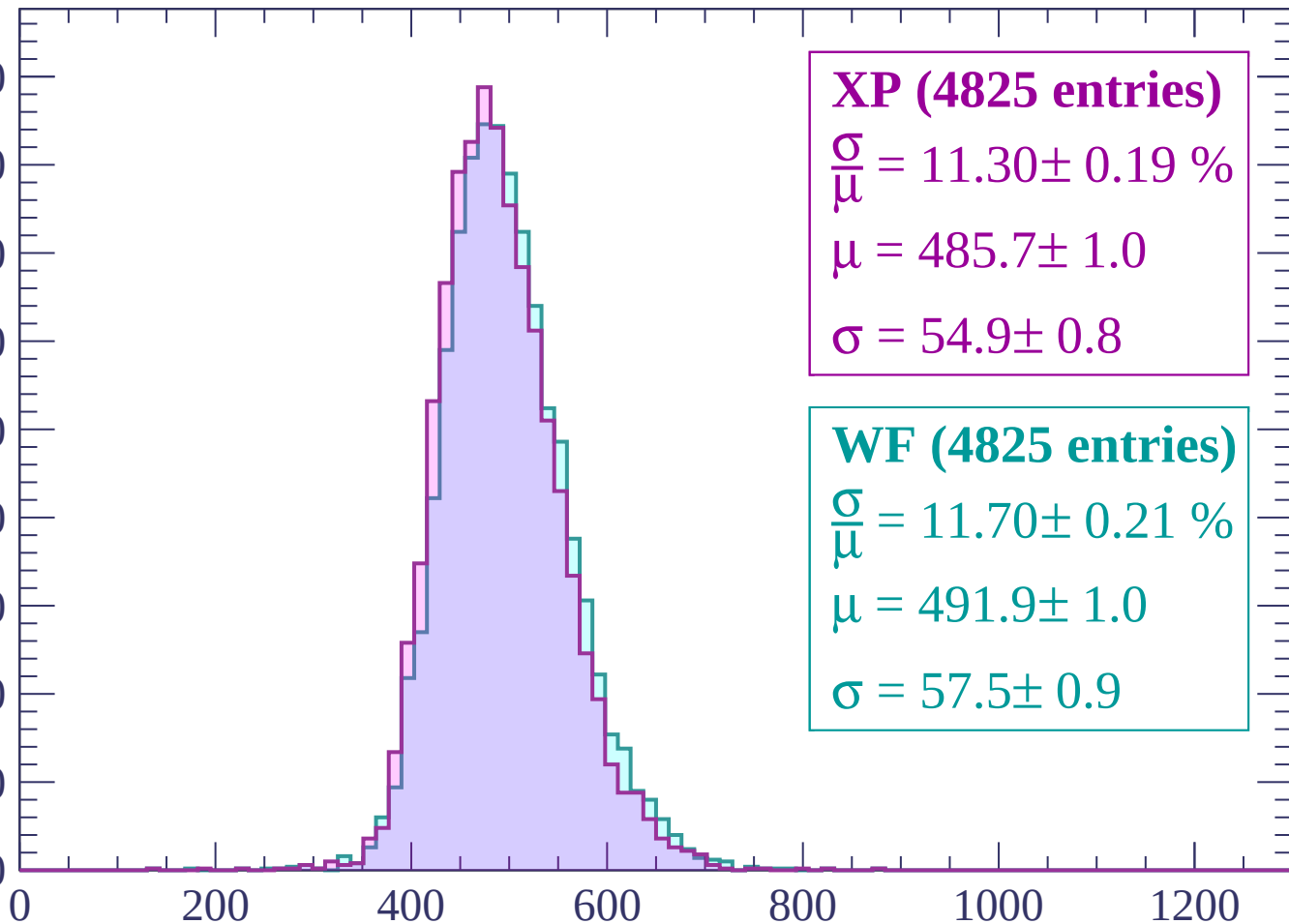
$$\sigma = 60.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1680 < p < -1620

Count

450
400
350
300
250
200
150
100
50
0



XP (4825 entries)

$\frac{\sigma}{\mu} = 11.30 \pm 0.19 \%$

$\mu = 485.7 \pm 1.0$

$\sigma = 54.9 \pm 0.8$

WF (4825 entries)

$\frac{\sigma}{\mu} = 11.70 \pm 0.21 \%$

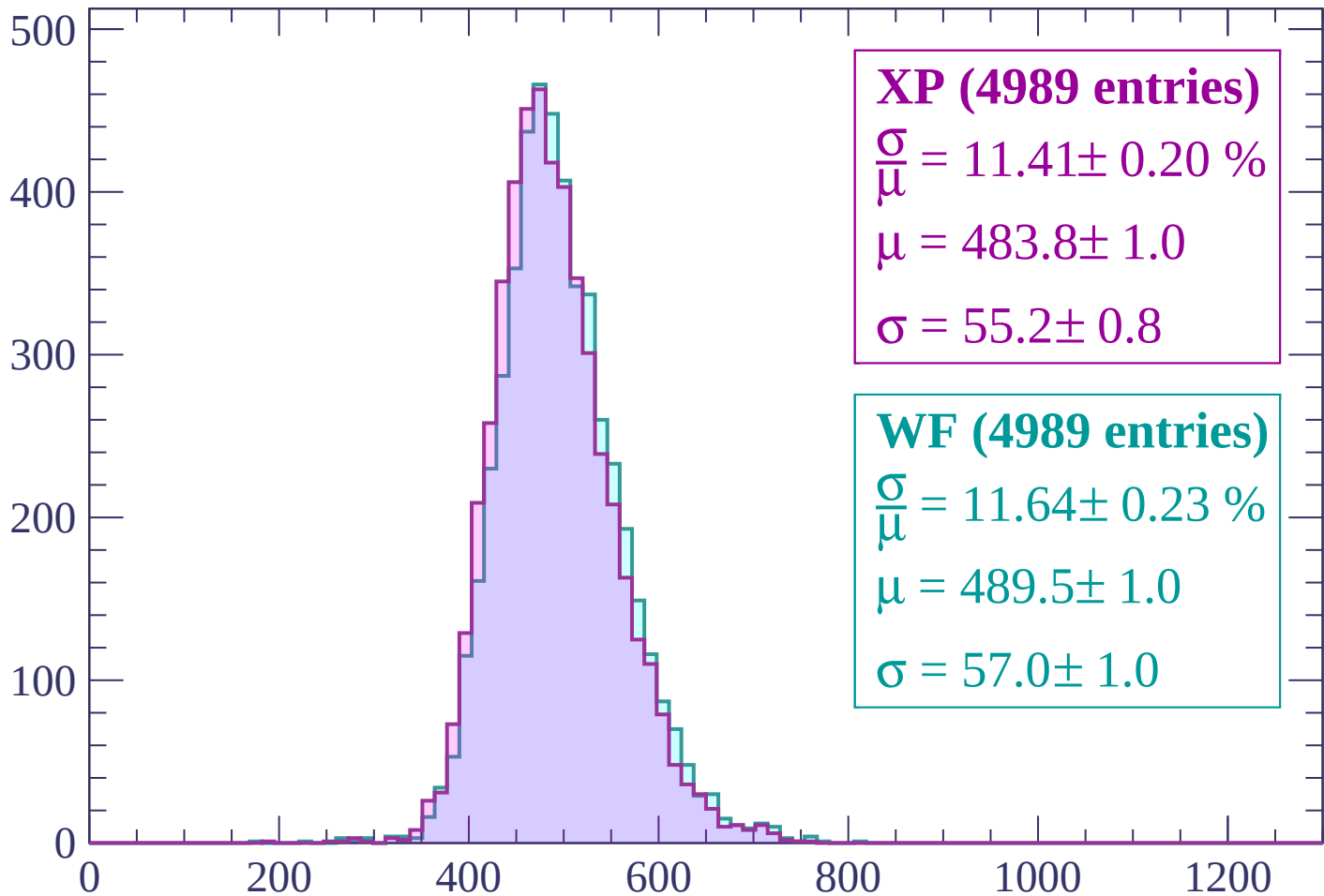
$\mu = 491.9 \pm 1.0$

$\sigma = 57.5 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | -1620 < p < -1560

Count



XP (4989 entries)

$\frac{\sigma}{\mu} = 11.41 \pm 0.20 \%$

$\mu = 483.8 \pm 1.0$

$\sigma = 55.2 \pm 0.8$

WF (4989 entries)

$\frac{\sigma}{\mu} = 11.64 \pm 0.23 \%$

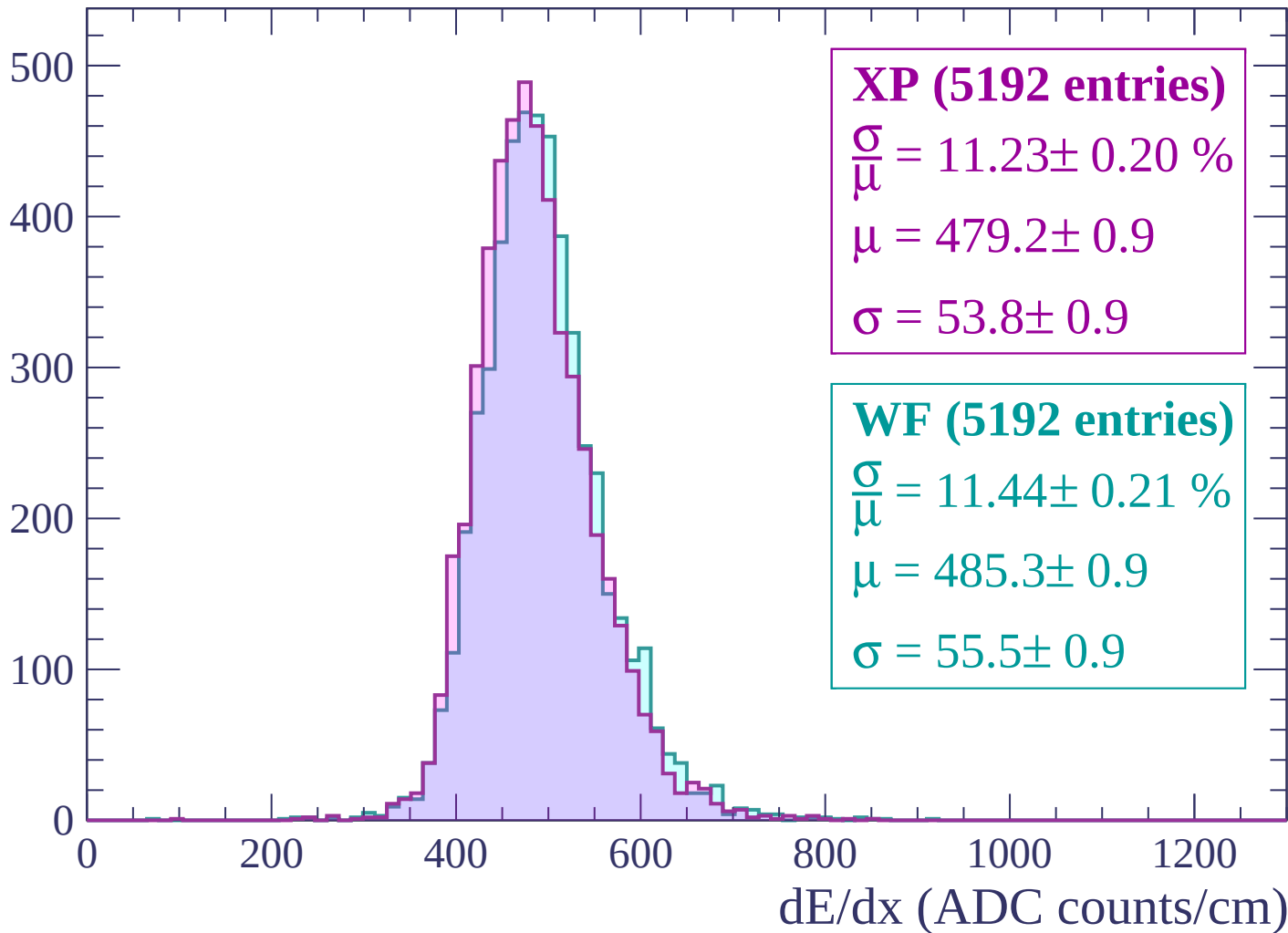
$\mu = 489.5 \pm 1.0$

$\sigma = 57.0 \pm 1.0$

dE/dx (ADC counts/cm)

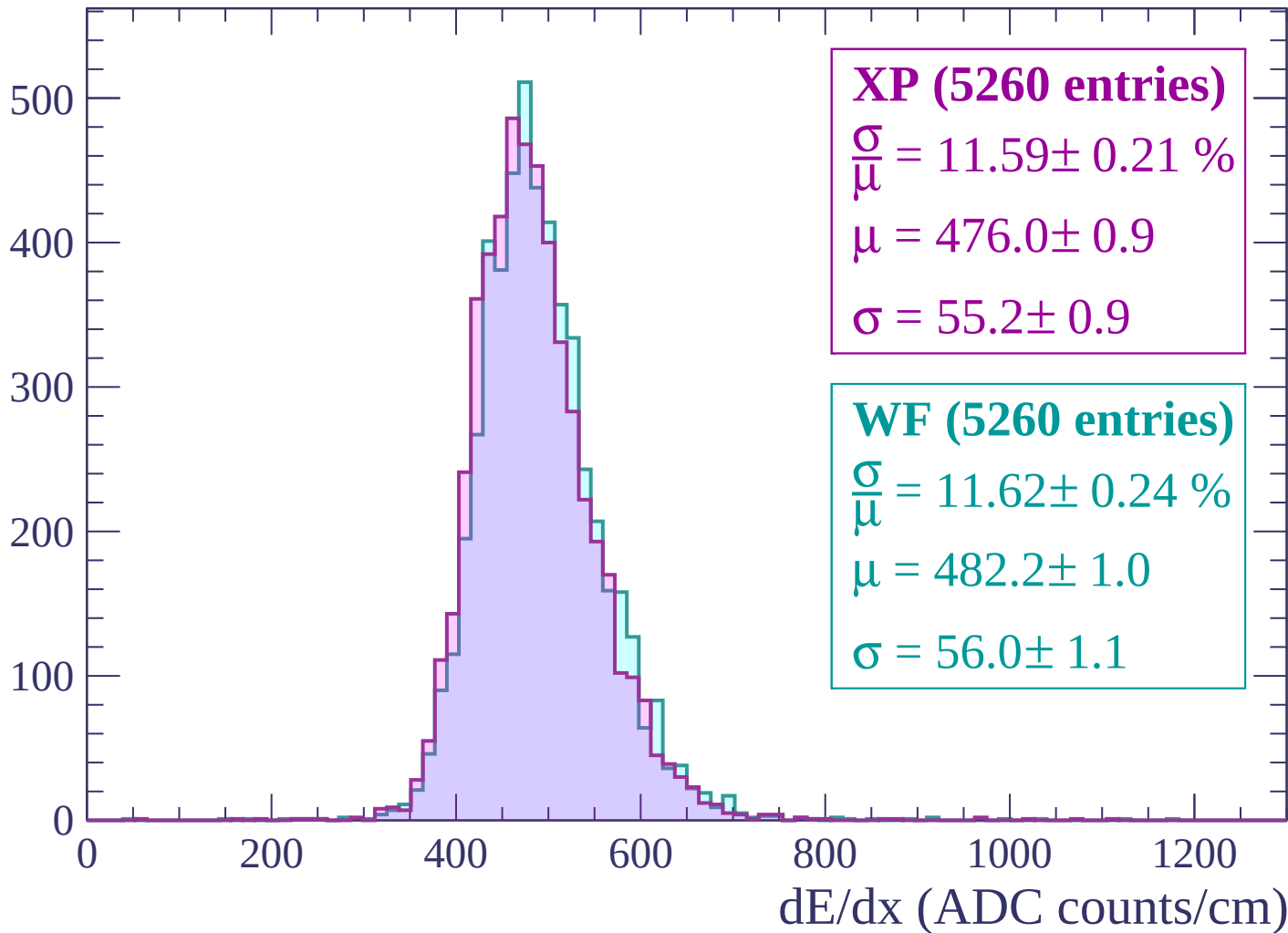
Energy loss | -1560 < p < -1500

Count



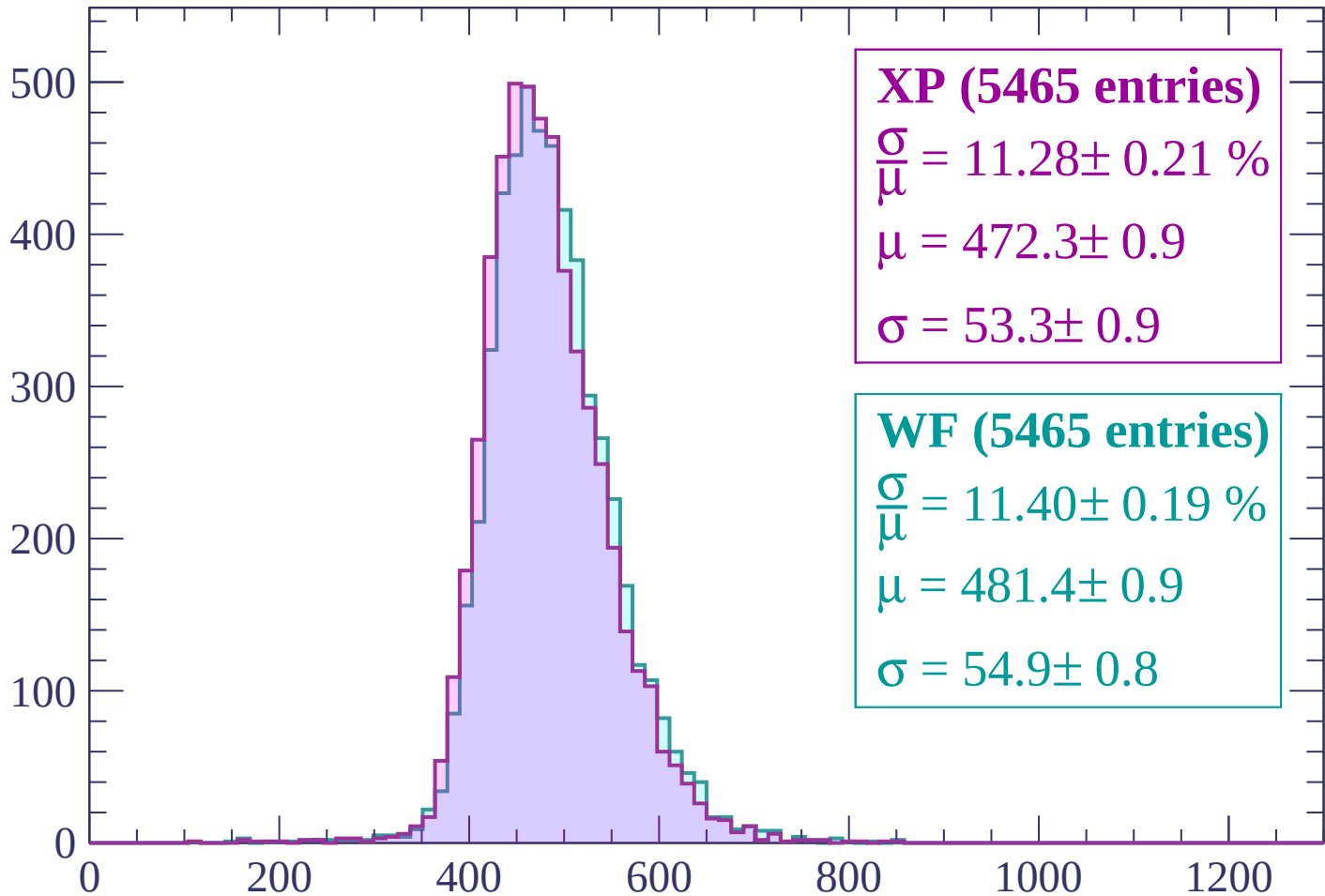
Energy loss | $-1500 < p < -1440$

Count



Energy loss | -1440 < p < -1380

Count



XP (5465 entries)

$$\frac{\sigma}{\mu} = 11.28 \pm 0.21 \%$$

$$\mu = 472.3 \pm 0.9$$

$$\sigma = 53.3 \pm 0.9$$

WF (5465 entries)

$$\frac{\sigma}{\mu} = 11.40 \pm 0.19 \%$$

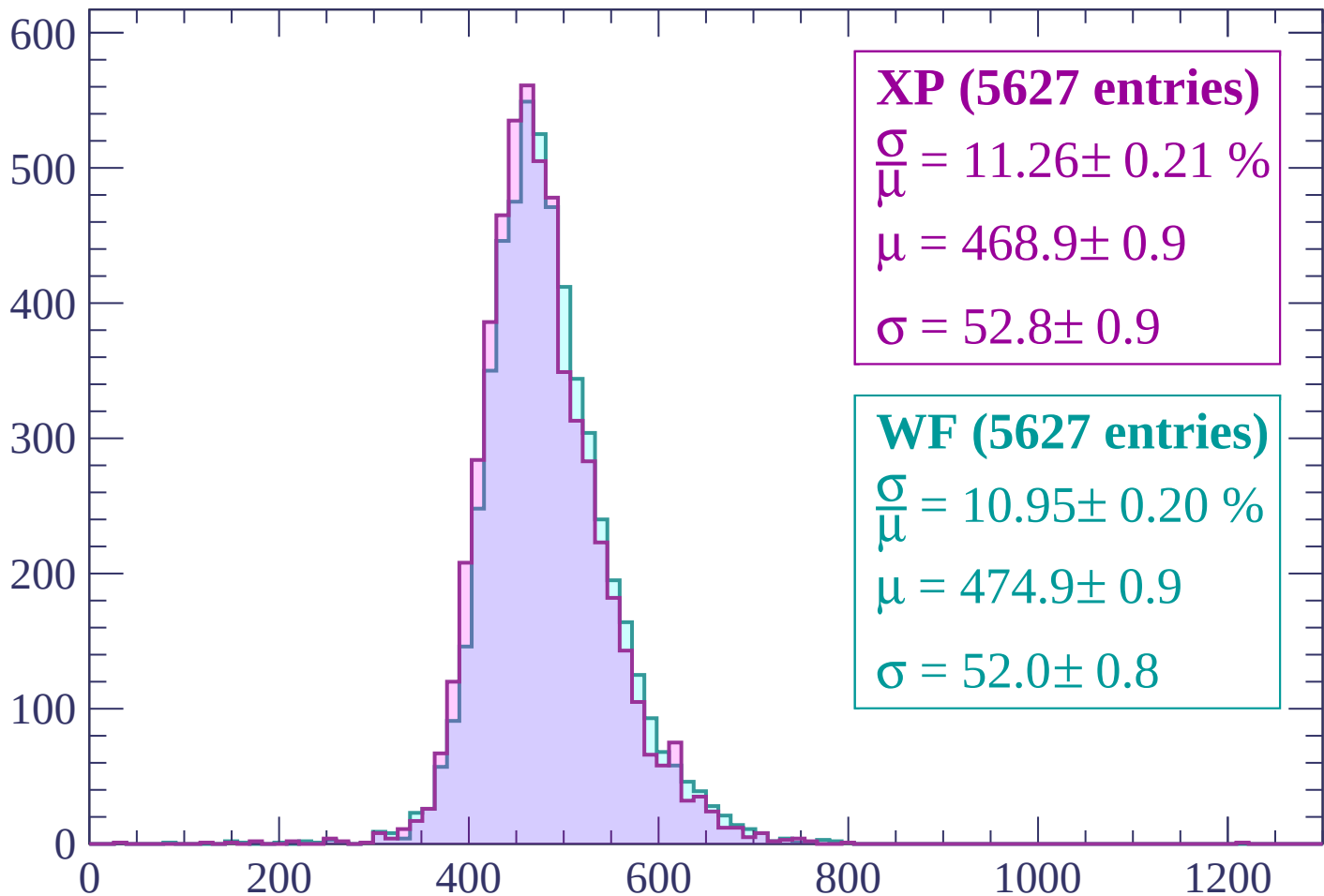
$$\mu = 481.4 \pm 0.9$$

$$\sigma = 54.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1380 < p < -1320

Count



XP (5627 entries)

$$\frac{\sigma}{\mu} = 11.26 \pm 0.21 \%$$

$$\mu = 468.9 \pm 0.9$$

$$\sigma = 52.8 \pm 0.9$$

WF (5627 entries)

$$\frac{\sigma}{\mu} = 10.95 \pm 0.20 \%$$

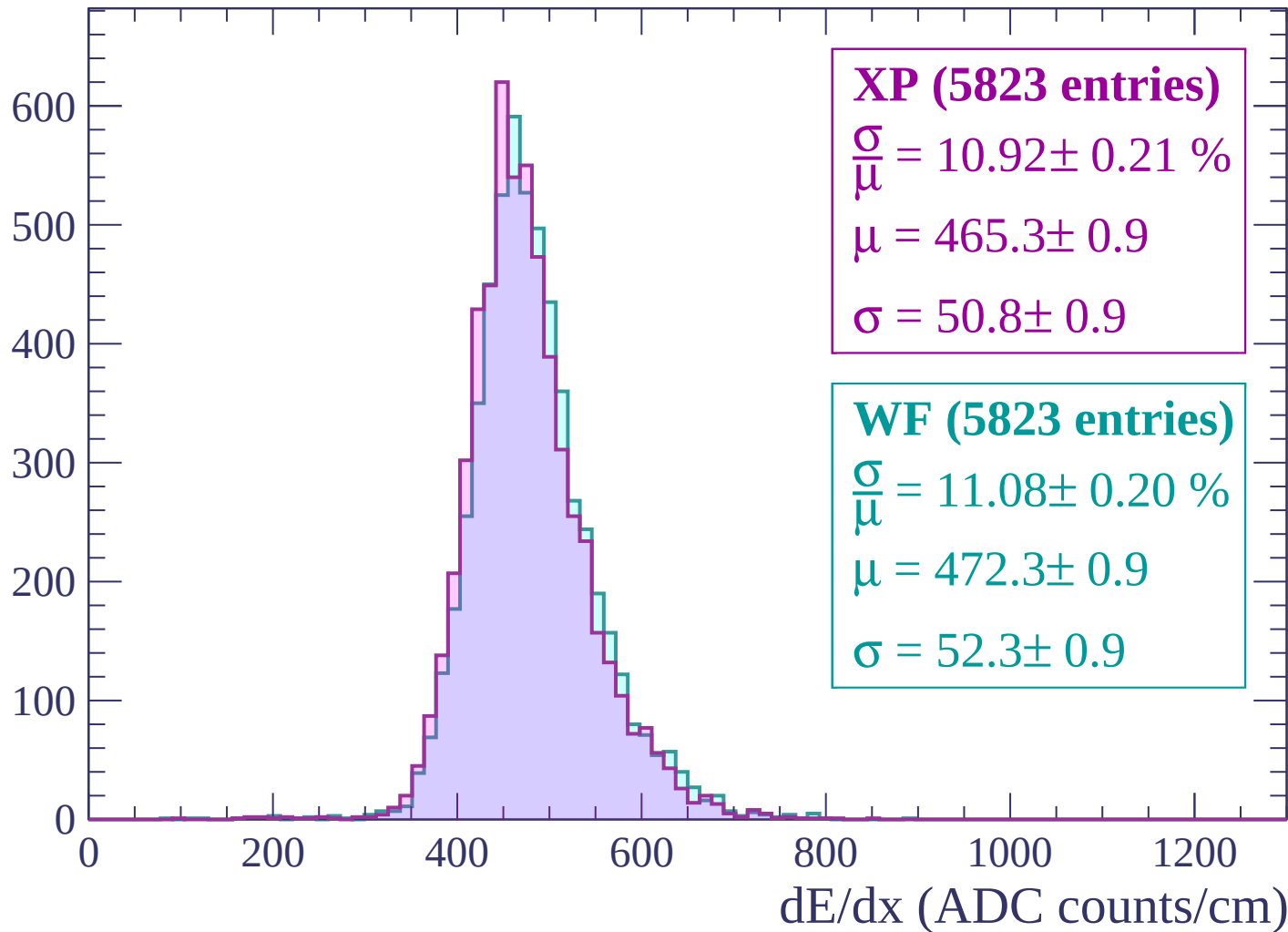
$$\mu = 474.9 \pm 0.9$$

$$\sigma = 52.0 \pm 0.8$$

dE/dx (ADC counts/cm)

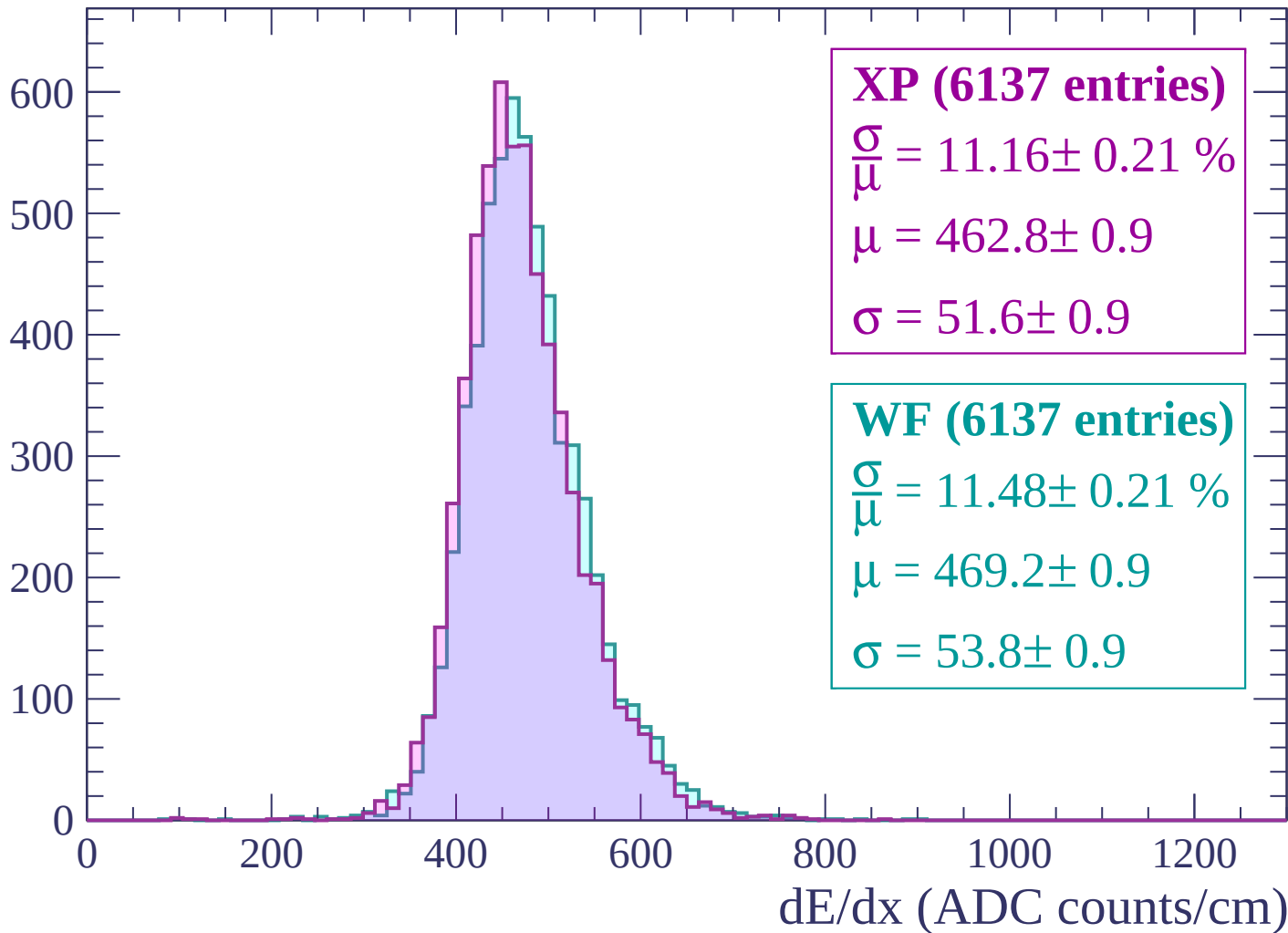
Energy loss | $-1320 < p < -1260$

Count



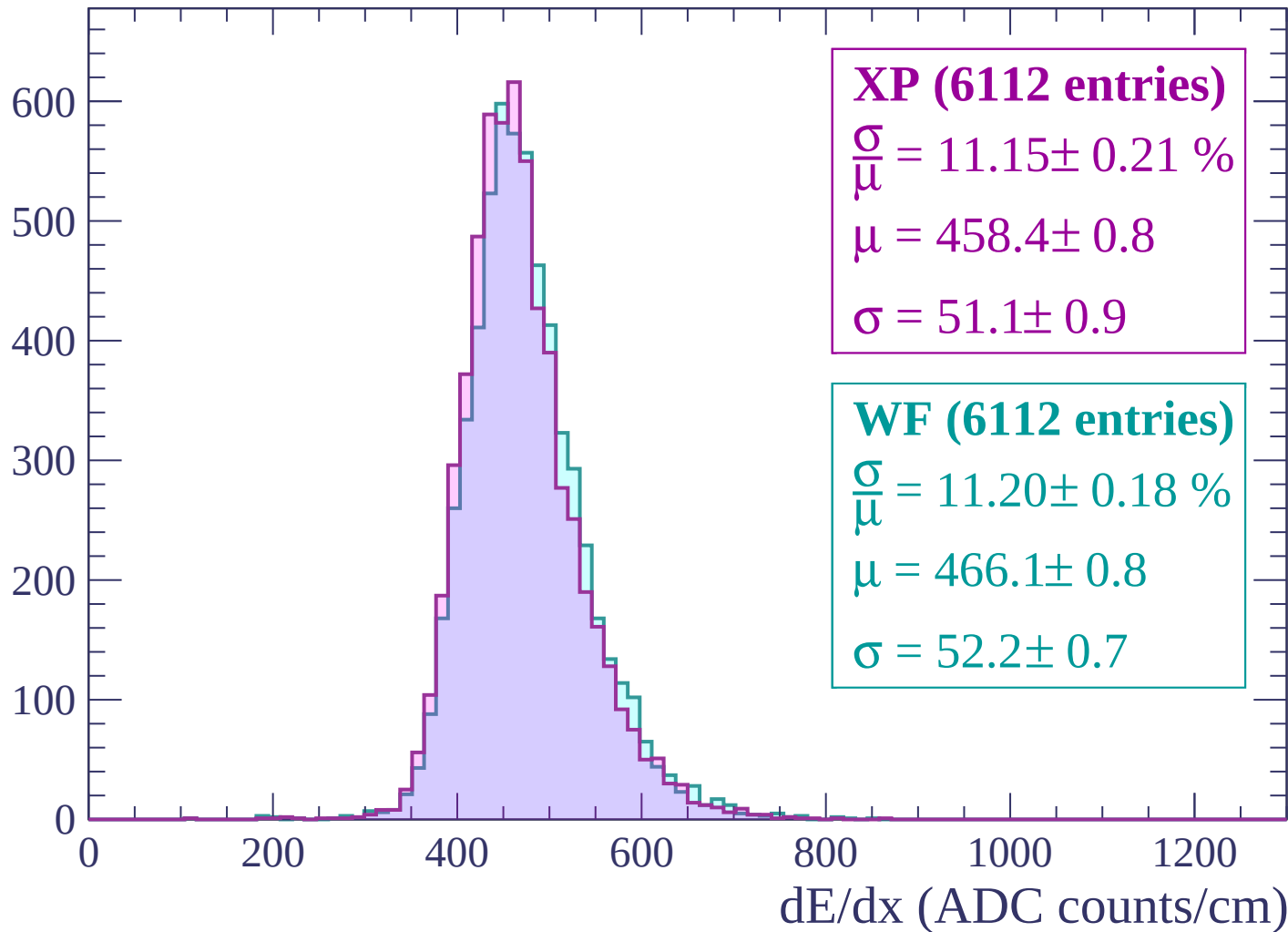
Energy loss | $-1260 < p < -1200$

Count



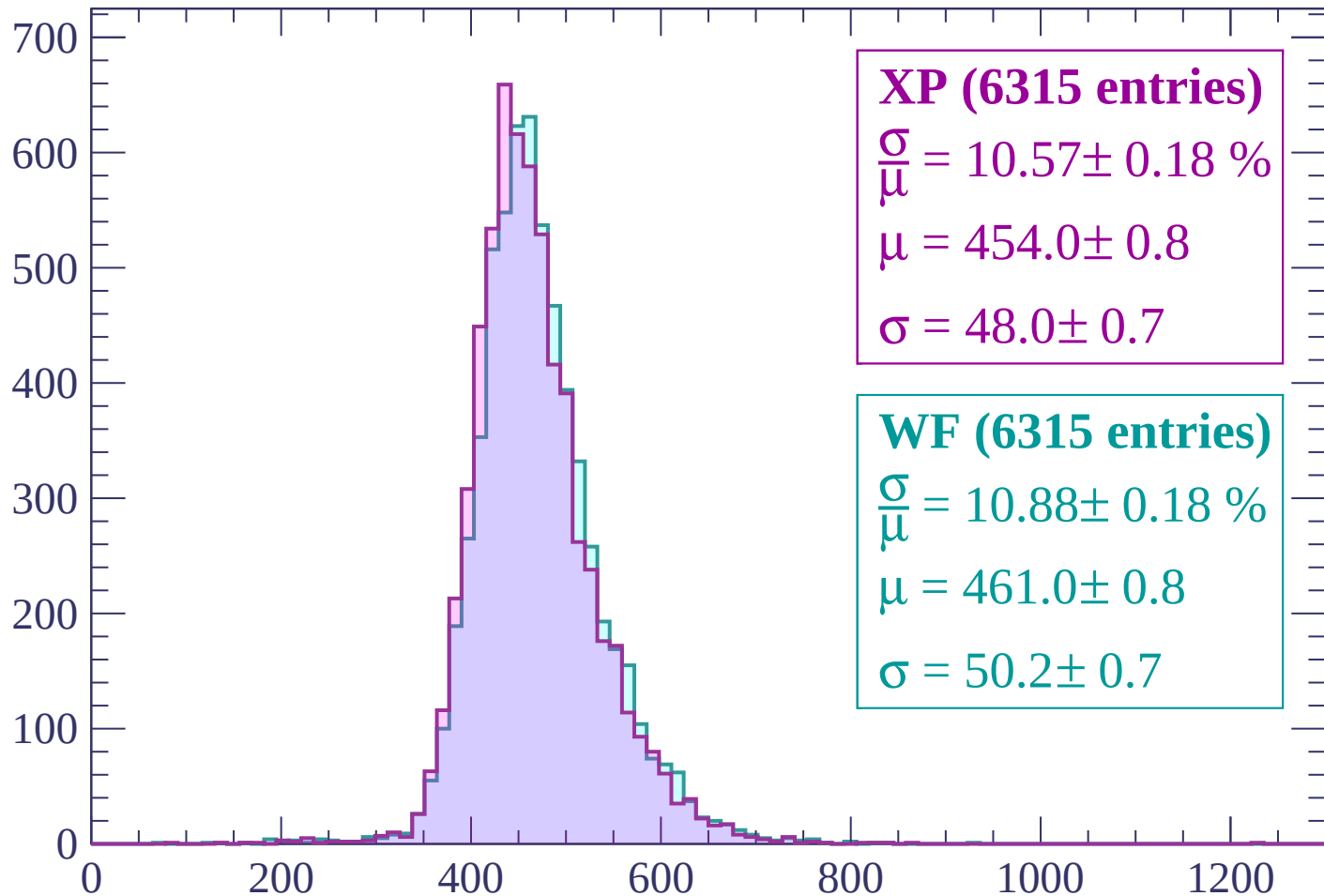
Energy loss | -1200 < p < -1140

Count



Energy loss | $-1140 < p < -1080$

Count



XP (6315 entries)

$$\frac{\sigma}{\mu} = 10.57 \pm 0.18 \%$$

$$\mu = 454.0 \pm 0.8$$

$$\sigma = 48.0 \pm 0.7$$

WF (6315 entries)

$$\frac{\sigma}{\mu} = 10.88 \pm 0.18 \%$$

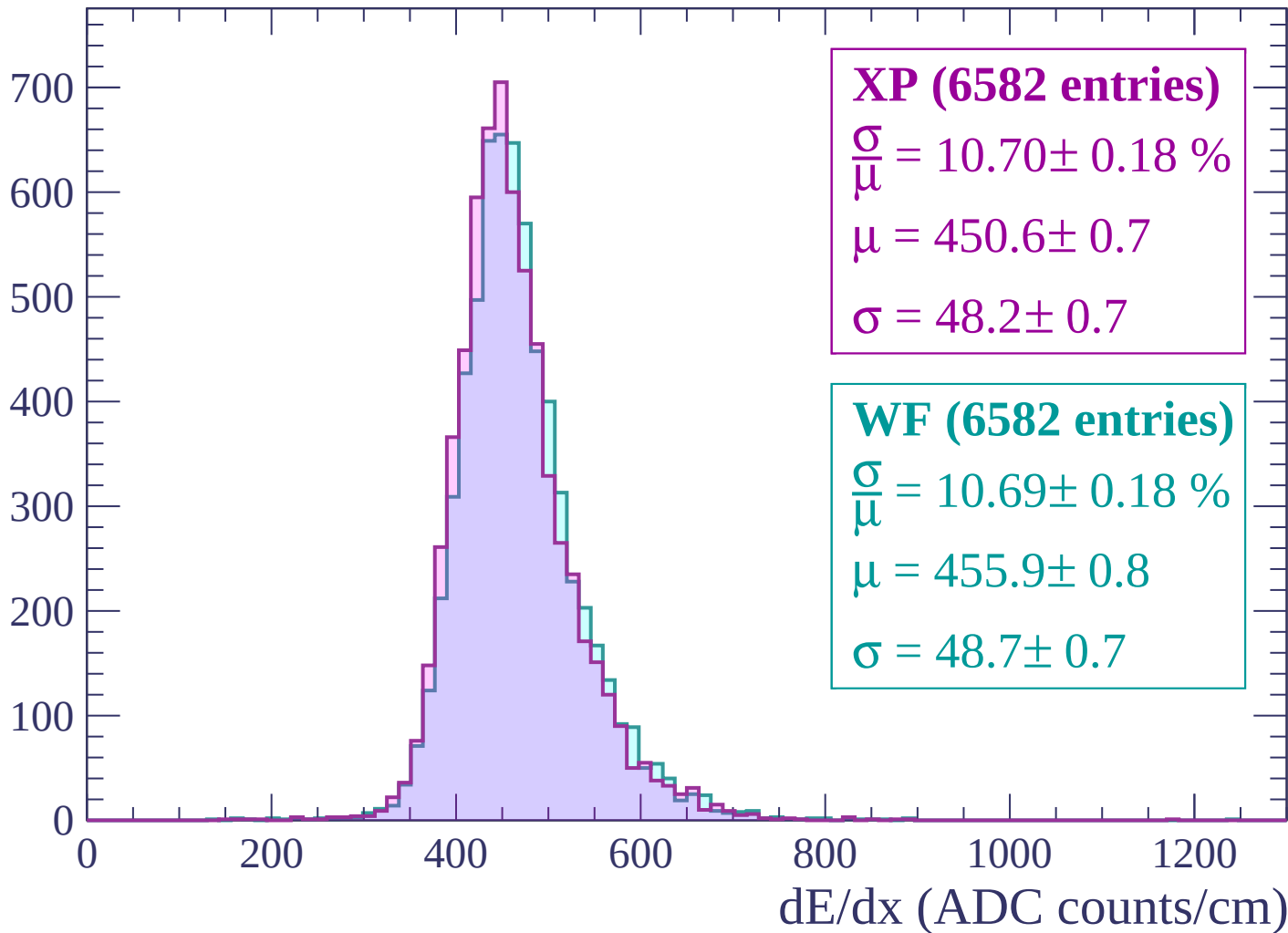
$$\mu = 461.0 \pm 0.8$$

$$\sigma = 50.2 \pm 0.7$$

dE/dx (ADC counts/cm)

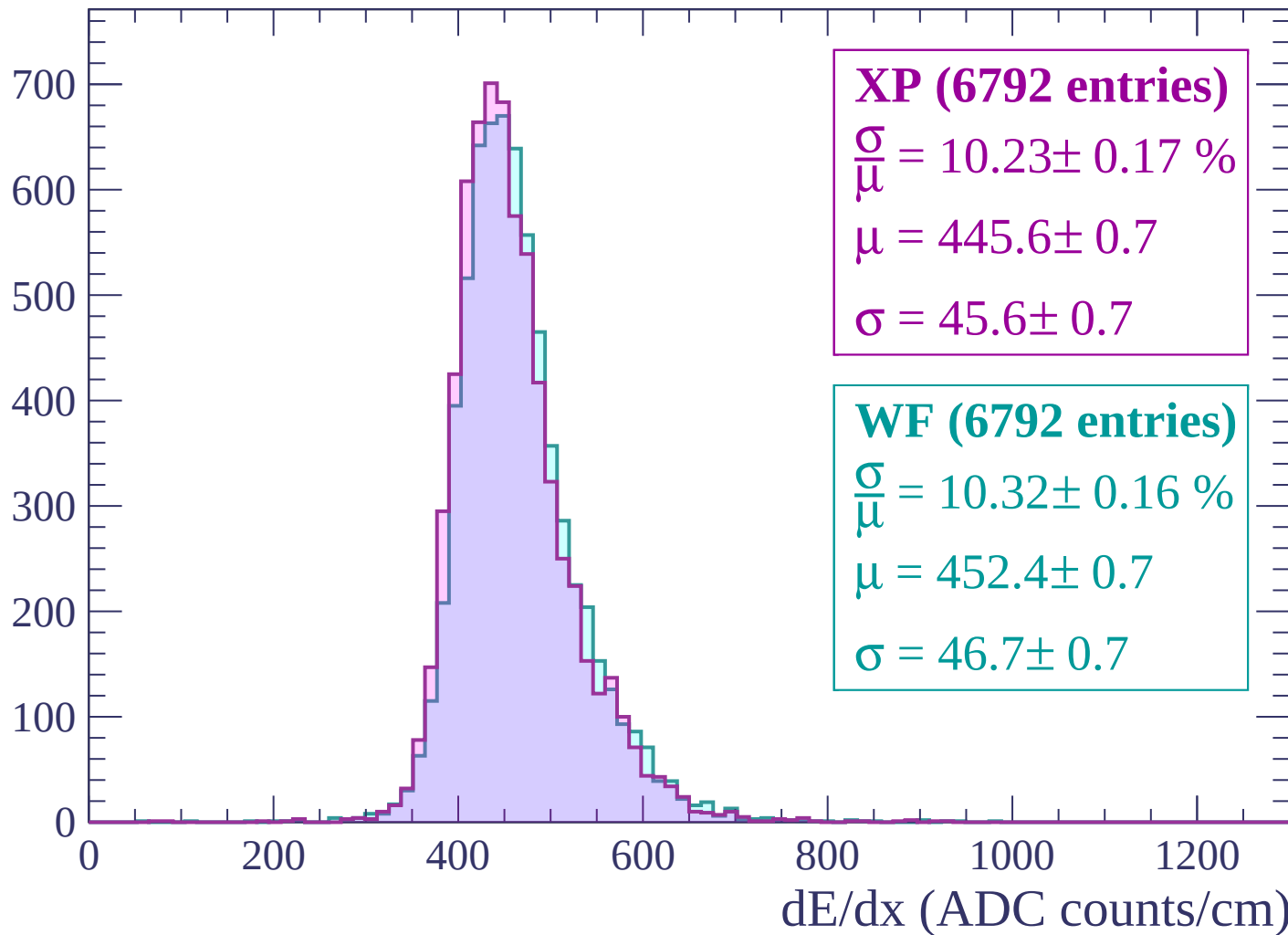
Energy loss | $-1080 < p < -1020$

Count



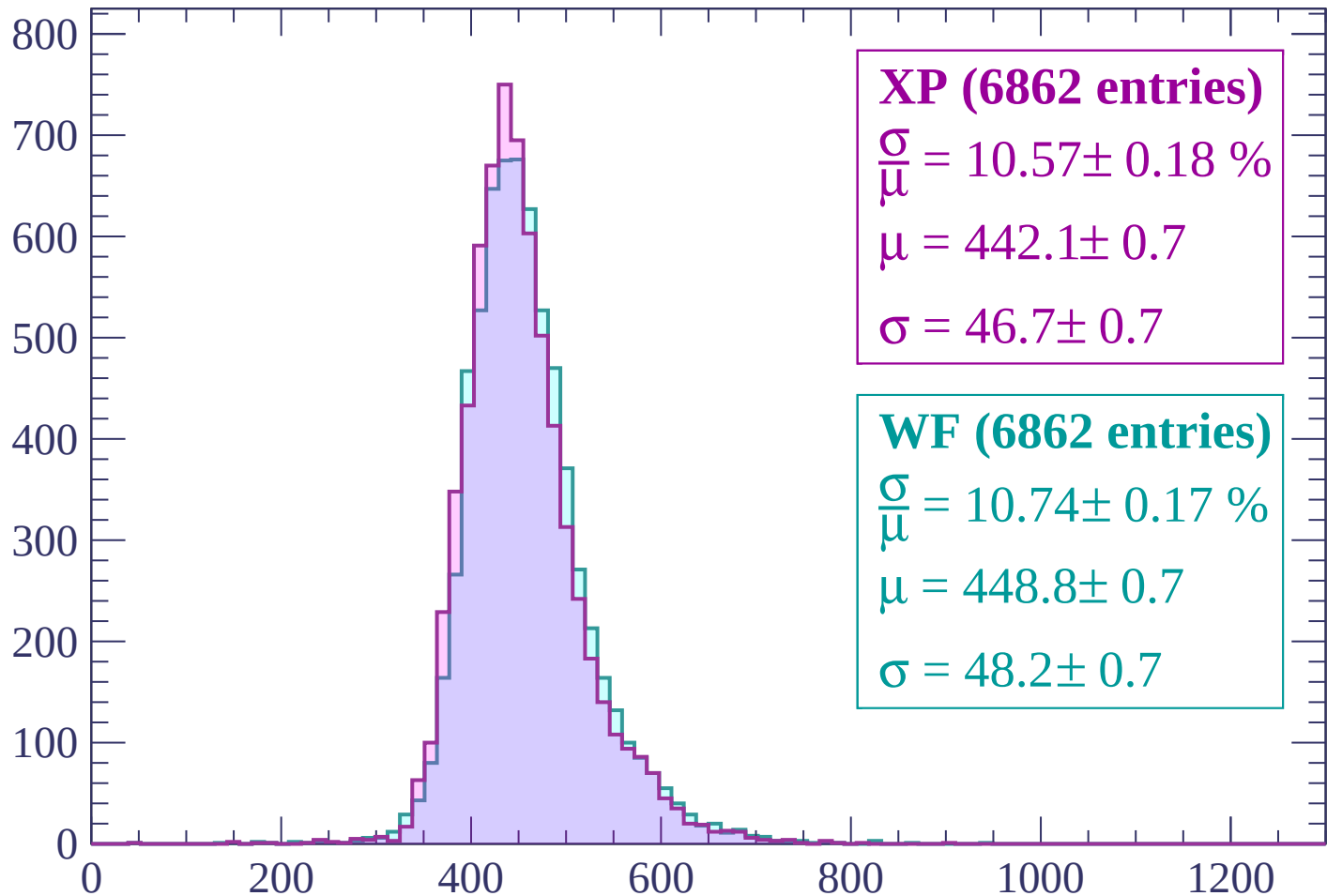
Energy loss | $-1020 < p < -960$

Count



Energy loss | $-960 < p < -900$

Count



XP (6862 entries)

$$\frac{\sigma}{\mu} = 10.57 \pm 0.18 \%$$

$$\mu = 442.1 \pm 0.7$$

$$\sigma = 46.7 \pm 0.7$$

WF (6862 entries)

$$\frac{\sigma}{\mu} = 10.74 \pm 0.17 \%$$

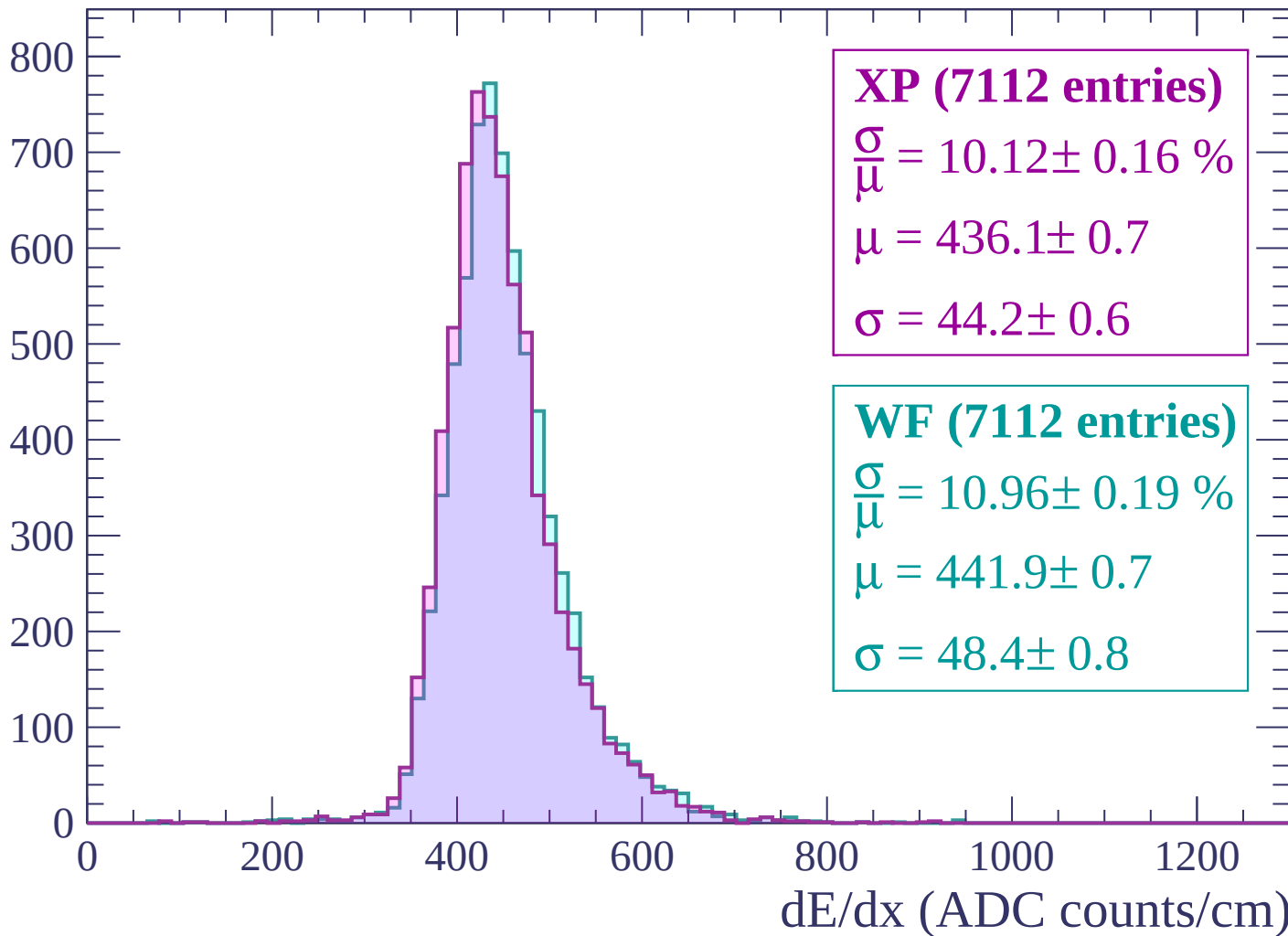
$$\mu = 448.8 \pm 0.7$$

$$\sigma = 48.2 \pm 0.7$$

dE/dx (ADC counts/cm)

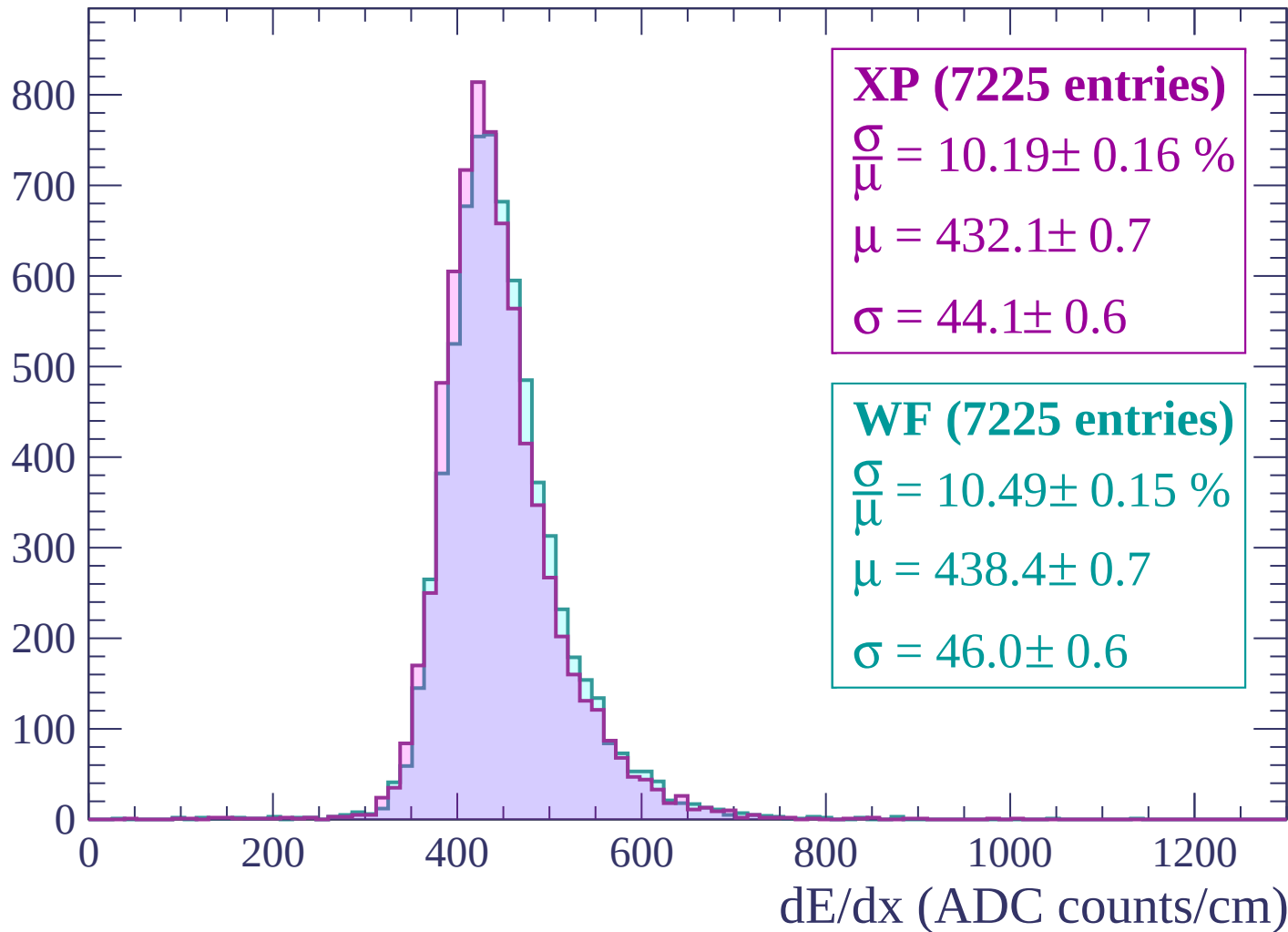
Energy loss | $-900 < p < -840$

Count



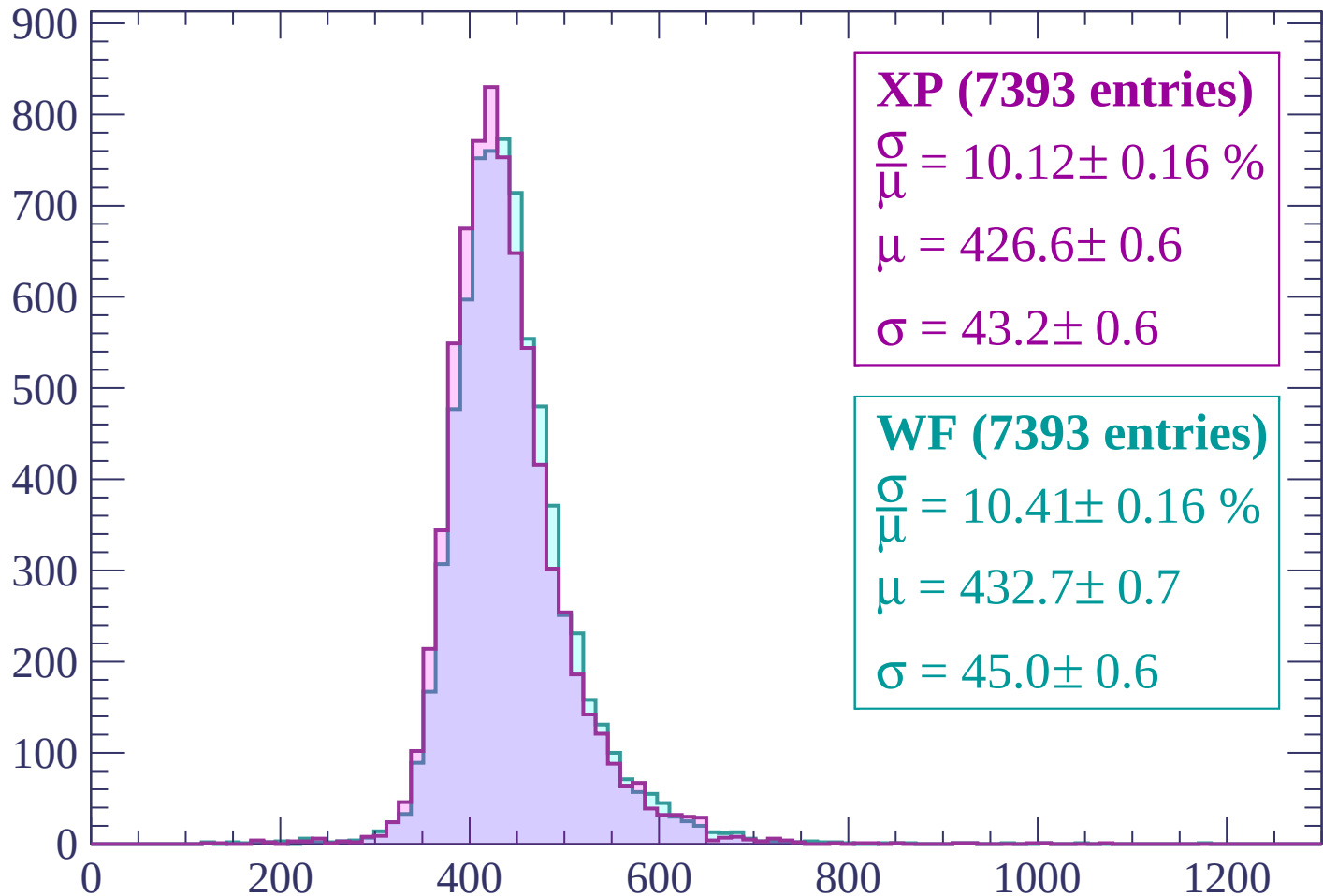
Energy loss | $-840 < p < -780$

Count



Energy loss | $-780 < p < -720$

Count



XP (7393 entries)

$$\frac{\sigma}{\mu} = 10.12 \pm 0.16 \%$$

$$\mu = 426.6 \pm 0.6$$

$$\sigma = 43.2 \pm 0.6$$

WF (7393 entries)

$$\frac{\sigma}{\mu} = 10.41 \pm 0.16 \%$$

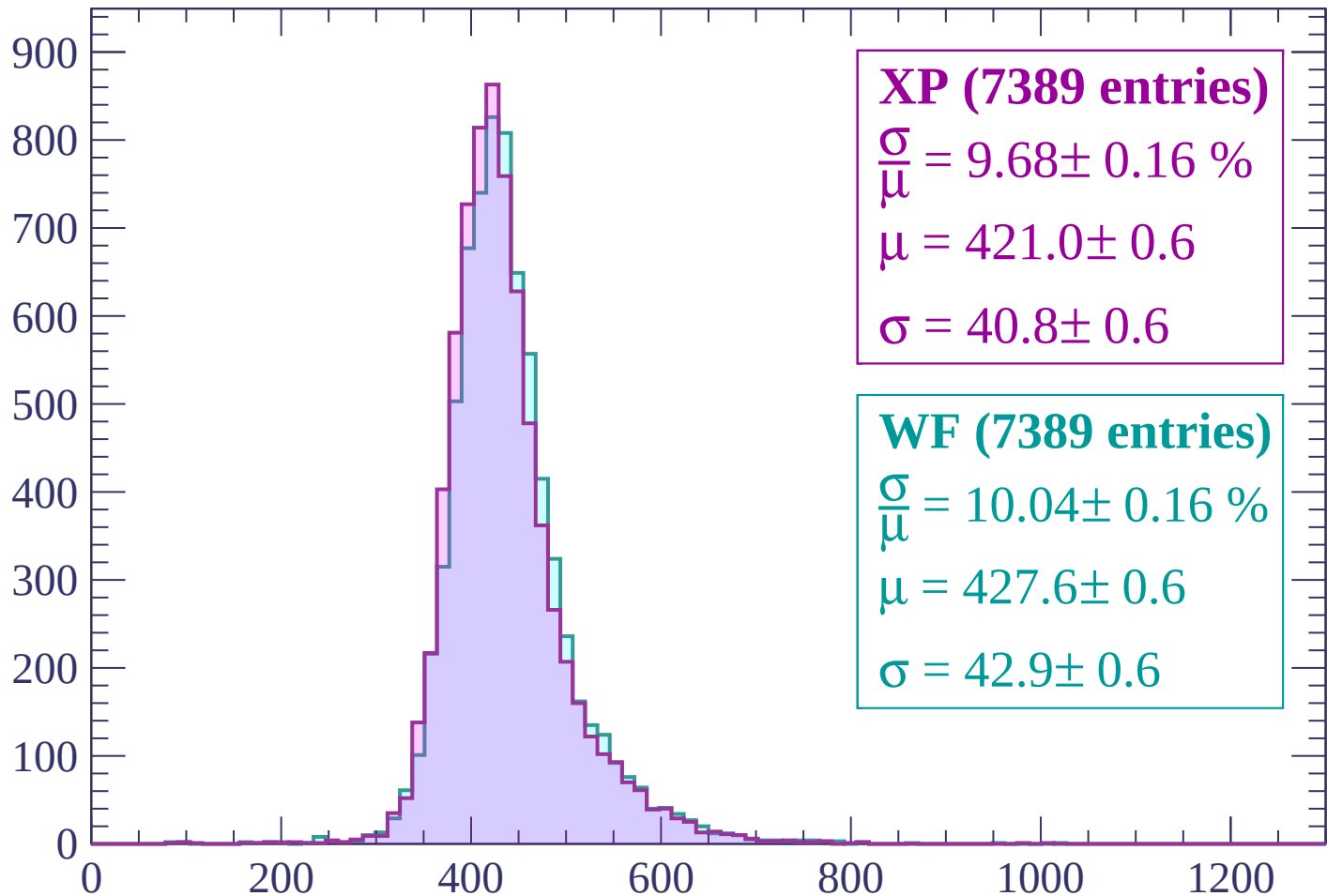
$$\mu = 432.7 \pm 0.7$$

$$\sigma = 45.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP (7389 entries)

$$\frac{\sigma}{\mu} = 9.68 \pm 0.16 \%$$

$$\mu = 421.0 \pm 0.6$$

$$\sigma = 40.8 \pm 0.6$$

WF (7389 entries)

$$\frac{\sigma}{\mu} = 10.04 \pm 0.16 \%$$

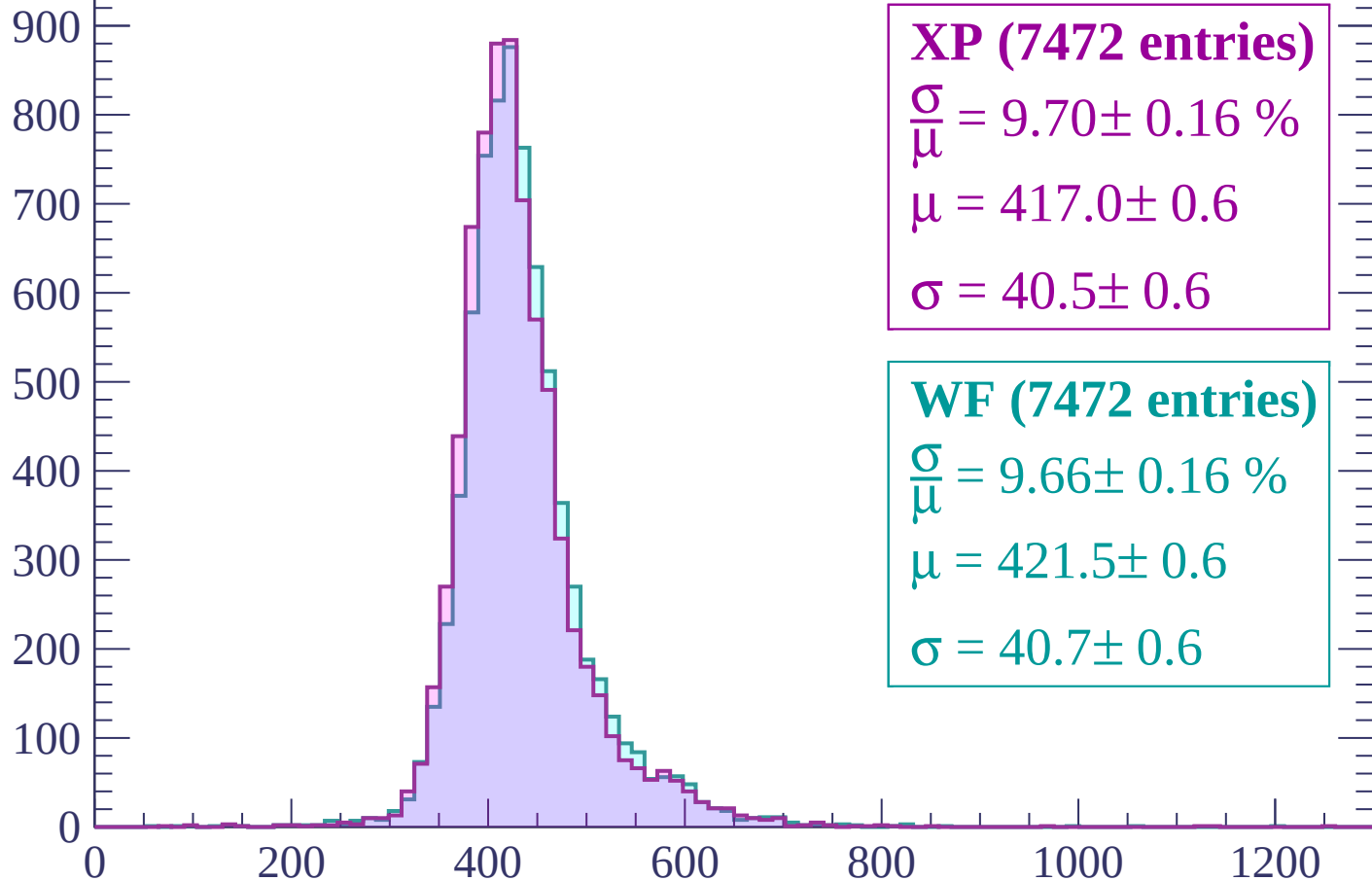
$$\mu = 427.6 \pm 0.6$$

$$\sigma = 42.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count



XP (7472 entries)

$$\frac{\sigma}{\mu} = 9.70 \pm 0.16 \%$$

$$\mu = 417.0 \pm 0.6$$

$$\sigma = 40.5 \pm 0.6$$

WF (7472 entries)

$$\frac{\sigma}{\mu} = 9.66 \pm 0.16 \%$$

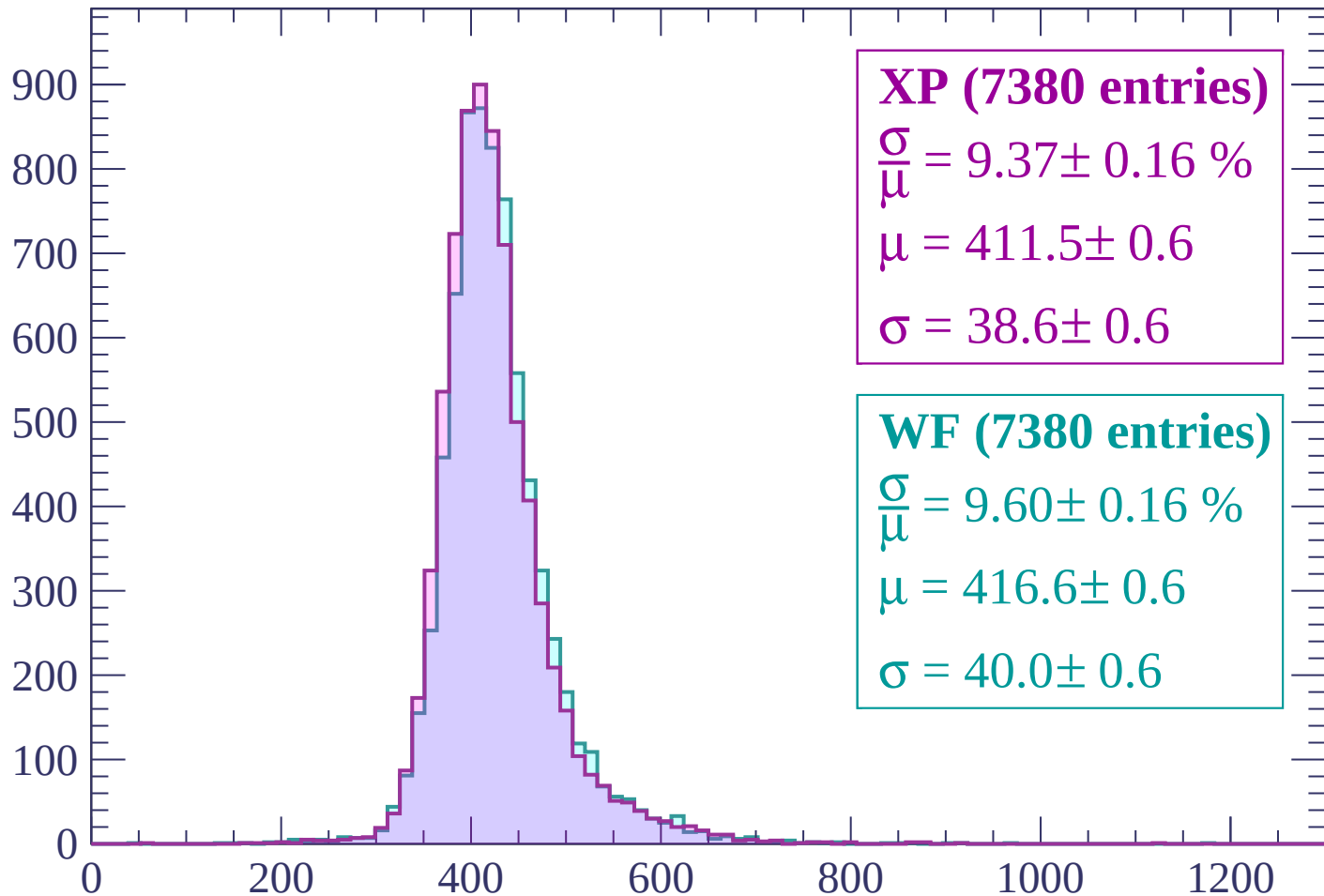
$$\mu = 421.5 \pm 0.6$$

$$\sigma = 40.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count



XP (7380 entries)

$\frac{\sigma}{\mu} = 9.37 \pm 0.16 \%$

$\mu = 411.5 \pm 0.6$

$\sigma = 38.6 \pm 0.6$

WF (7380 entries)

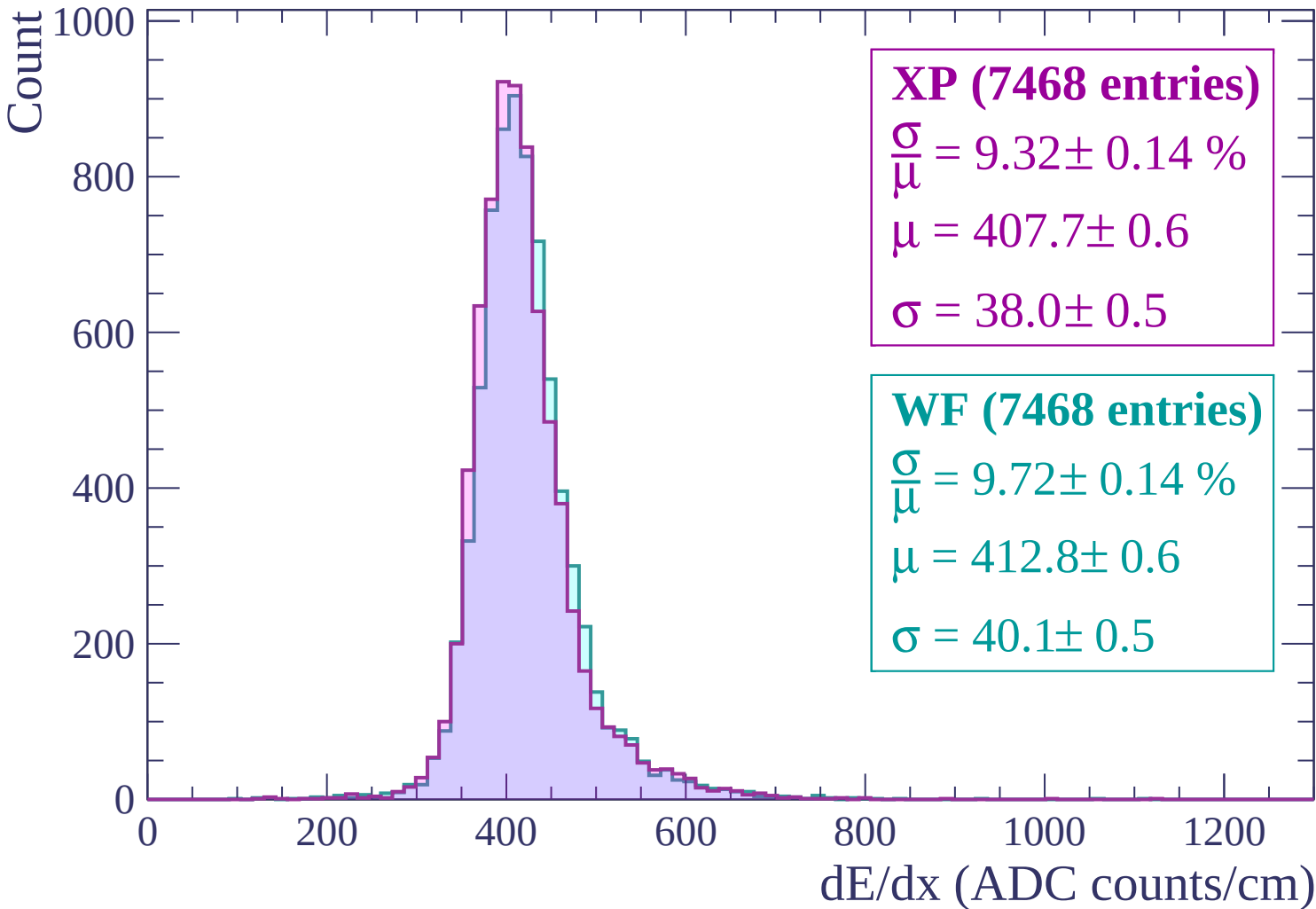
$\frac{\sigma}{\mu} = 9.60 \pm 0.16 \%$

$\mu = 416.6 \pm 0.6$

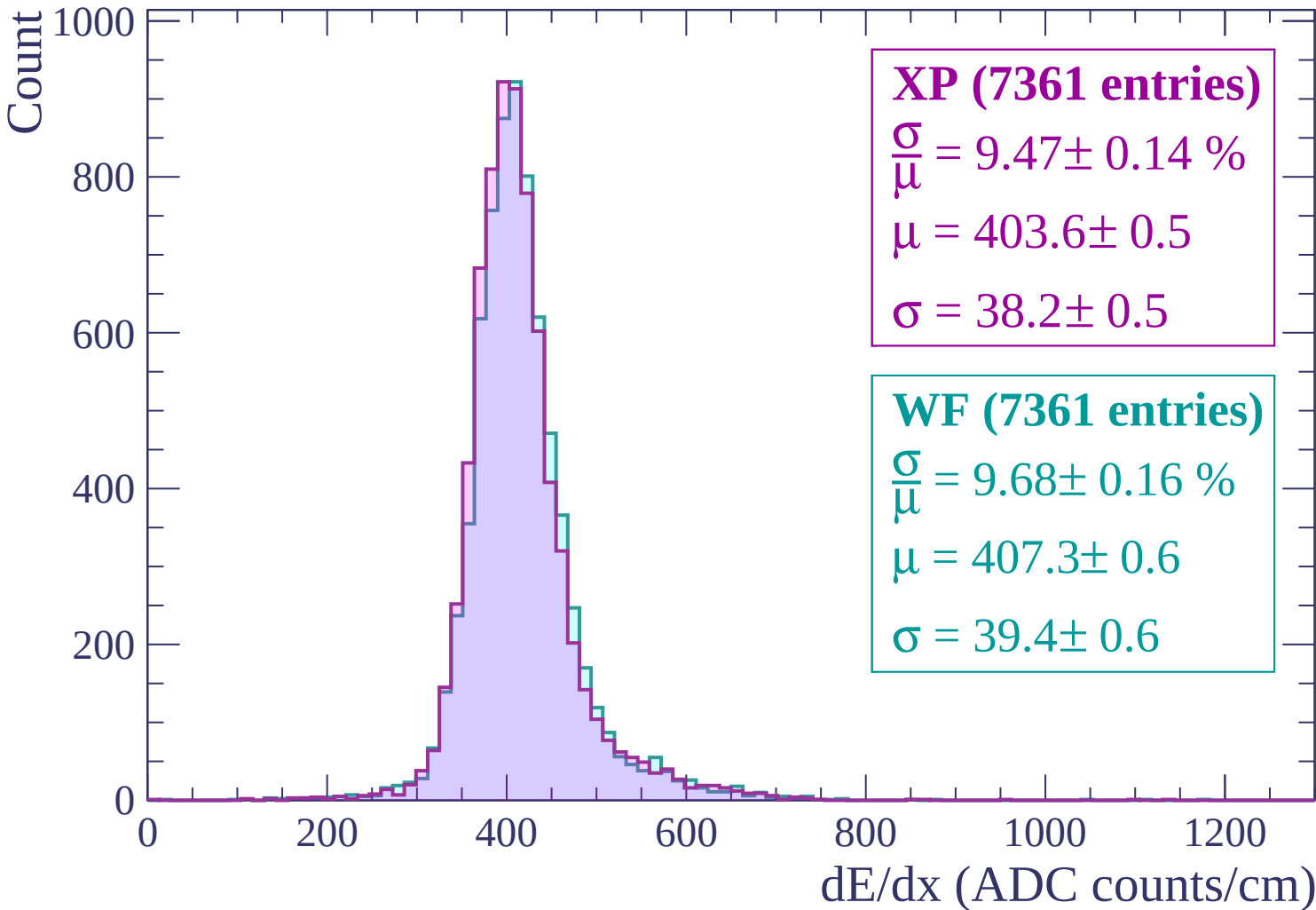
$\sigma = 40.0 \pm 0.6$

dE/dx (ADC counts/cm)

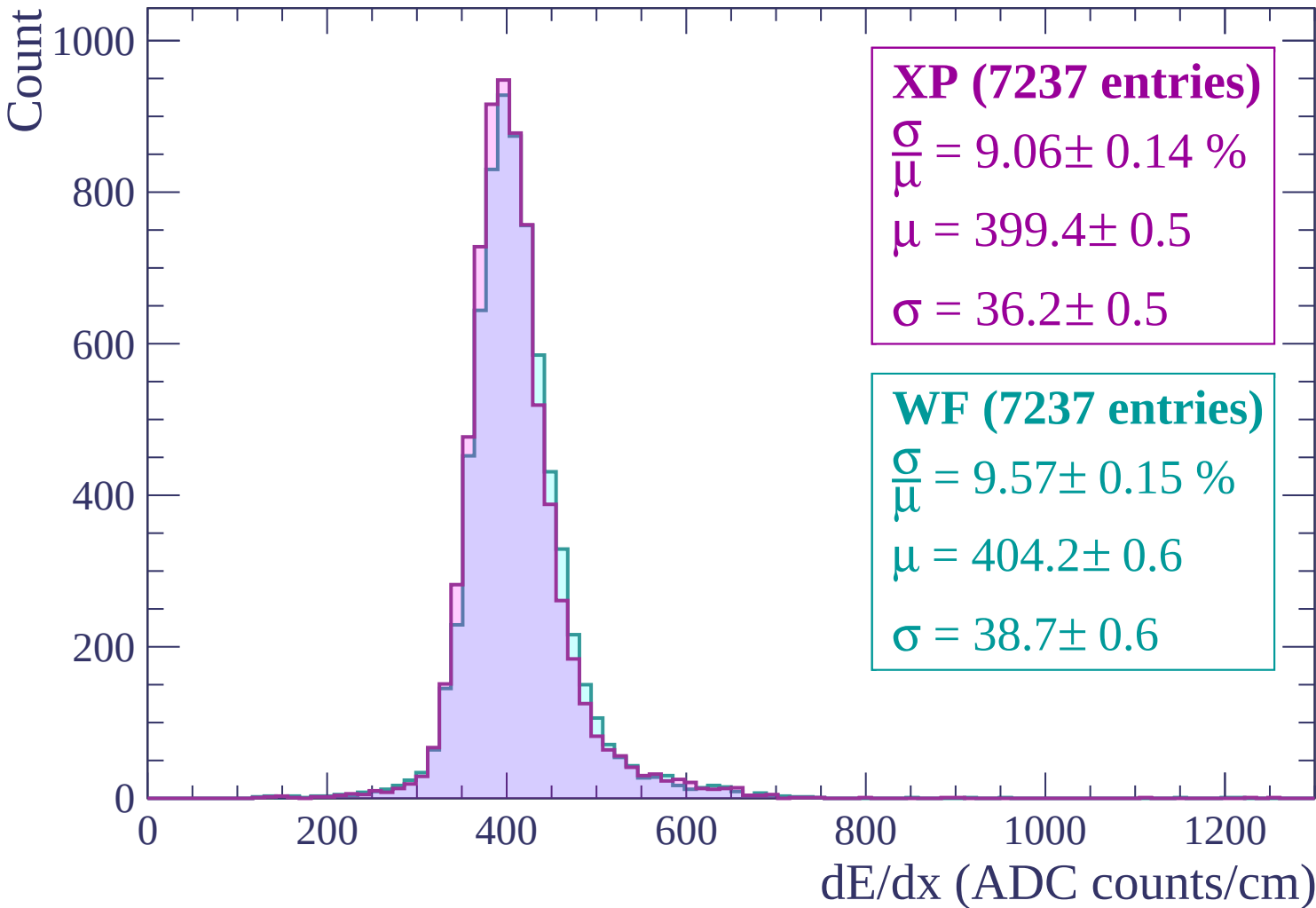
Energy loss | $-540 < p < -480$



Energy loss | $-480 < p < -420$

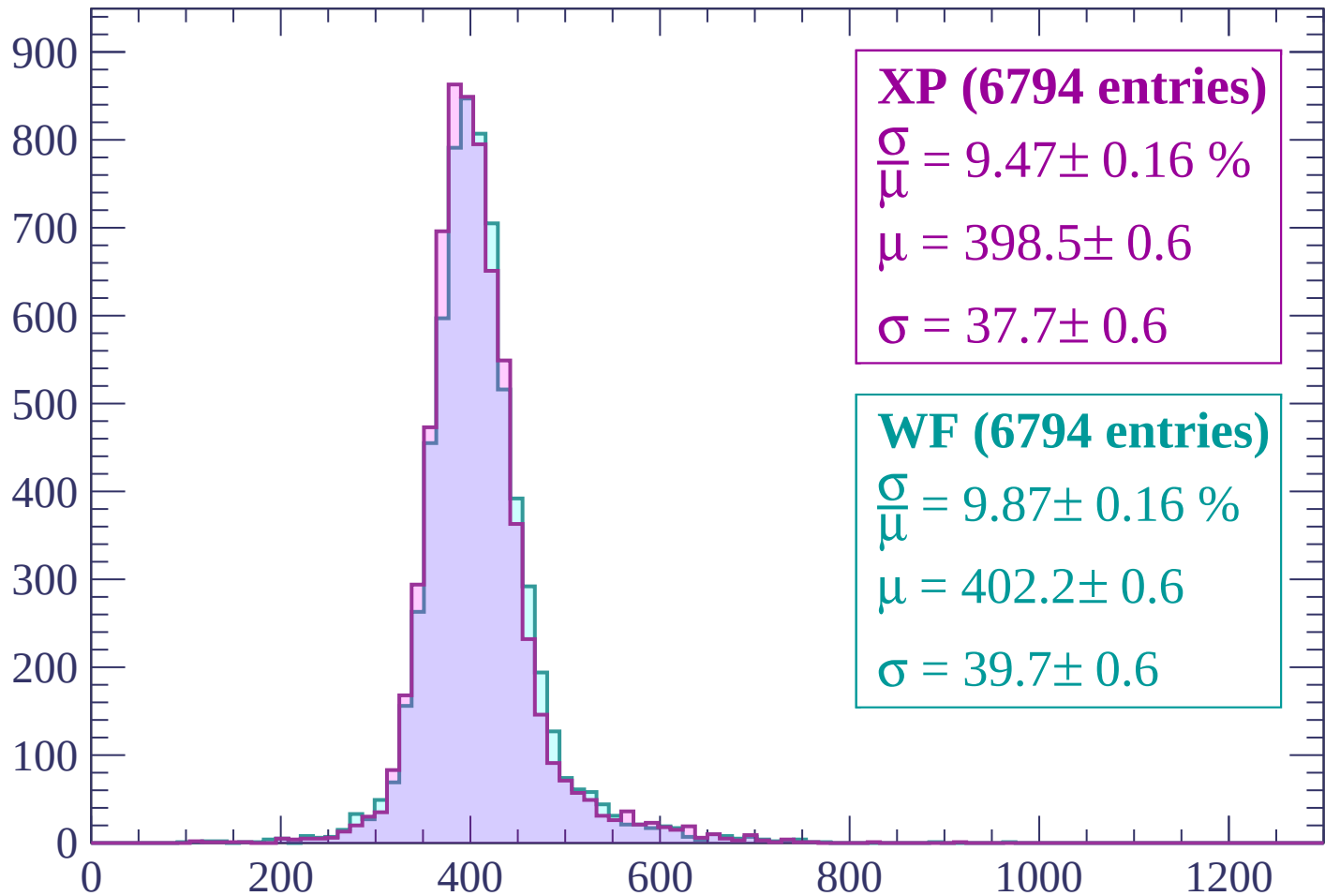


Energy loss | $-420 < p < -360$



Energy loss | $-360 < p < -300$

Count



XP (6794 entries)

$$\frac{\sigma}{\mu} = 9.47 \pm 0.16 \%$$

$$\mu = 398.5 \pm 0.6$$

$$\sigma = 37.7 \pm 0.6$$

WF (6794 entries)

$$\frac{\sigma}{\mu} = 9.87 \pm 0.16 \%$$

$$\mu = 402.2 \pm 0.6$$

$$\sigma = 39.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count

800
700
600
500
400
300
200
100
0

XP (6018 entries)

$$\frac{\sigma}{\mu} = 9.27 \pm 0.17 \%$$

$$\mu = 399.1 \pm 0.6$$

$$\sigma = 37.0 \pm 0.6$$

WF (6018 entries)

$$\frac{\sigma}{\mu} = 9.89 \pm 0.18 \%$$

$$\mu = 402.1 \pm 0.6$$

$$\sigma = 39.8 \pm 0.6$$

0

200

400

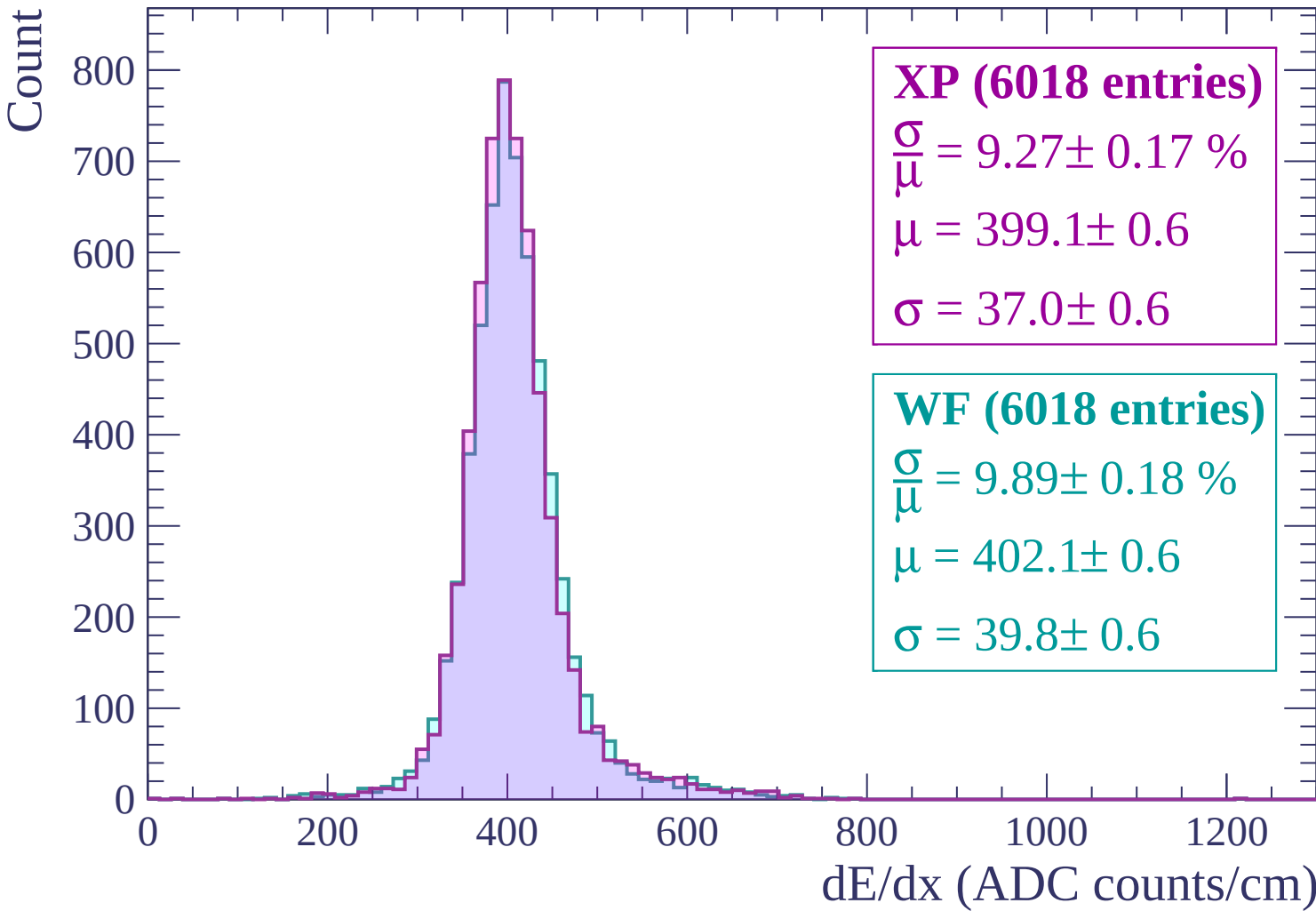
600

800

1000

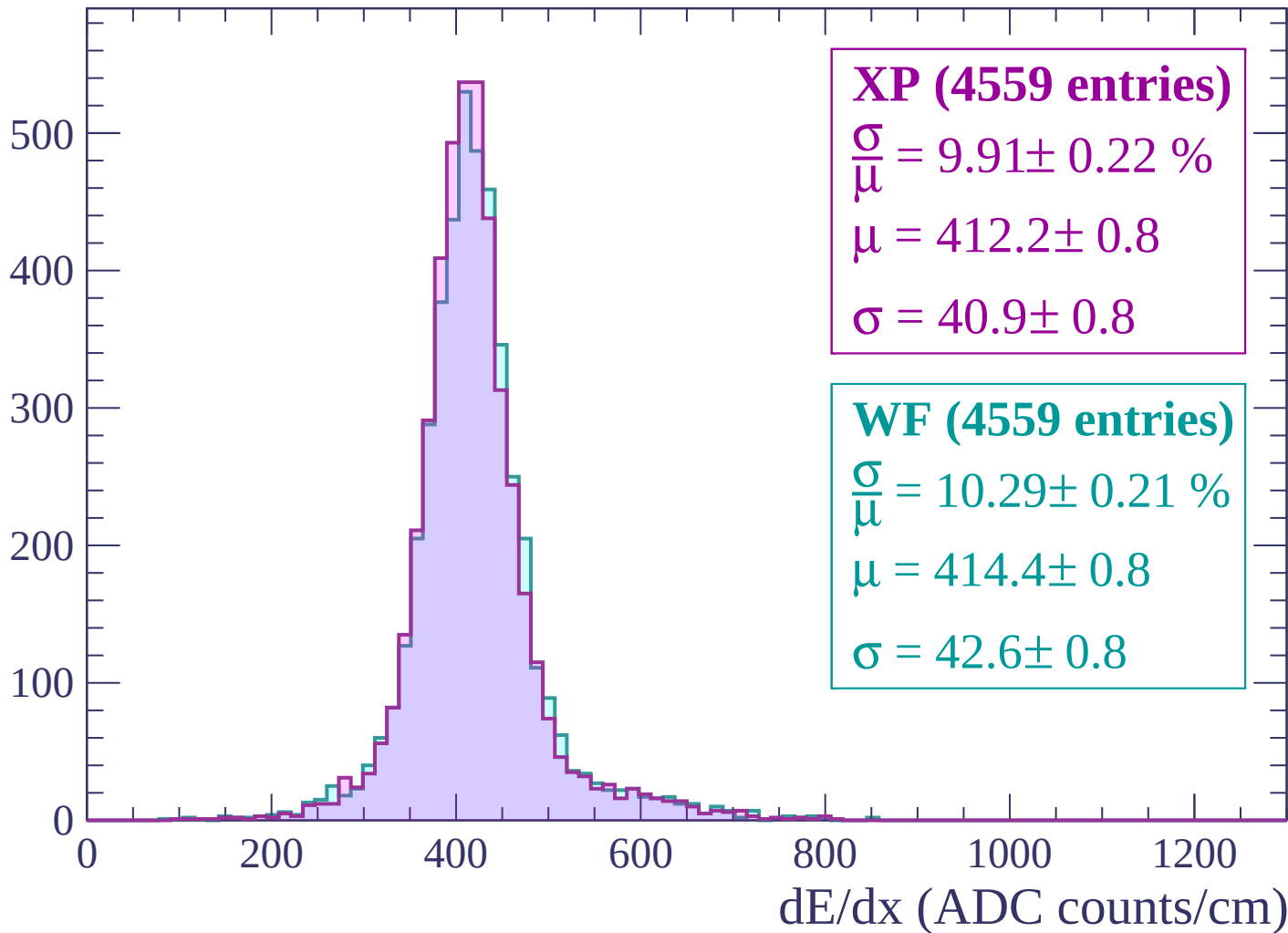
1200

dE/dx (ADC counts/cm)



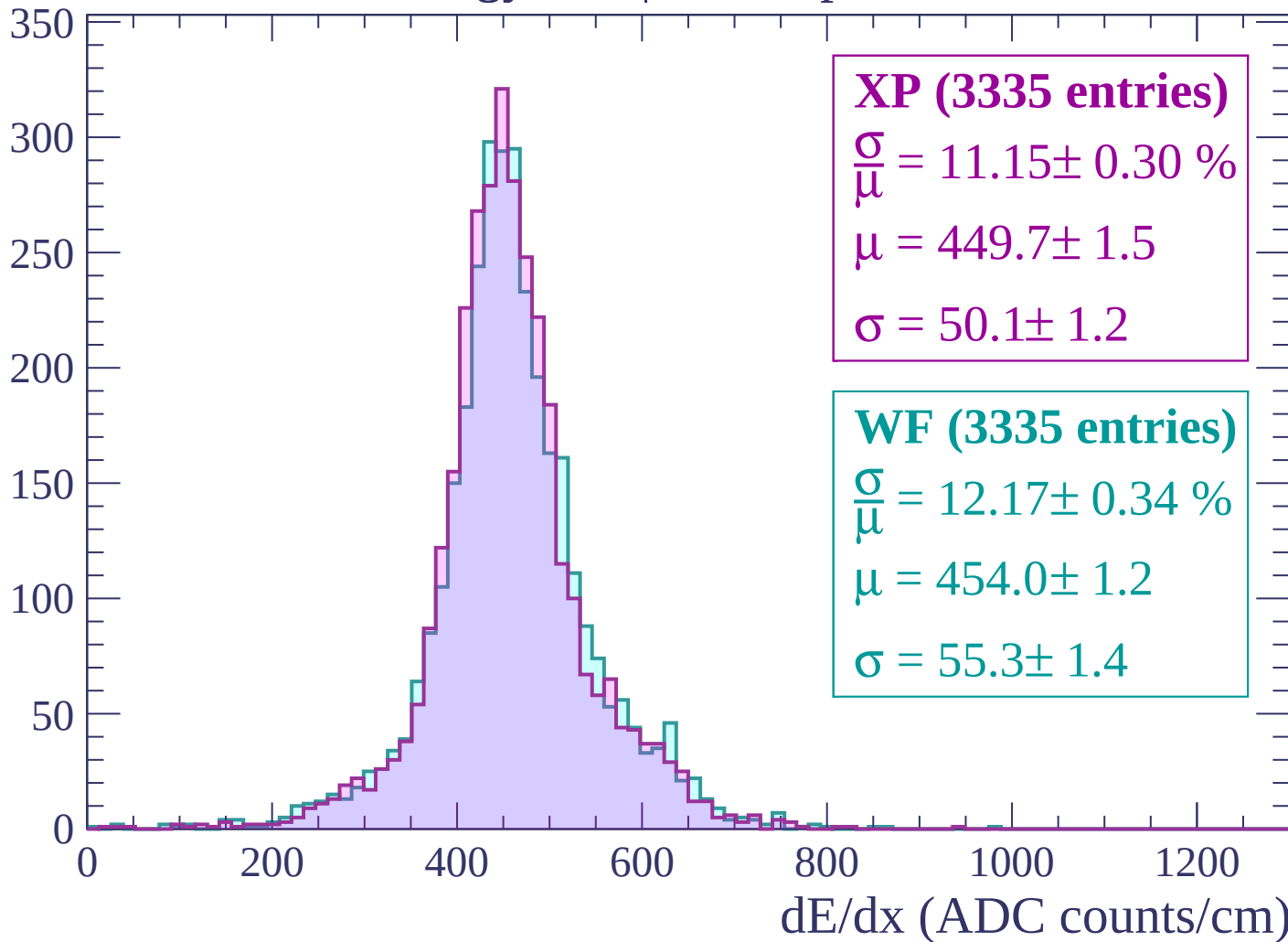
Energy loss | $-240 < p < -180$

Count



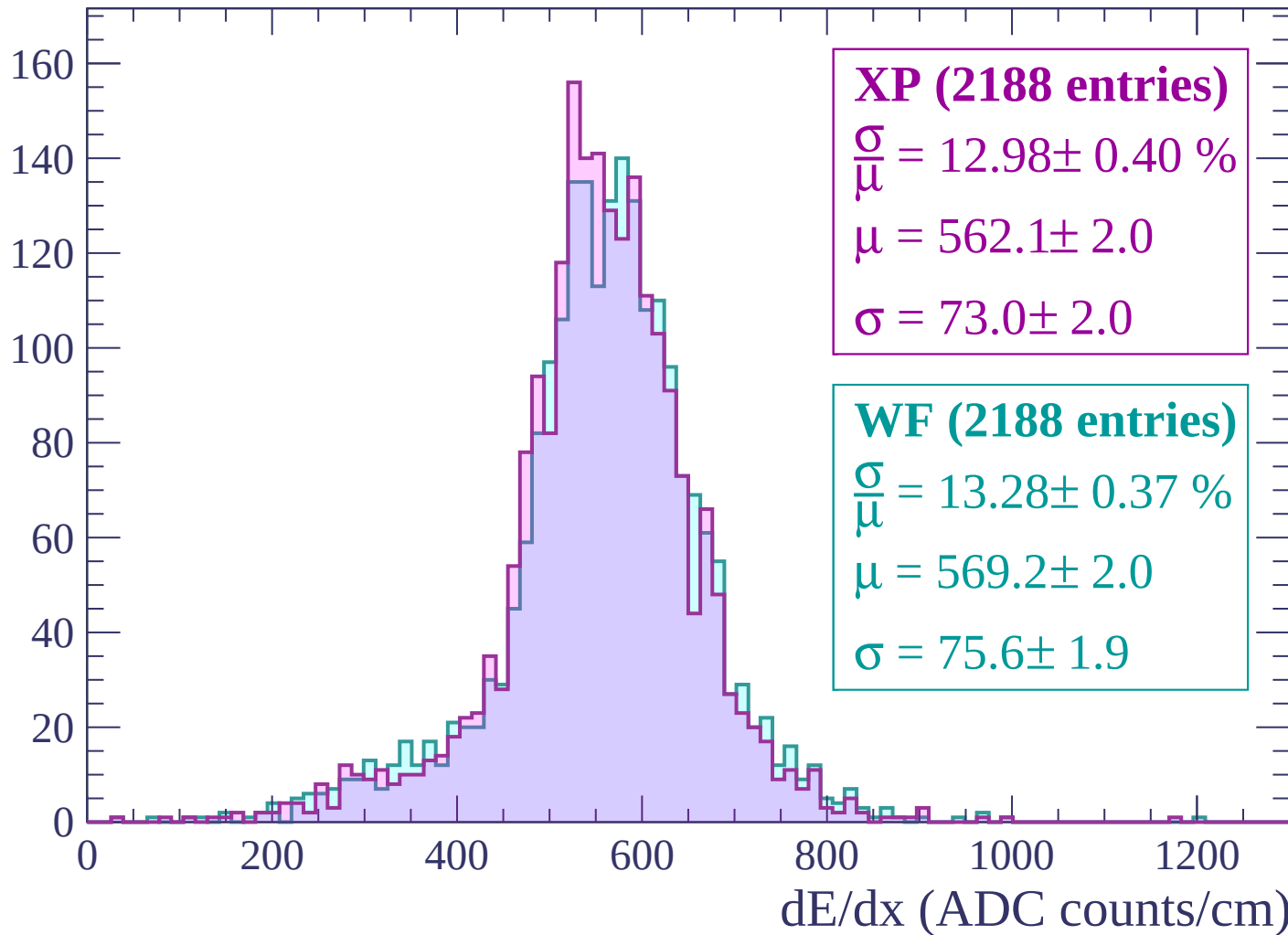
Energy loss | -180 < p < -120

Count



Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count

350
300
250
200
150
100
50
0

XP (3755 entries)

$$\frac{\sigma}{\mu} = 9.65 \pm 0.21 \%$$

$$\mu = 554.9 \pm 1.1$$

$$\sigma = 53.6 \pm 1.1$$

WF (3755 entries)

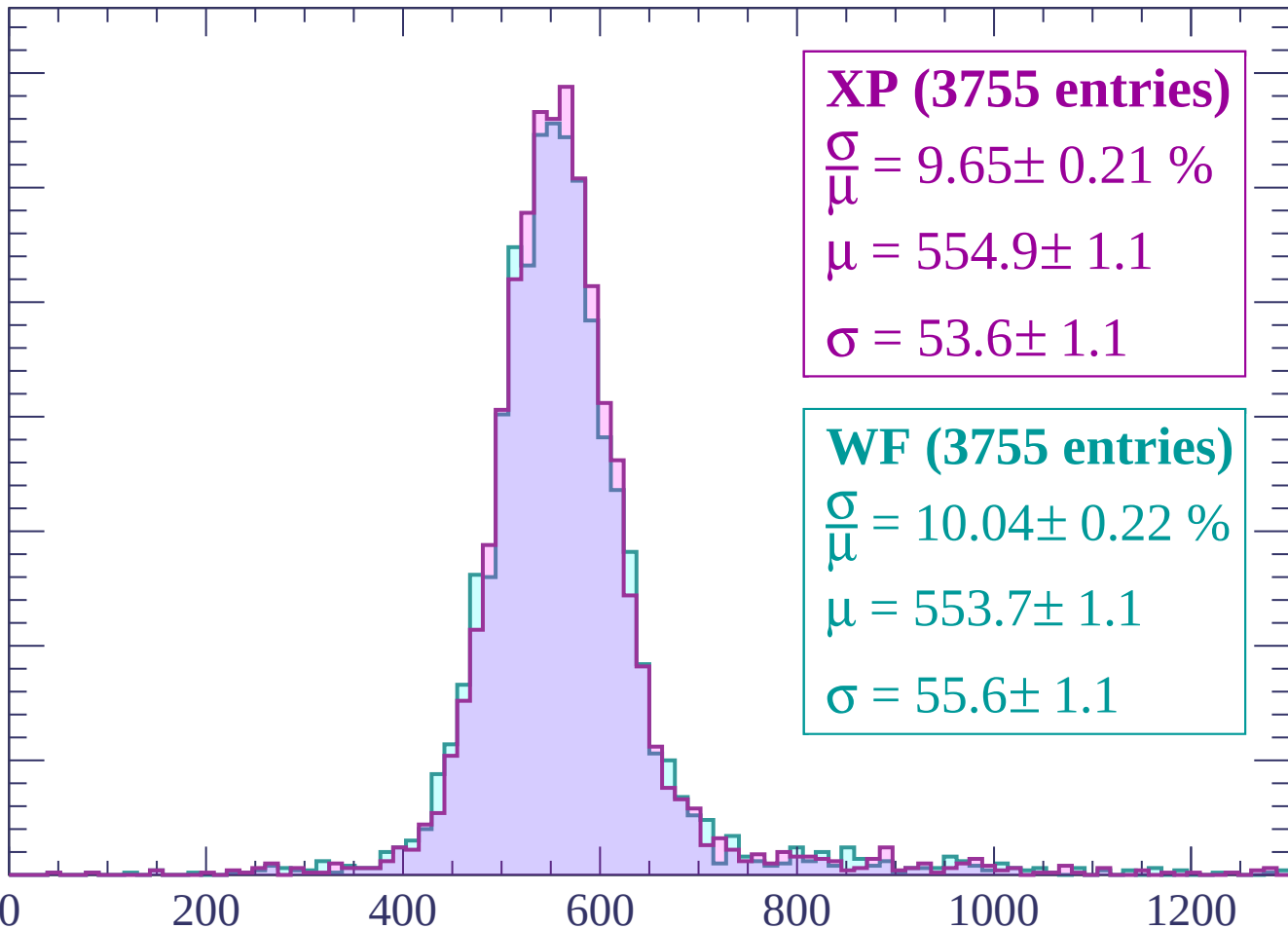
$$\frac{\sigma}{\mu} = 10.04 \pm 0.22 \%$$

$$\mu = 553.7 \pm 1.1$$

$$\sigma = 55.6 \pm 1.1$$

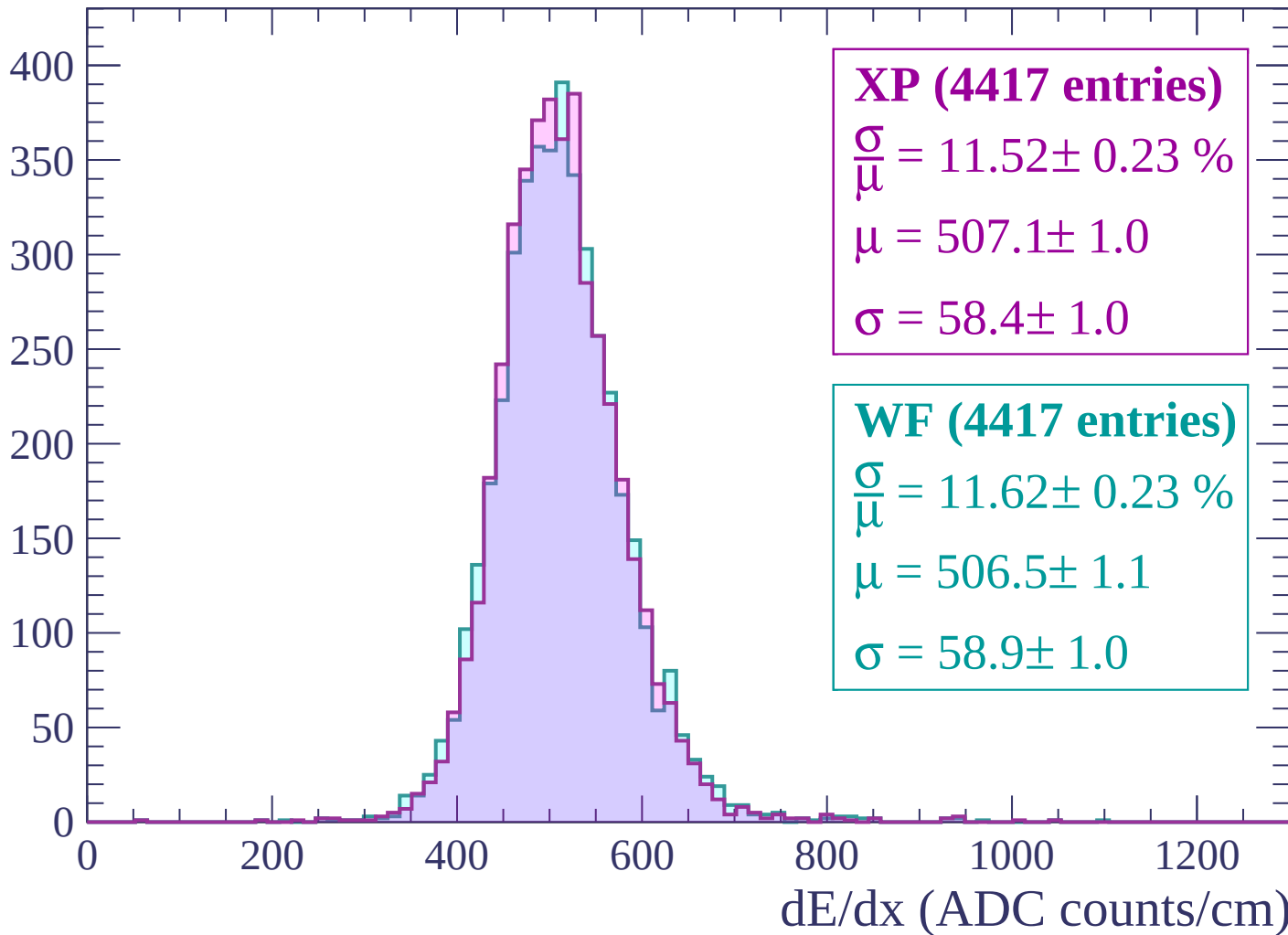
0 200 400 600 800 1000 1200

dE/dx (ADC counts/cm)



Energy loss | $0 < p < 60$

Count



Energy loss | $60 < p < 120$

Count

300

250

200

150

100

50

0

0

200

400

600

800

1000

1200

XP (2948 entries)

$$\frac{\sigma}{\mu} = 9.47 \pm 0.23 \%$$

$$\mu = 546.4 \pm 1.2$$

$$\sigma = 51.7 \pm 1.1$$

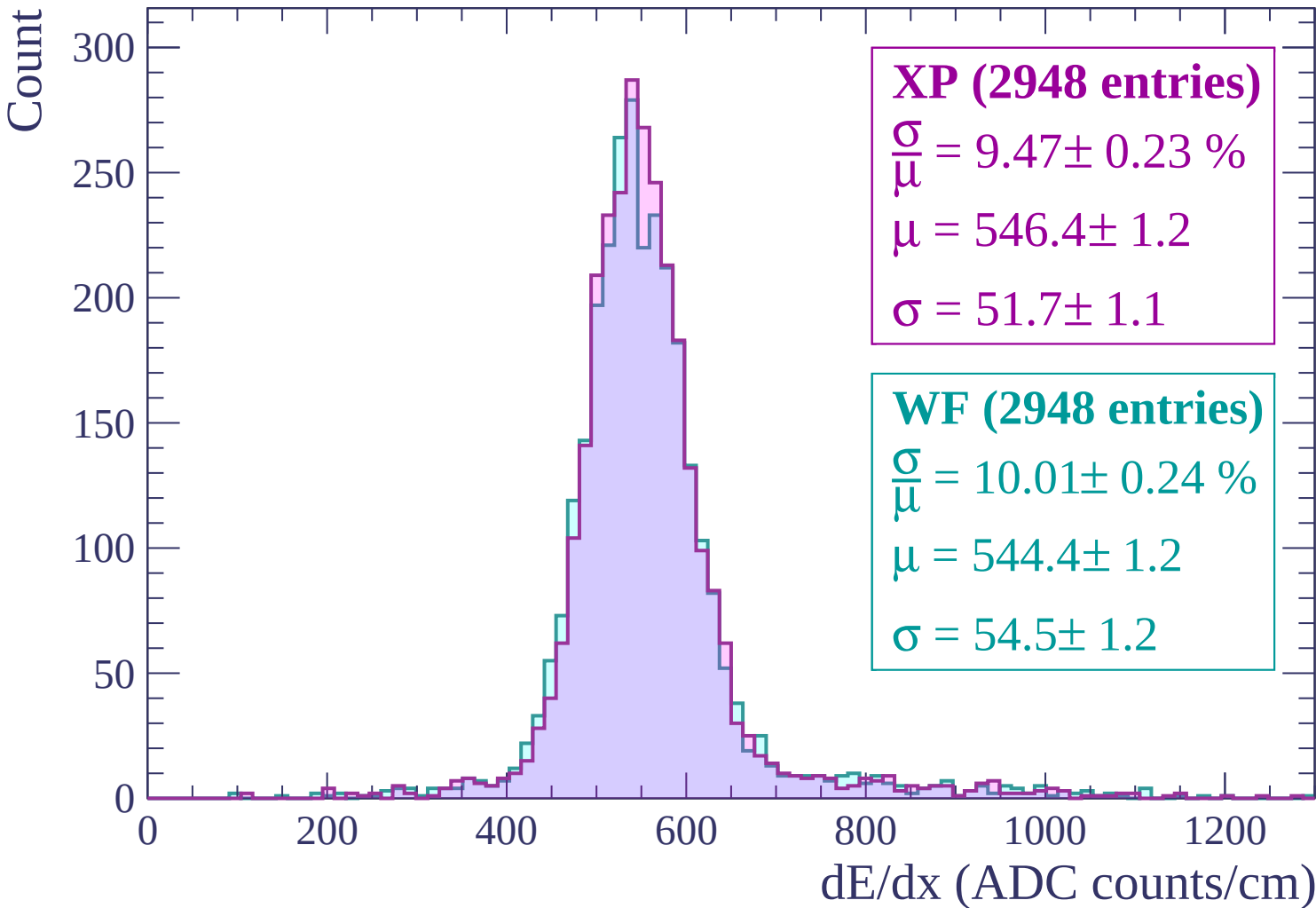
WF (2948 entries)

$$\frac{\sigma}{\mu} = 10.01 \pm 0.24 \%$$

$$\mu = 544.4 \pm 1.2$$

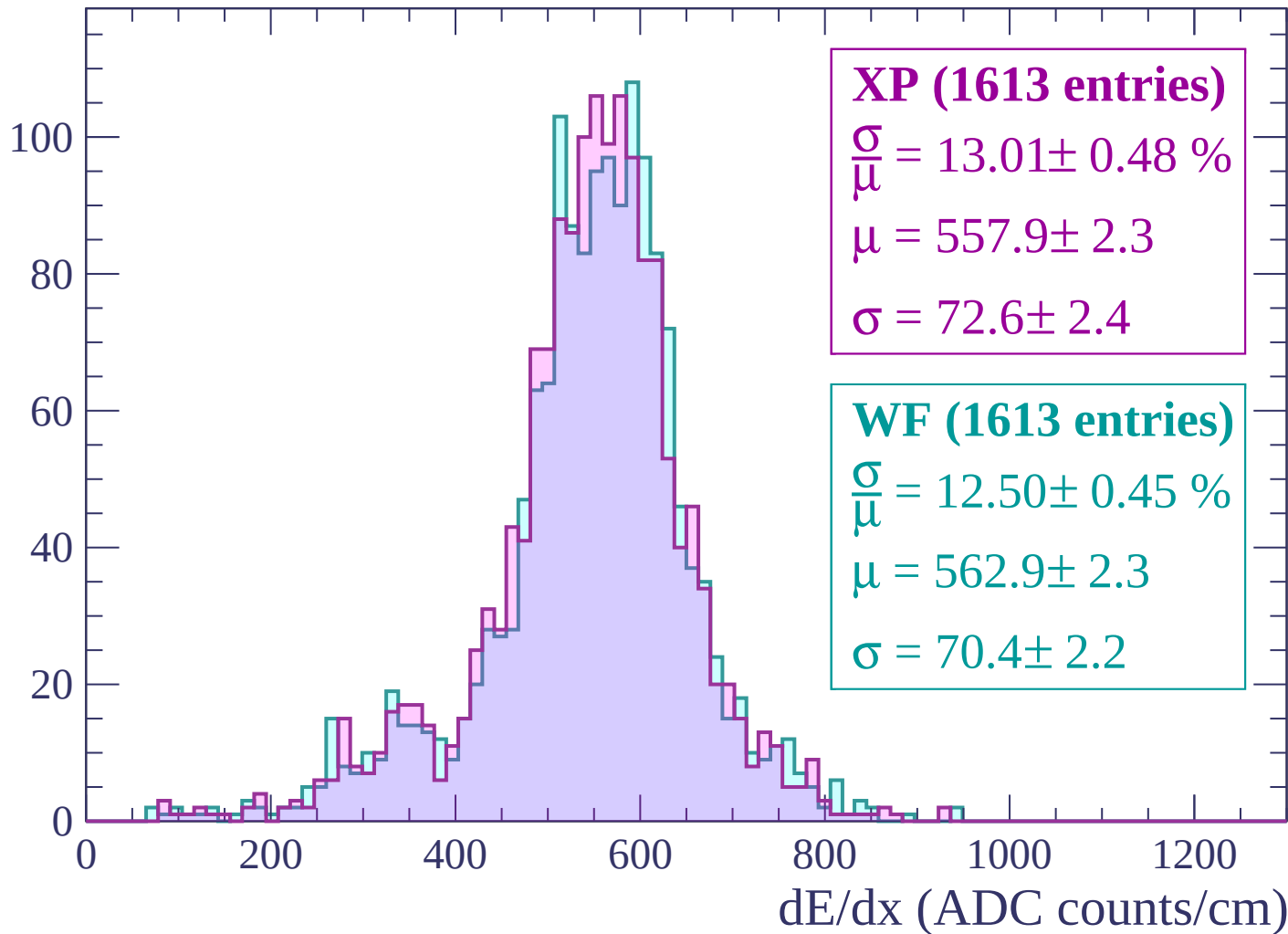
$$\sigma = 54.5 \pm 1.2$$

dE/dx (ADC counts/cm)



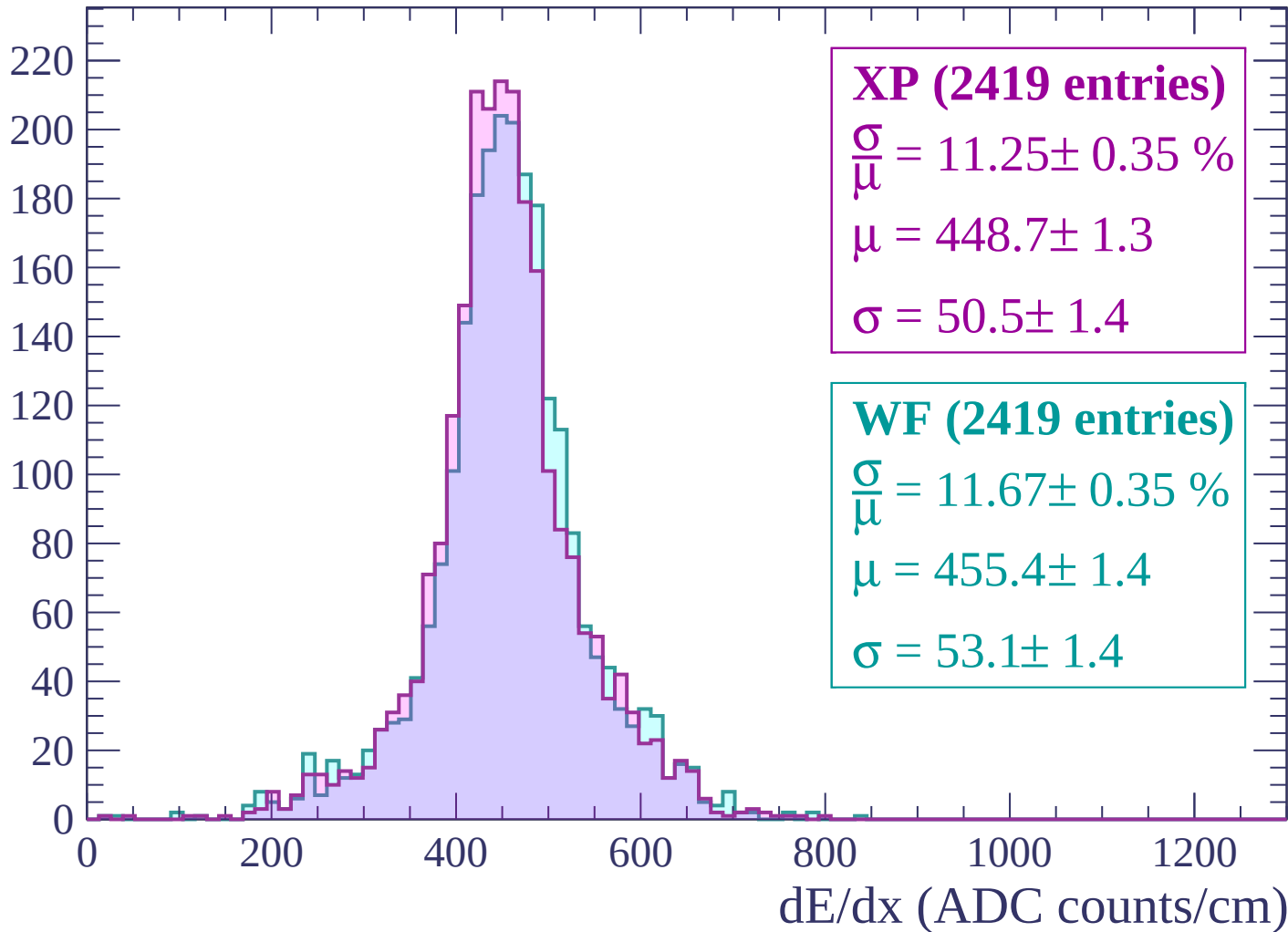
Energy loss | $120 < p < 180$

Count



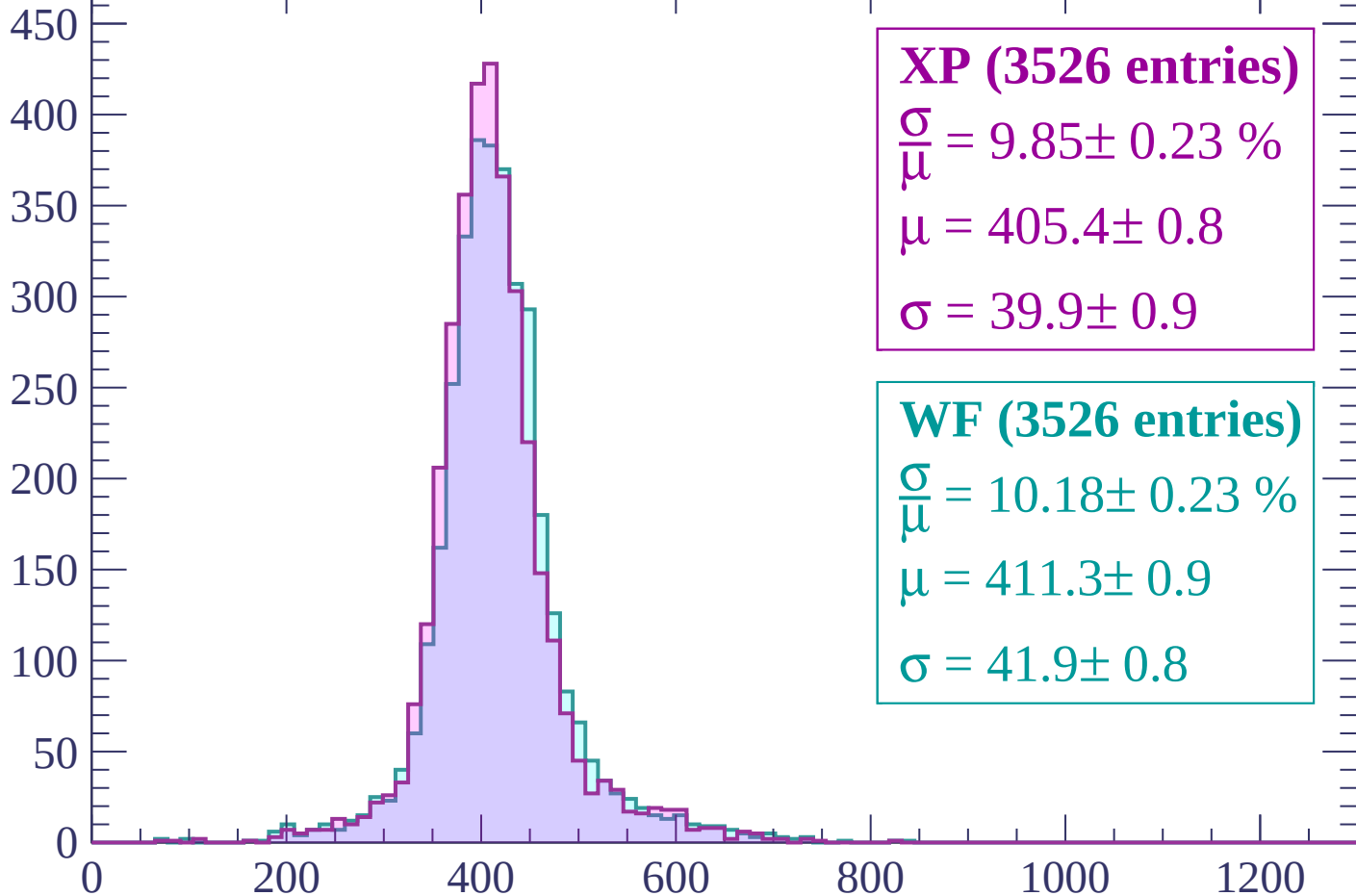
Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count



XP (3526 entries)

$\frac{\sigma}{\mu} = 9.85 \pm 0.23 \%$

$\mu = 405.4 \pm 0.8$

$\sigma = 39.9 \pm 0.9$

WF (3526 entries)

$\frac{\sigma}{\mu} = 10.18 \pm 0.23 \%$

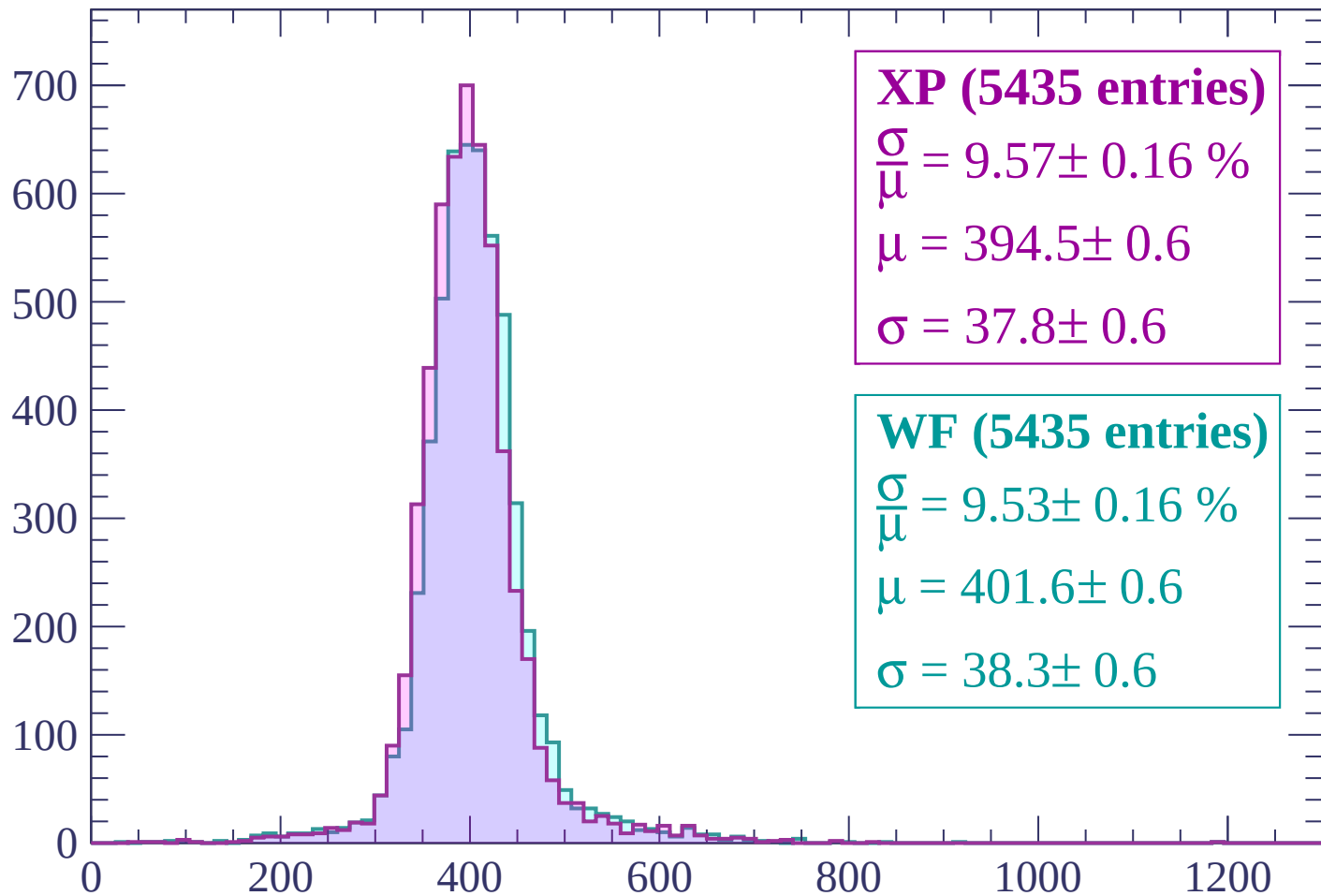
$\mu = 411.3 \pm 0.9$

$\sigma = 41.9 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (5435 entries)

$\frac{\sigma}{\mu} = 9.57 \pm 0.16 \%$

$\mu = 394.5 \pm 0.6$

$\sigma = 37.8 \pm 0.6$

WF (5435 entries)

$\frac{\sigma}{\mu} = 9.53 \pm 0.16 \%$

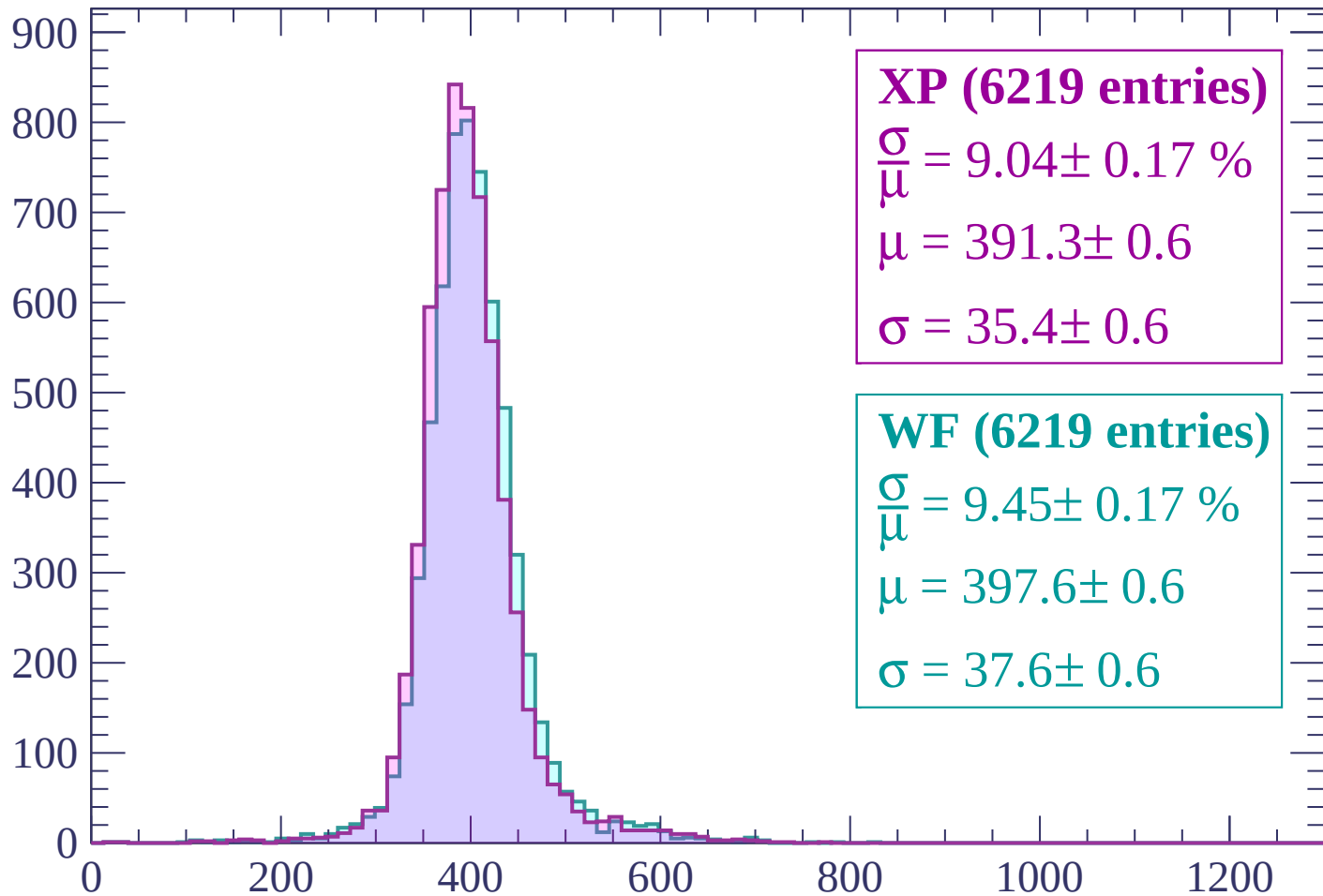
$\mu = 401.6 \pm 0.6$

$\sigma = 38.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (6219 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.17 \%$$

$$\mu = 391.3 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

WF (6219 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.17 \%$$

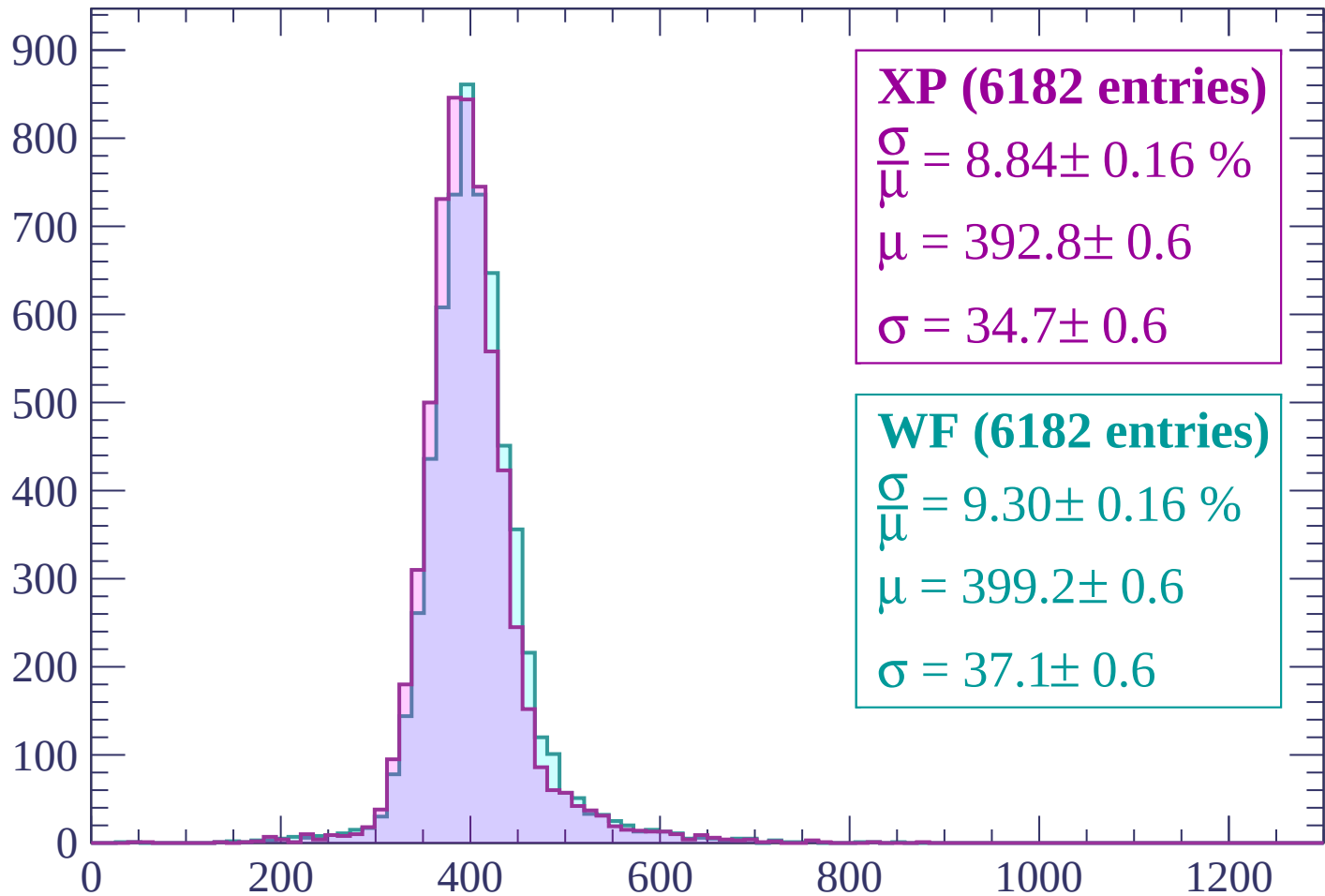
$$\mu = 397.6 \pm 0.6$$

$$\sigma = 37.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (6182 entries)

$$\frac{\sigma}{\mu} = 8.84 \pm 0.16 \%$$

$$\mu = 392.8 \pm 0.6$$

$$\sigma = 34.7 \pm 0.6$$

WF (6182 entries)

$$\frac{\sigma}{\mu} = 9.30 \pm 0.16 \%$$

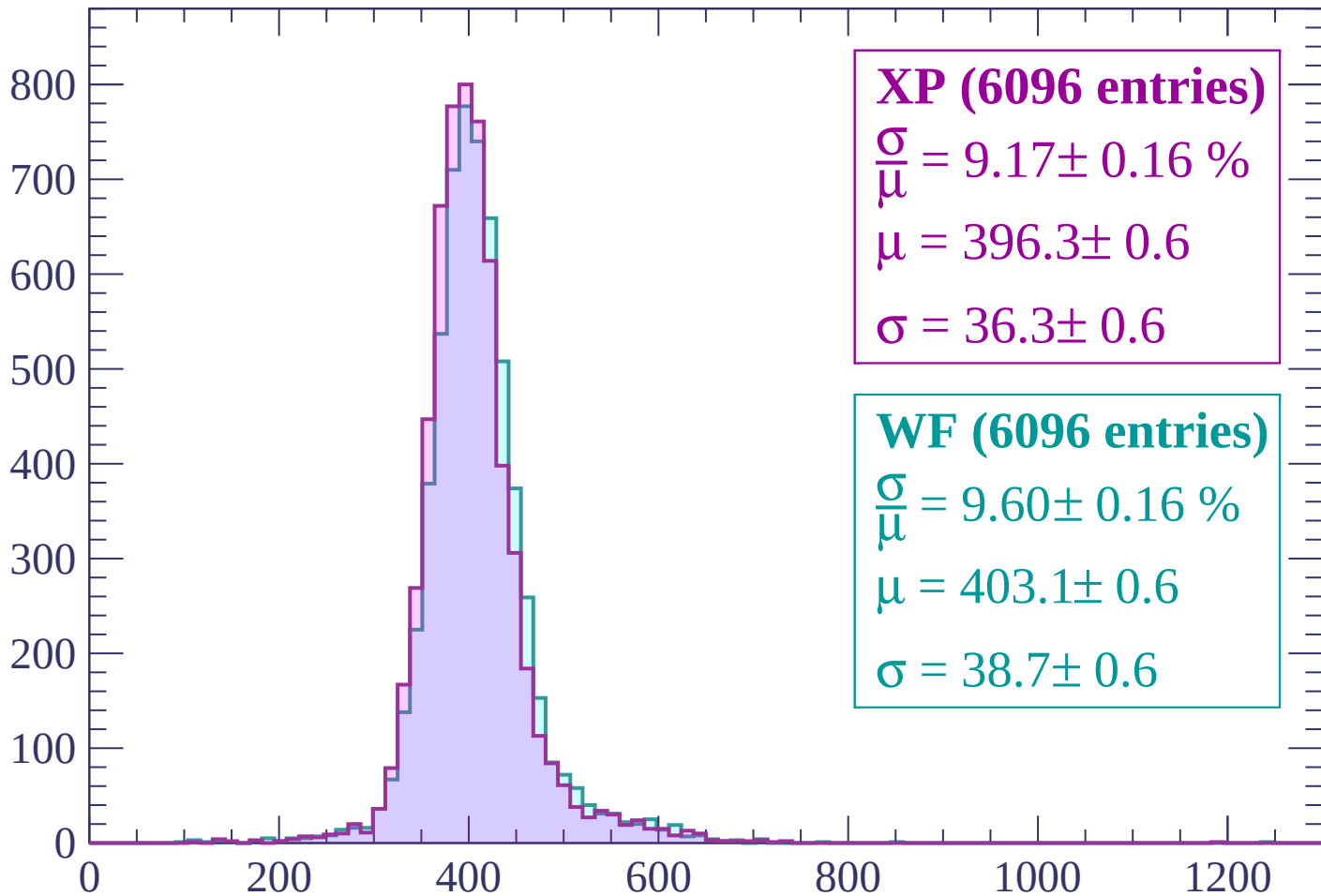
$$\mu = 399.2 \pm 0.6$$

$$\sigma = 37.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (6096 entries)

$$\frac{\sigma}{\mu} = 9.17 \pm 0.16 \%$$

$$\mu = 396.3 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

WF (6096 entries)

$$\frac{\sigma}{\mu} = 9.60 \pm 0.16 \%$$

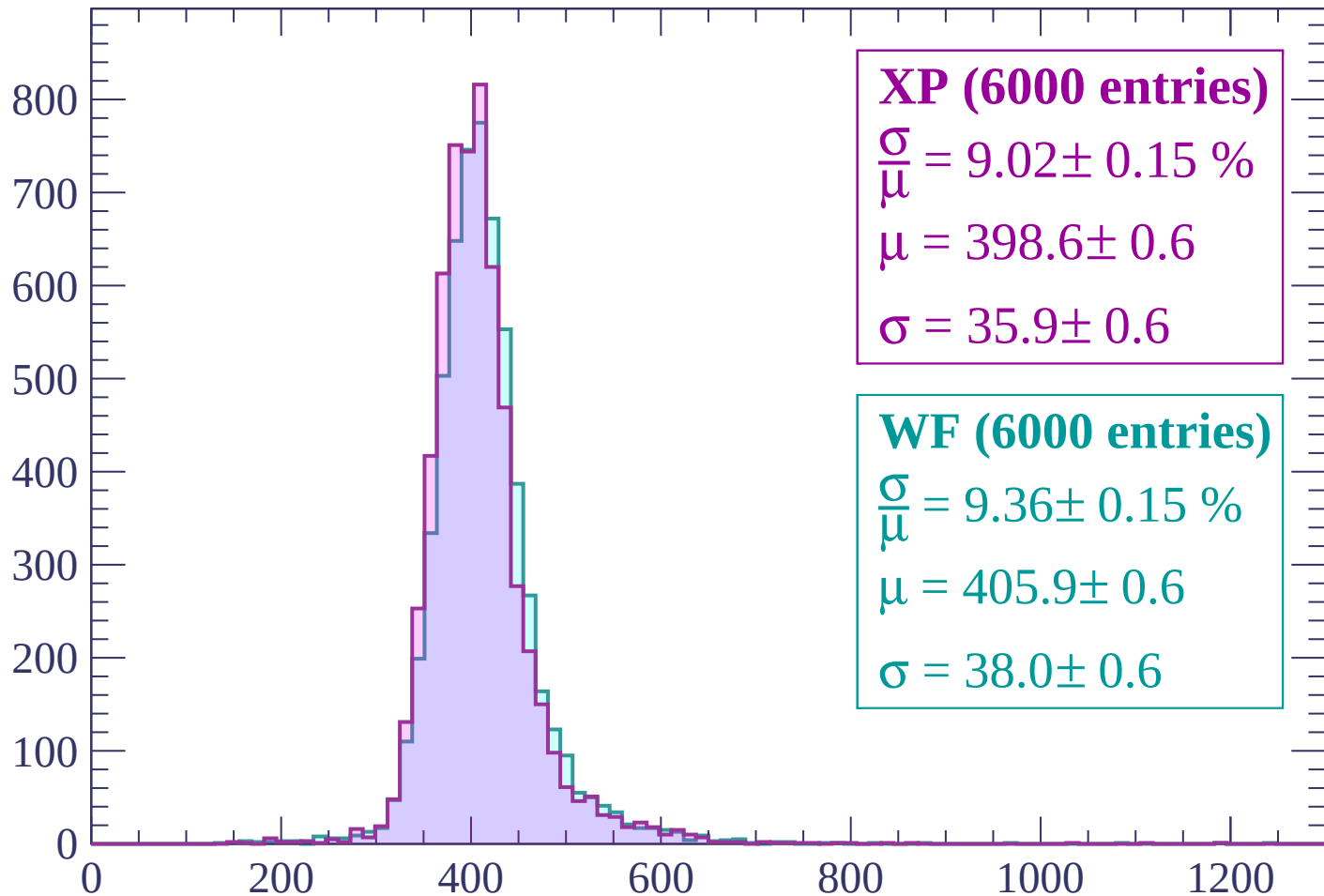
$$\mu = 403.1 \pm 0.6$$

$$\sigma = 38.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (6000 entries)

$$\frac{\sigma}{\mu} = 9.02 \pm 0.15 \%$$

$$\mu = 398.6 \pm 0.6$$

$$\sigma = 35.9 \pm 0.6$$

WF (6000 entries)

$$\frac{\sigma}{\mu} = 9.36 \pm 0.15 \%$$

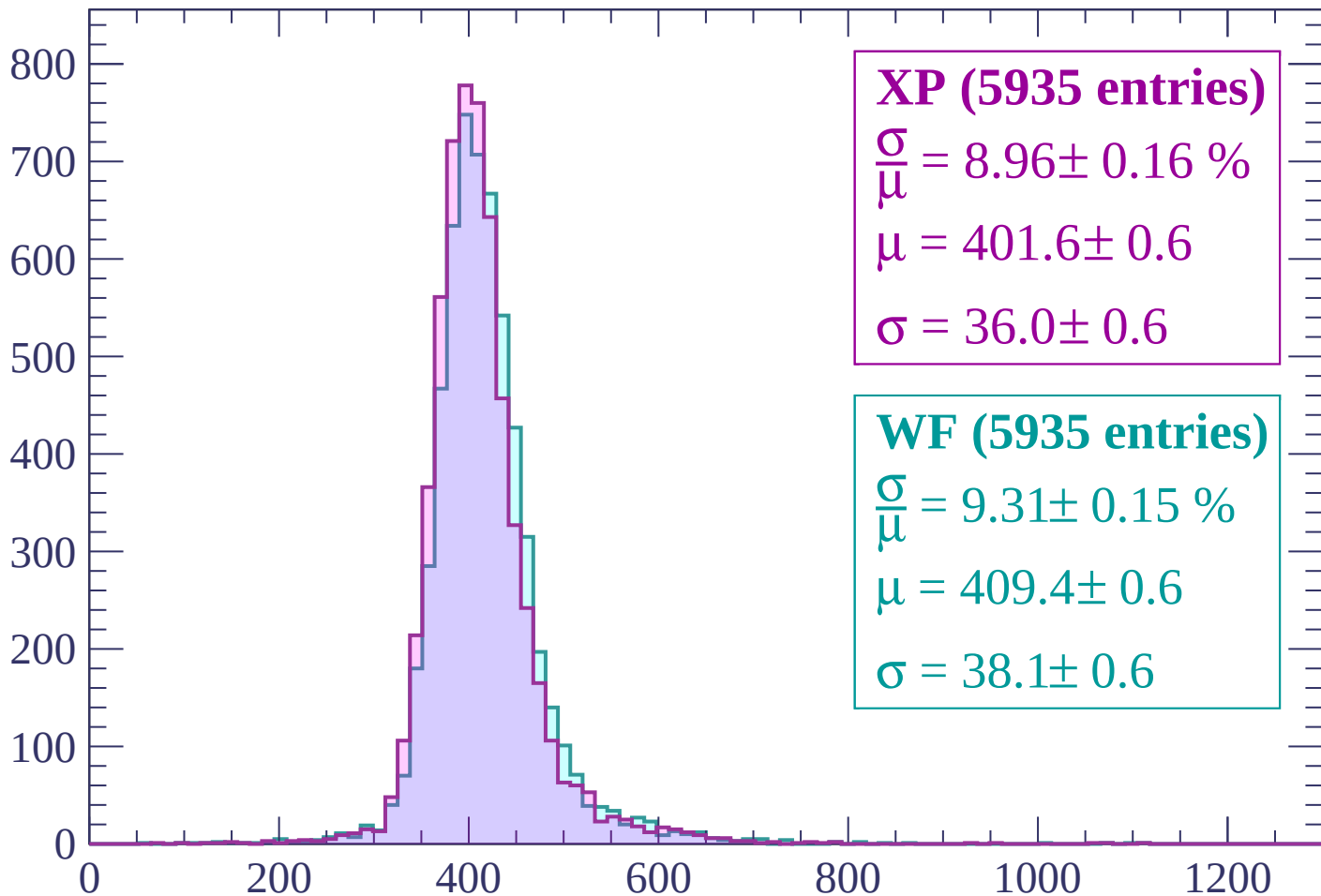
$$\mu = 405.9 \pm 0.6$$

$$\sigma = 38.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (5935 entries)

$$\frac{\sigma}{\mu} = 8.96 \pm 0.16 \%$$

$$\mu = 401.6 \pm 0.6$$

$$\sigma = 36.0 \pm 0.6$$

WF (5935 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.15 \%$$

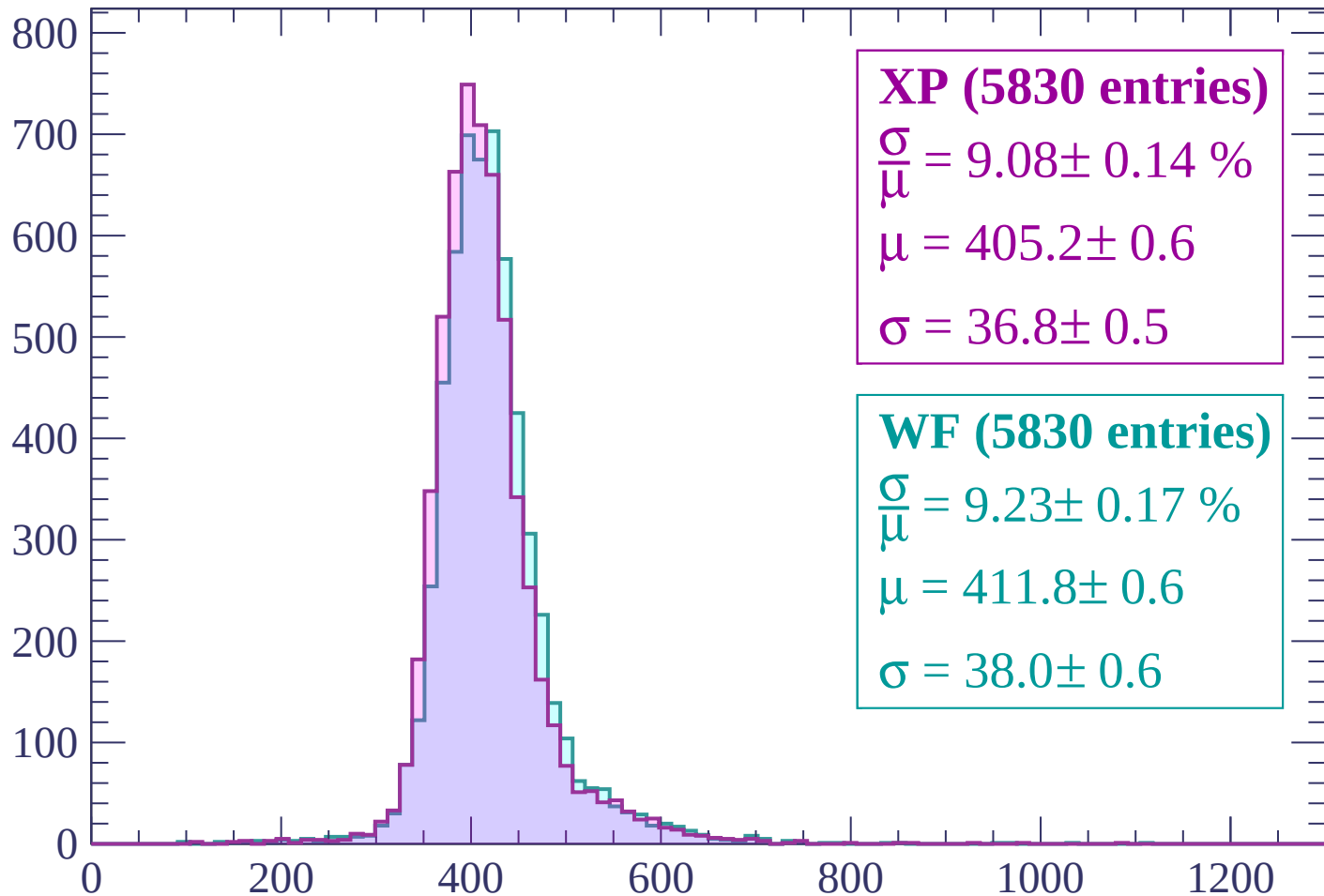
$$\mu = 409.4 \pm 0.6$$

$$\sigma = 38.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (5830 entries)

$\frac{\sigma}{\mu} = 9.08 \pm 0.14 \%$

$\mu = 405.2 \pm 0.6$

$\sigma = 36.8 \pm 0.5$

WF (5830 entries)

$\frac{\sigma}{\mu} = 9.23 \pm 0.17 \%$

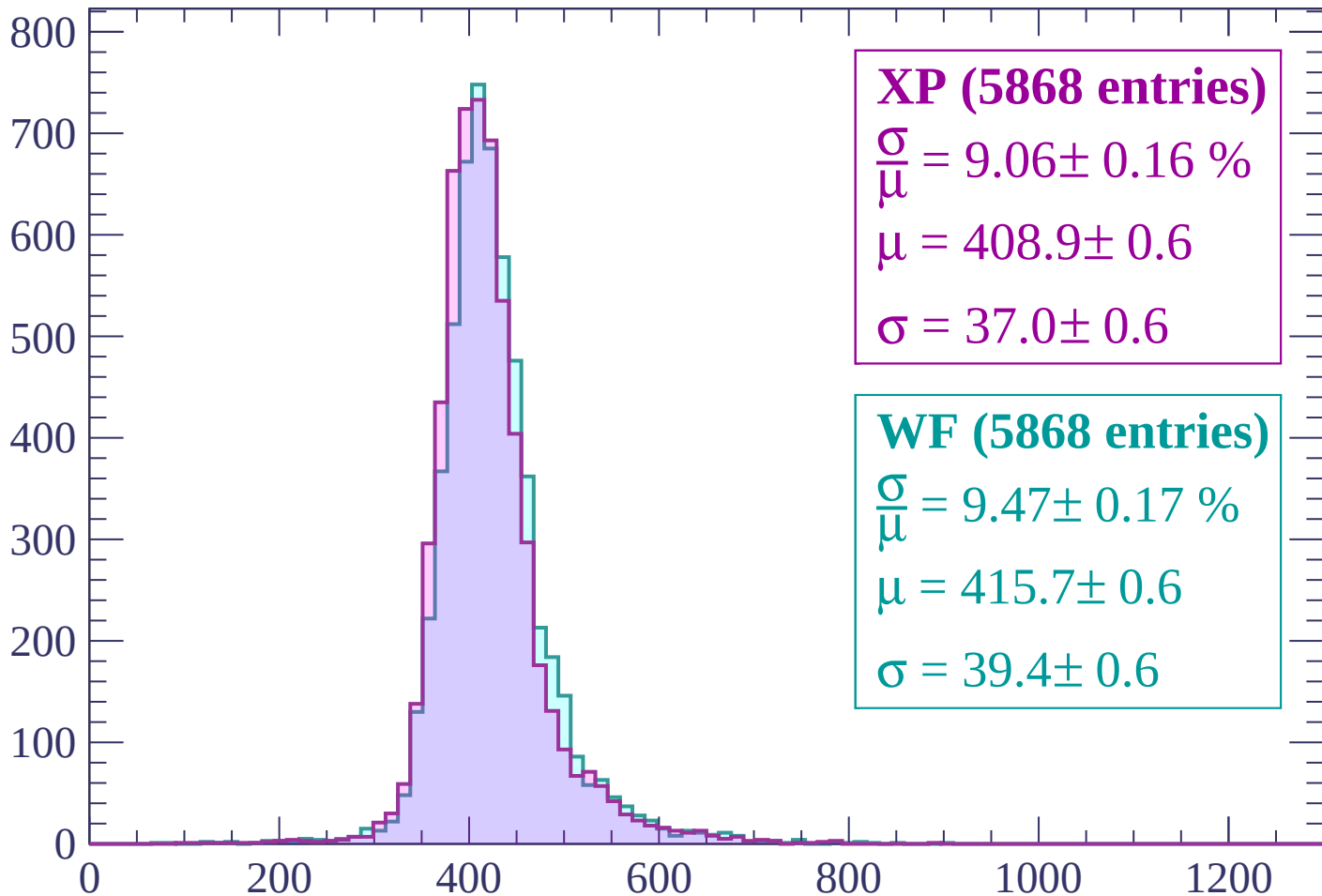
$\mu = 411.8 \pm 0.6$

$\sigma = 38.0 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (5868 entries)

$\frac{\sigma}{\mu} = 9.06 \pm 0.16 \%$

$\mu = 408.9 \pm 0.6$

$\sigma = 37.0 \pm 0.6$

WF (5868 entries)

$\frac{\sigma}{\mu} = 9.47 \pm 0.17 \%$

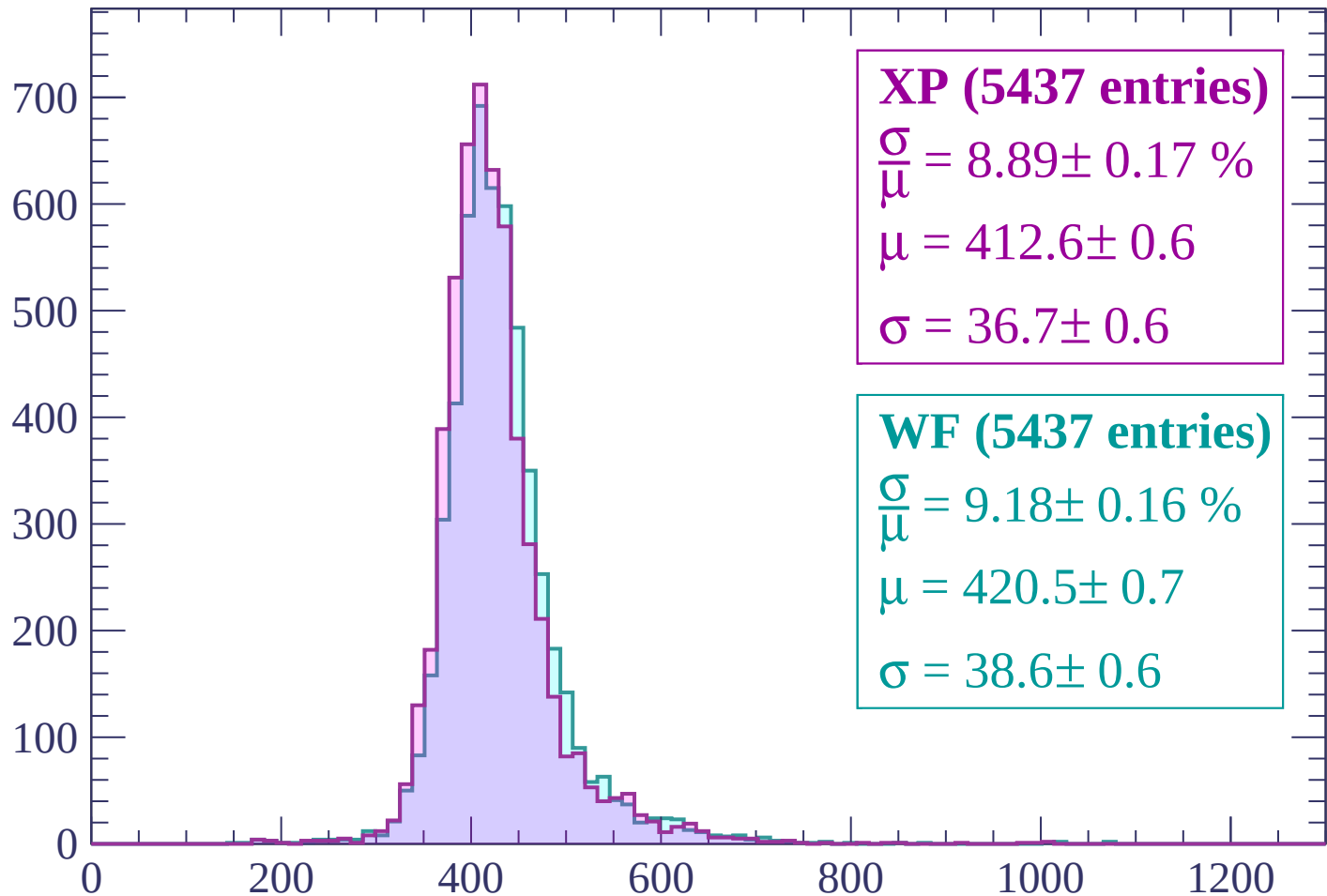
$\mu = 415.7 \pm 0.6$

$\sigma = 39.4 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (5437 entries)

$$\frac{\sigma}{\mu} = 8.89 \pm 0.17 \%$$

$$\mu = 412.6 \pm 0.6$$

$$\sigma = 36.7 \pm 0.6$$

WF (5437 entries)

$$\frac{\sigma}{\mu} = 9.18 \pm 0.16 \%$$

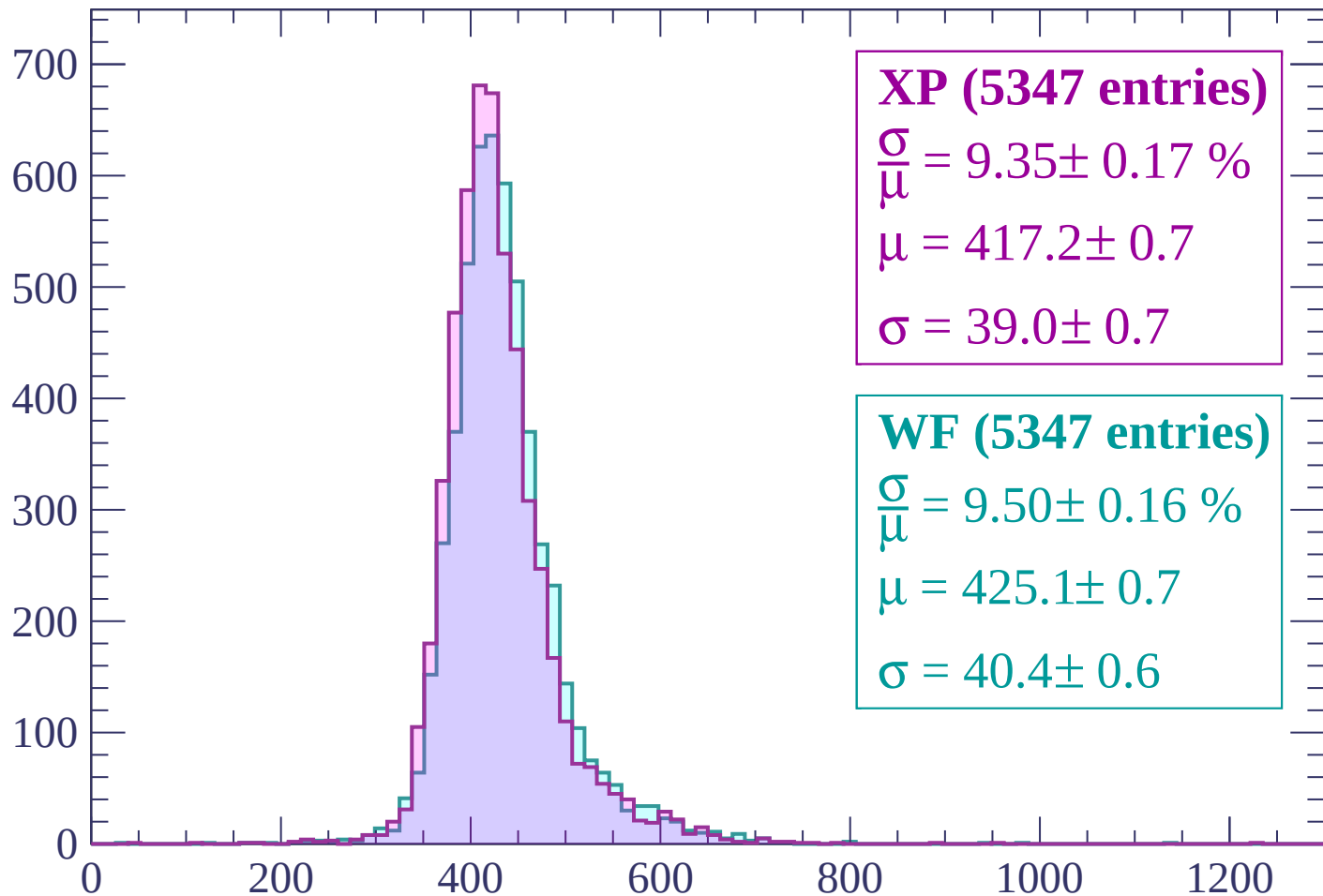
$$\mu = 420.5 \pm 0.7$$

$$\sigma = 38.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP (5347 entries)

$$\frac{\sigma}{\mu} = 9.35 \pm 0.17 \%$$

$$\mu = 417.2 \pm 0.7$$

$$\sigma = 39.0 \pm 0.7$$

WF (5347 entries)

$$\frac{\sigma}{\mu} = 9.50 \pm 0.16 \%$$

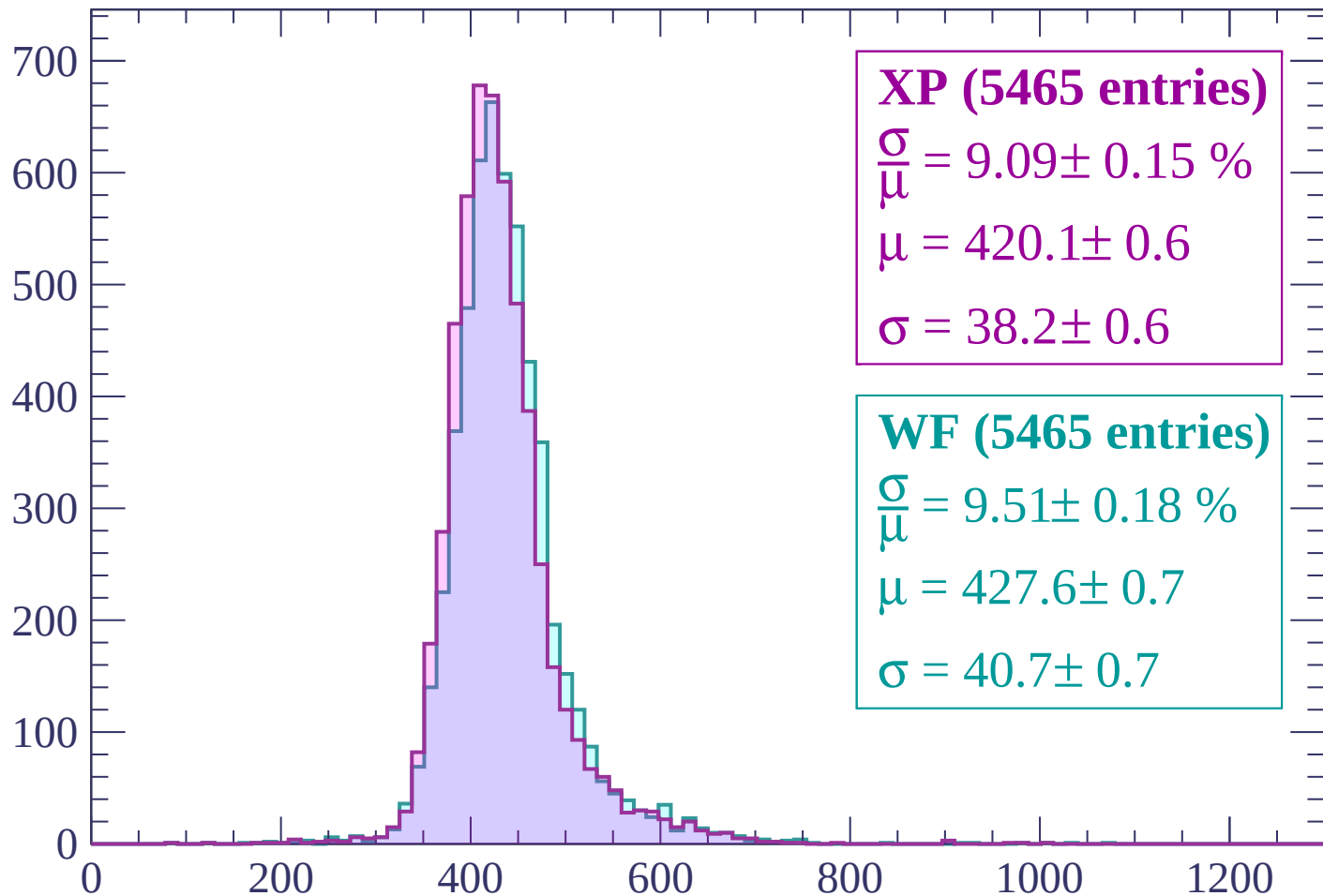
$$\mu = 425.1 \pm 0.7$$

$$\sigma = 40.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (5465 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.15 \%$$

$$\mu = 420.1 \pm 0.6$$

$$\sigma = 38.2 \pm 0.6$$

WF (5465 entries)

$$\frac{\sigma}{\mu} = 9.51 \pm 0.18 \%$$

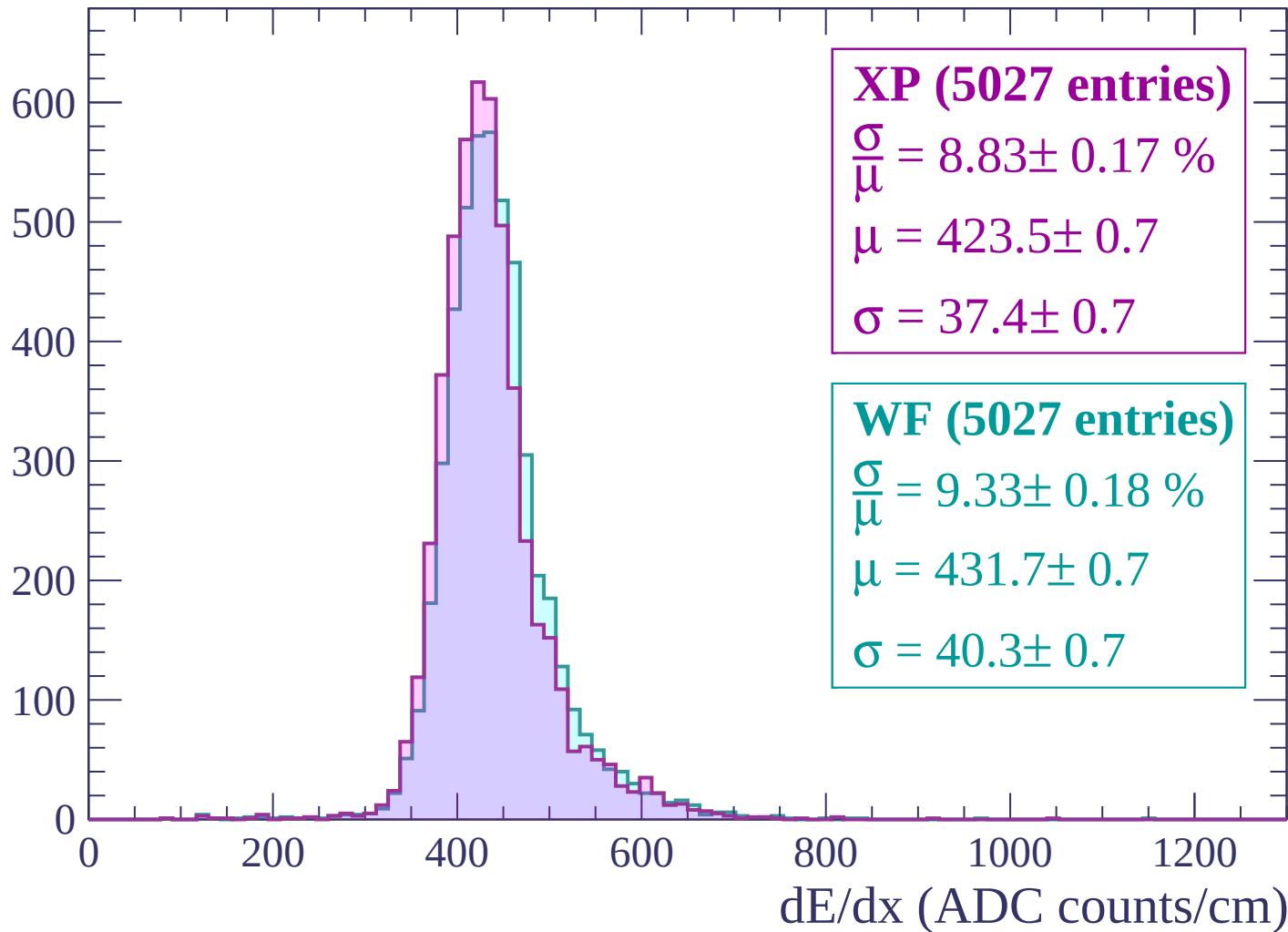
$$\mu = 427.6 \pm 0.7$$

$$\sigma = 40.7 \pm 0.7$$

dE/dx (ADC counts/cm)

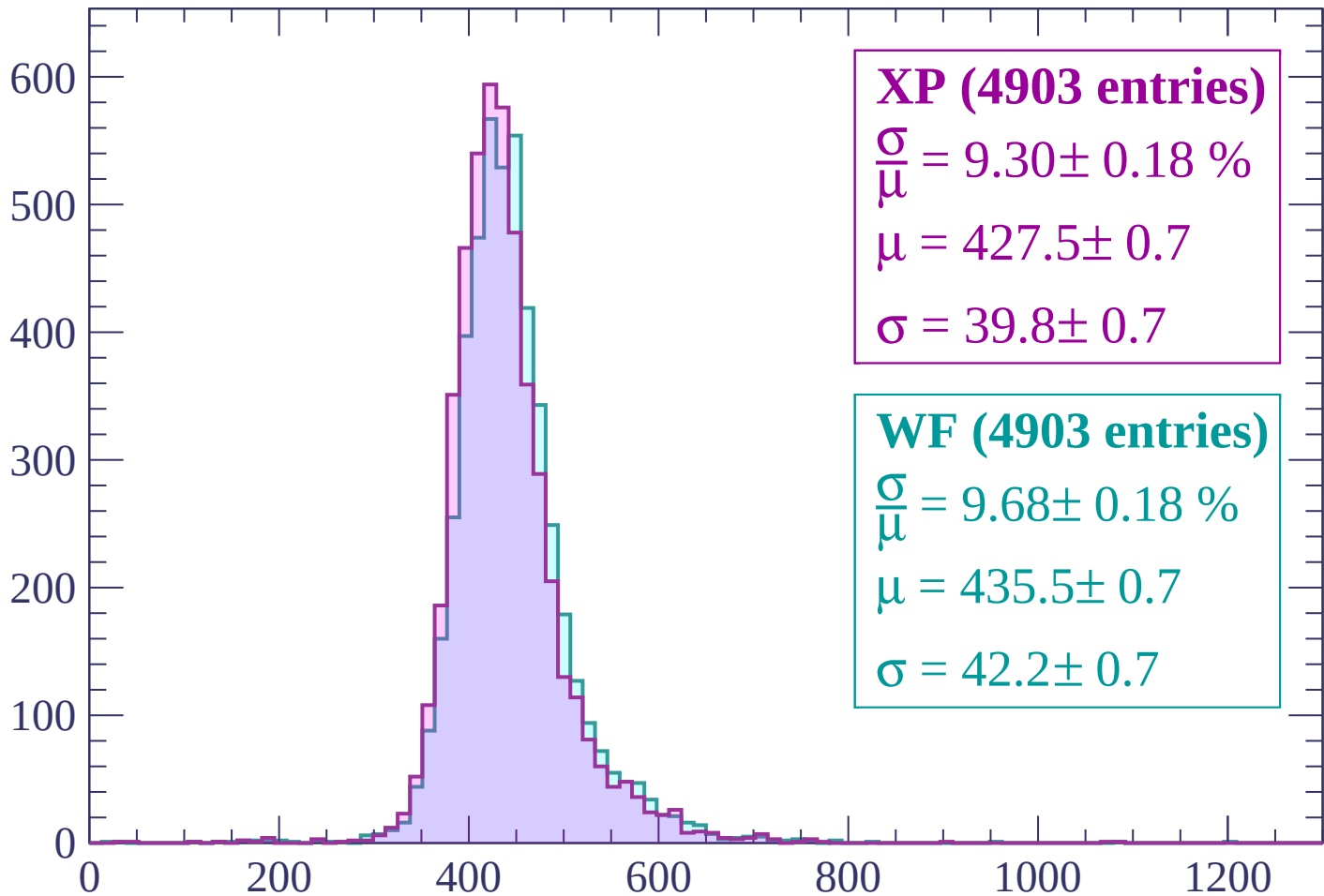
Energy loss | $960 < p < 1020$

Count



Energy loss | $1020 < p < 1080$

Count



XP (4903 entries)

$$\frac{\sigma}{\mu} = 9.30 \pm 0.18 \%$$

$$\mu = 427.5 \pm 0.7$$

$$\sigma = 39.8 \pm 0.7$$

WF (4903 entries)

$$\frac{\sigma}{\mu} = 9.68 \pm 0.18 \%$$

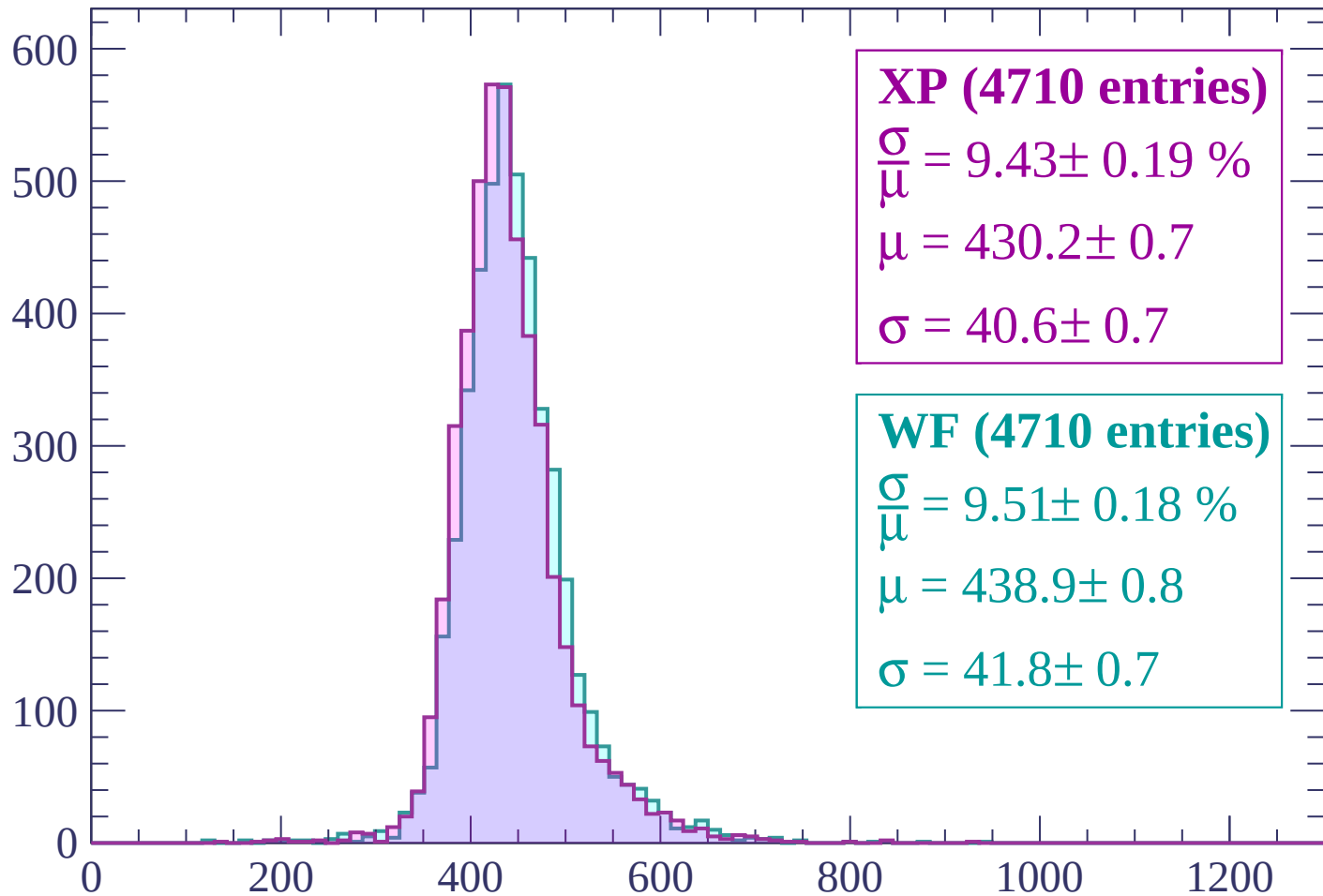
$$\mu = 435.5 \pm 0.7$$

$$\sigma = 42.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP (4710 entries)

$\frac{\sigma}{\mu} = 9.43 \pm 0.19 \%$

$\mu = 430.2 \pm 0.7$

$\sigma = 40.6 \pm 0.7$

WF (4710 entries)

$\frac{\sigma}{\mu} = 9.51 \pm 0.18 \%$

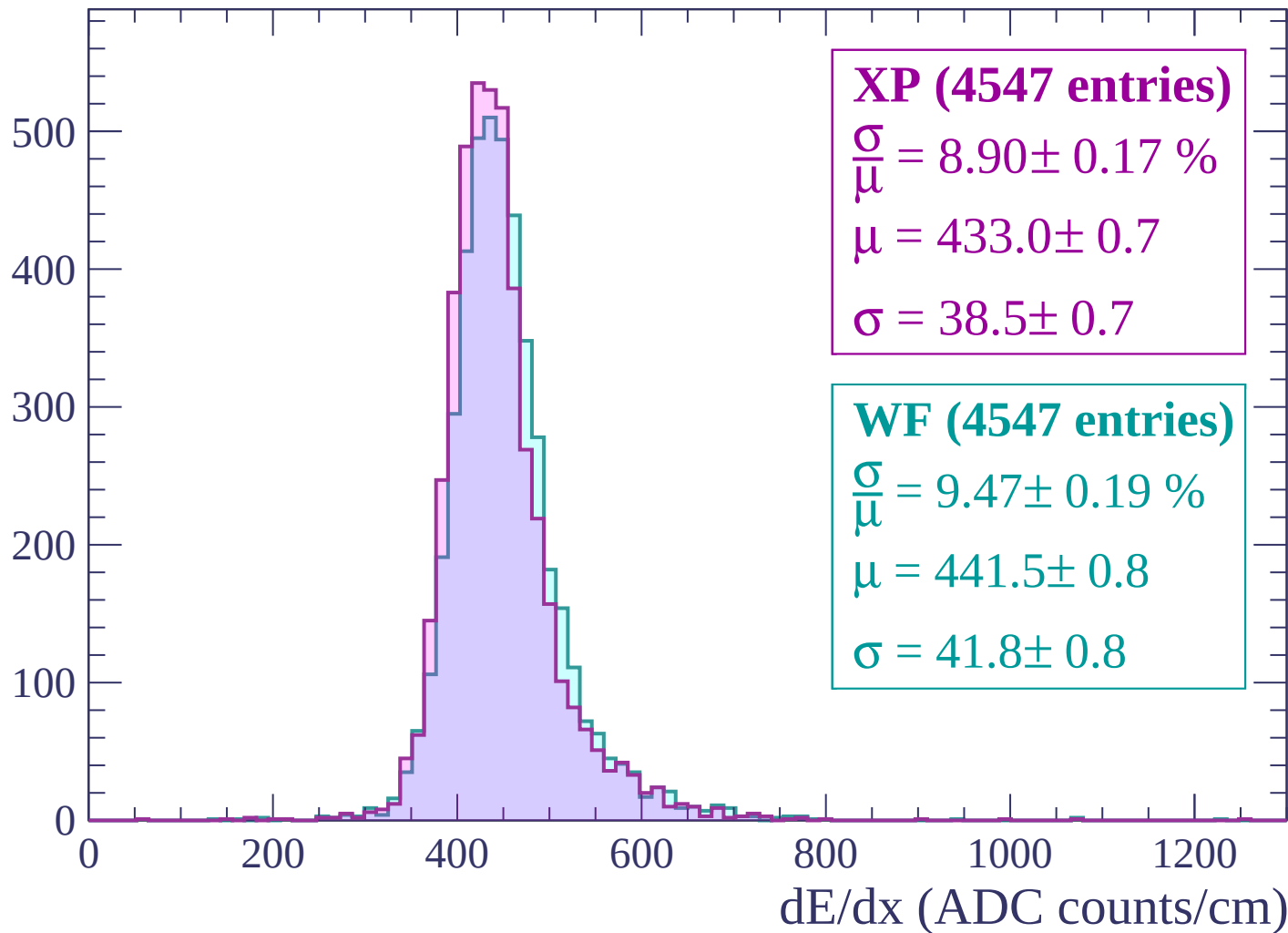
$\mu = 438.9 \pm 0.8$

$\sigma = 41.8 \pm 0.7$

dE/dx (ADC counts/cm)

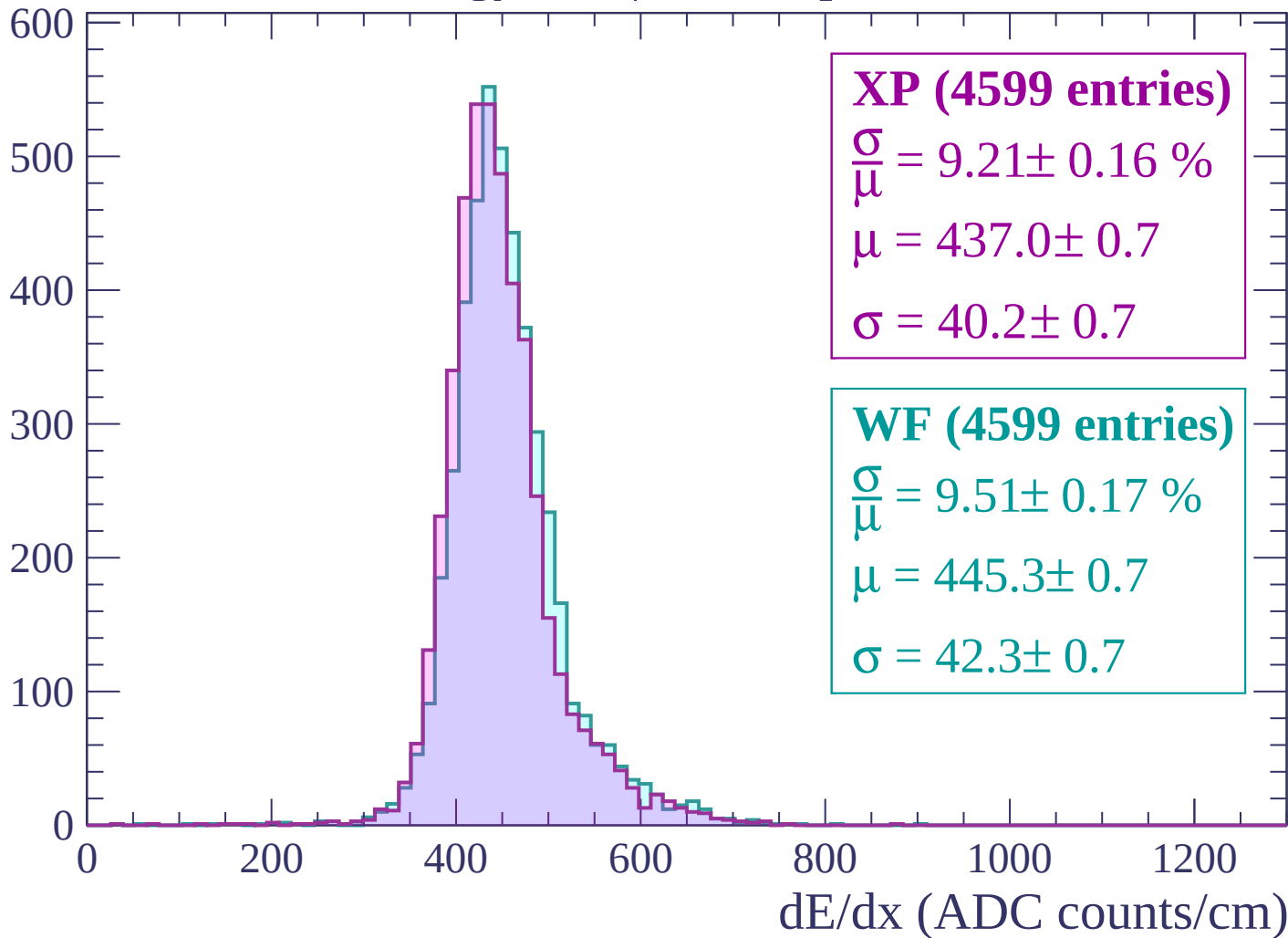
Energy loss | $1140 < p < 1200$

Count



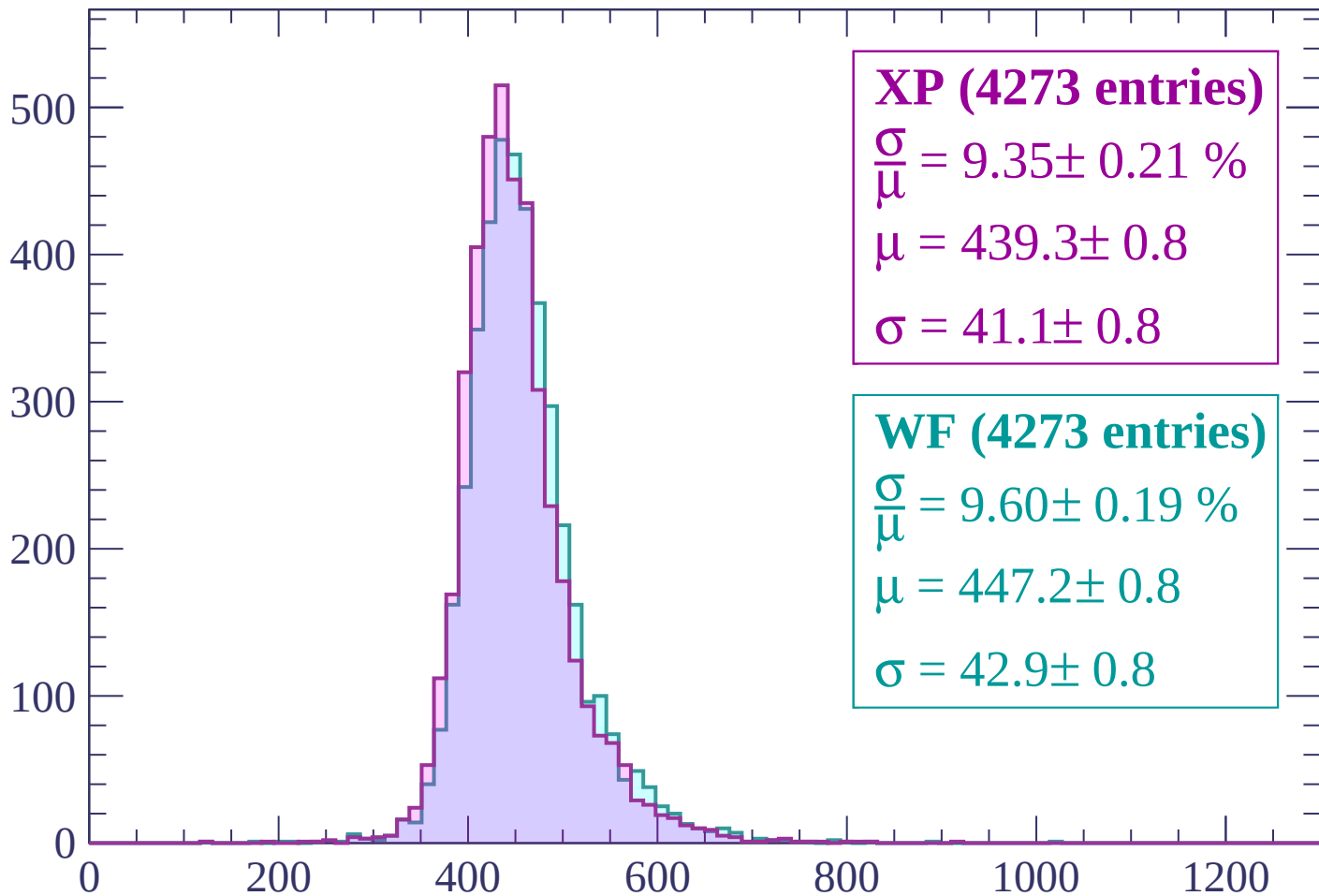
Energy loss | $1200 < p < 1260$

Count



Energy loss | $1260 < p < 1320$

Count



XP (4273 entries)

$$\frac{\sigma}{\mu} = 9.35 \pm 0.21 \%$$

$$\mu = 439.3 \pm 0.8$$

$$\sigma = 41.1 \pm 0.8$$

WF (4273 entries)

$$\frac{\sigma}{\mu} = 9.60 \pm 0.19 \%$$

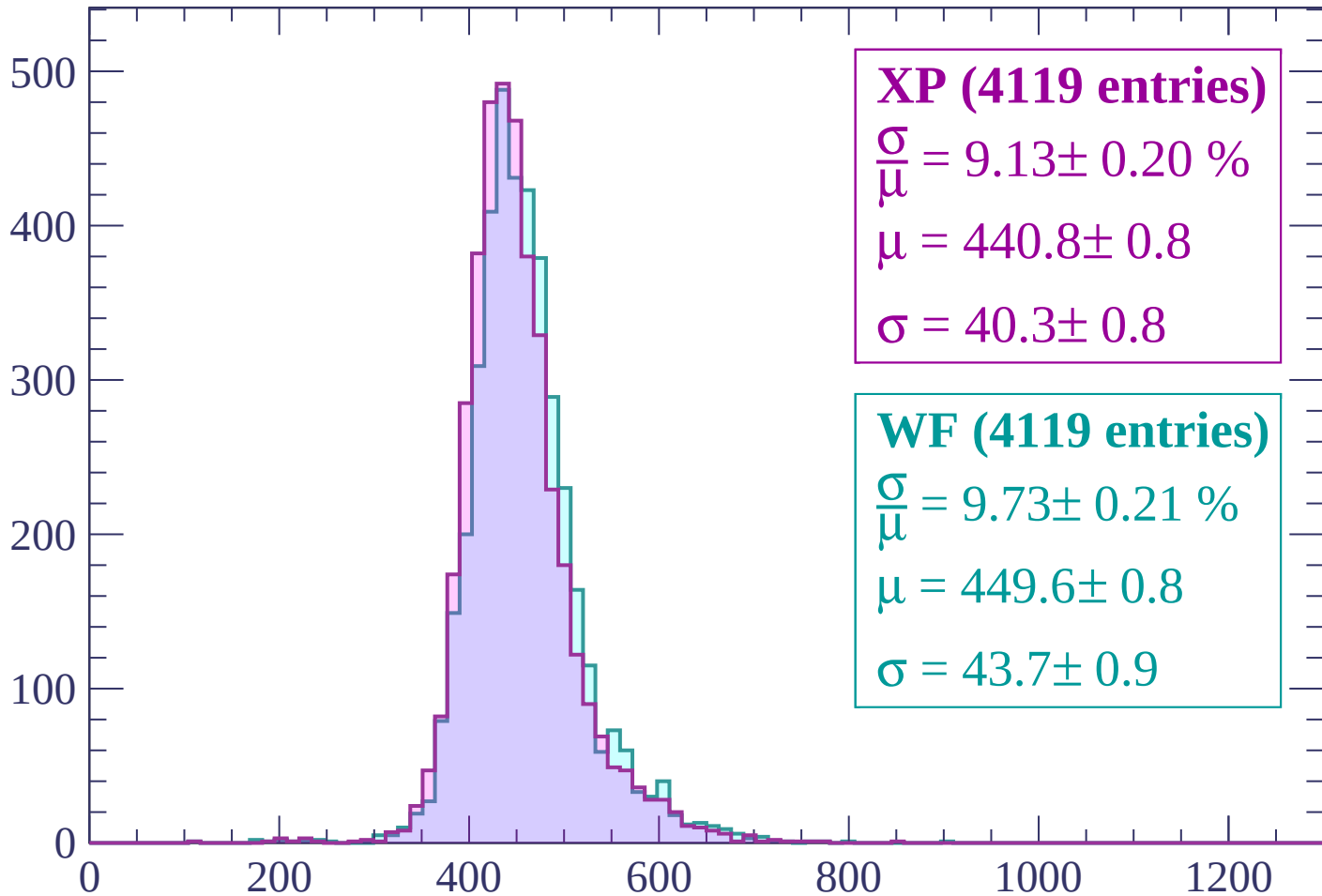
$$\mu = 447.2 \pm 0.8$$

$$\sigma = 42.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (4119 entries)

$\frac{\sigma}{\mu} = 9.13 \pm 0.20 \%$

$\mu = 440.8 \pm 0.8$

$\sigma = 40.3 \pm 0.8$

WF (4119 entries)

$\frac{\sigma}{\mu} = 9.73 \pm 0.21 \%$

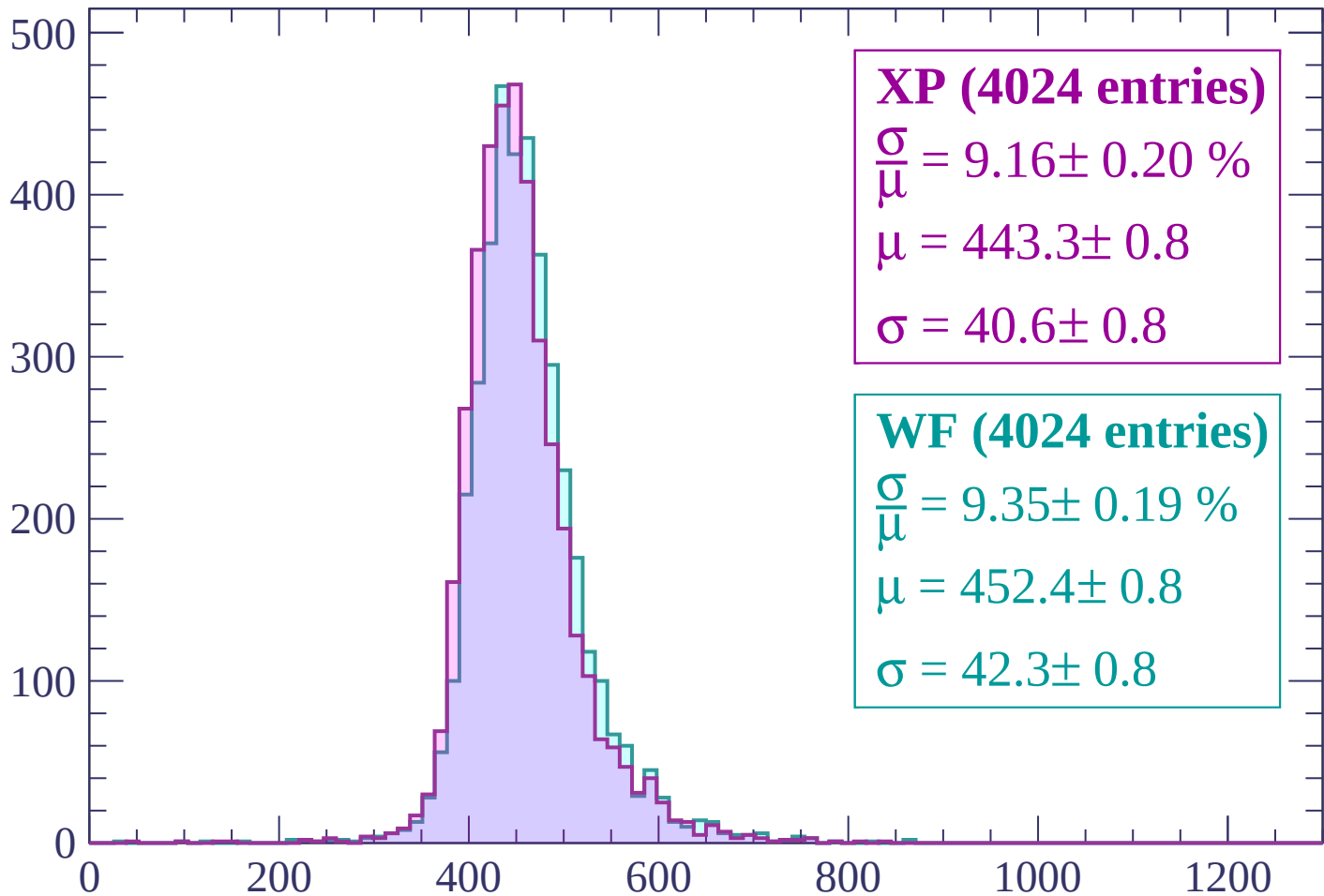
$\mu = 449.6 \pm 0.8$

$\sigma = 43.7 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $1380 < p < 1440$

Count



XP (4024 entries)

$$\frac{\sigma}{\mu} = 9.16 \pm 0.20 \%$$

$$\mu = 443.3 \pm 0.8$$

$$\sigma = 40.6 \pm 0.8$$

WF (4024 entries)

$$\frac{\sigma}{\mu} = 9.35 \pm 0.19 \%$$

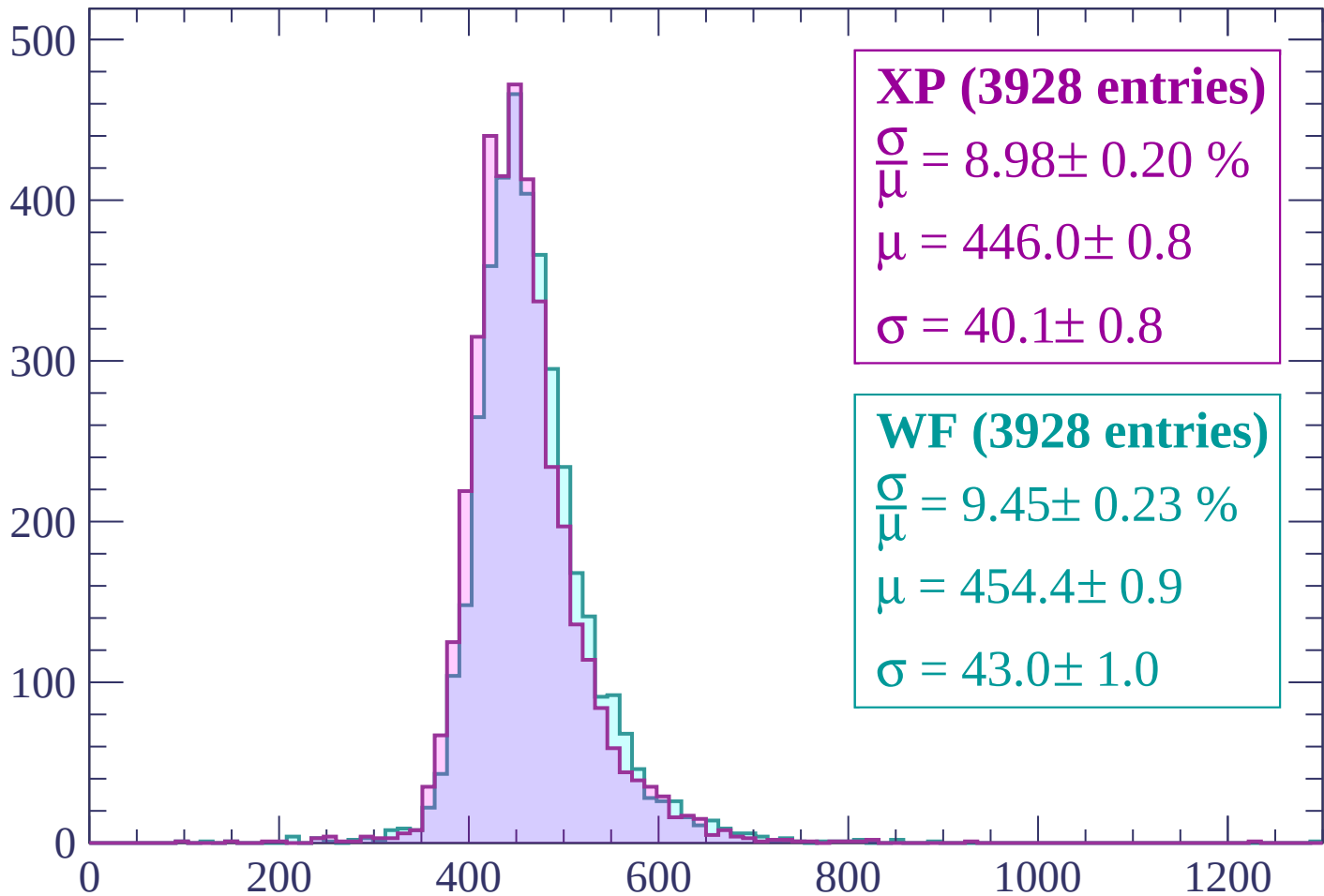
$$\mu = 452.4 \pm 0.8$$

$$\sigma = 42.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP (3928 entries)

$$\frac{\sigma}{\mu} = 8.98 \pm 0.20 \%$$

$$\mu = 446.0 \pm 0.8$$

$$\sigma = 40.1 \pm 0.8$$

WF (3928 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.23 \%$$

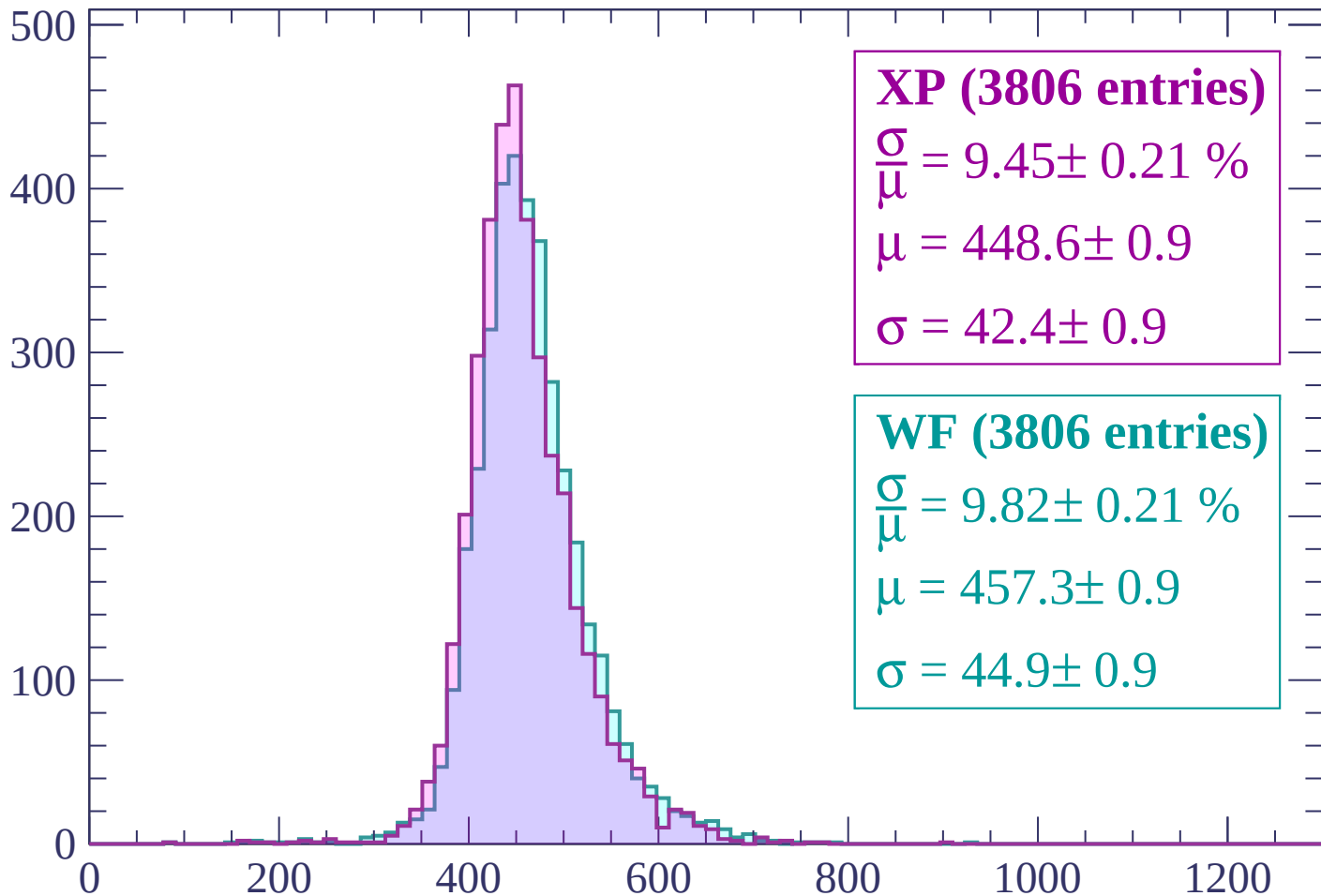
$$\mu = 454.4 \pm 0.9$$

$$\sigma = 43.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



XP (3806 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.21 \%$$

$$\mu = 448.6 \pm 0.9$$

$$\sigma = 42.4 \pm 0.9$$

WF (3806 entries)

$$\frac{\sigma}{\mu} = 9.82 \pm 0.21 \%$$

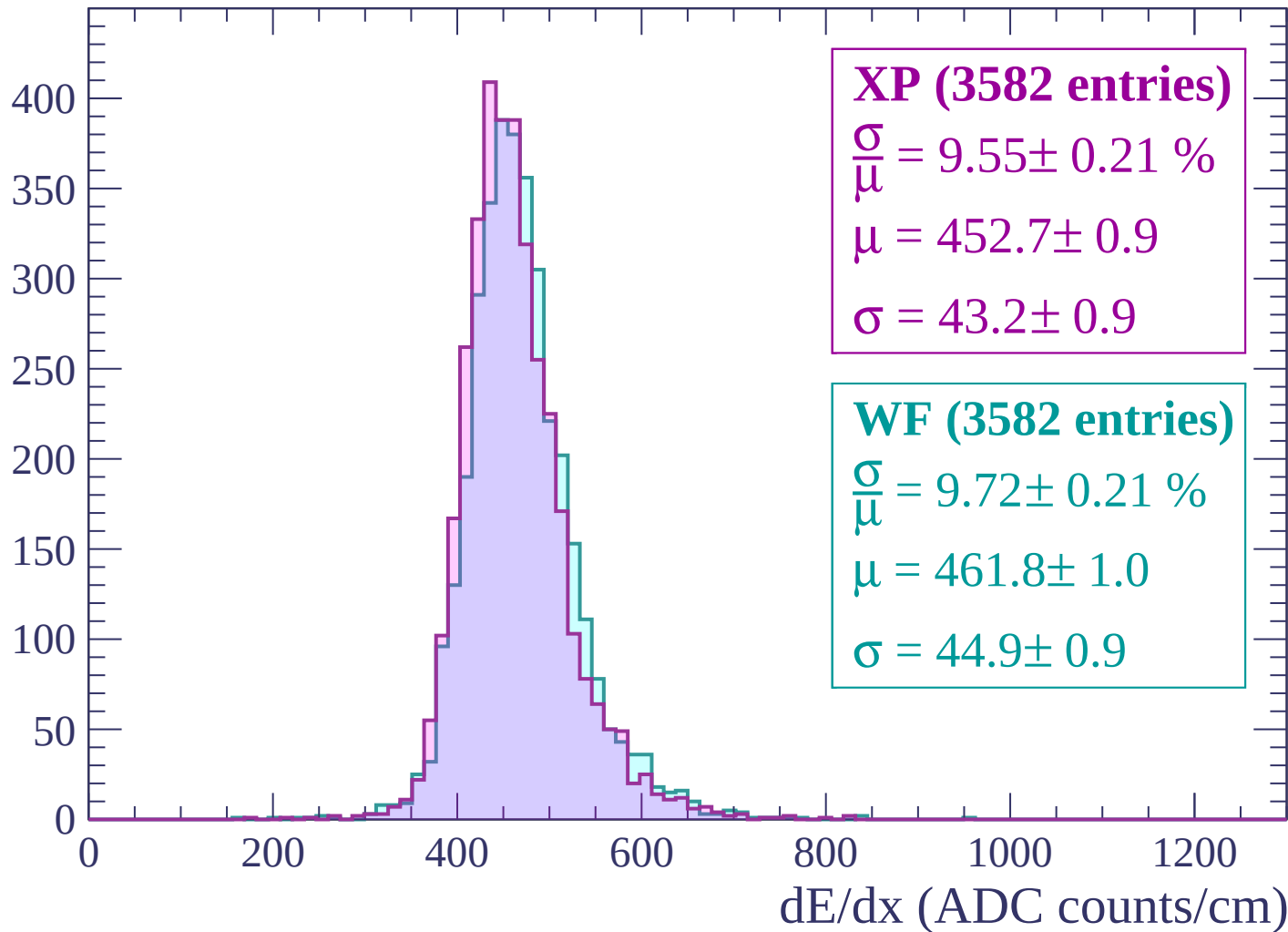
$$\mu = 457.3 \pm 0.9$$

$$\sigma = 44.9 \pm 0.9$$

dE/dx (ADC counts/cm)

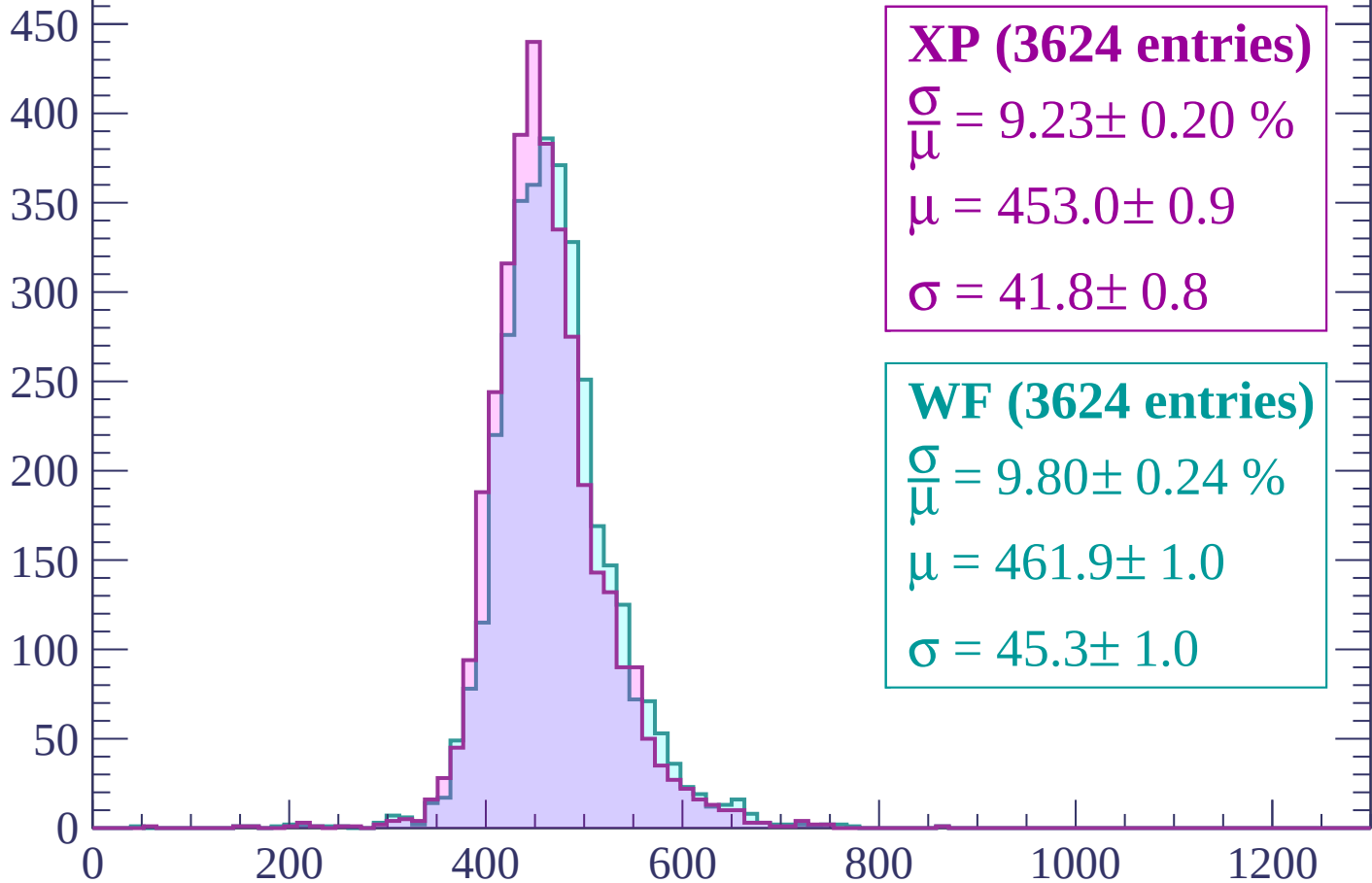
Energy loss | $1560 < p < 1620$

Count



Energy loss | $1620 < p < 1680$

Count



XP (3624 entries)

$$\frac{\sigma}{\mu} = 9.23 \pm 0.20 \%$$

$$\mu = 453.0 \pm 0.9$$

$$\sigma = 41.8 \pm 0.8$$

WF (3624 entries)

$$\frac{\sigma}{\mu} = 9.80 \pm 0.24 \%$$

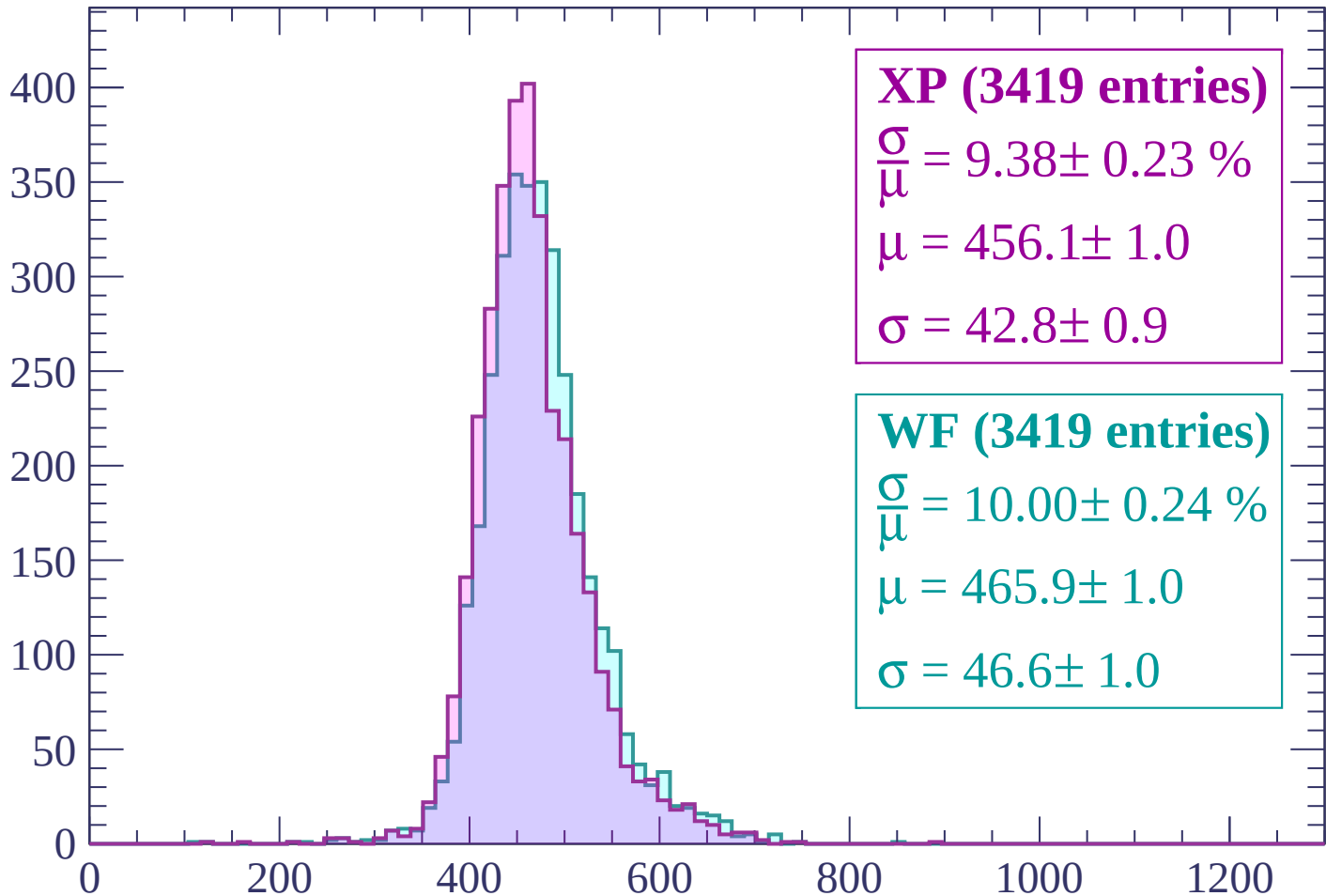
$$\mu = 461.9 \pm 1.0$$

$$\sigma = 45.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (3419 entries)

$$\frac{\sigma}{\mu} = 9.38 \pm 0.23 \%$$

$$\mu = 456.1 \pm 1.0$$

$$\sigma = 42.8 \pm 0.9$$

WF (3419 entries)

$$\frac{\sigma}{\mu} = 10.00 \pm 0.24 \%$$

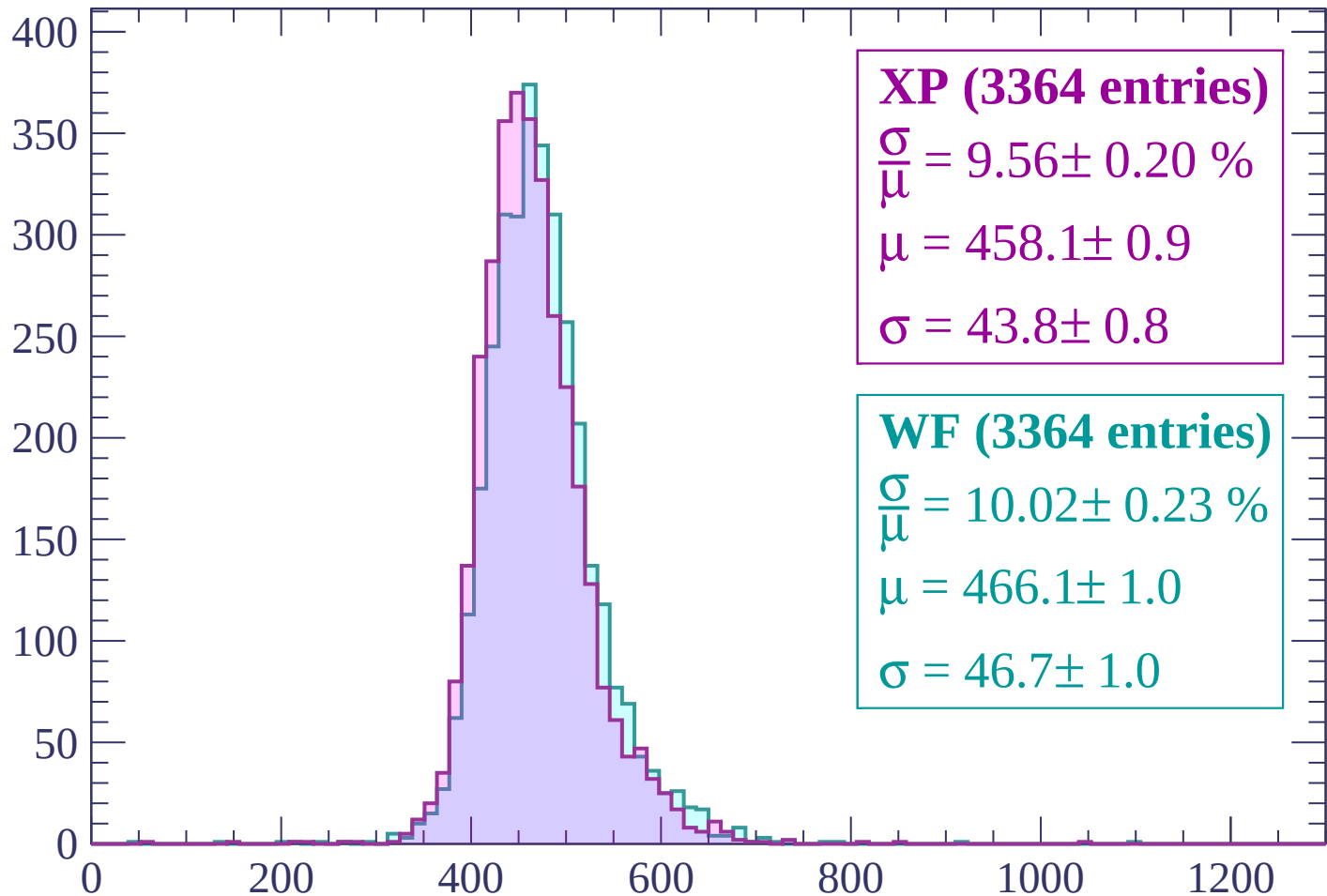
$$\mu = 465.9 \pm 1.0$$

$$\sigma = 46.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (3364 entries)

$$\frac{\sigma}{\mu} = 9.56 \pm 0.20 \%$$

$$\mu = 458.1 \pm 0.9$$

$$\sigma = 43.8 \pm 0.8$$

WF (3364 entries)

$$\frac{\sigma}{\mu} = 10.02 \pm 0.23 \%$$

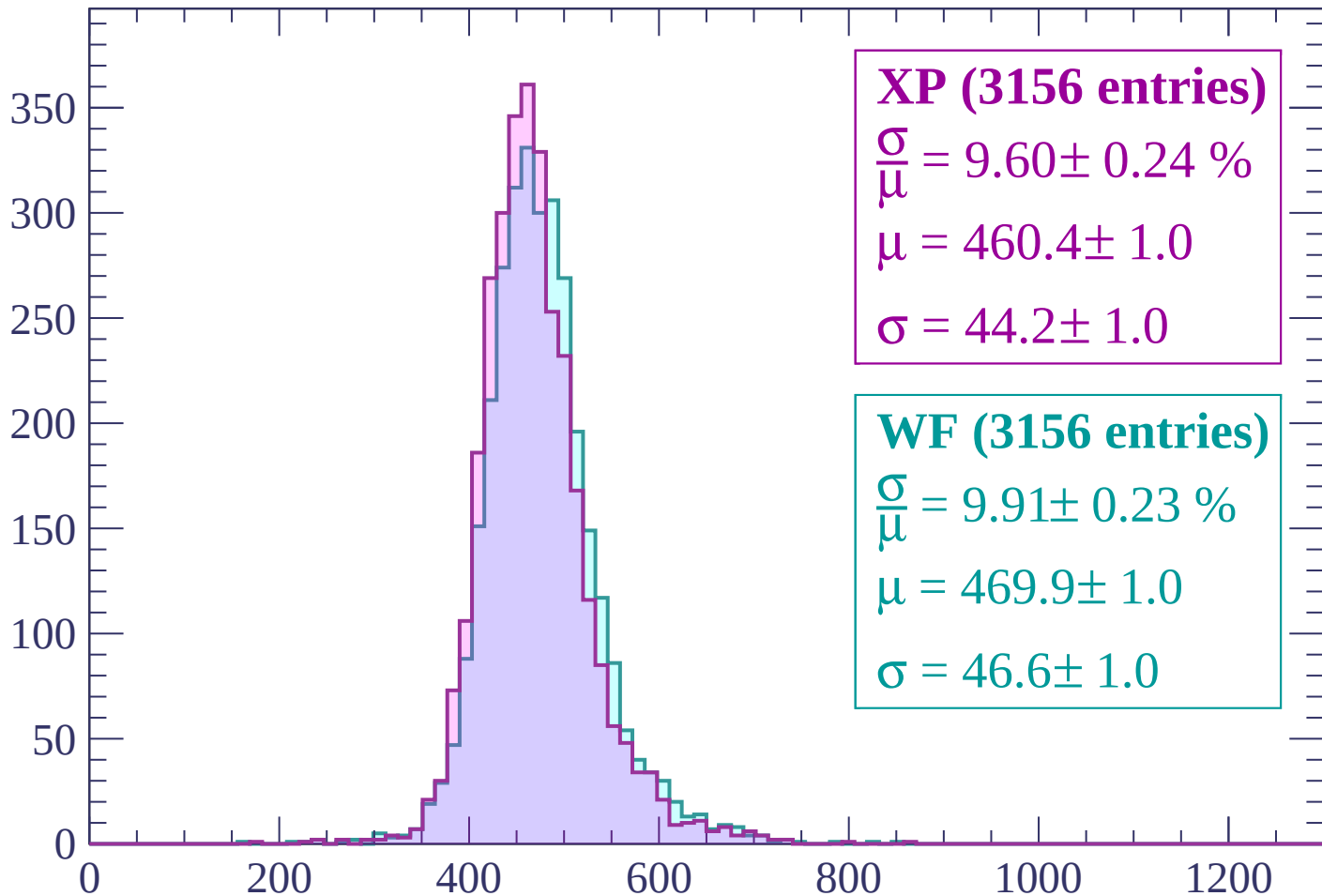
$$\mu = 466.1 \pm 1.0$$

$$\sigma = 46.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1860$

Count



XP (3156 entries)

$$\frac{\sigma}{\mu} = 9.60 \pm 0.24 \%$$

$$\mu = 460.4 \pm 1.0$$

$$\sigma = 44.2 \pm 1.0$$

WF (3156 entries)

$$\frac{\sigma}{\mu} = 9.91 \pm 0.23 \%$$

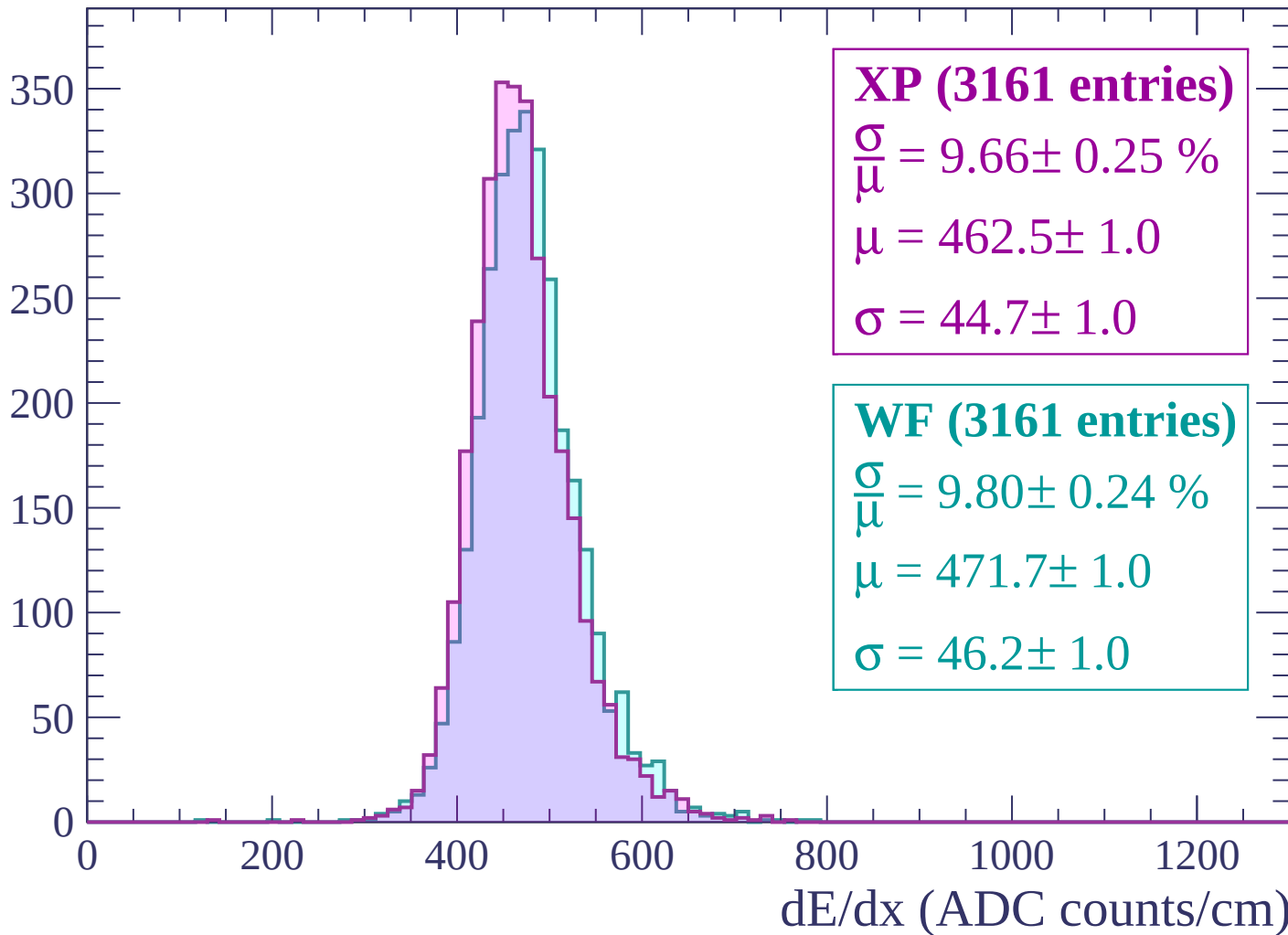
$$\mu = 469.9 \pm 1.0$$

$$\sigma = 46.6 \pm 1.0$$

dE/dx (ADC counts/cm)

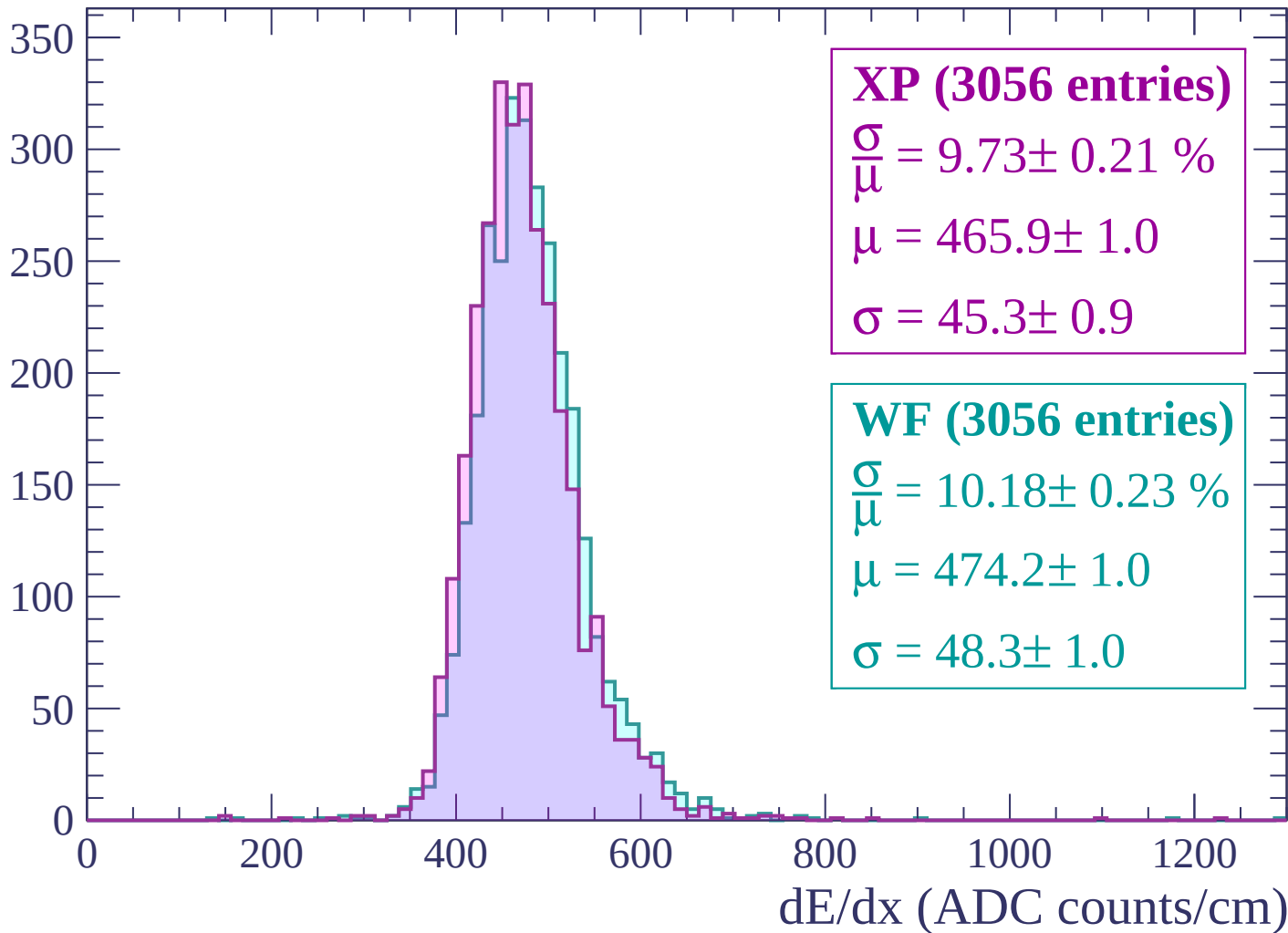
Energy loss | $1860 < p < 1920$

Count



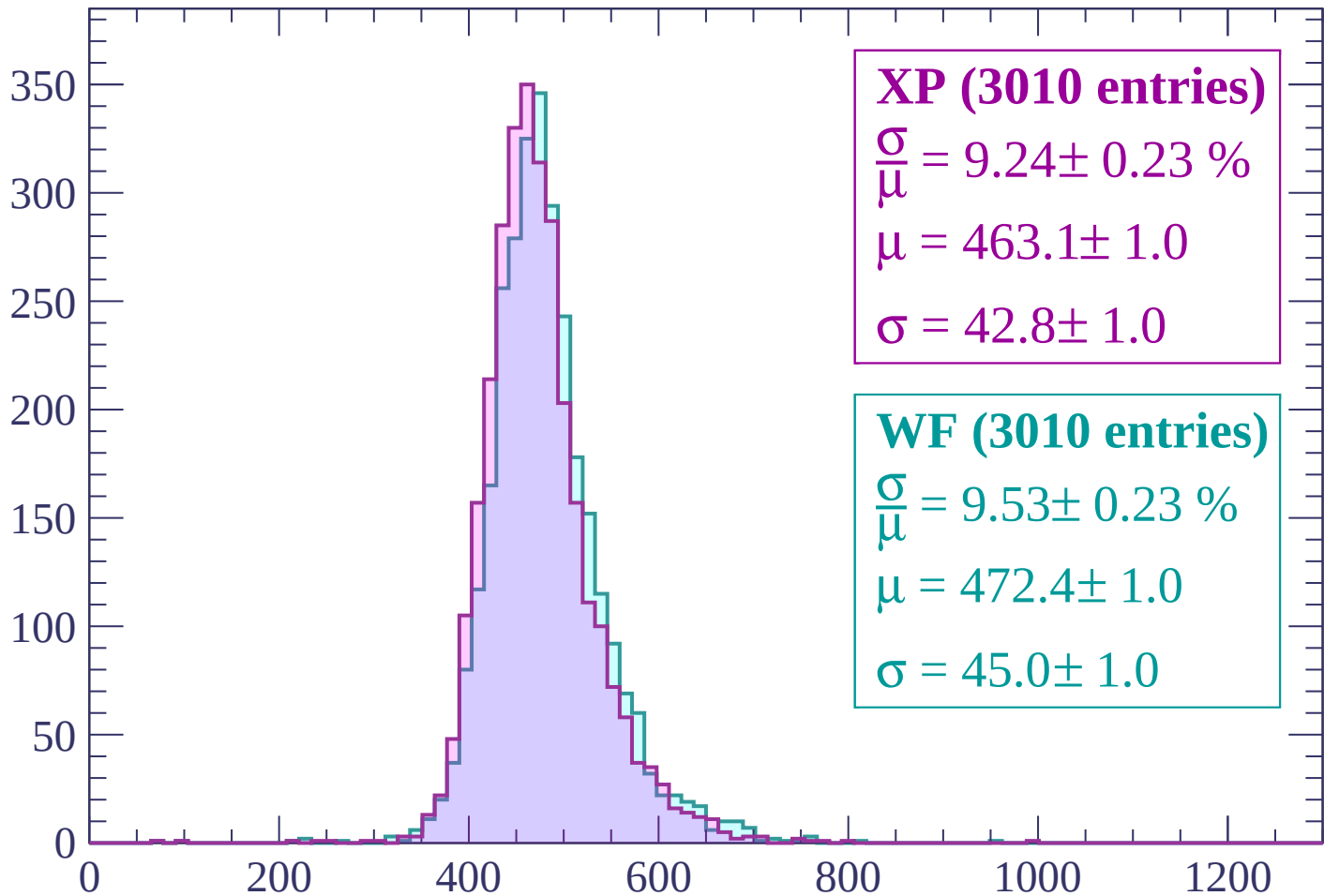
Energy loss | $1920 < p < 1980$

Count



Energy loss | $1980 < p < 2040$

Count



XP (3010 entries)

$$\frac{\sigma}{\mu} = 9.24 \pm 0.23 \%$$

$$\mu = 463.1 \pm 1.0$$

$$\sigma = 42.8 \pm 1.0$$

WF (3010 entries)

$$\frac{\sigma}{\mu} = 9.53 \pm 0.23 \%$$

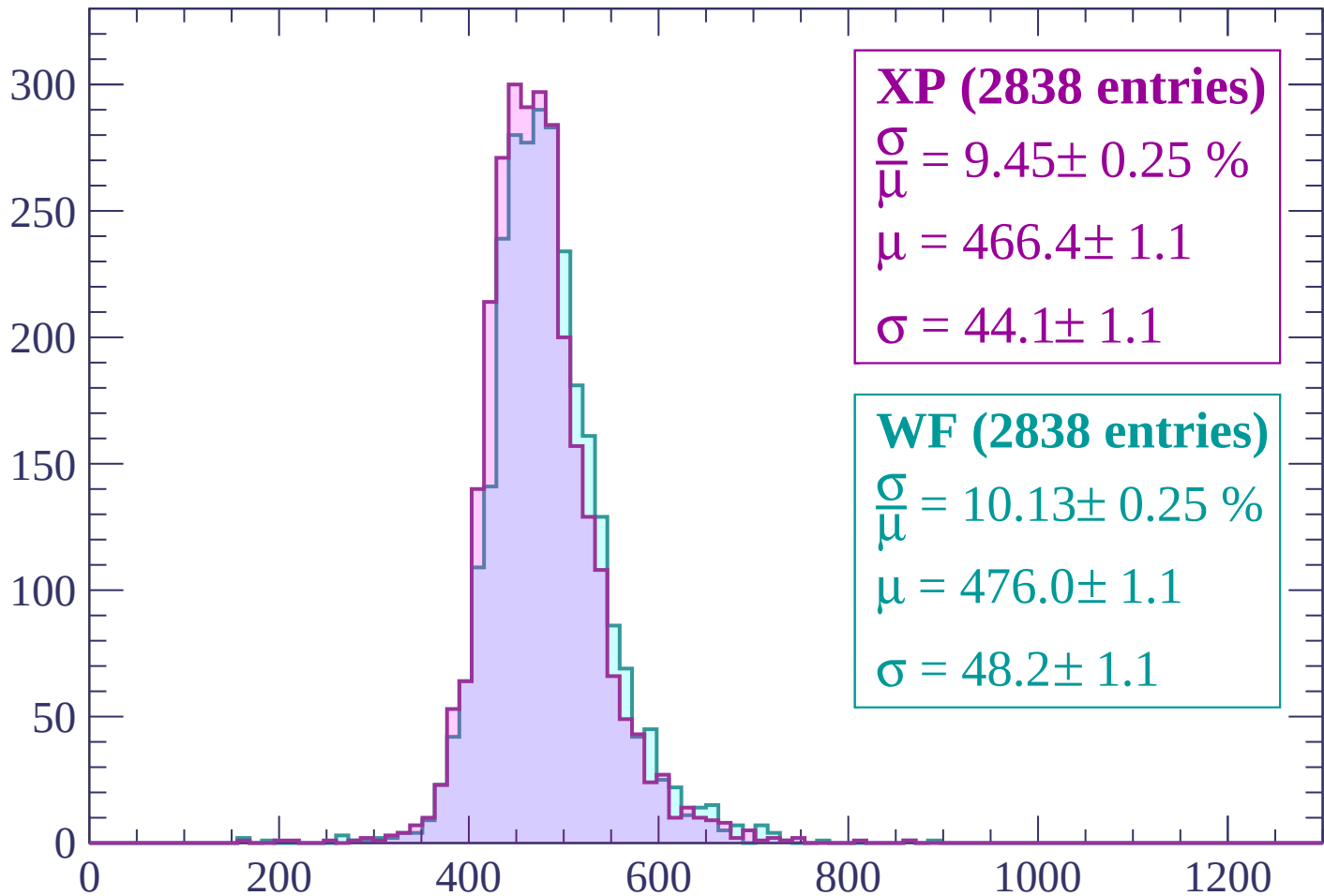
$$\mu = 472.4 \pm 1.0$$

$$\sigma = 45.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2040 < p < 2100$

Count



XP (2838 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.25 \%$$

$$\mu = 466.4 \pm 1.1$$

$$\sigma = 44.1 \pm 1.1$$

WF (2838 entries)

$$\frac{\sigma}{\mu} = 10.13 \pm 0.25 \%$$

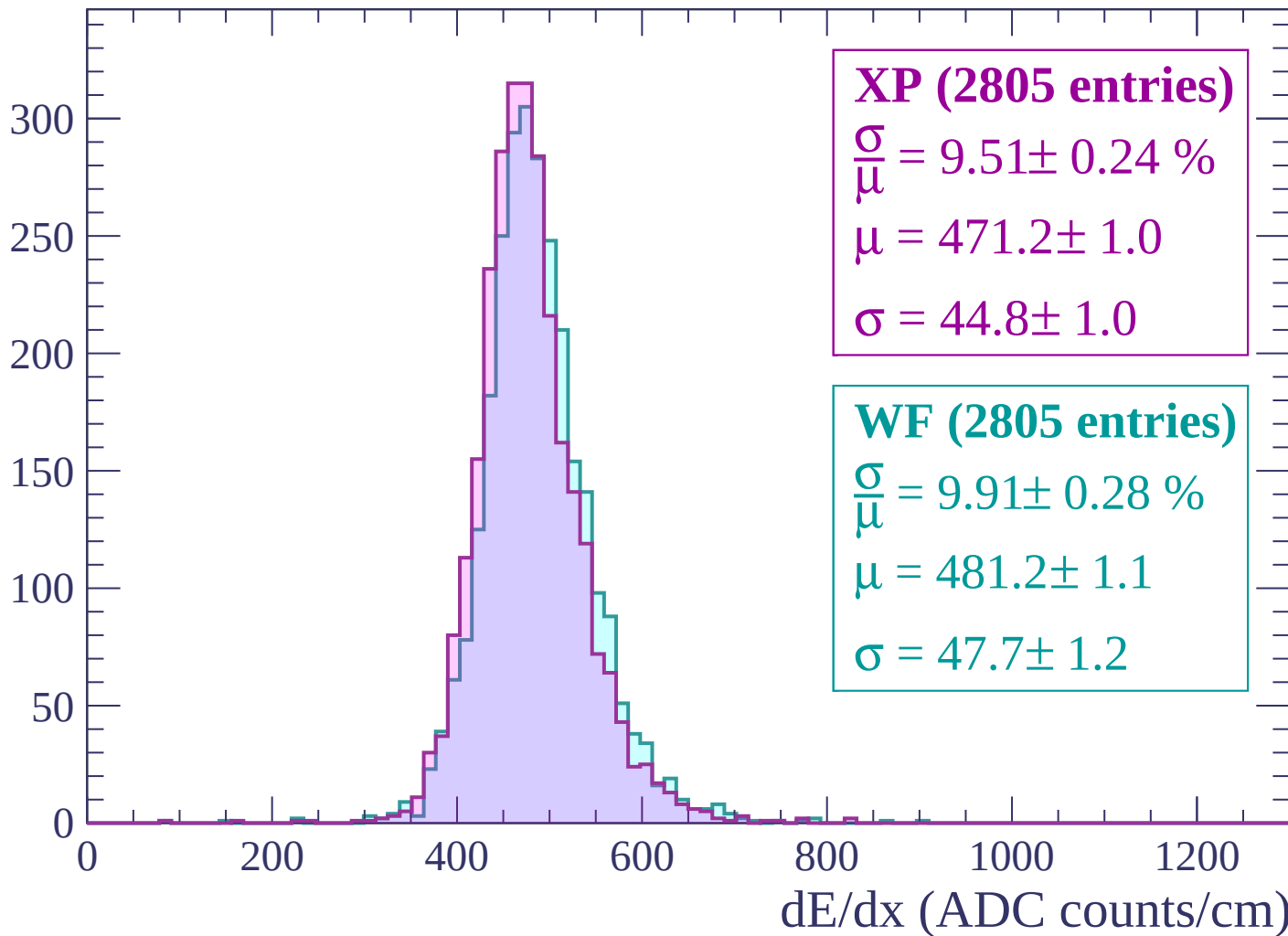
$$\mu = 476.0 \pm 1.1$$

$$\sigma = 48.2 \pm 1.1$$

dE/dx (ADC counts/cm)

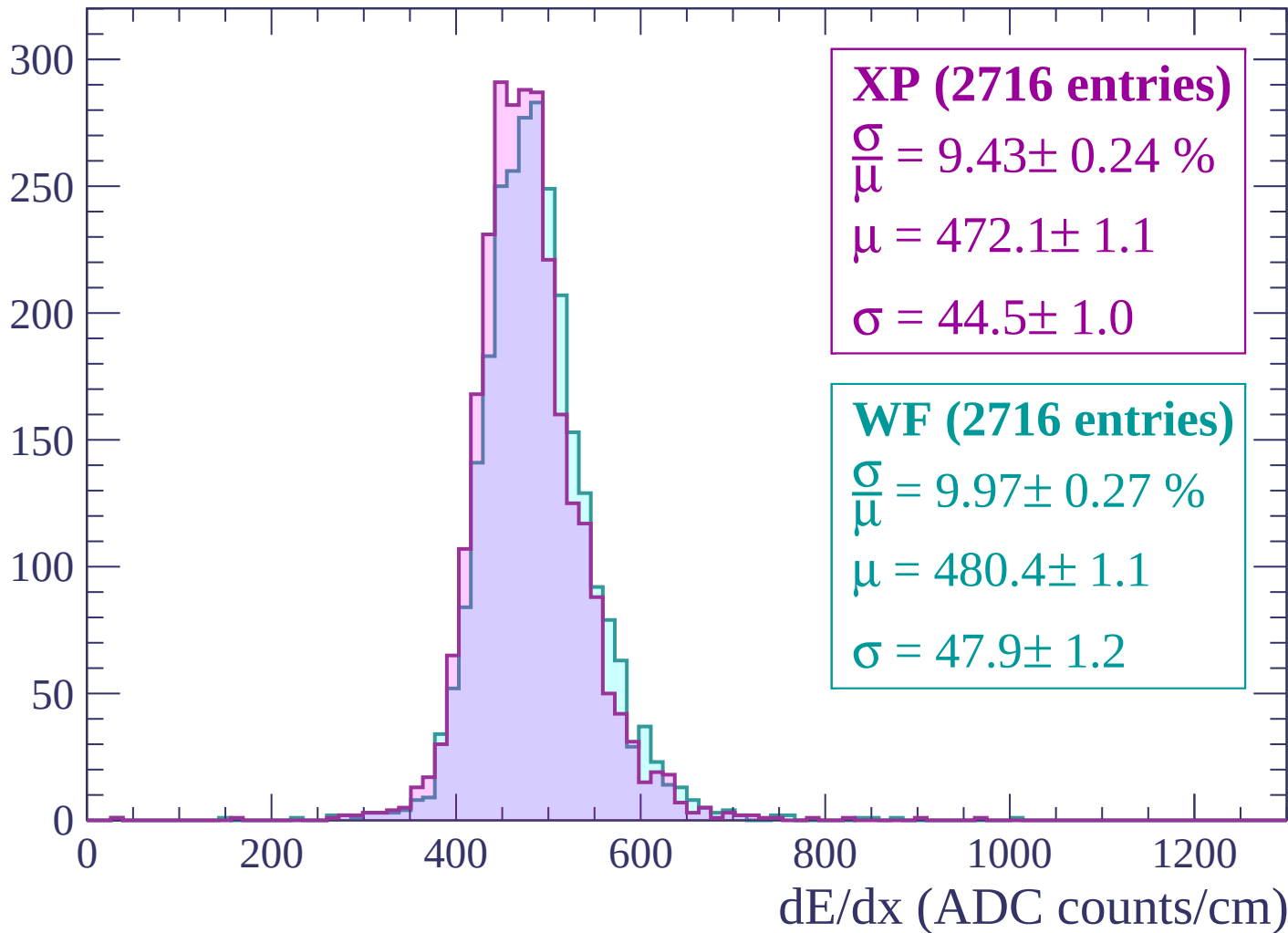
Energy loss | $2100 < p < 2160$

Count



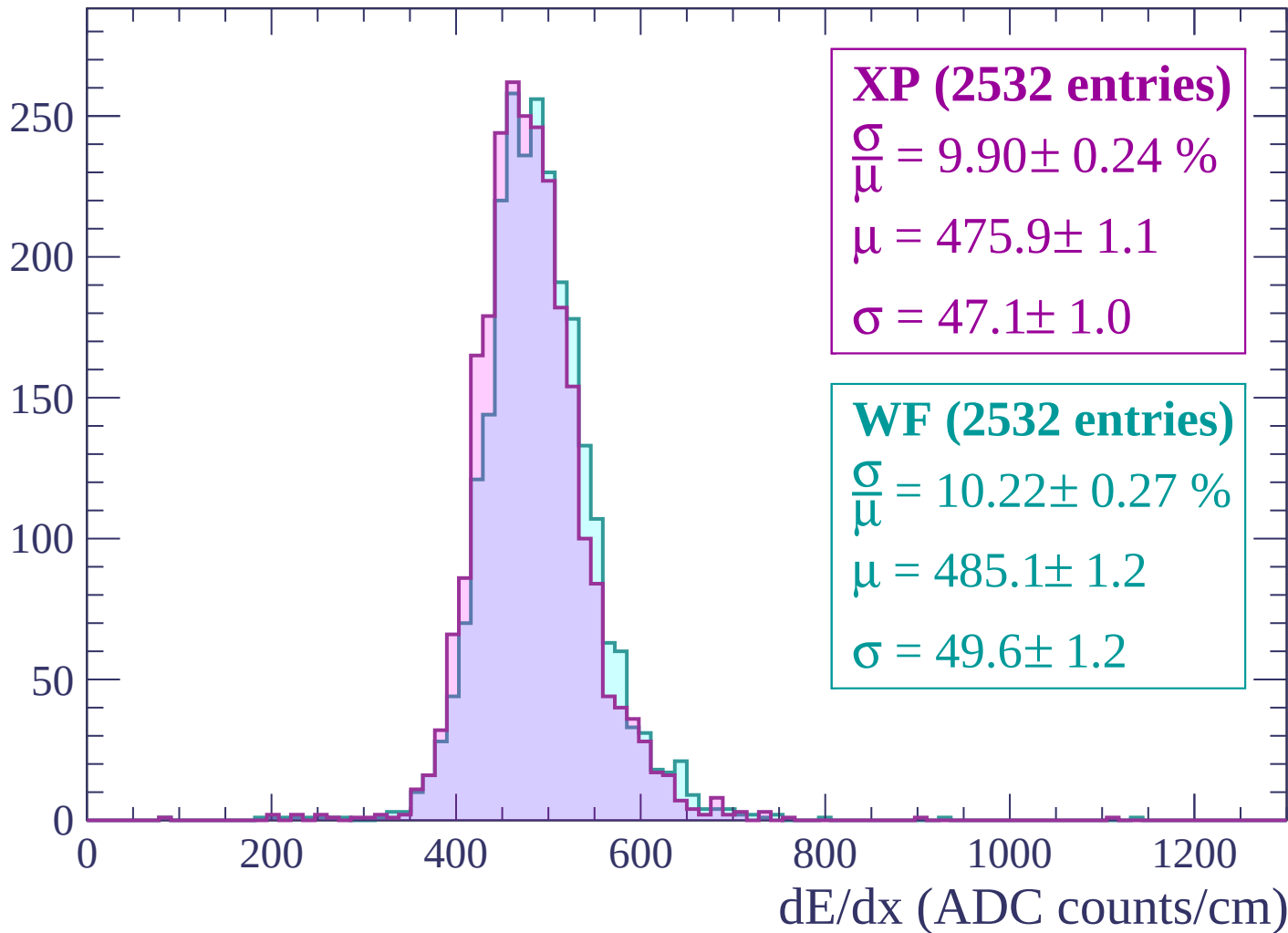
Energy loss | $2160 < p < 2220$

Count



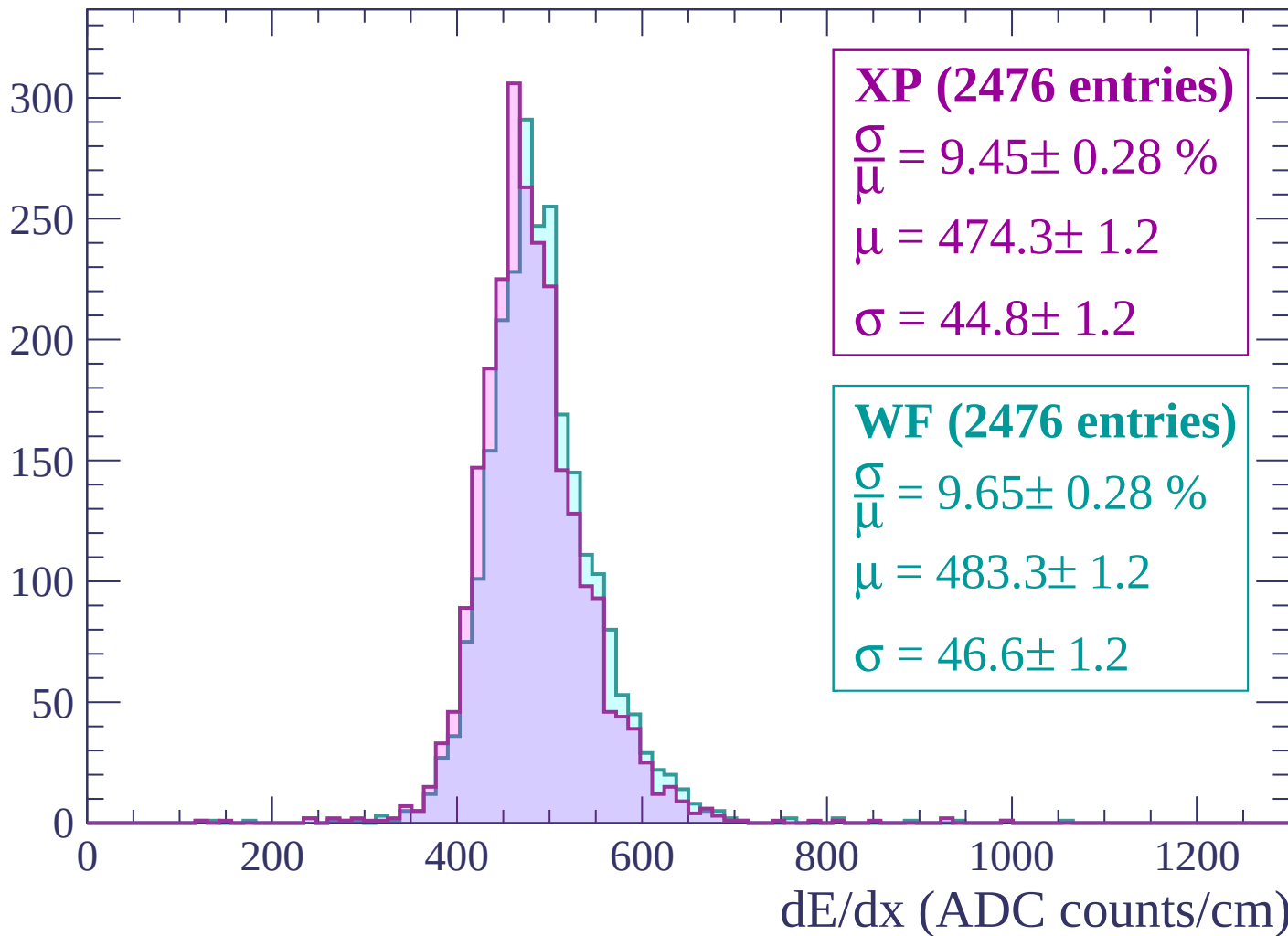
Energy loss | $2220 < p < 2280$

Count



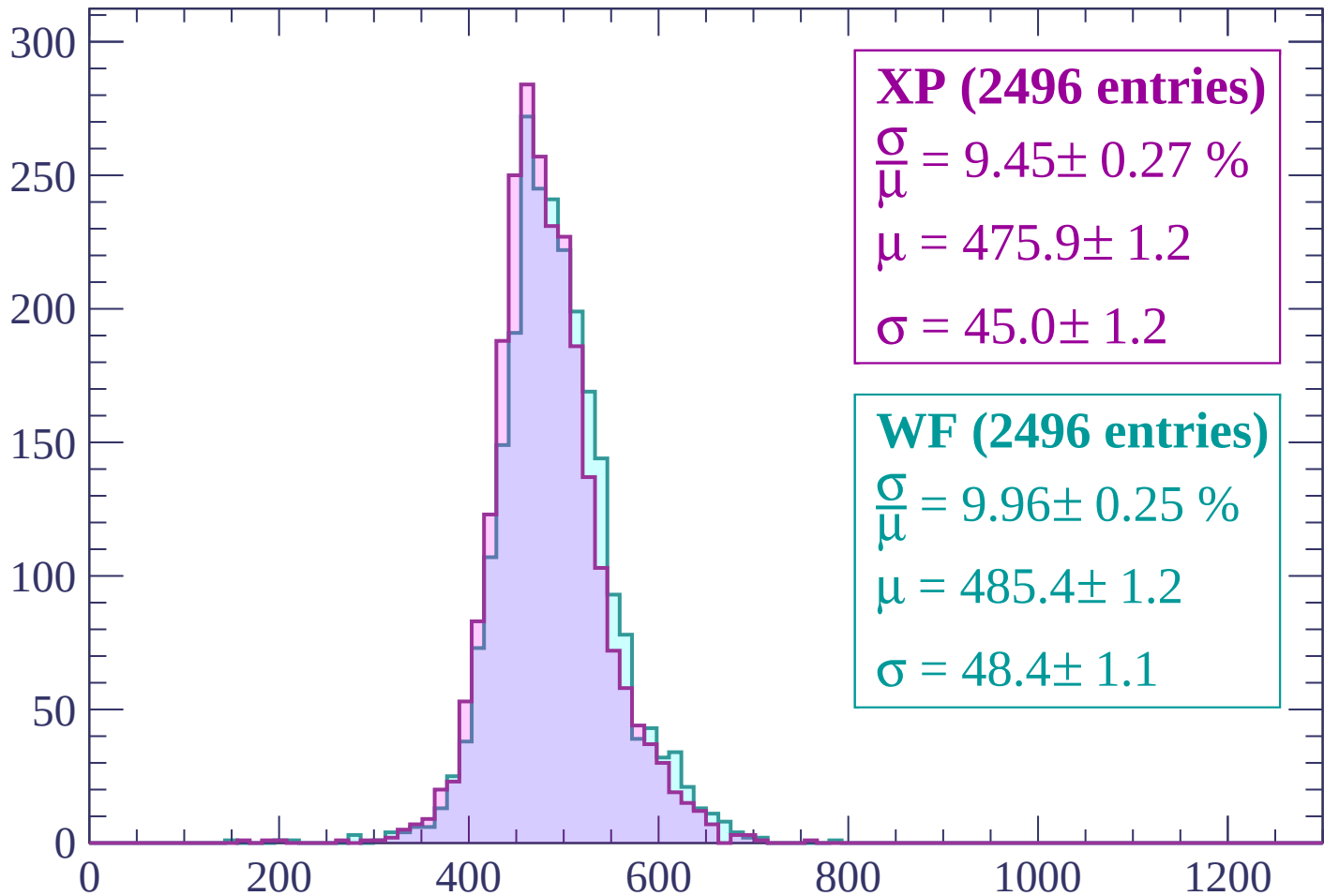
Energy loss | $2280 < p < 2340$

Count



Energy loss | $2340 < p < 2400$

Count



XP (2496 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.27 \%$$

$$\mu = 475.9 \pm 1.2$$

$$\sigma = 45.0 \pm 1.2$$

WF (2496 entries)

$$\frac{\sigma}{\mu} = 9.96 \pm 0.25 \%$$

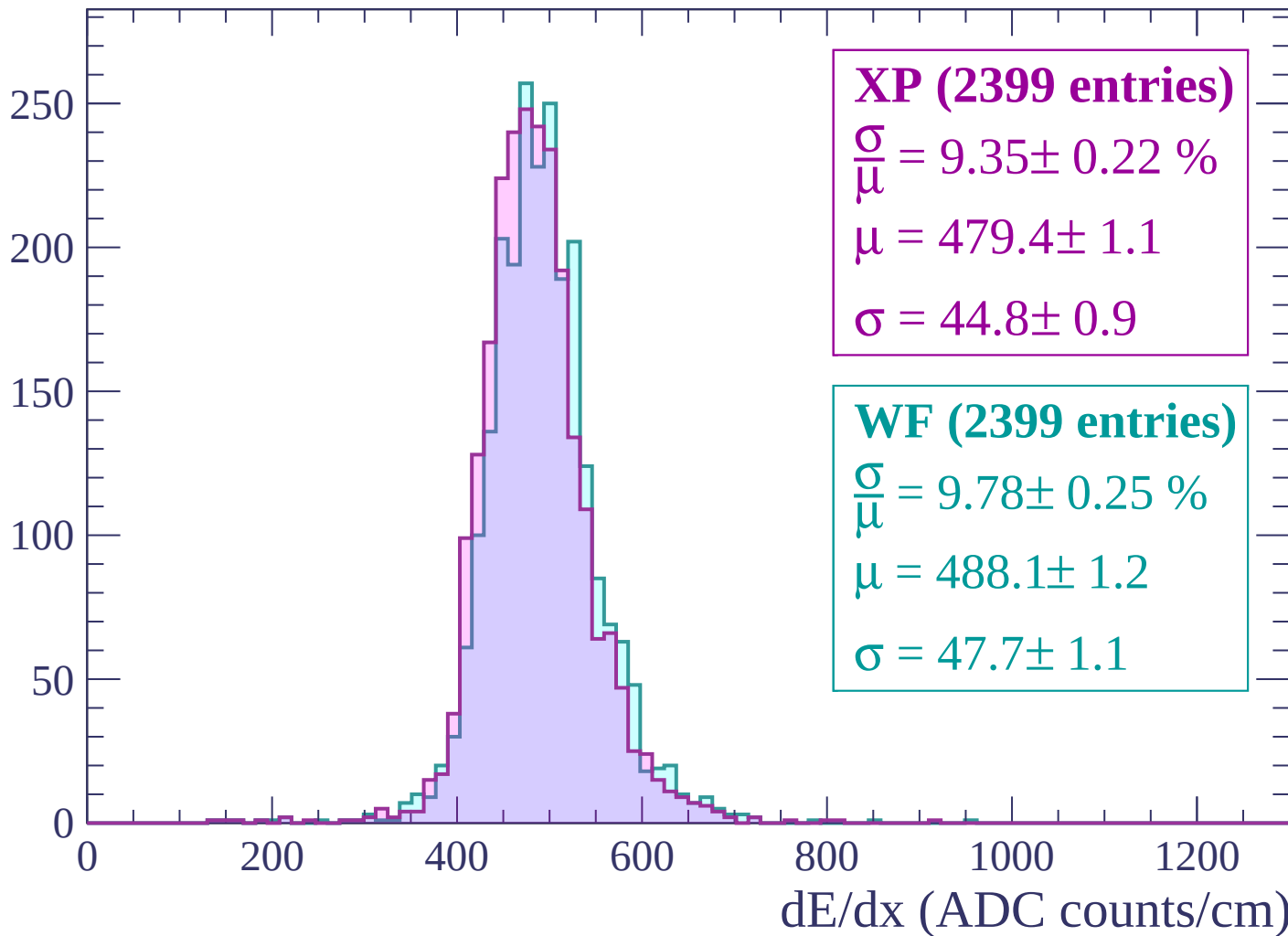
$$\mu = 485.4 \pm 1.2$$

$$\sigma = 48.4 \pm 1.1$$

dE/dx (ADC counts/cm)

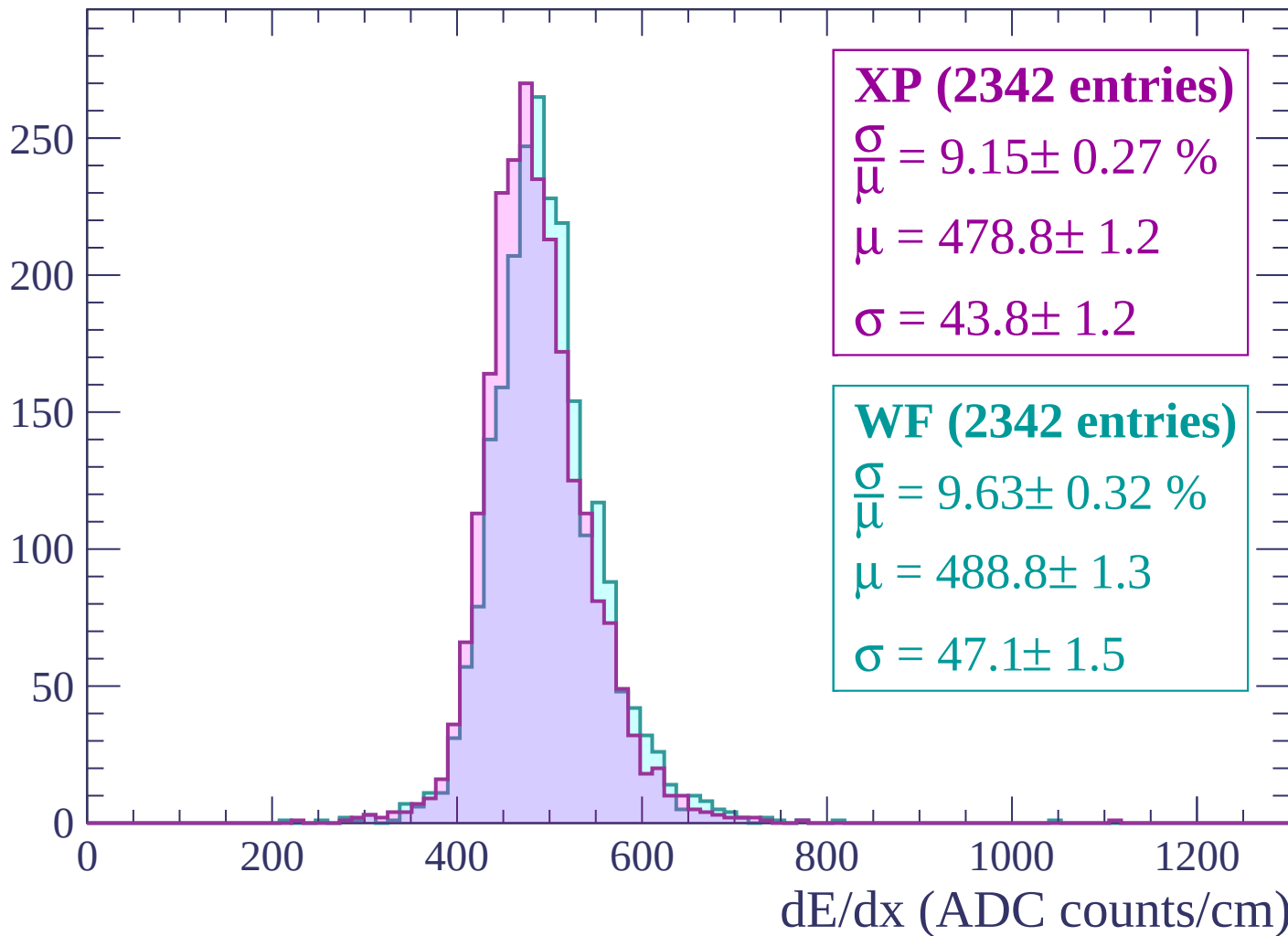
Energy loss | $2400 < p < 2460$

Count



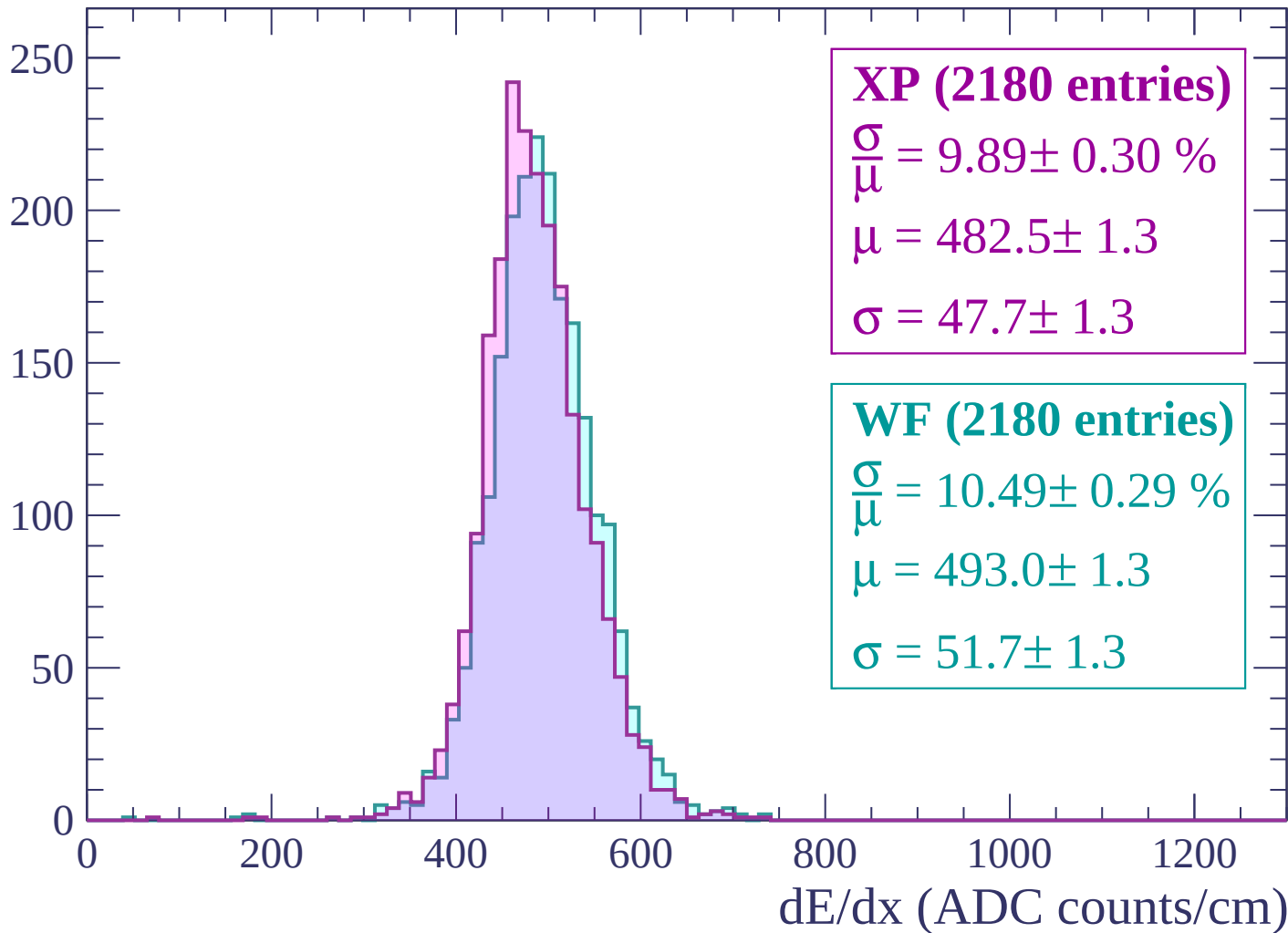
Energy loss | $2460 < p < 2520$

Count



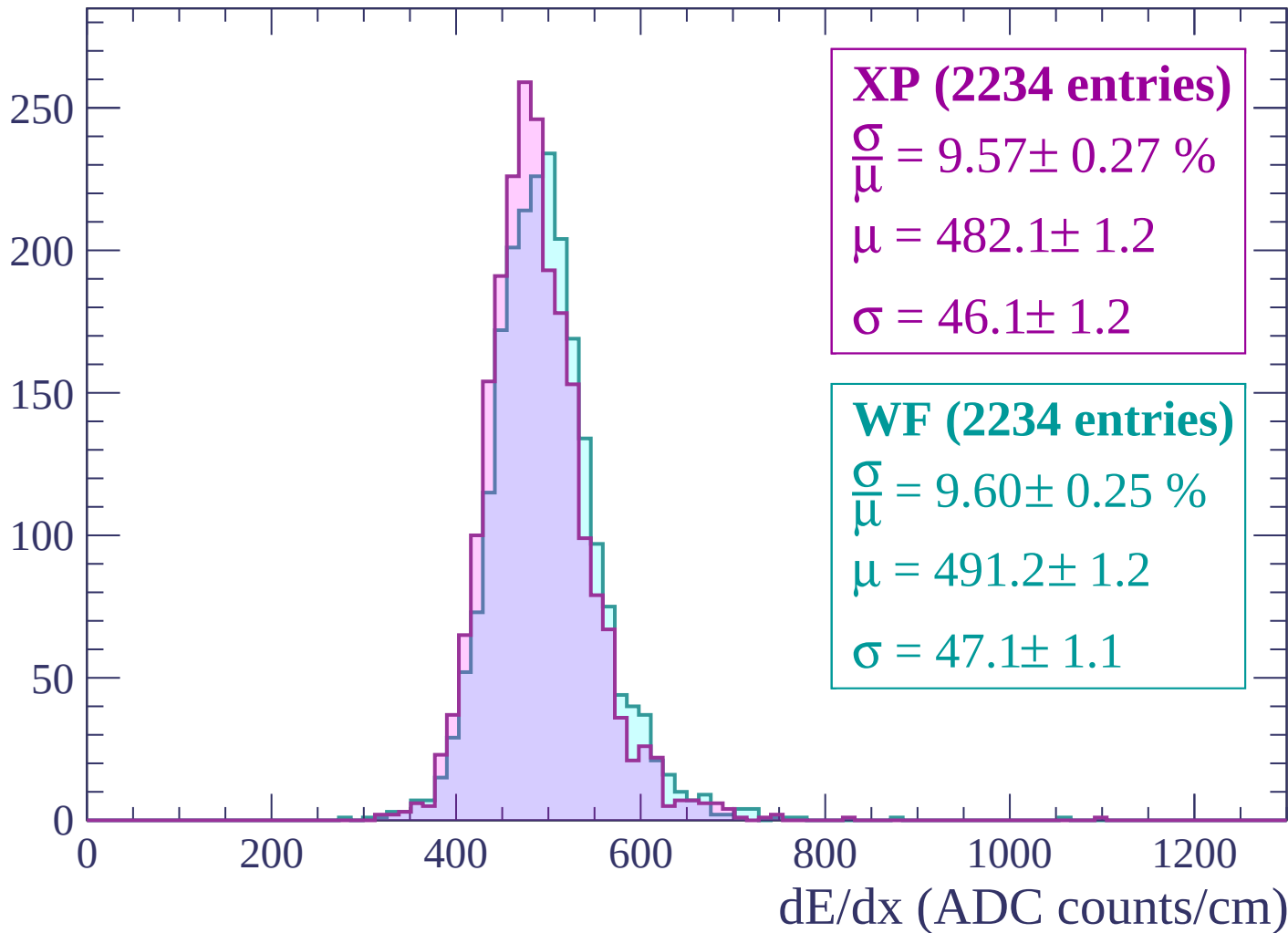
Energy loss | $2520 < p < 2580$

Count



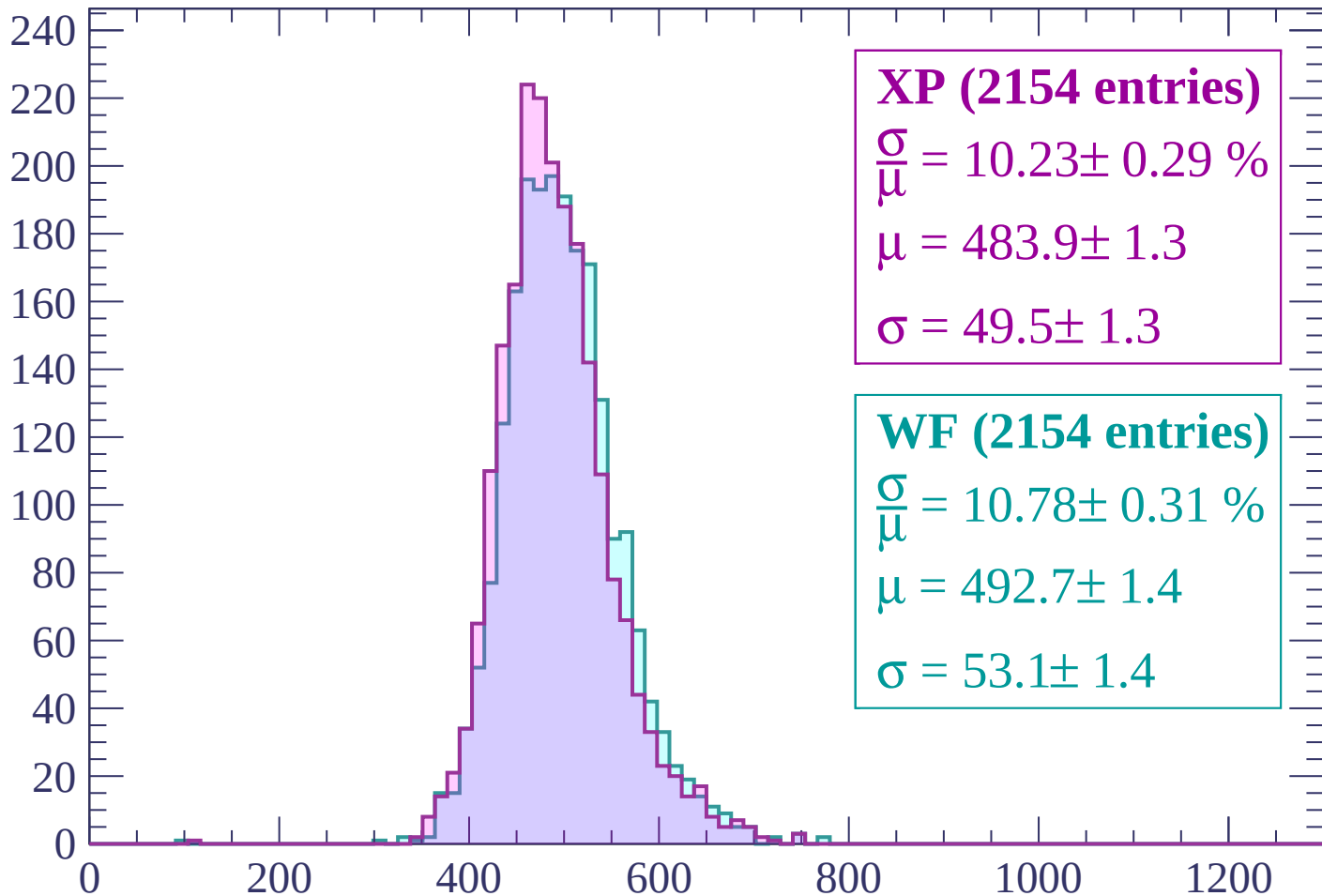
Energy loss | $2580 < p < 2640$

Count



Energy loss | $2640 < p < 2700$

Count



XP (2154 entries)

$\frac{\sigma}{\mu} = 10.23 \pm 0.29 \%$

$\mu = 483.9 \pm 1.3$

$\sigma = 49.5 \pm 1.3$

WF (2154 entries)

$\frac{\sigma}{\mu} = 10.78 \pm 0.31 \%$

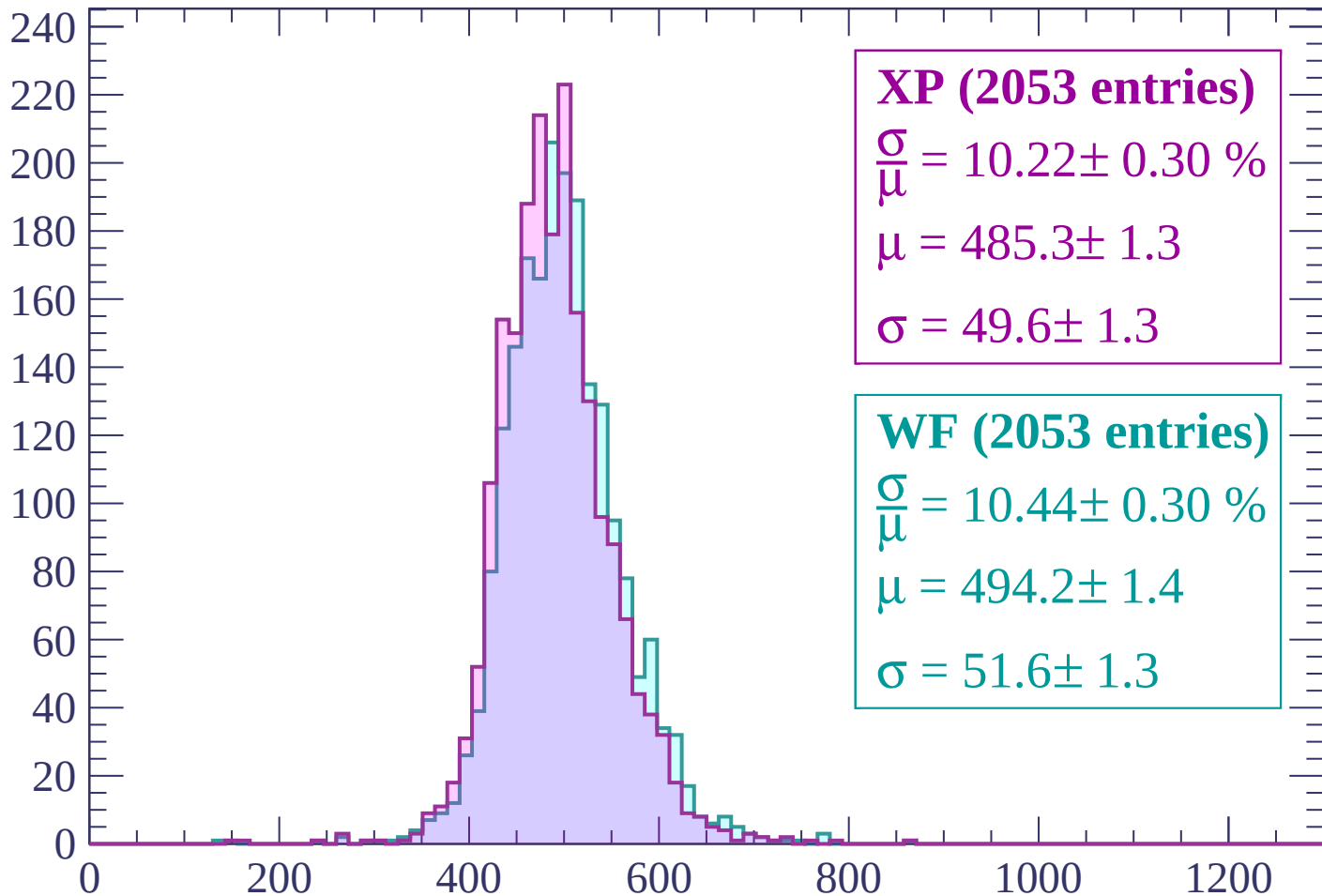
$\mu = 492.7 \pm 1.4$

$\sigma = 53.1 \pm 1.4$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP (2053 entries)

$$\frac{\sigma}{\mu} = 10.22 \pm 0.30 \%$$

$$\mu = 485.3 \pm 1.3$$

$$\sigma = 49.6 \pm 1.3$$

WF (2053 entries)

$$\frac{\sigma}{\mu} = 10.44 \pm 0.30 \%$$

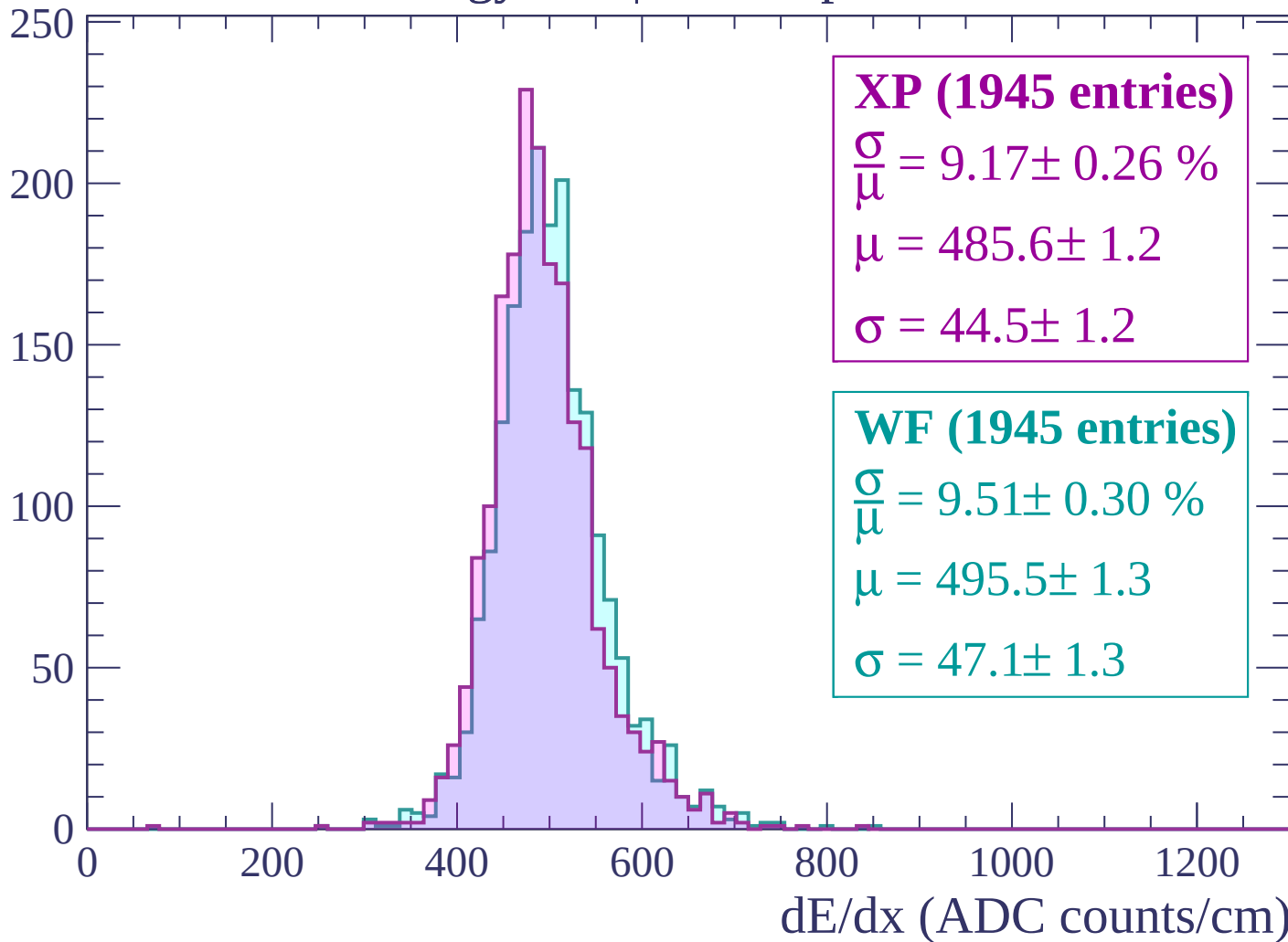
$$\mu = 494.2 \pm 1.4$$

$$\sigma = 51.6 \pm 1.3$$

dE/dx (ADC counts/cm)

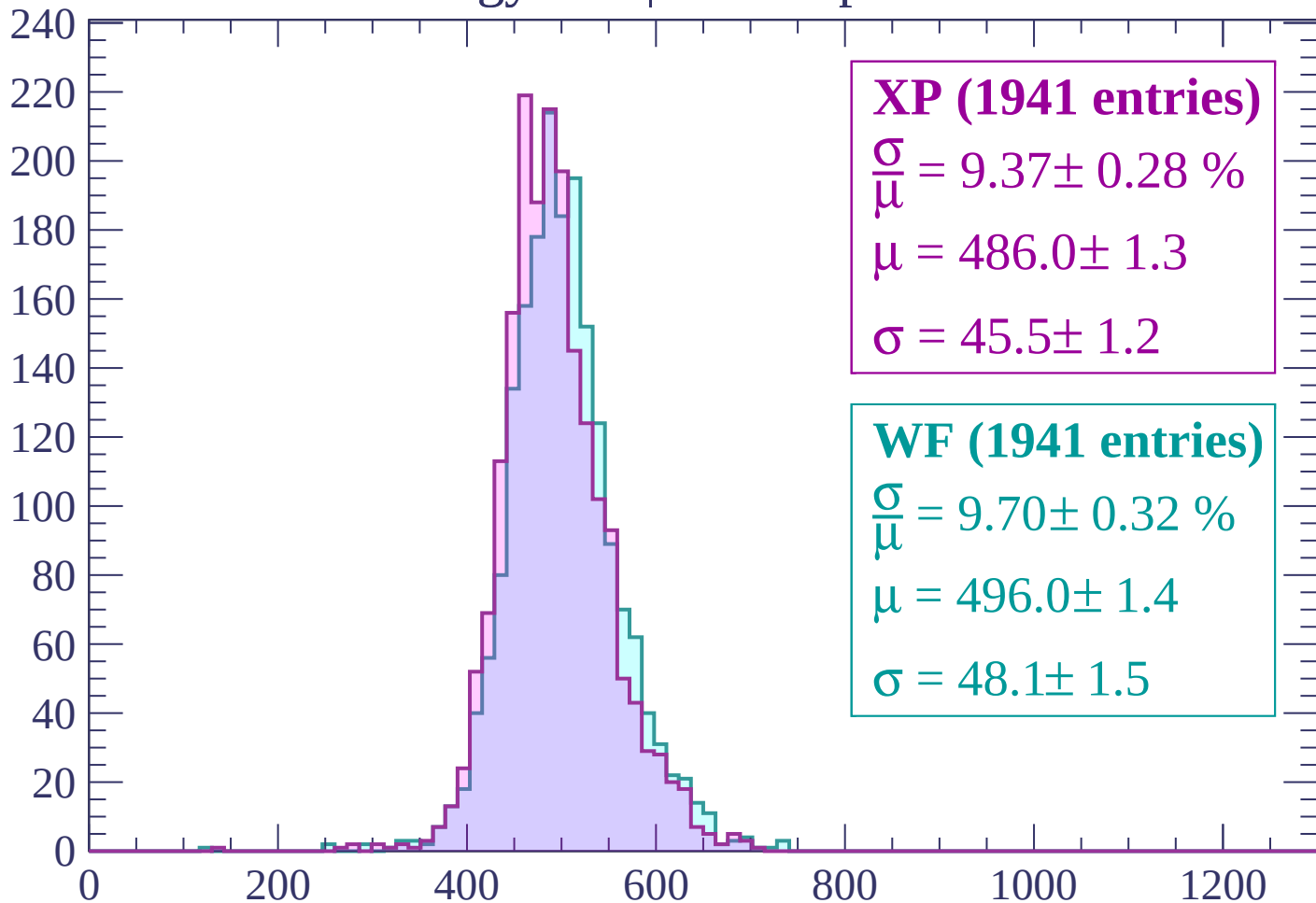
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP (1941 entries)

$$\frac{\sigma}{\mu} = 9.37 \pm 0.28 \%$$

$$\mu = 486.0 \pm 1.3$$

$$\sigma = 45.5 \pm 1.2$$

WF (1941 entries)

$$\frac{\sigma}{\mu} = 9.70 \pm 0.32 \%$$

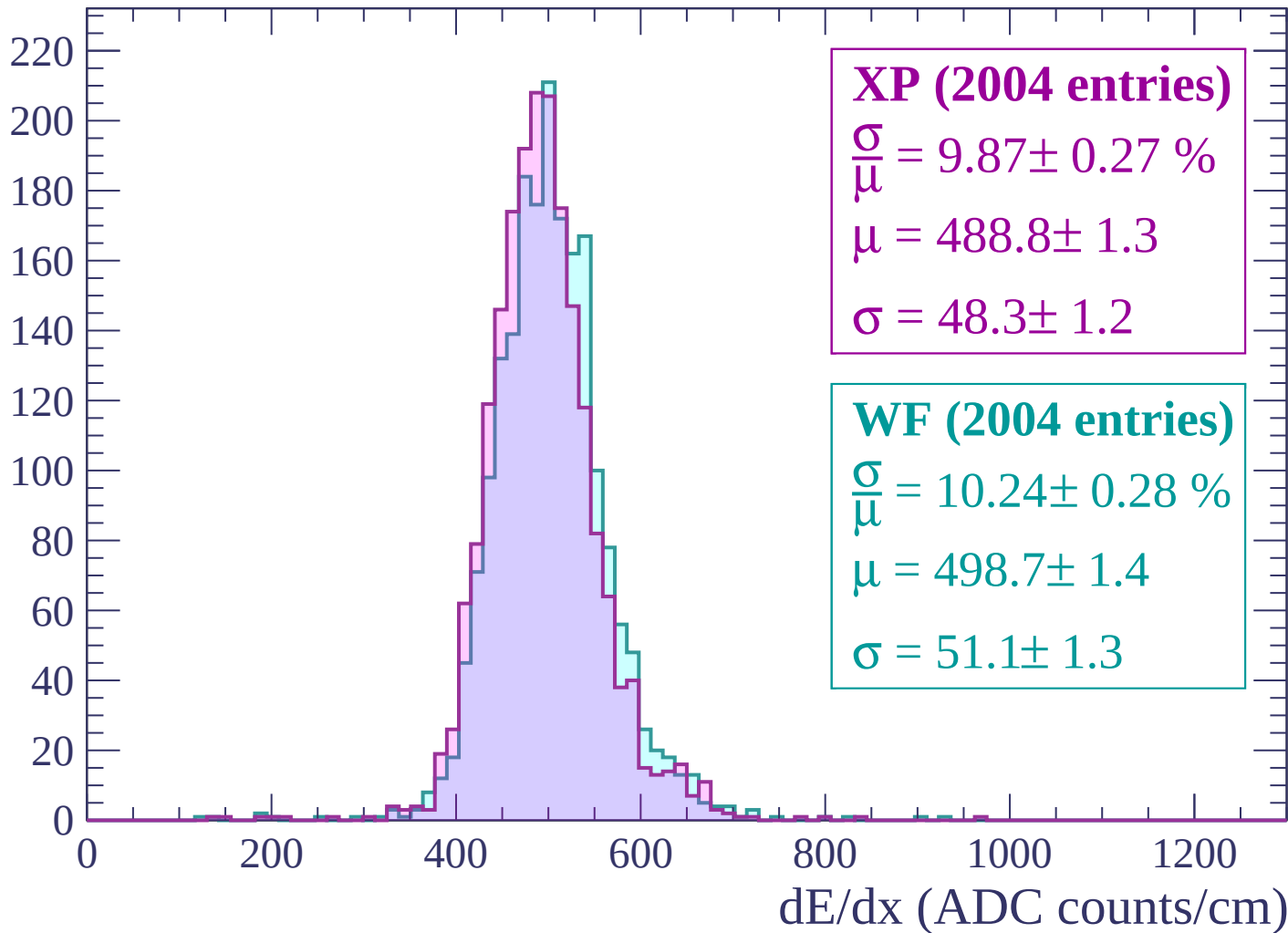
$$\mu = 496.0 \pm 1.4$$

$$\sigma = 48.1 \pm 1.5$$

dE/dx (ADC counts/cm)

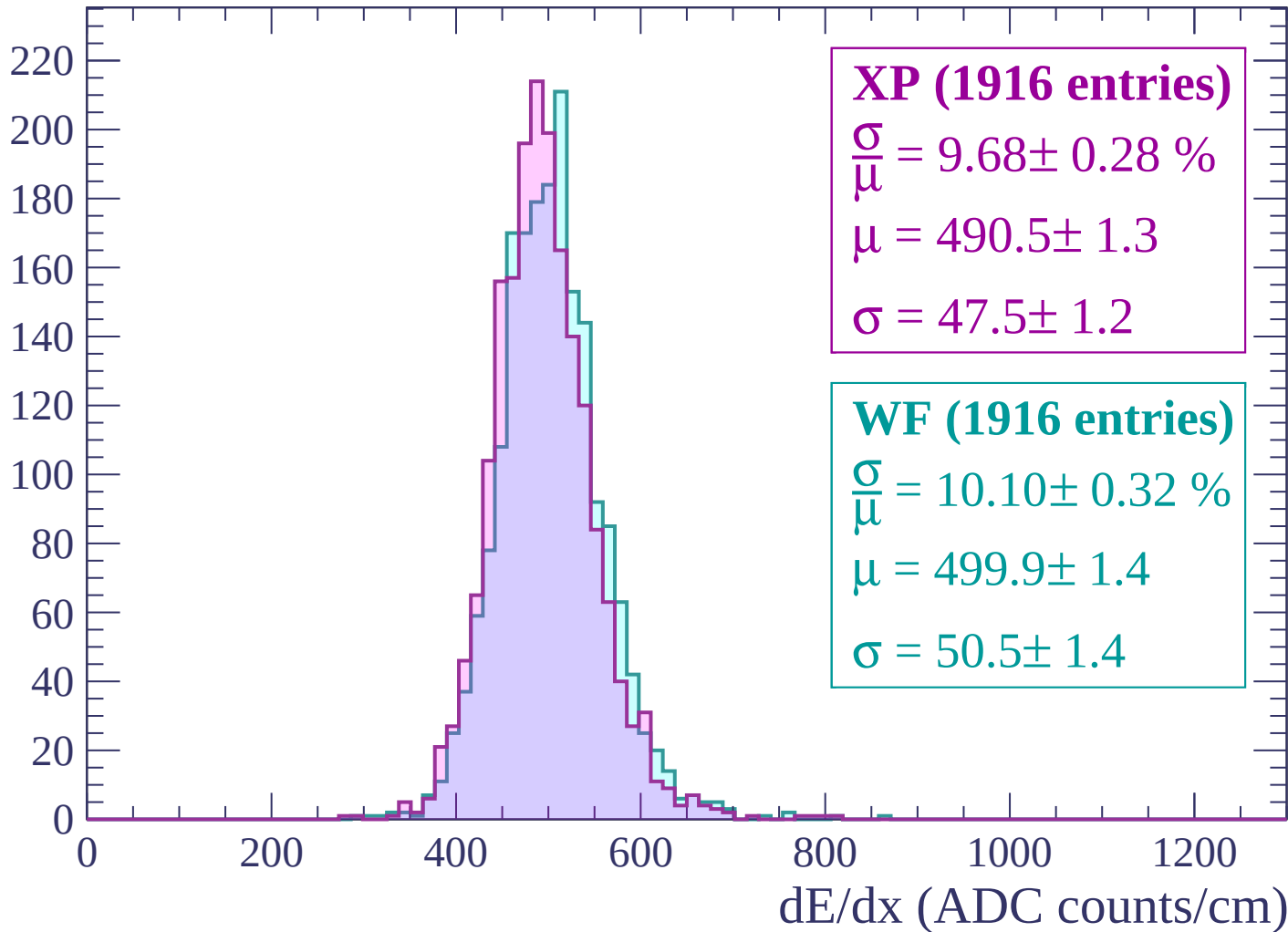
Energy loss | $2880 < p < 2940$

Count



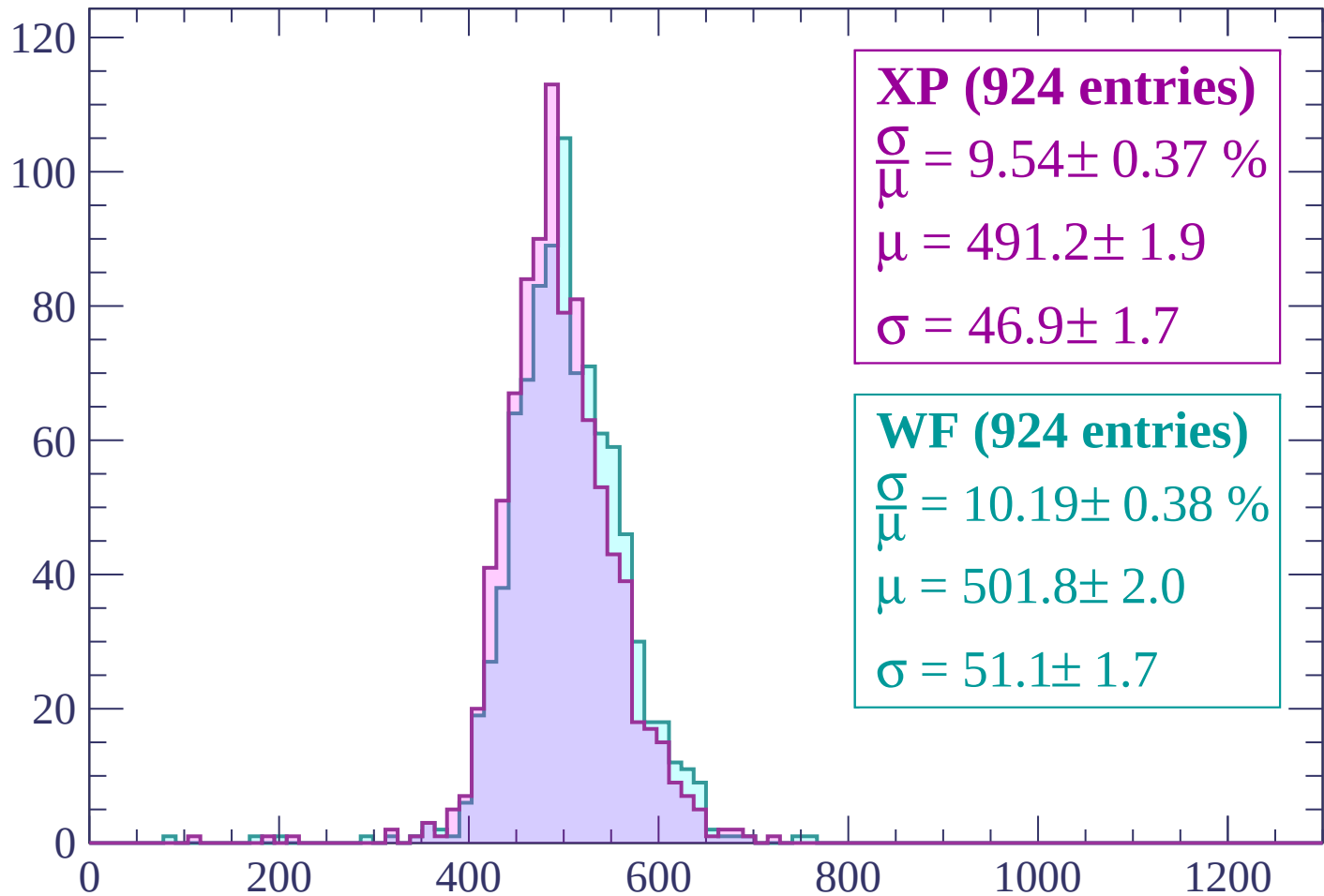
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



XP (924 entries)

$\frac{\sigma}{\mu} = 9.54 \pm 0.37 \%$

$\mu = 491.2 \pm 1.9$

$\sigma = 46.9 \pm 1.7$

WF (924 entries)

$\frac{\sigma}{\mu} = 10.19 \pm 0.38 \%$

$\mu = 501.8 \pm 2.0$

$\sigma = 51.1 \pm 1.7$

dE/dx (ADC counts/cm)

Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

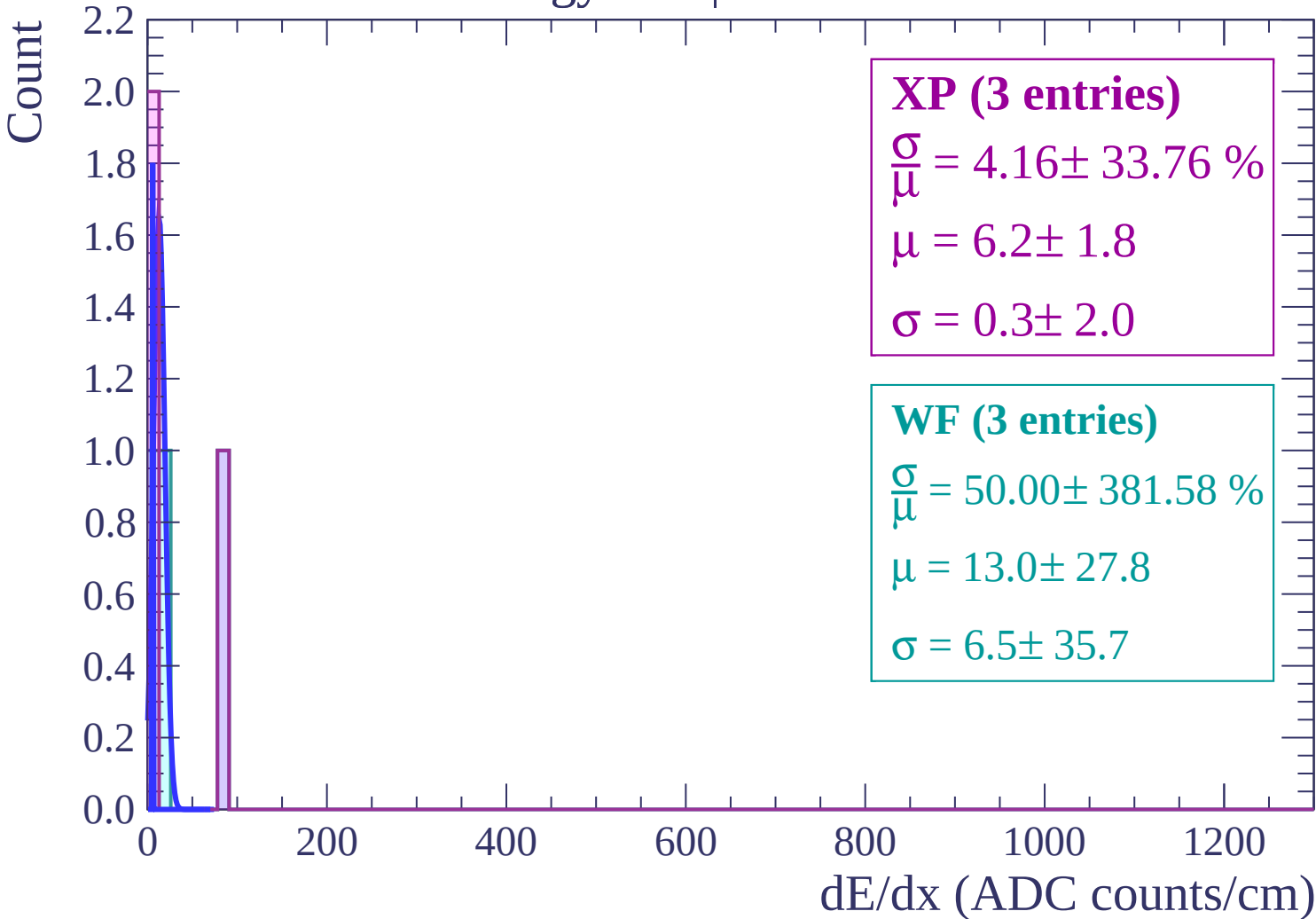
WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

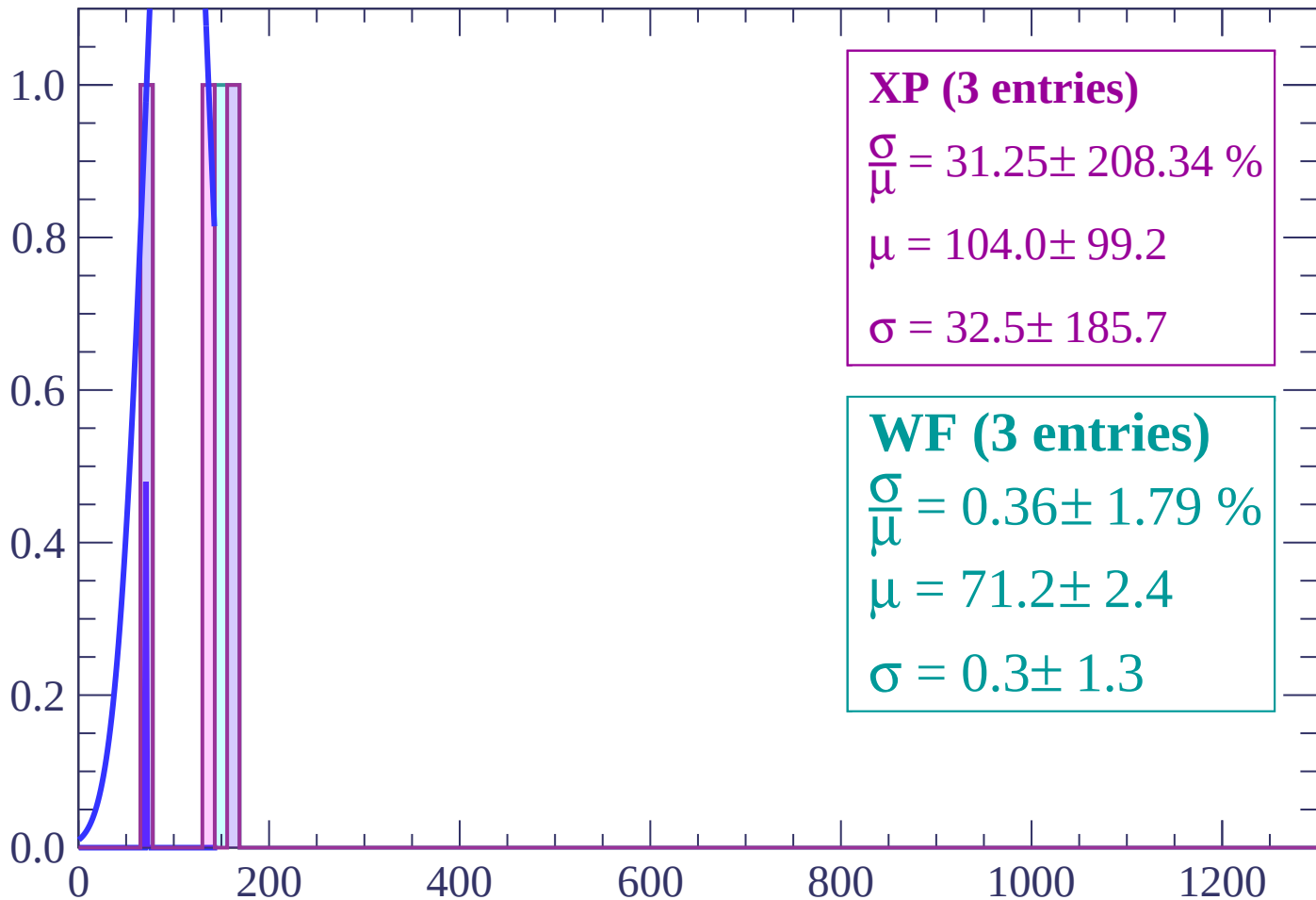
$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$



Energy loss | $-78 < \theta < -76$

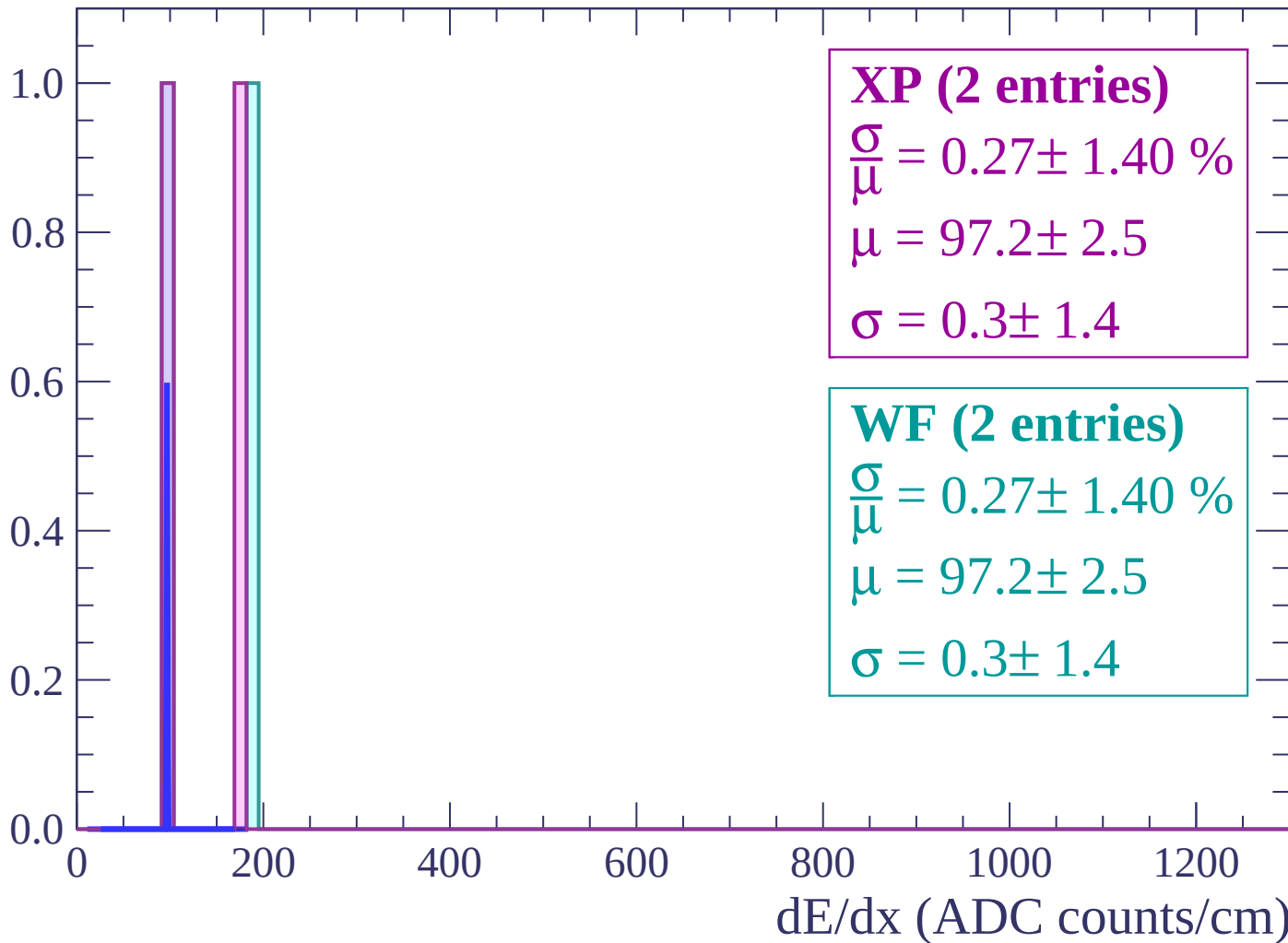
Count



dE/dx (ADC counts/cm)

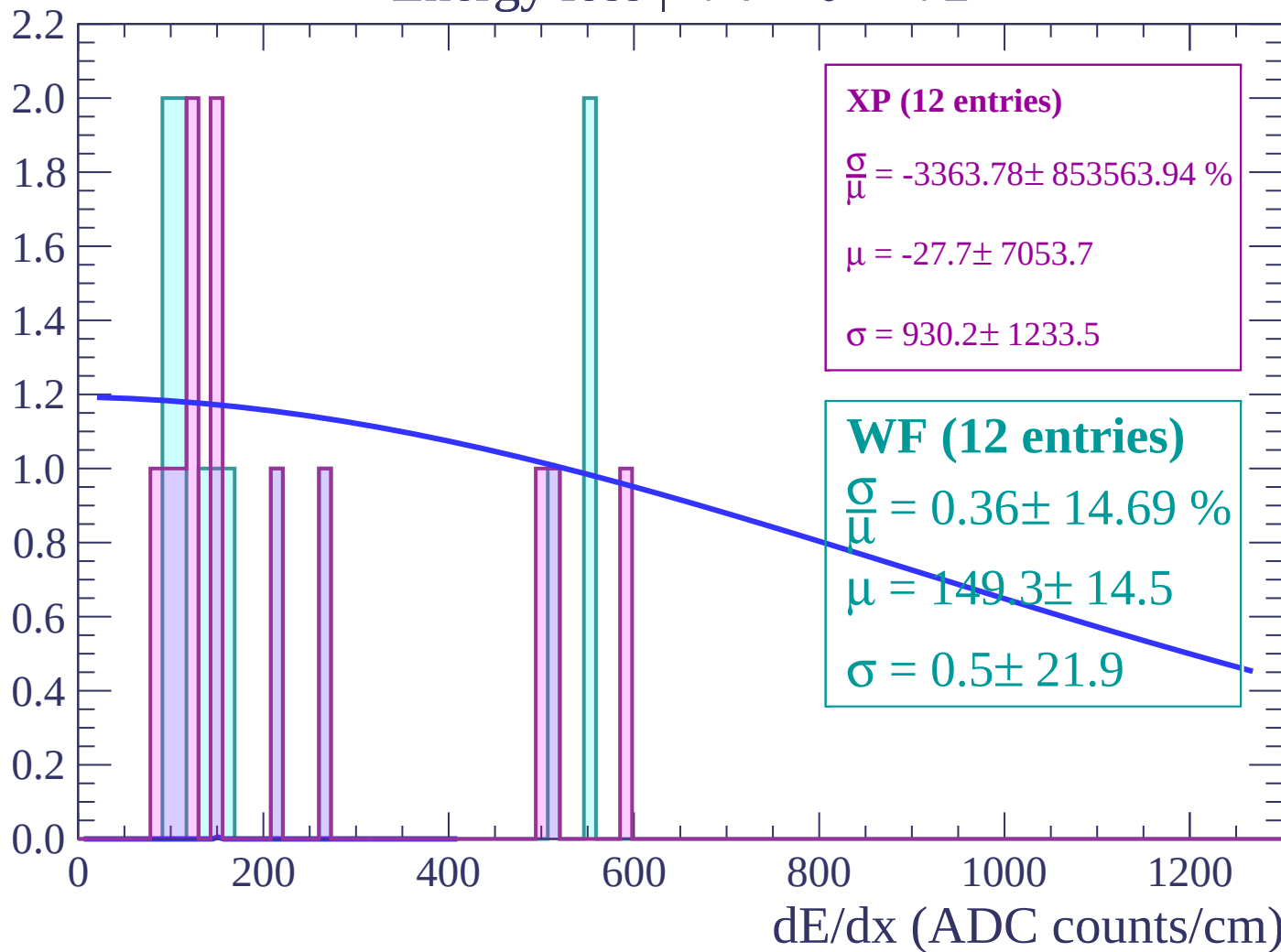
Energy loss | $-76 < \theta < -74$

Count



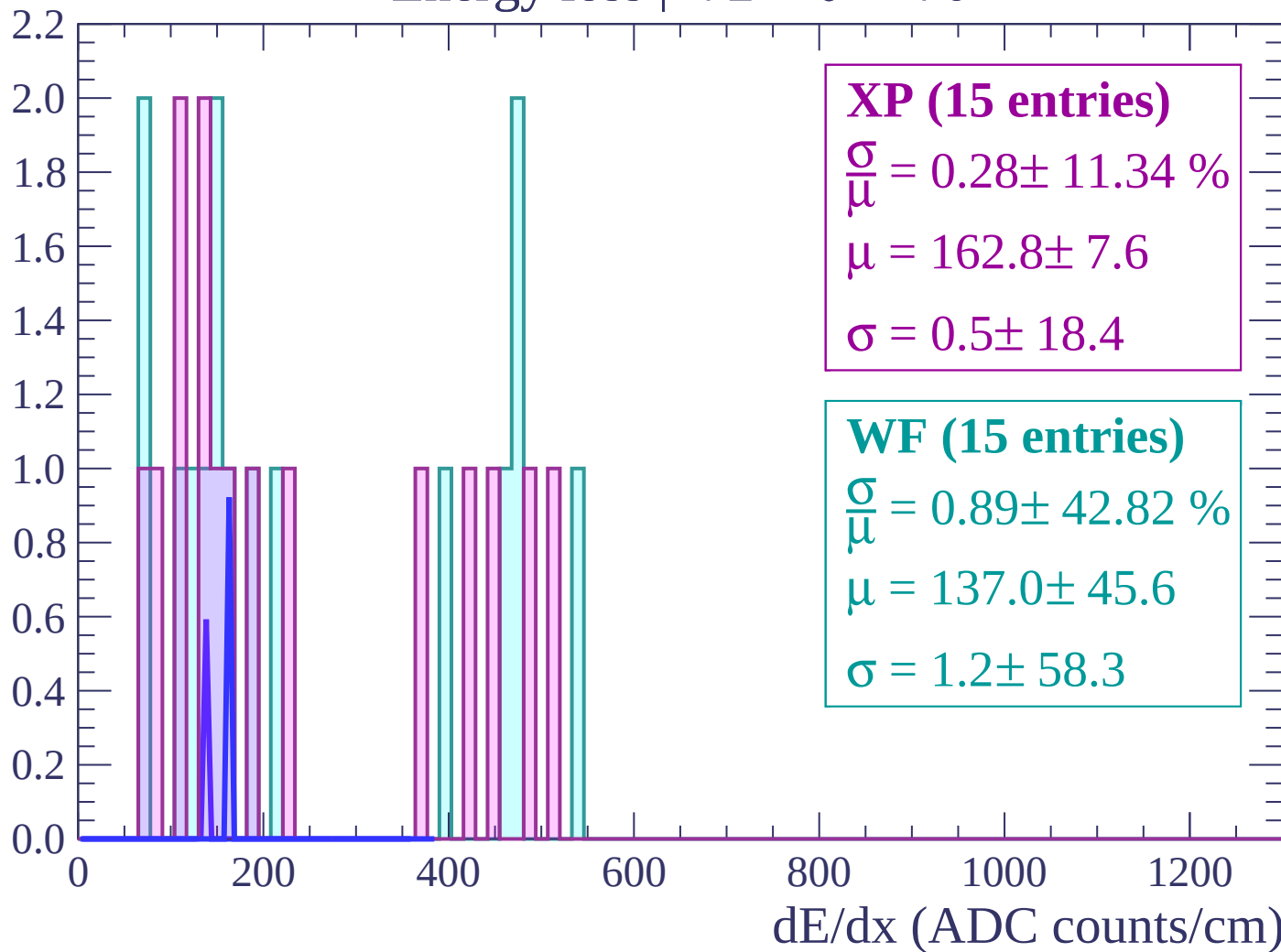
Energy loss | $-74 < \theta < -72$

Count



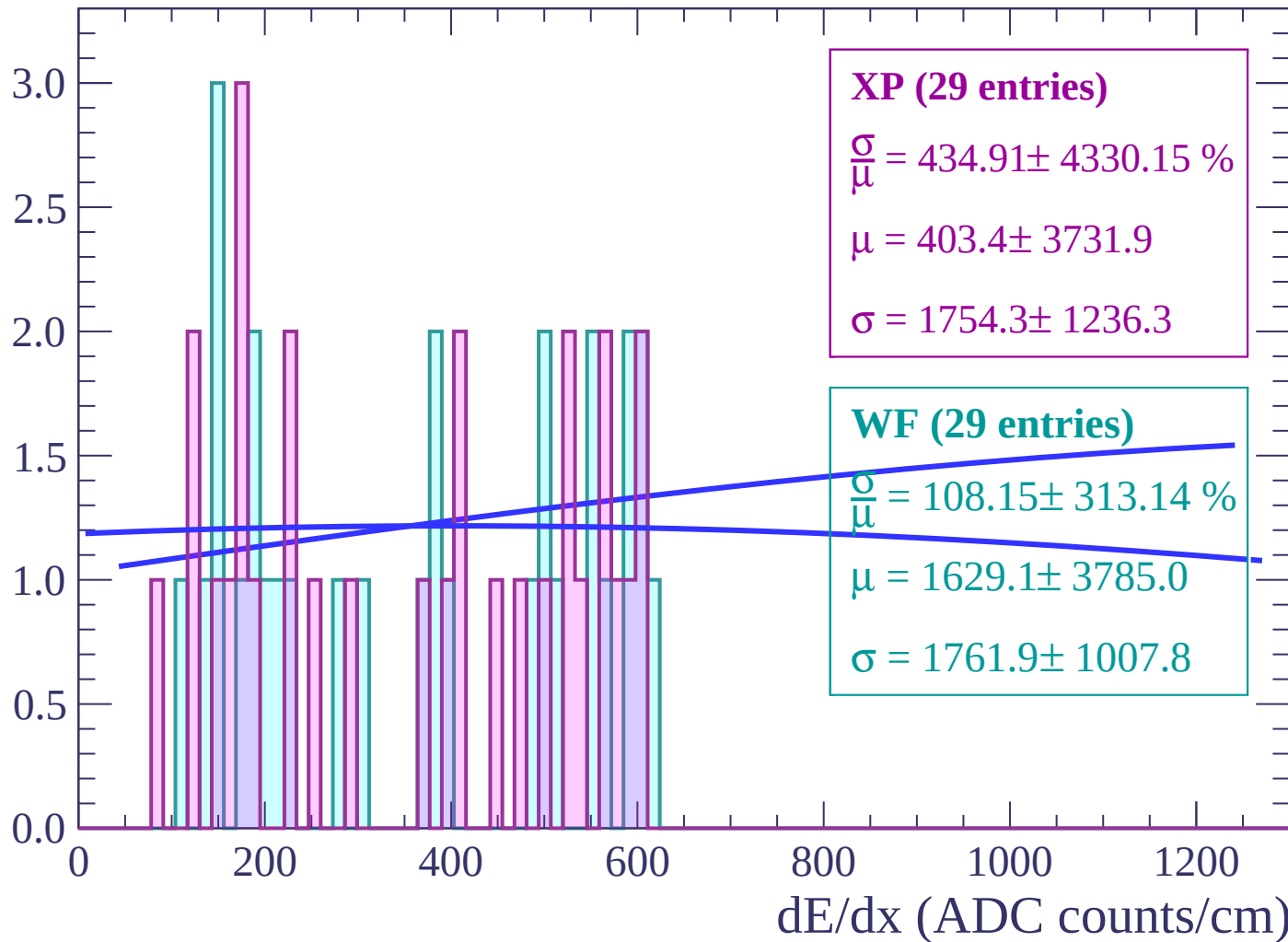
Energy loss | $-72 < \theta < -70$

Count



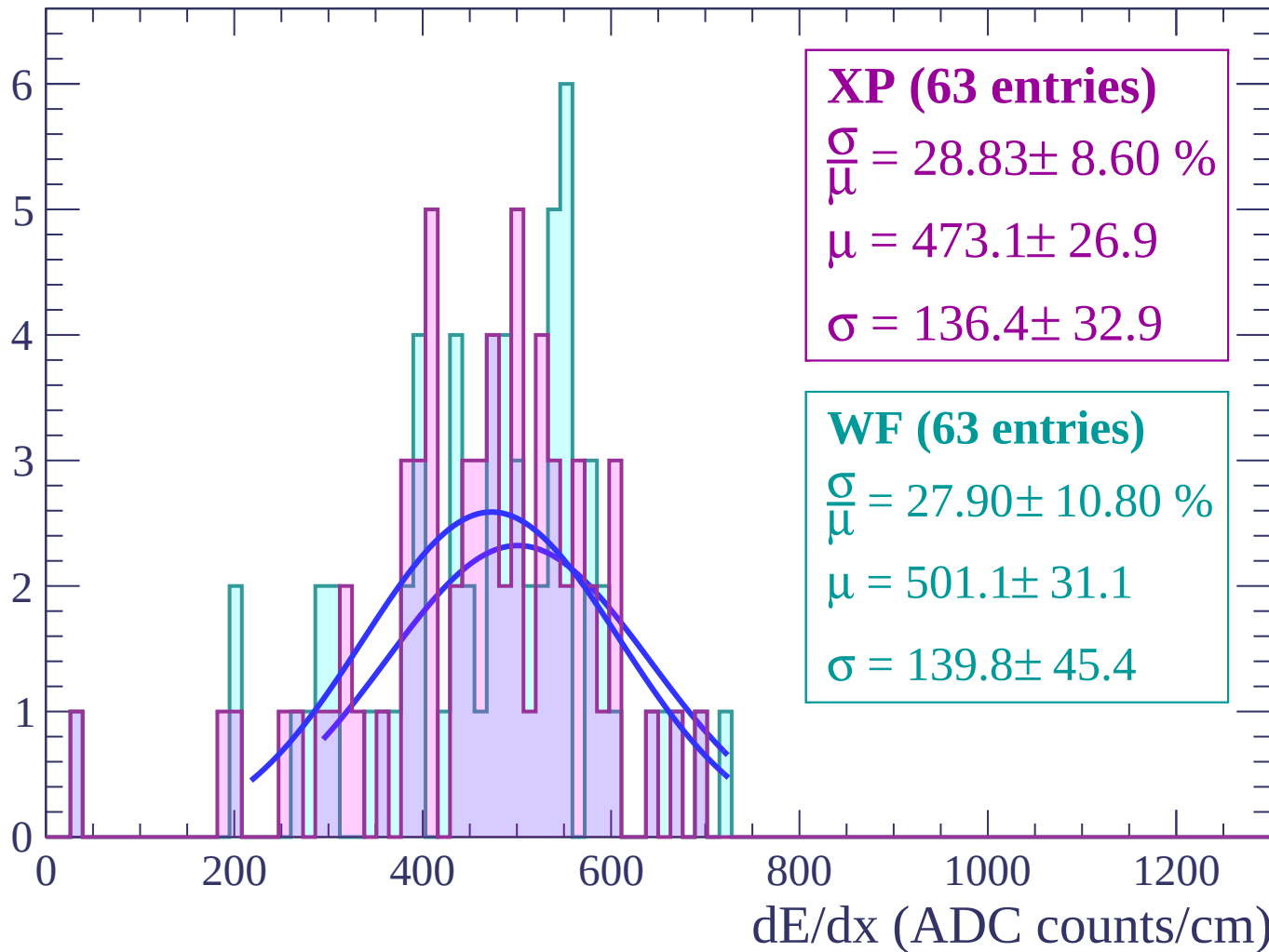
Energy loss | $-70 < \theta < -68$

Count



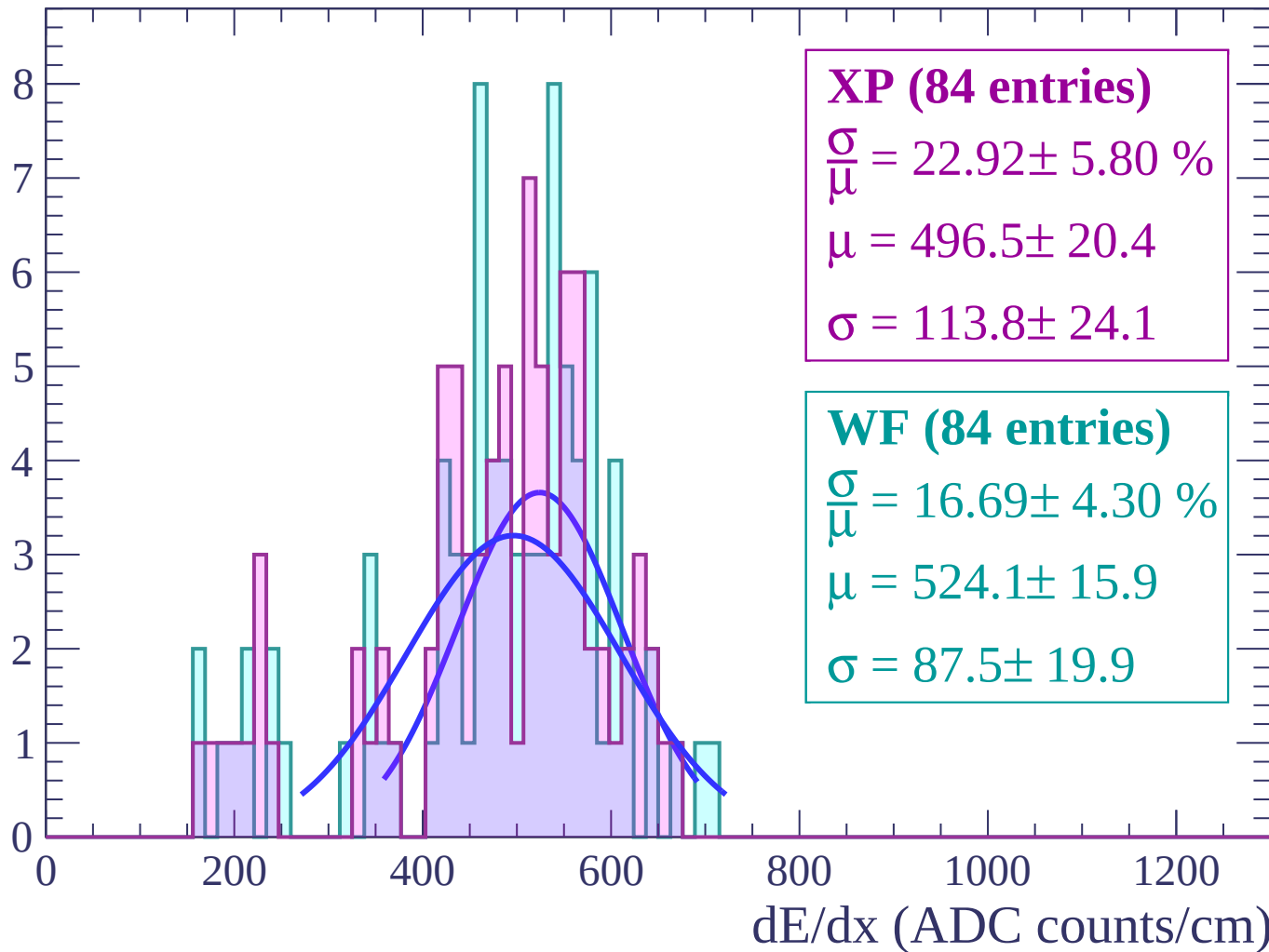
Energy loss | $-68 < \theta < -66$

Count



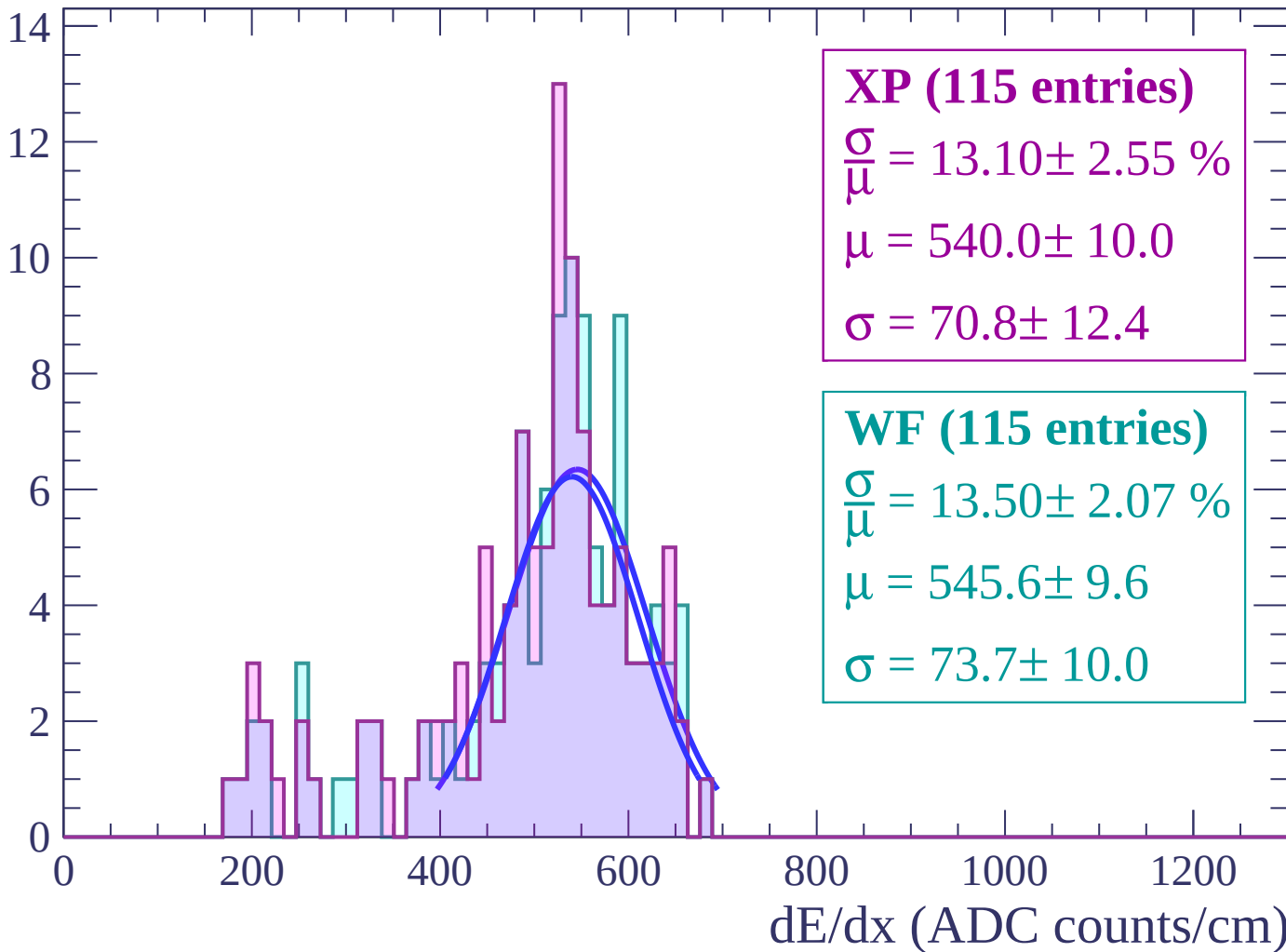
Energy loss | $-66 < \theta < -64$

Count



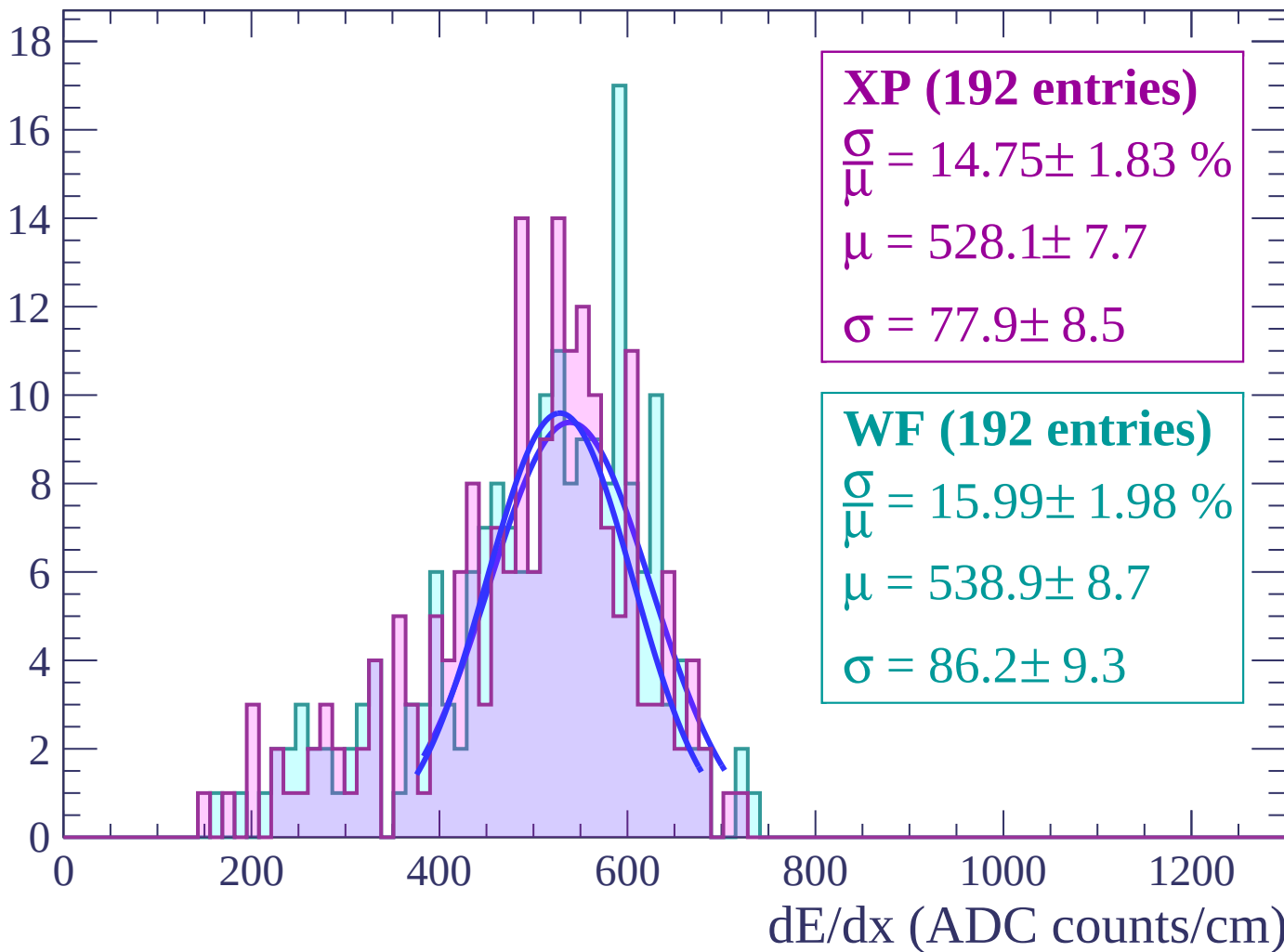
Energy loss | $-64 < \theta < -62$

Count



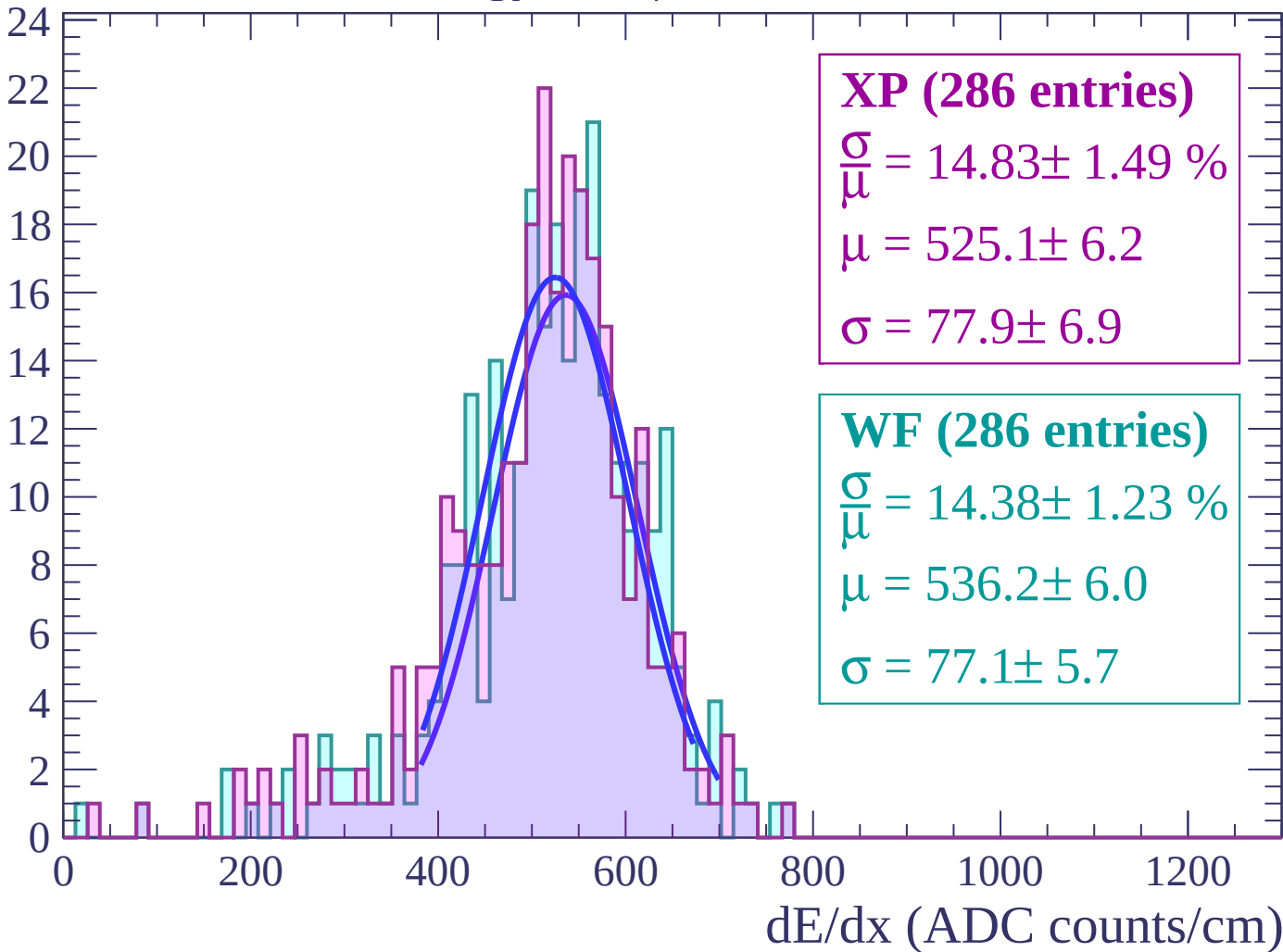
Energy loss | $-62 < \theta < -60$

Count



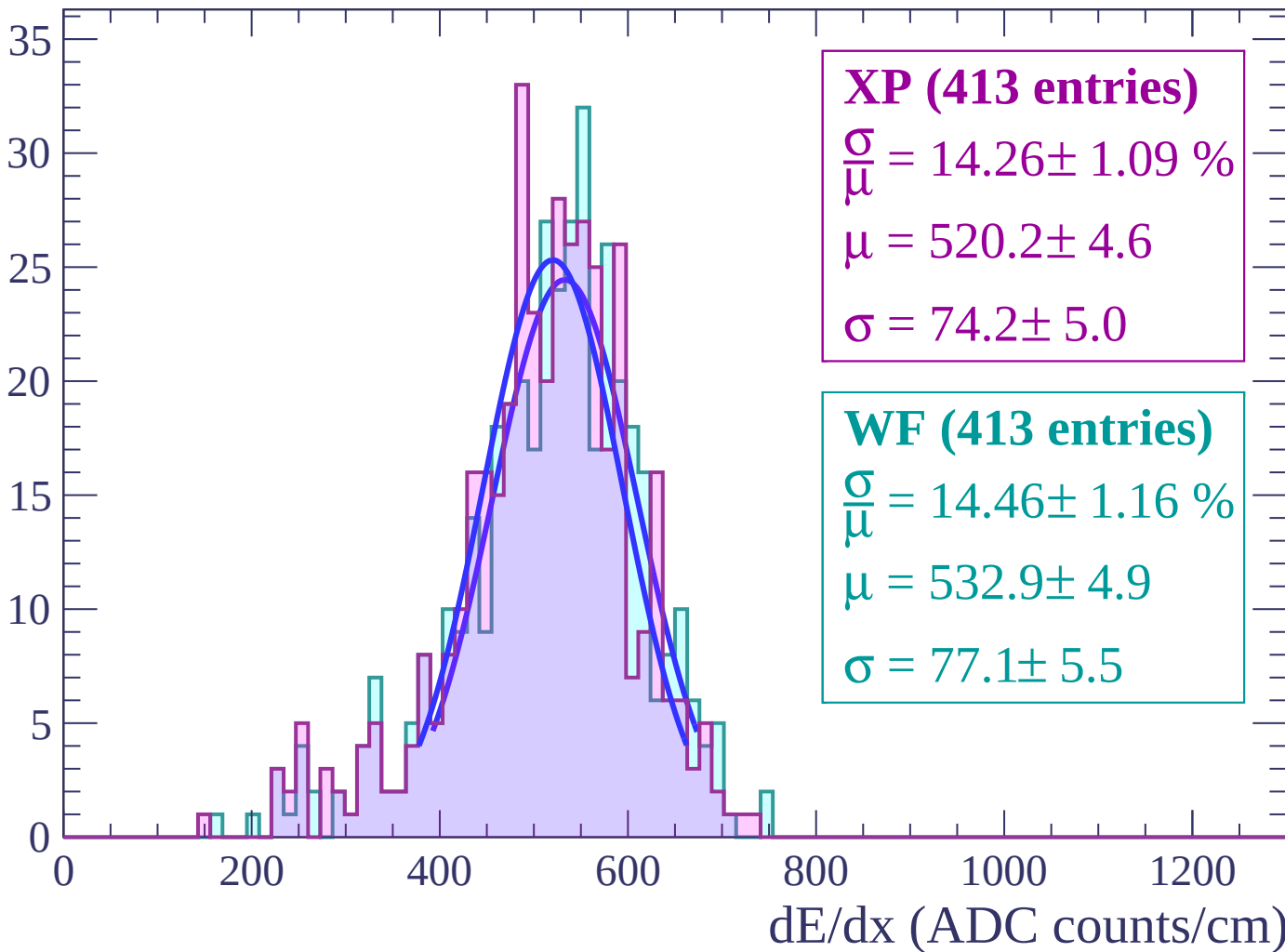
Energy loss | $-60 < \theta < -58$

Count



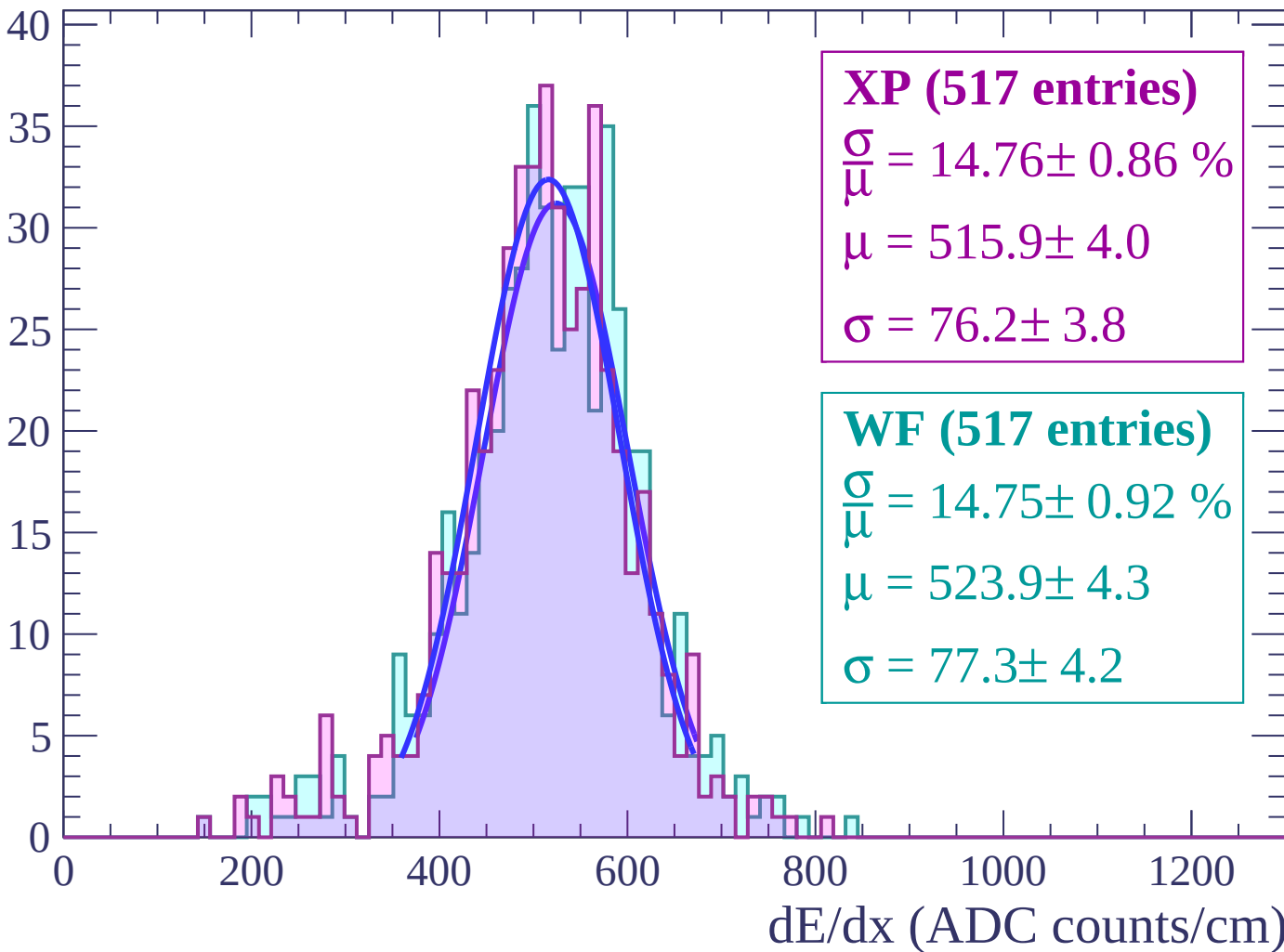
Energy loss | $-58 < \theta < -56$

Count



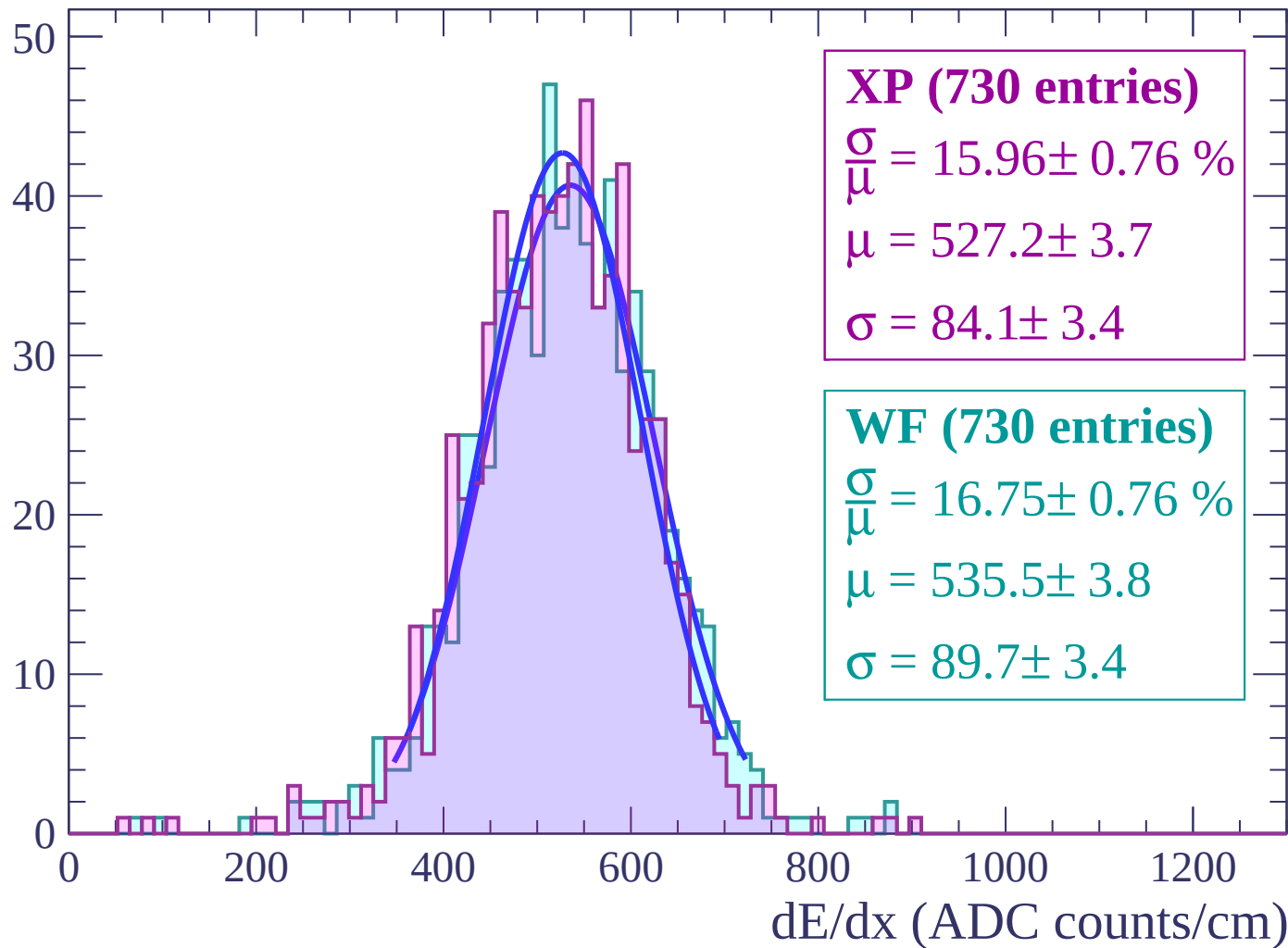
Energy loss | $-56 < \theta < -54$

Count



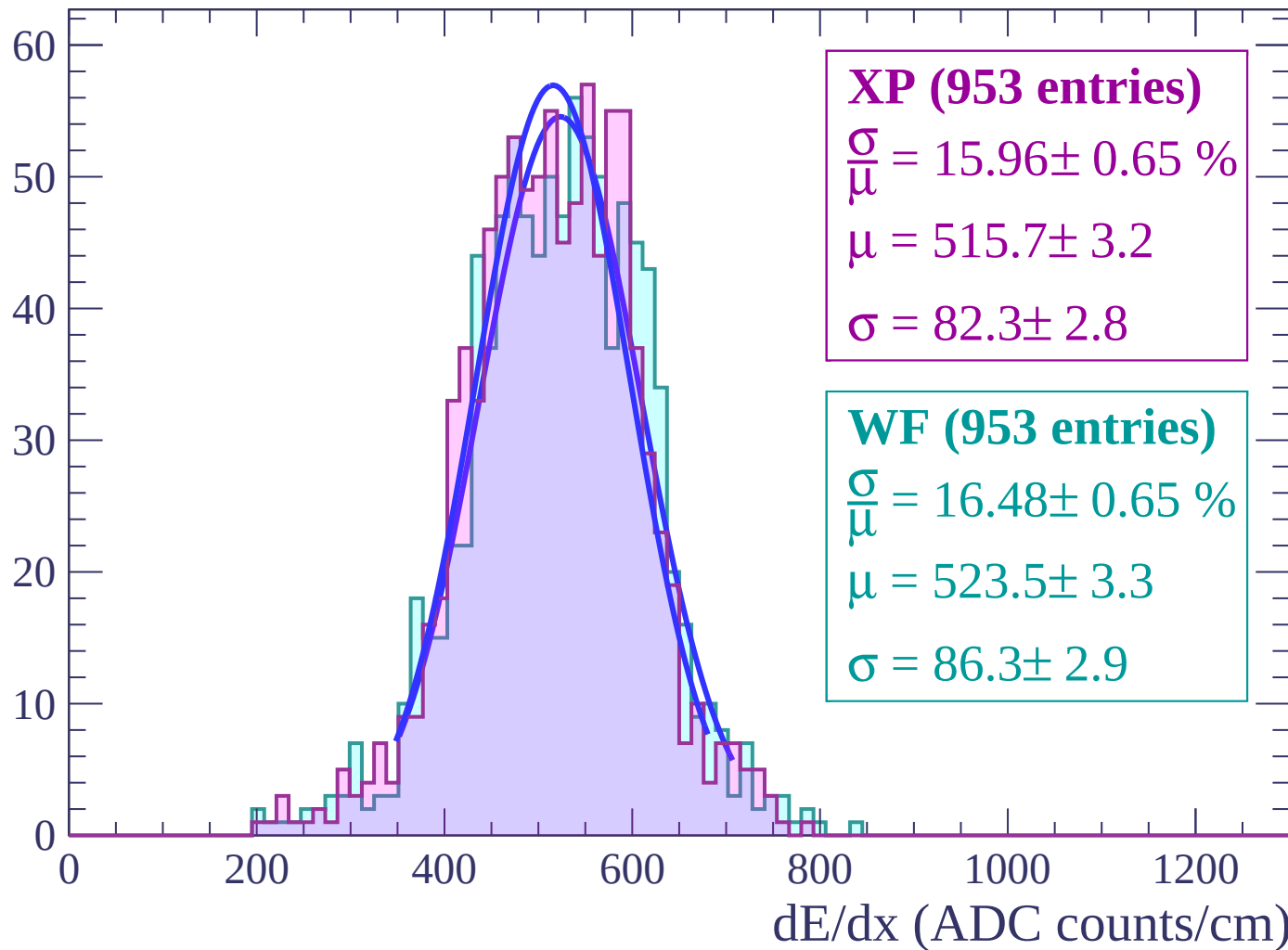
Energy loss | $-54 < \theta < -52$

Count



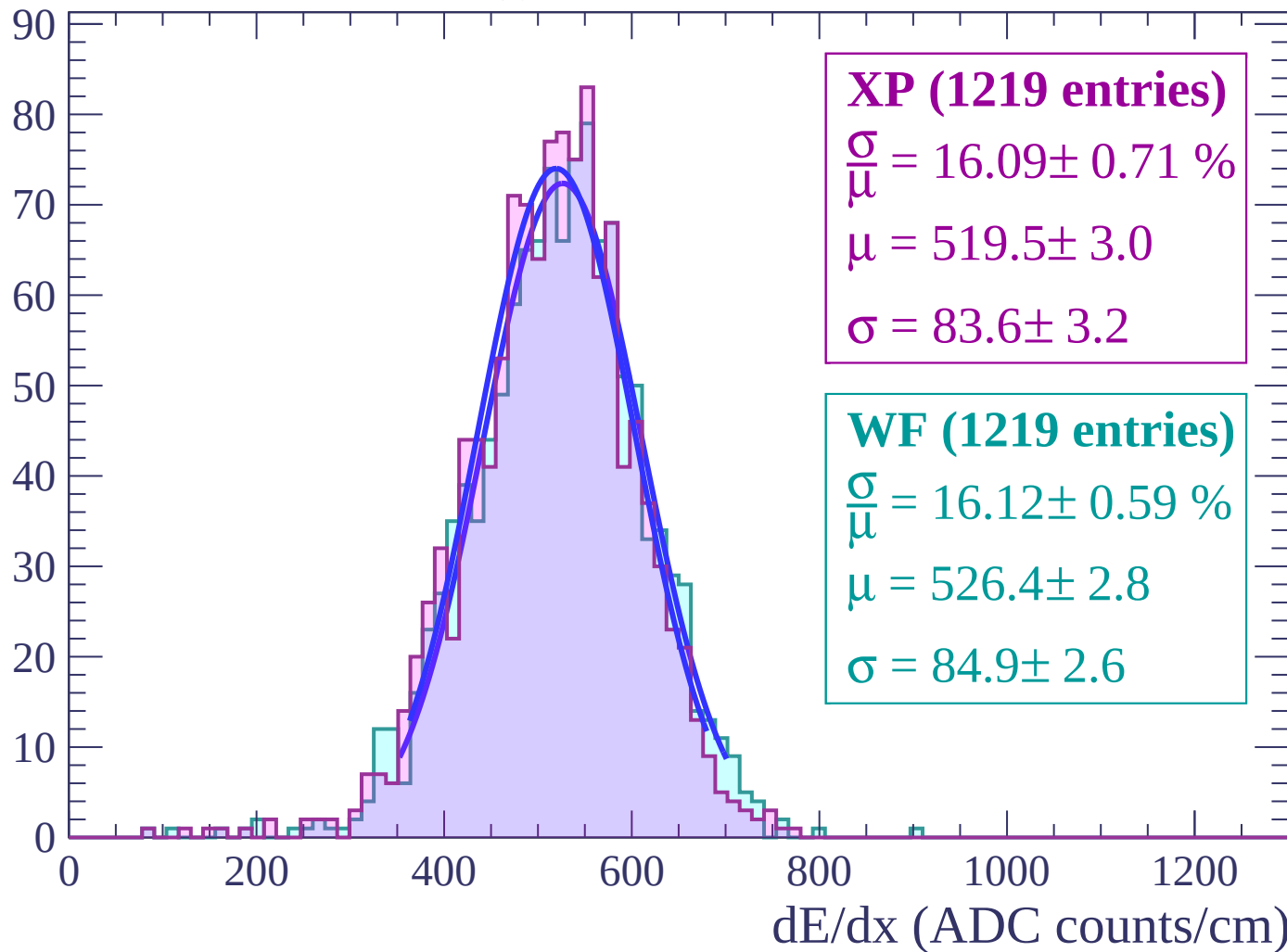
Energy loss | $-52 < \theta < -50$

Count



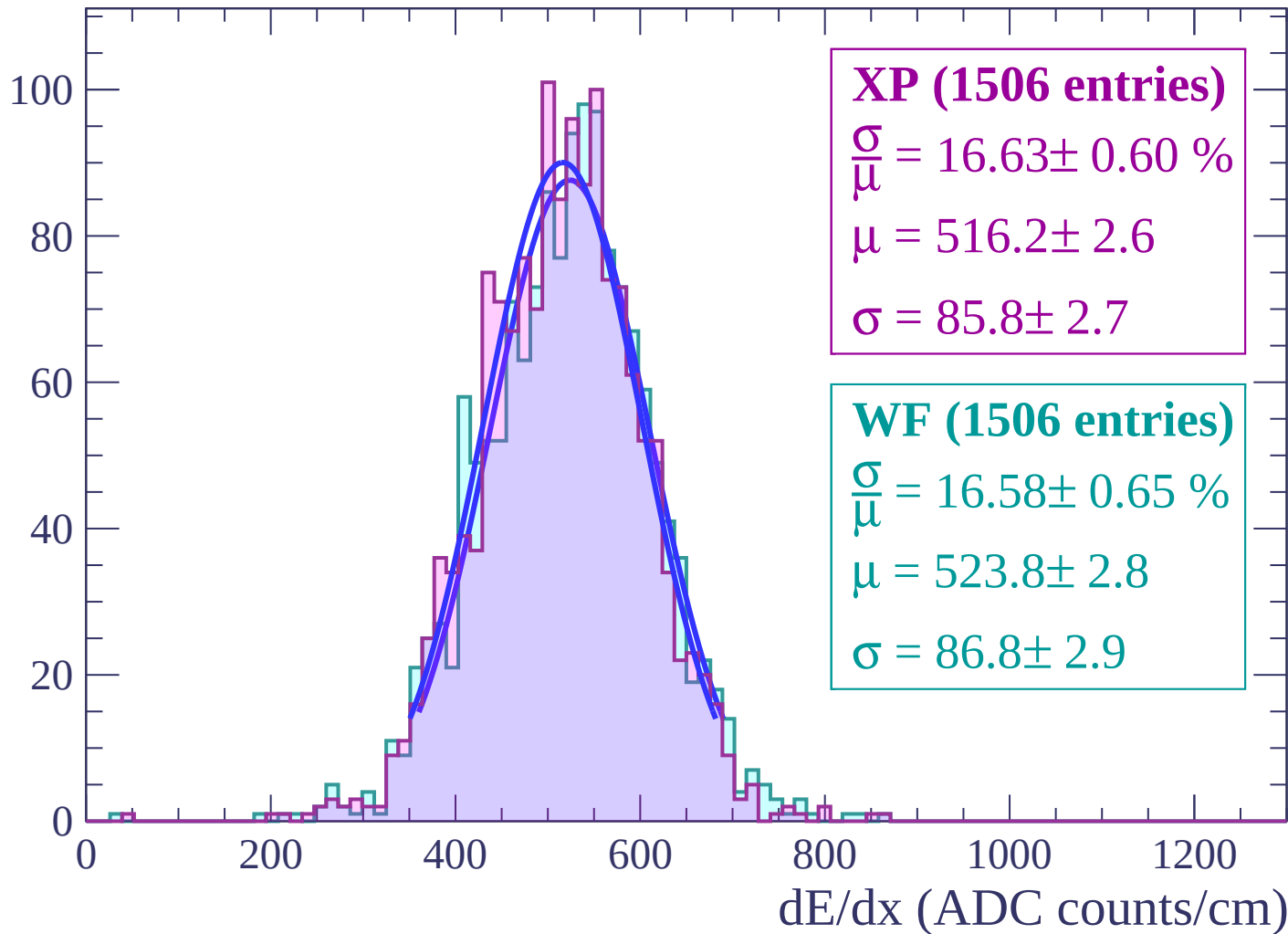
Energy loss | $-50 < \theta < -48$

Count



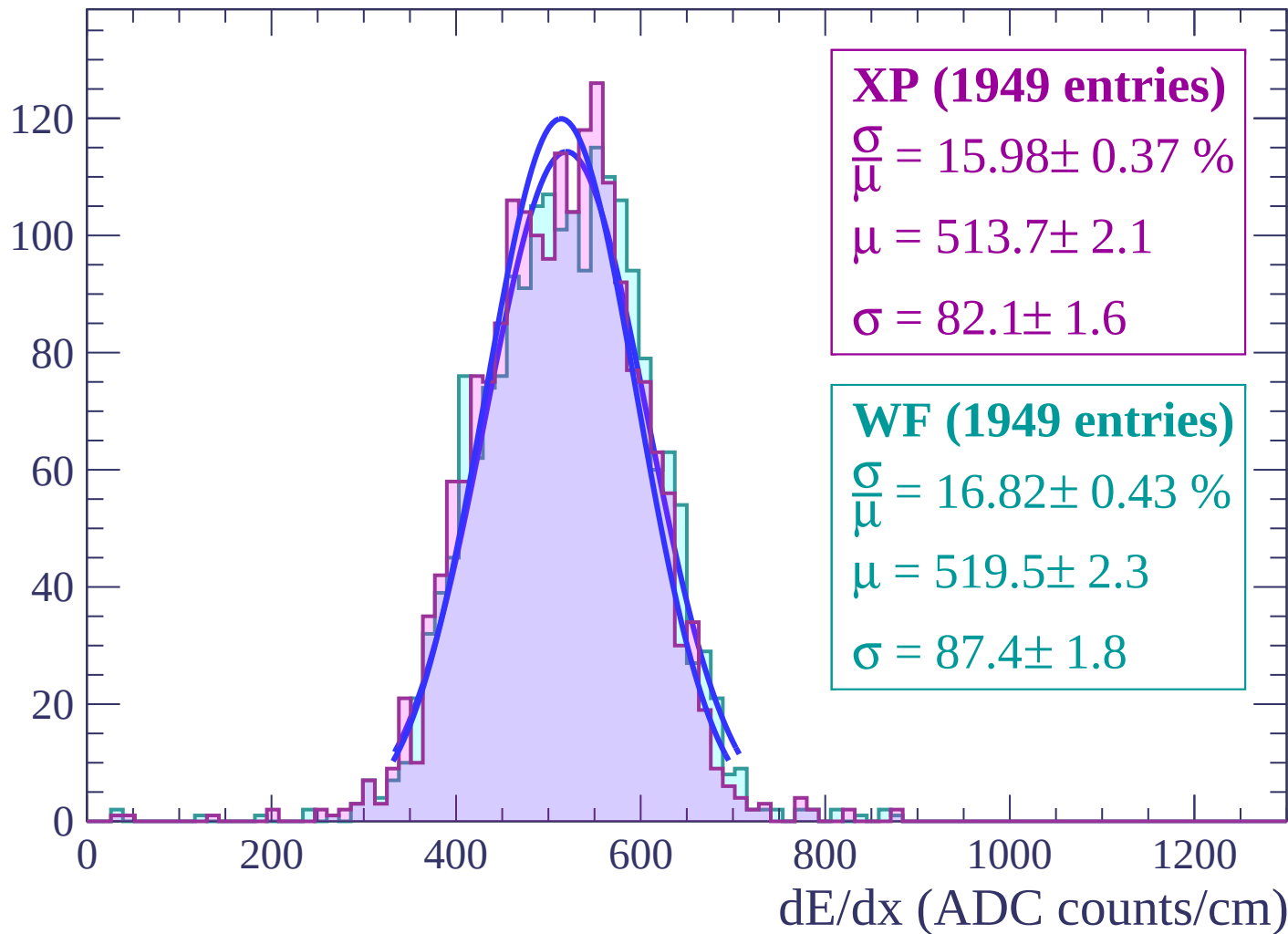
Energy loss | $-48 < \theta < -46$

Count



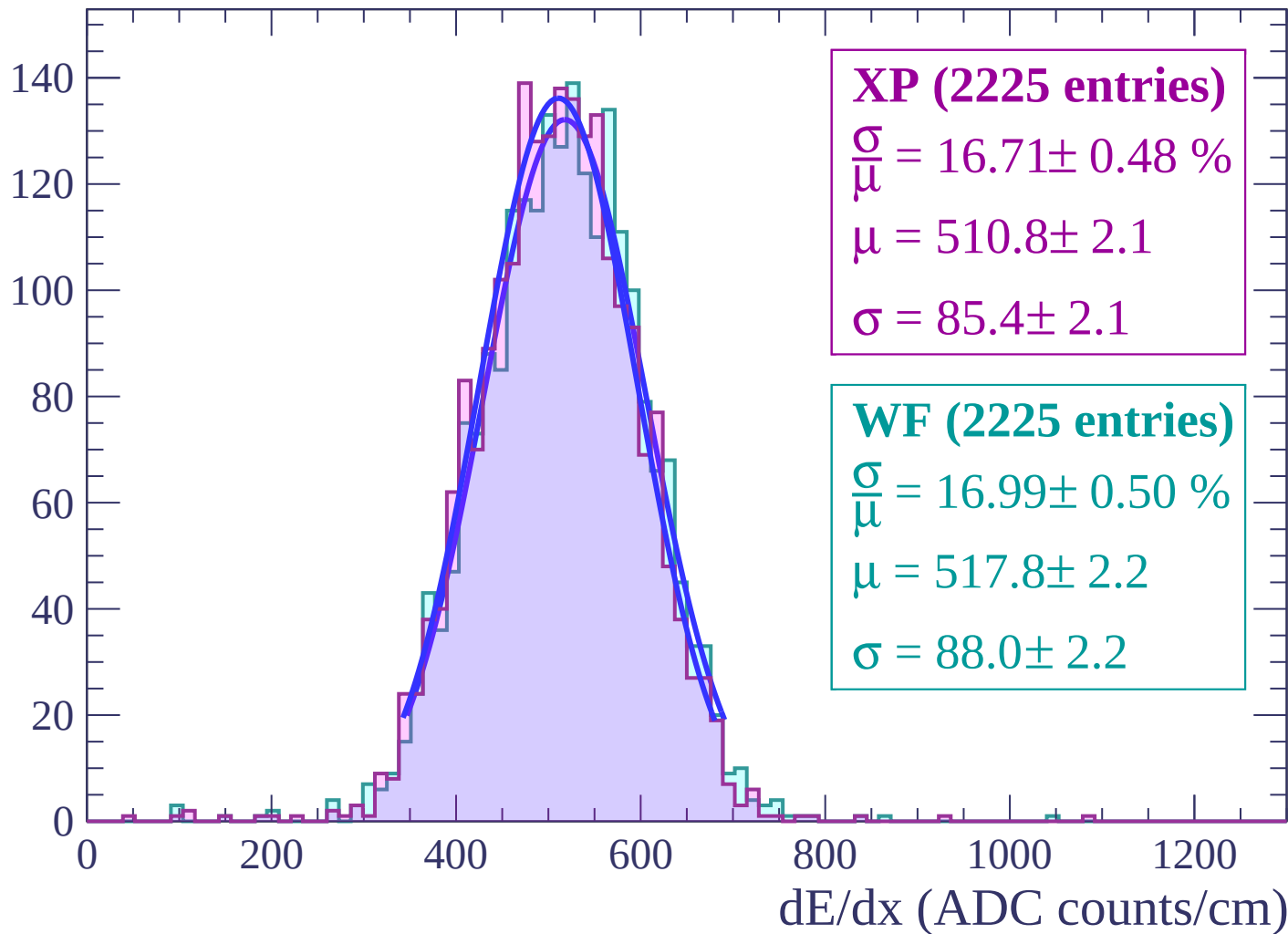
Energy loss | $-46 < \theta < -44$

Count



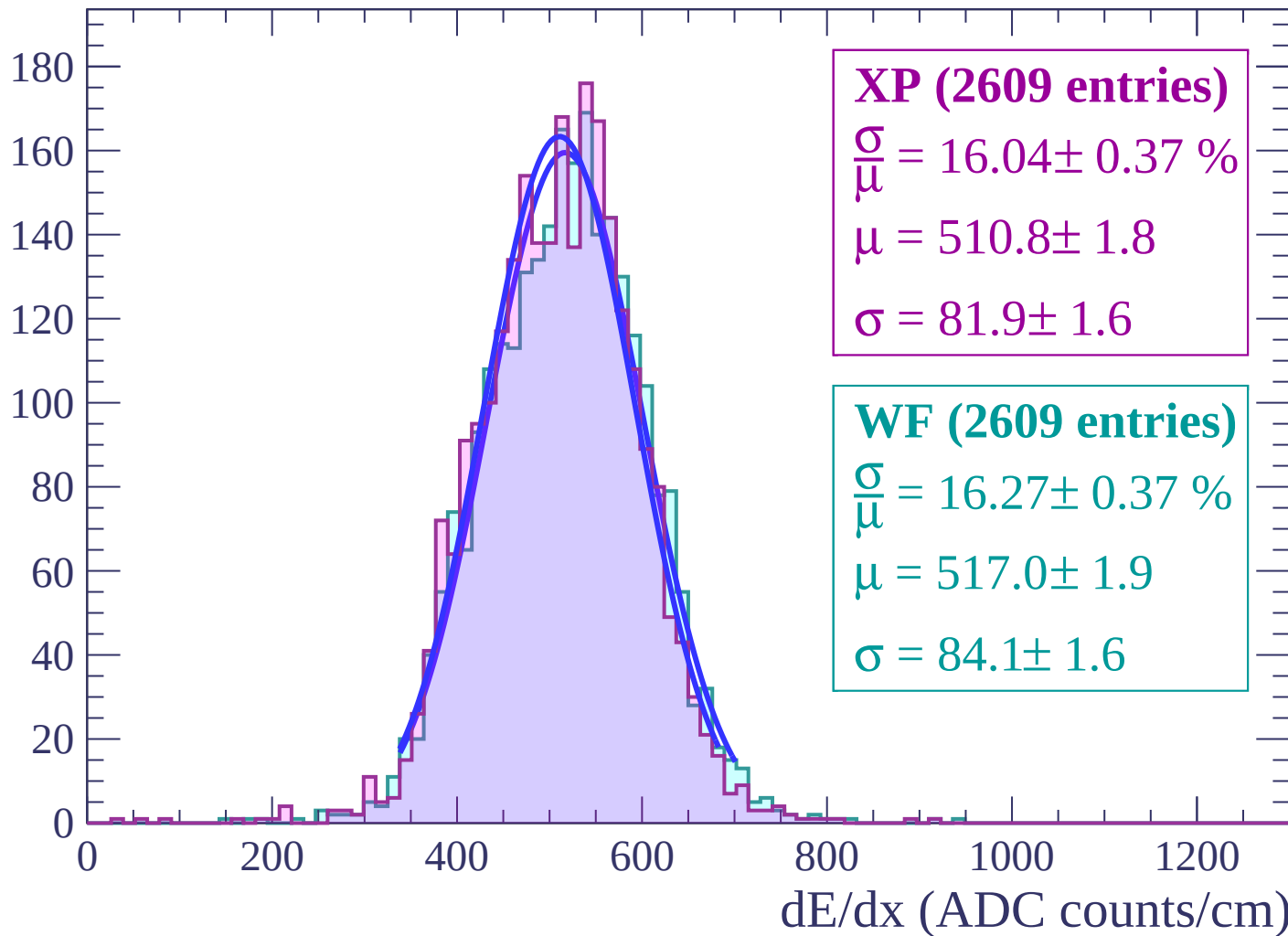
Energy loss | $-44 < \theta < -42$

Count



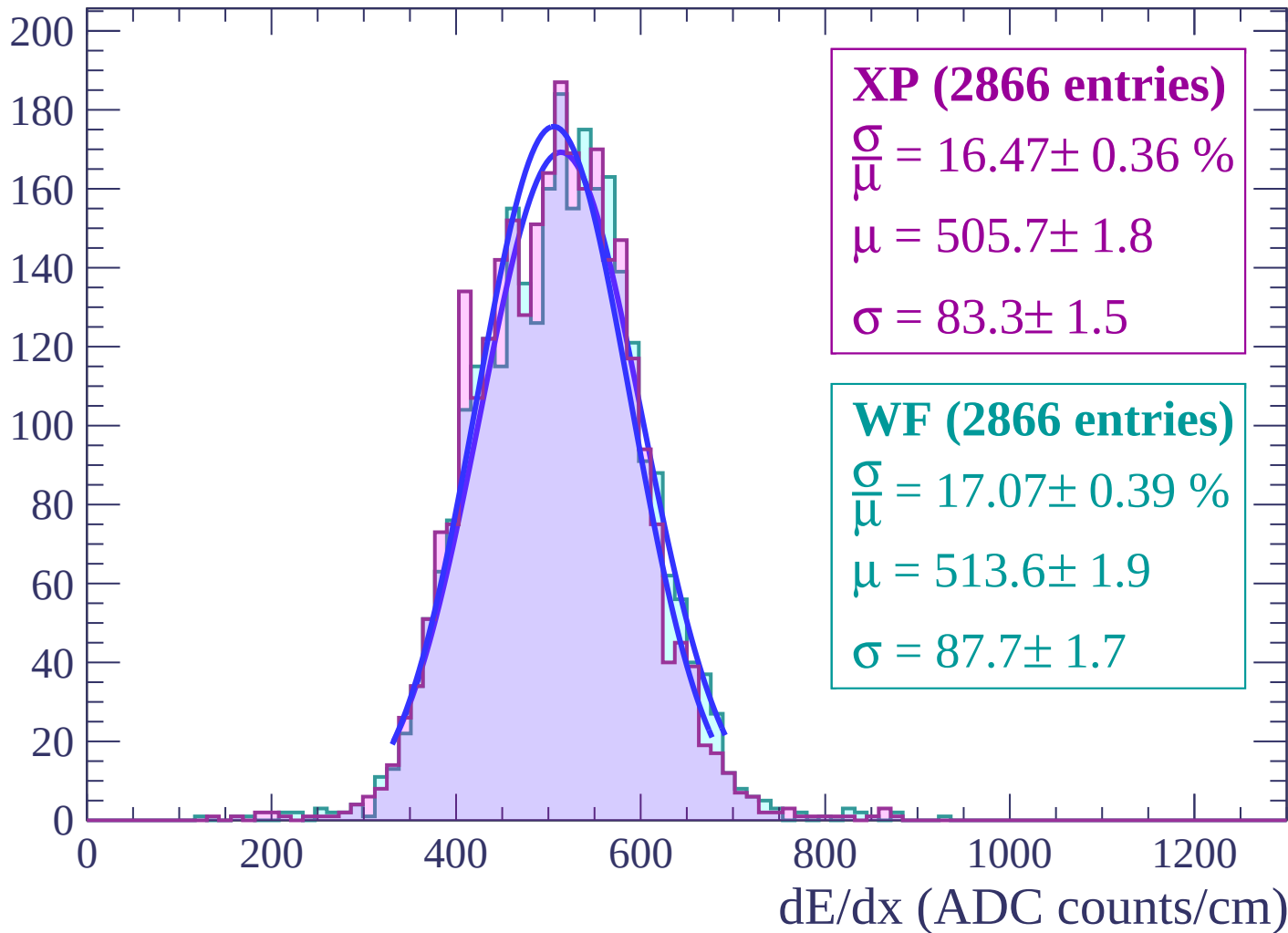
Energy loss | $-42 < \theta < -40$

Count



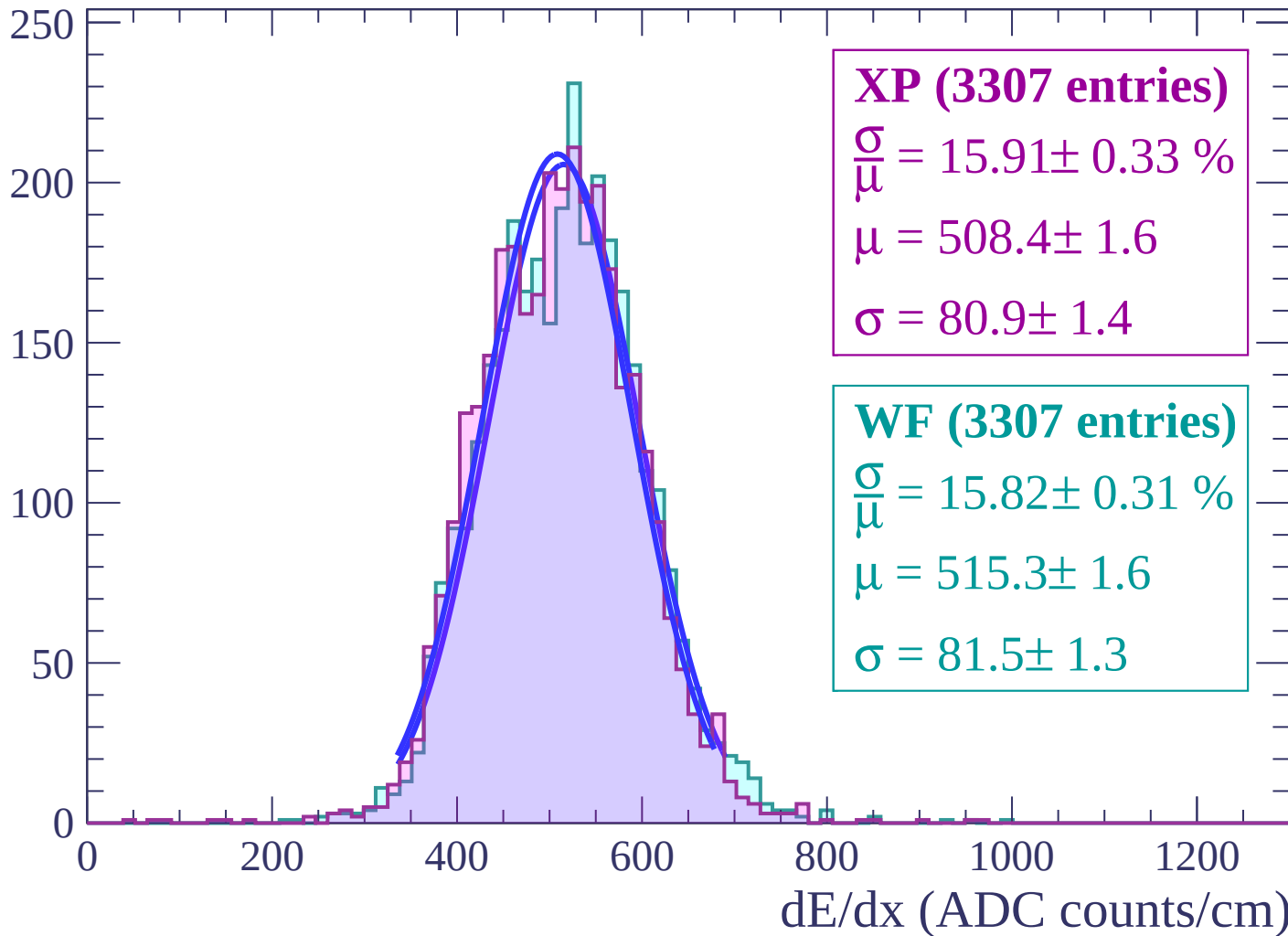
Energy loss | $-40 < \theta < -38$

Count



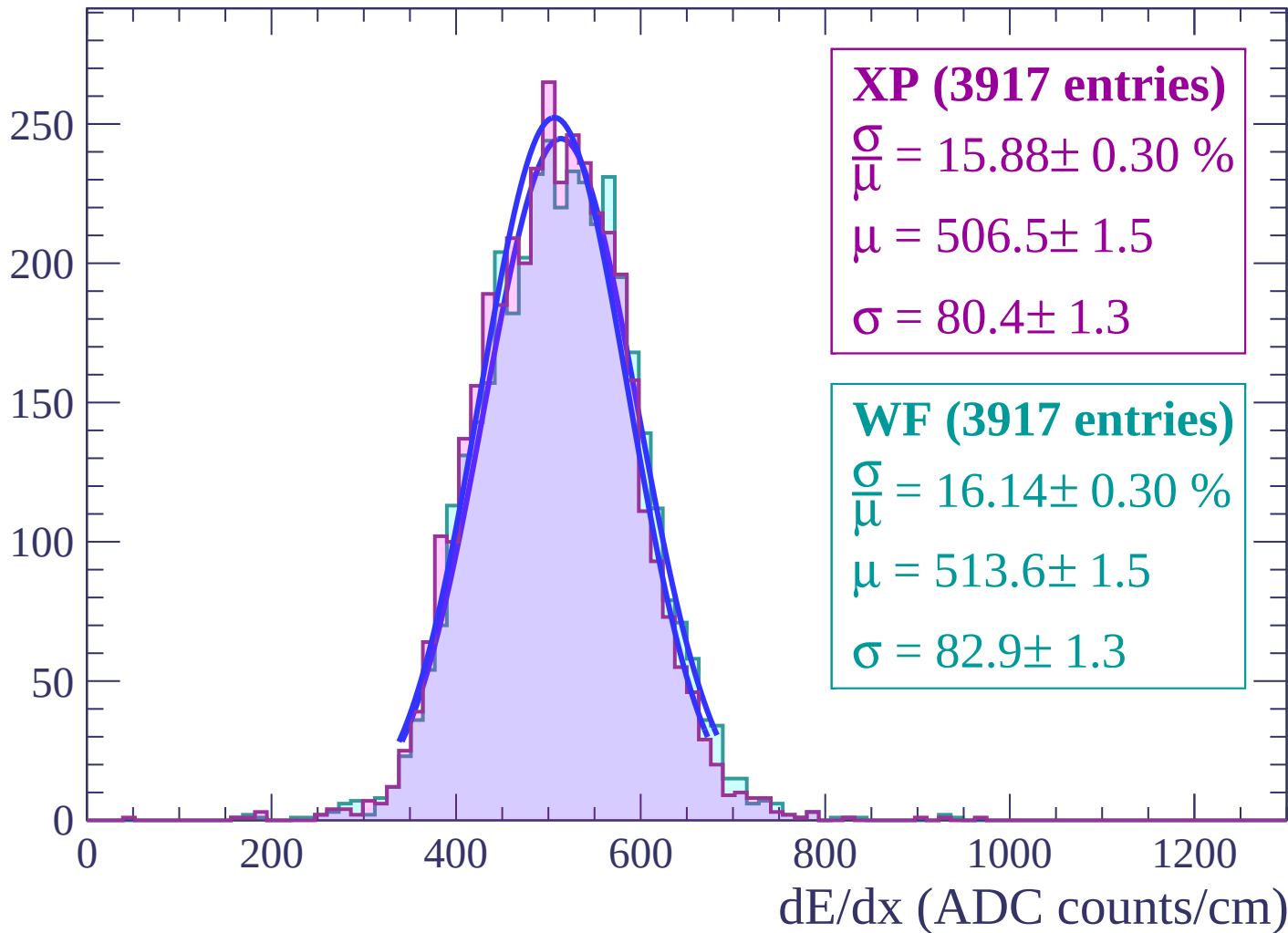
Energy loss | $-38 < \theta < -36$

Count



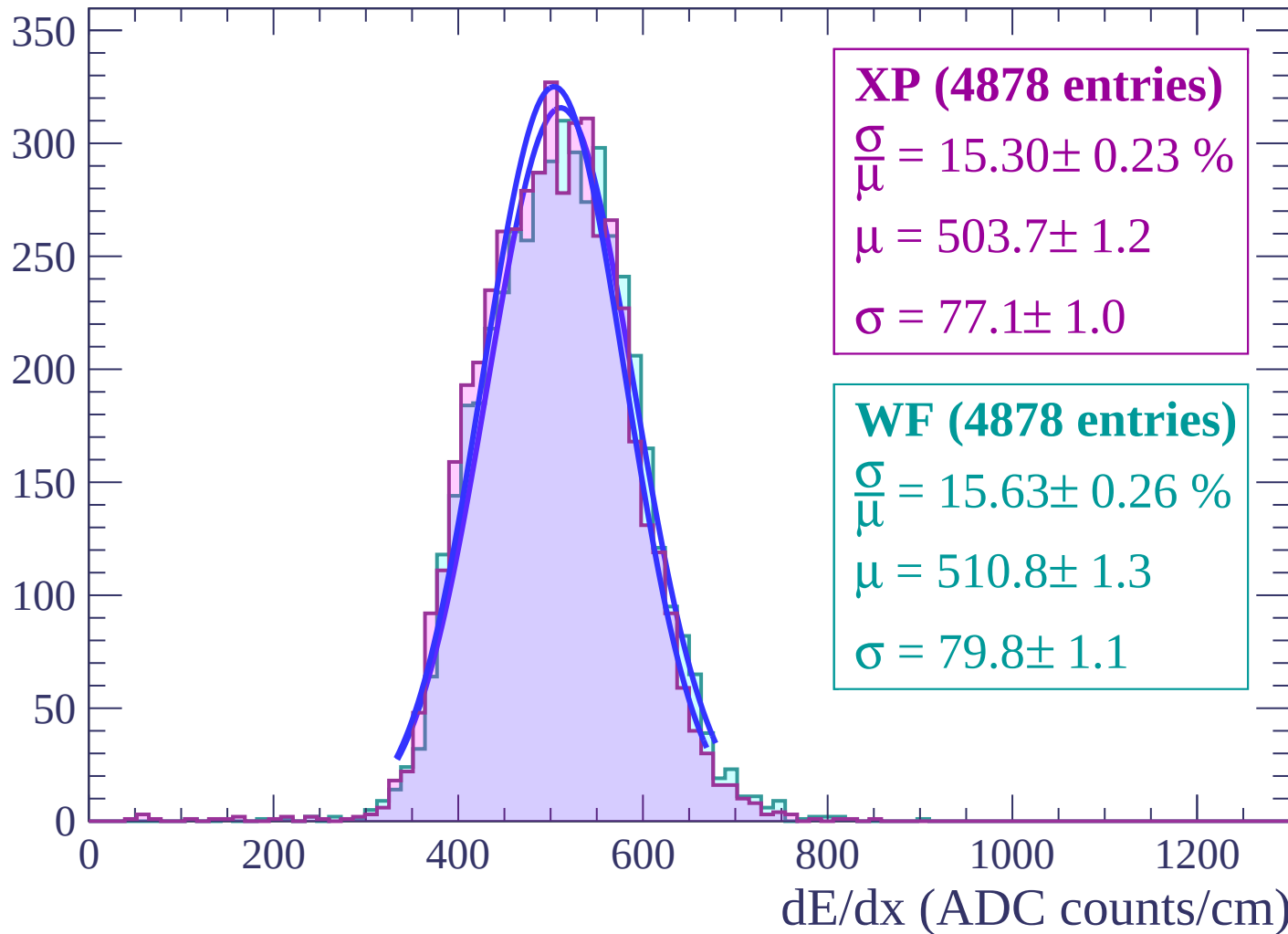
Energy loss | $-36 < \theta < -34$

Count



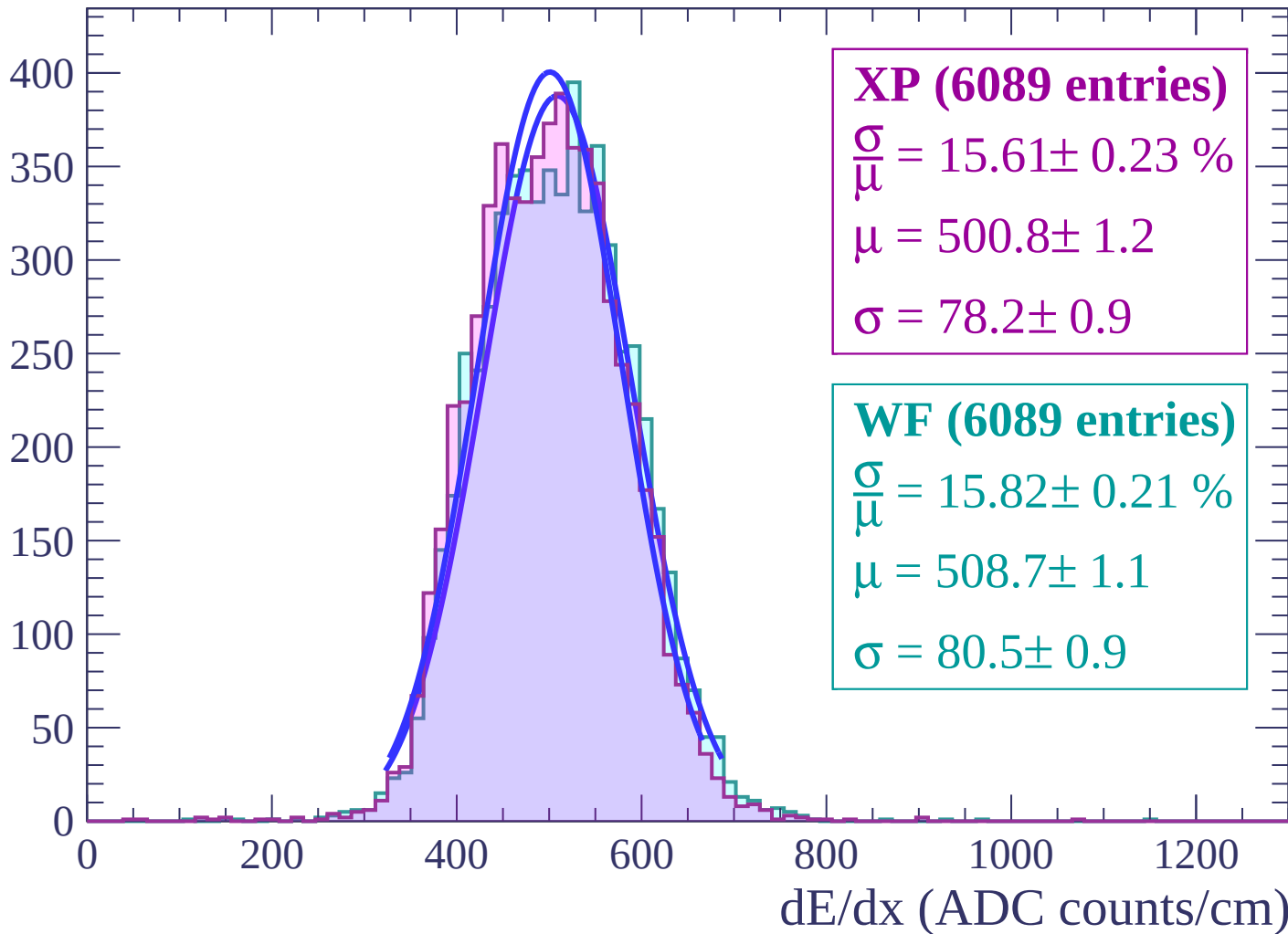
Energy loss | $-34 < \theta < -32$

Count



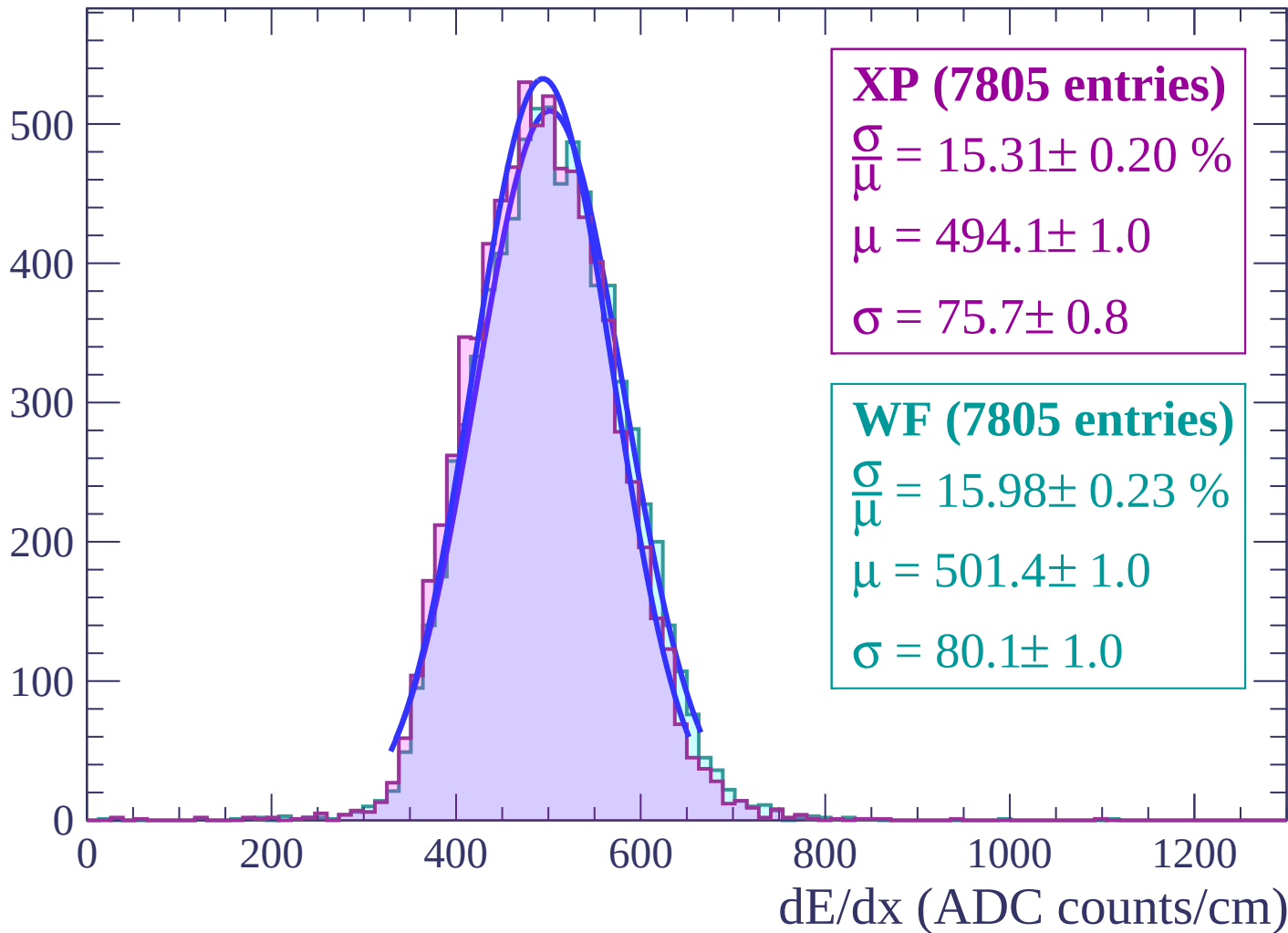
Energy loss | $-32 < \theta < -30$

Count



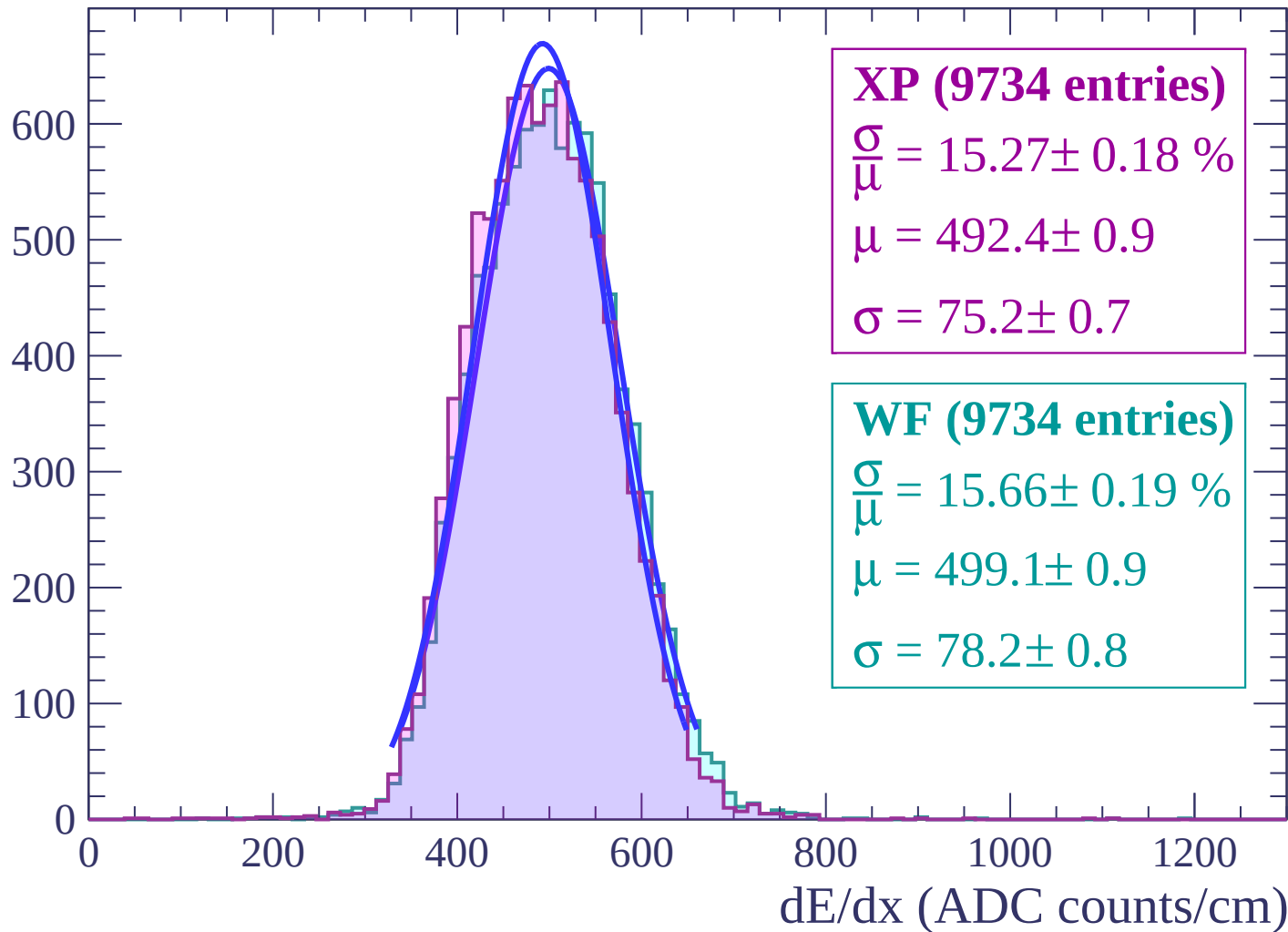
Energy loss | $-30 < \theta < -28$

Count



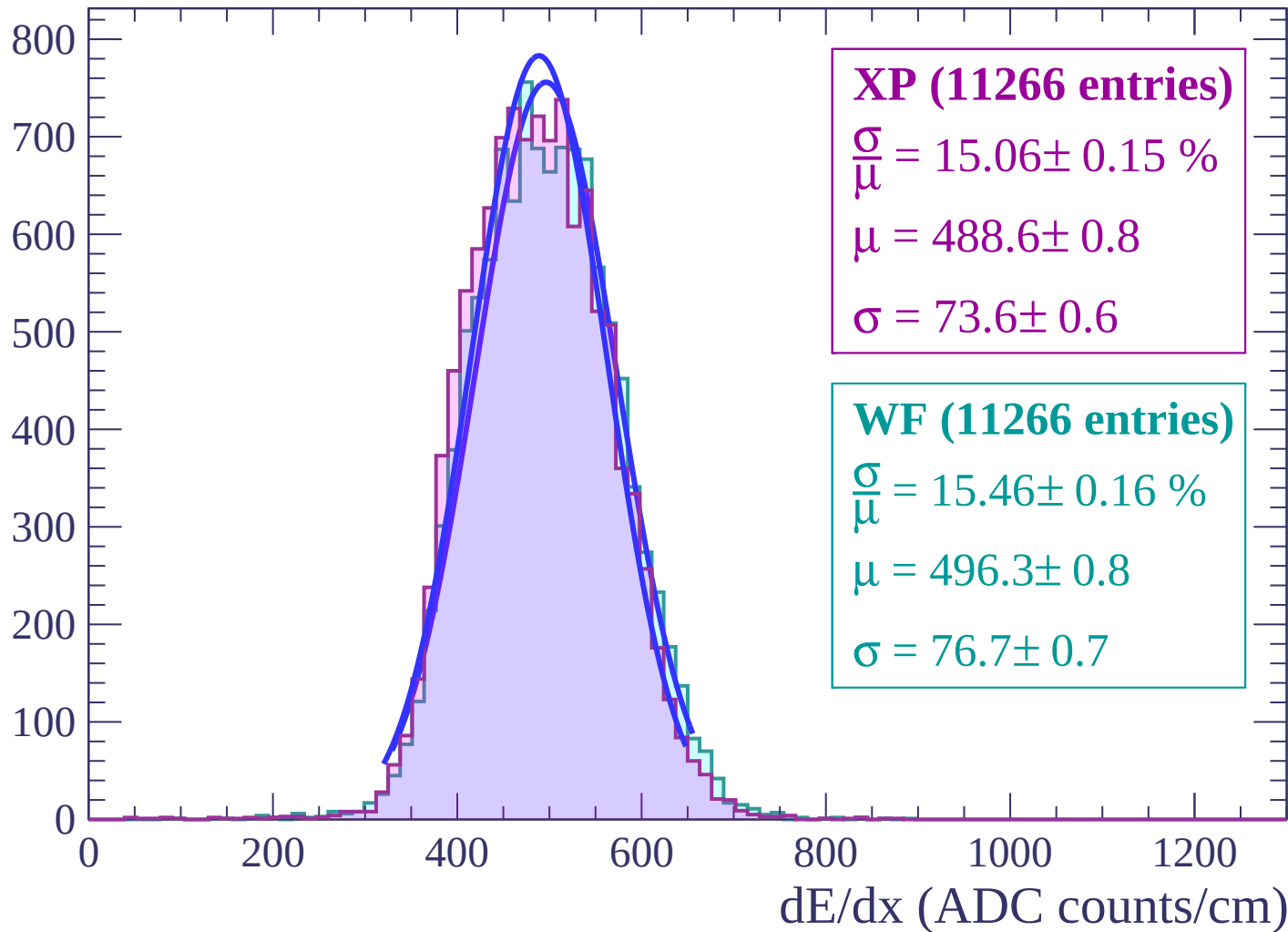
Energy loss | $-28 < \theta < -26$

Count



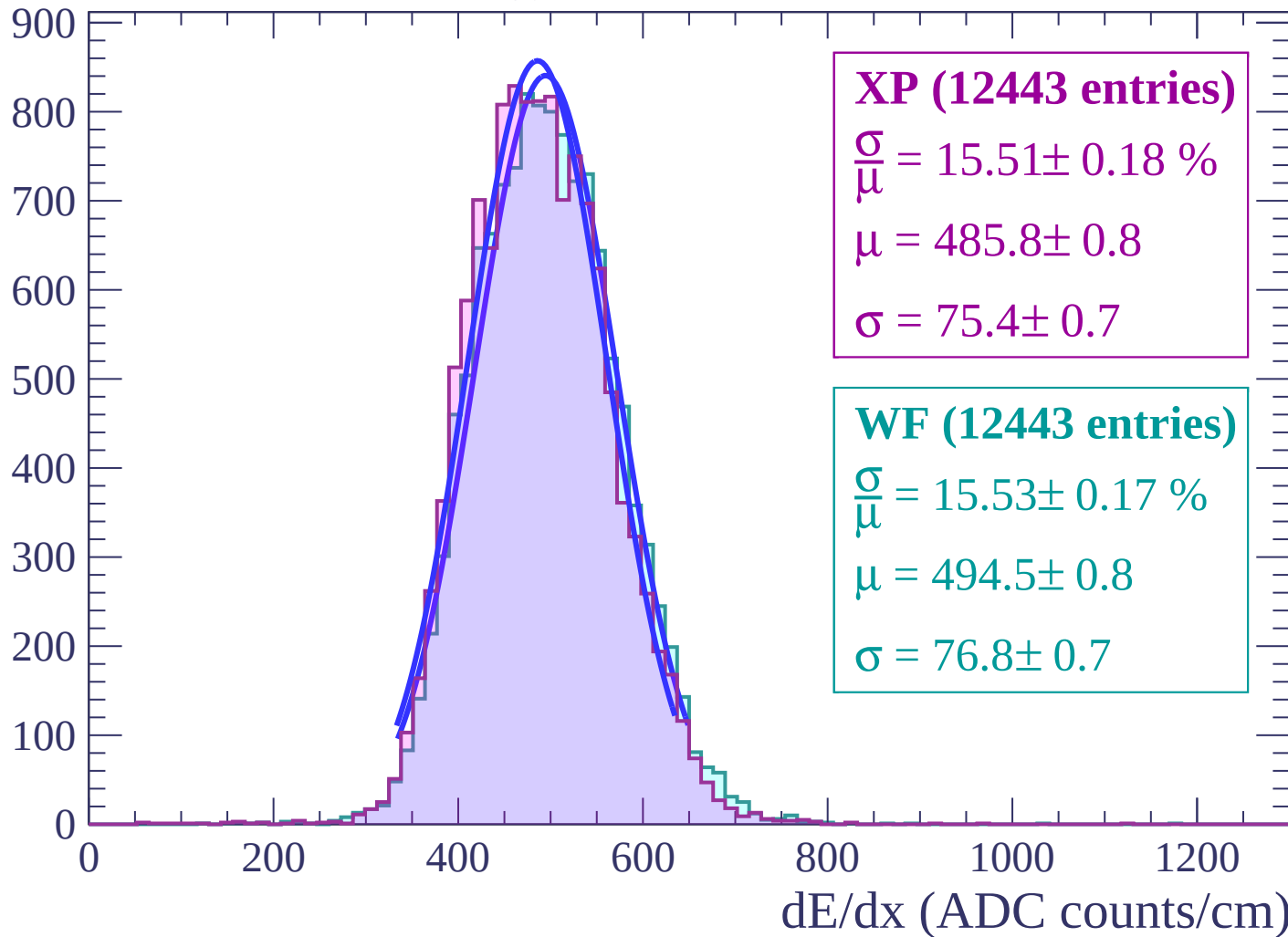
Energy loss | $-26 < \theta < -24$

Count



Energy loss | $-24 < \theta < -22$

Count



Energy loss | $-22 < \theta < -20$

Count

1000

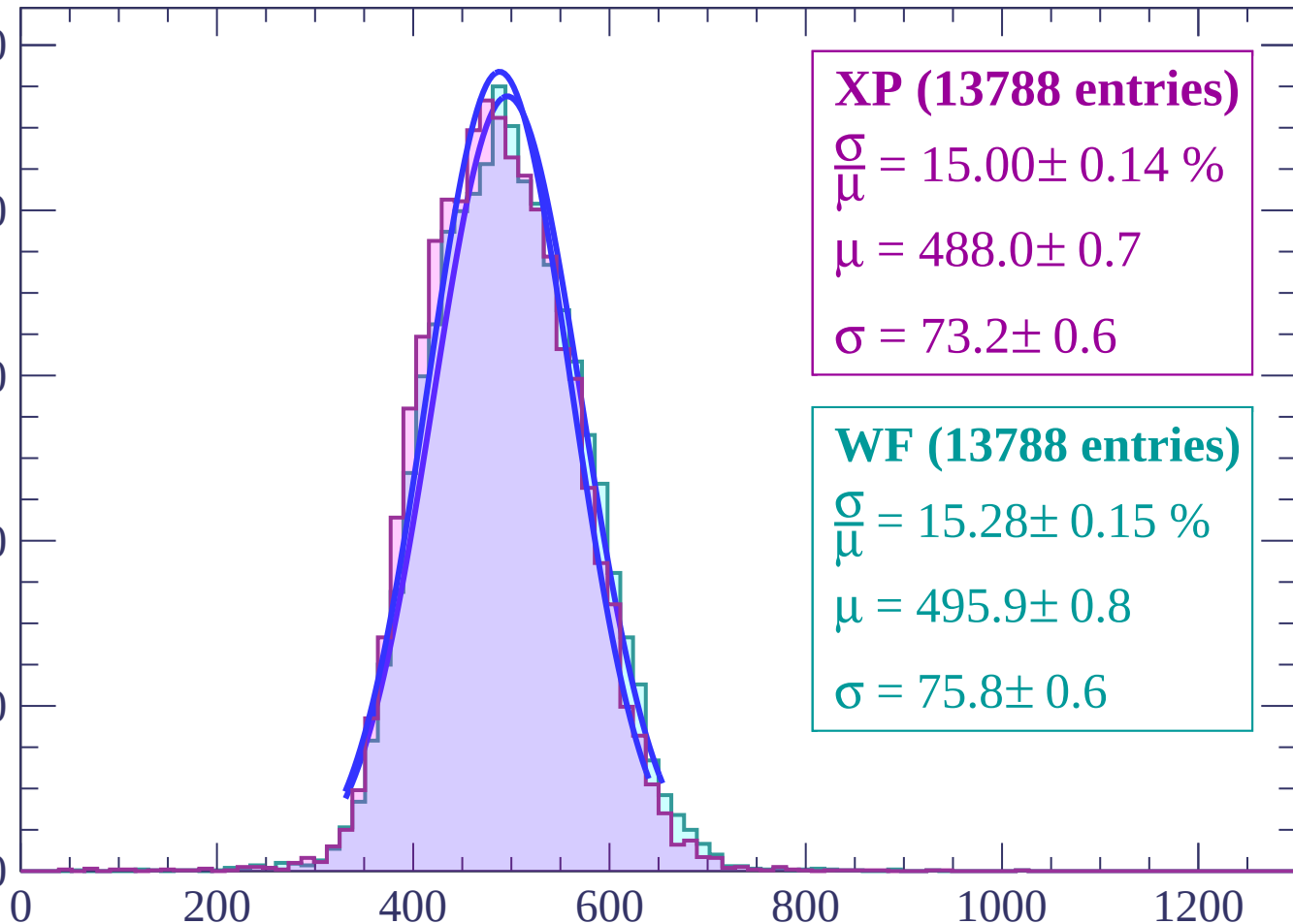
800

600

400

200

0



XP (13788 entries)

$\frac{\sigma}{\mu} = 15.00 \pm 0.14 \%$

$\mu = 488.0 \pm 0.7$

$\sigma = 73.2 \pm 0.6$

WF (13788 entries)

$\frac{\sigma}{\mu} = 15.28 \pm 0.15 \%$

$\mu = 495.9 \pm 0.8$

$\sigma = 75.8 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-20 < \theta < -18$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (15201 entries)

$\frac{\sigma}{\mu} = 15.02 \pm 0.14 \%$

$\mu = 486.9 \pm 0.7$

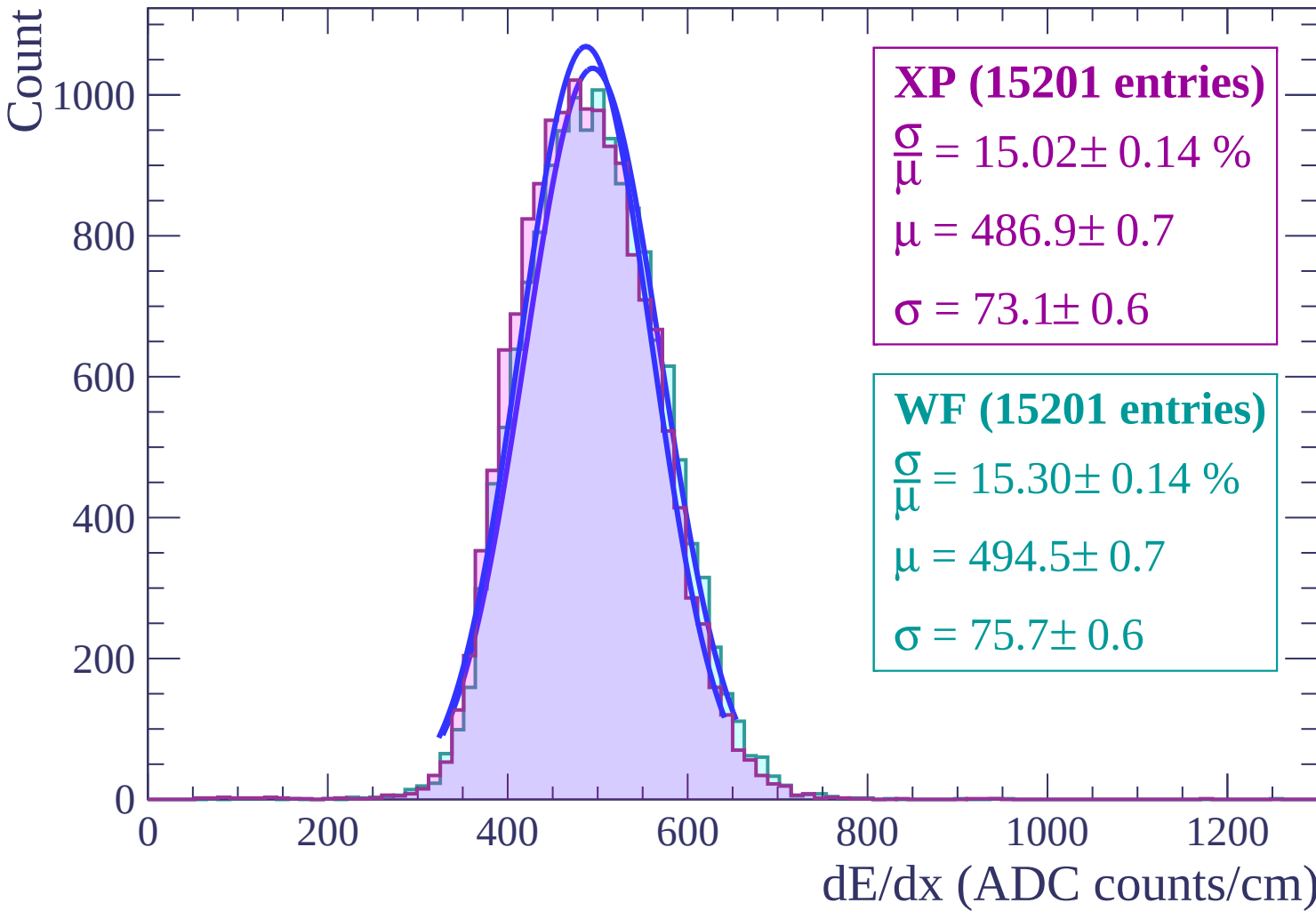
$\sigma = 73.1 \pm 0.6$

WF (15201 entries)

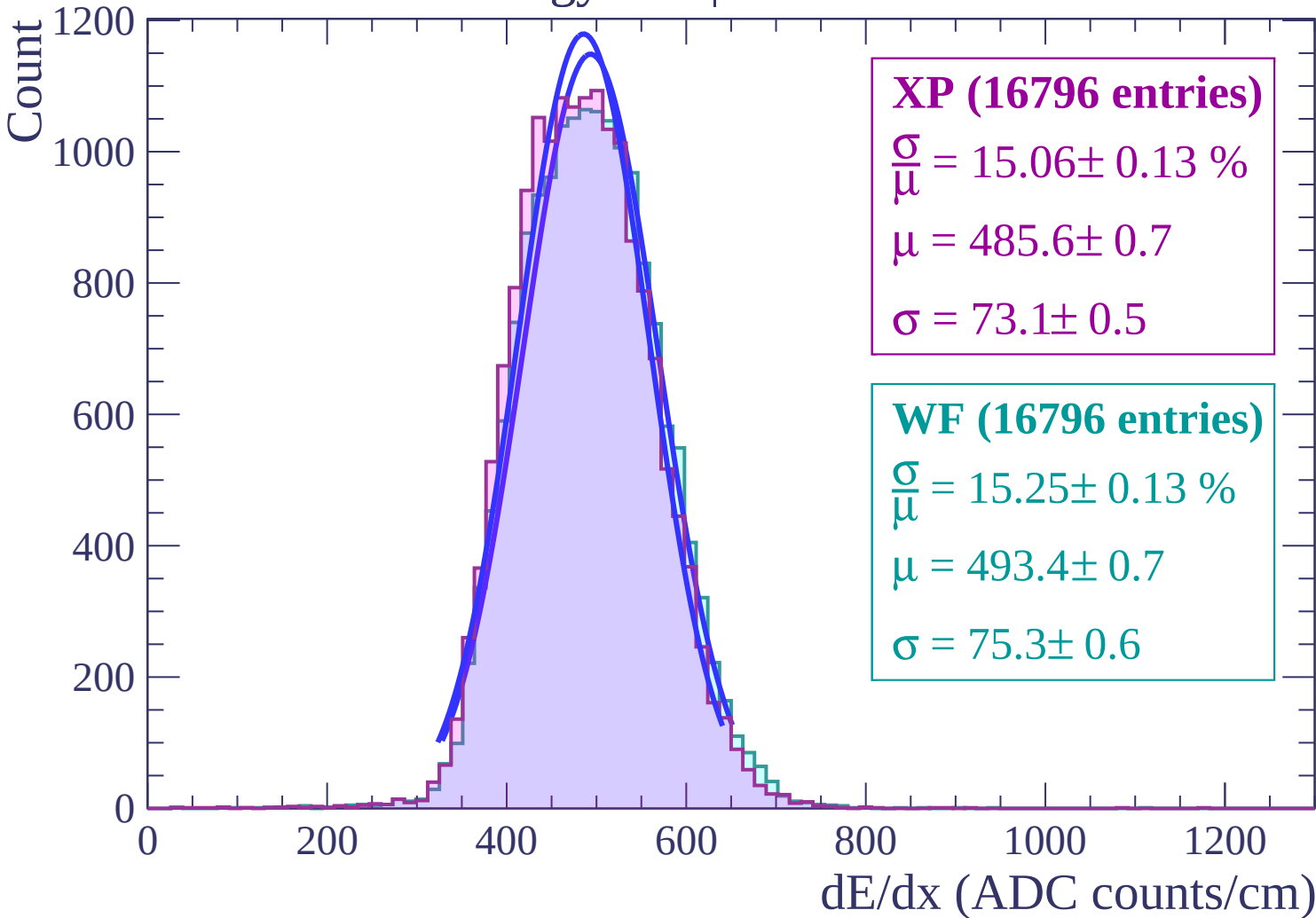
$\frac{\sigma}{\mu} = 15.30 \pm 0.14 \%$

$\mu = 494.5 \pm 0.7$

$\sigma = 75.7 \pm 0.6$

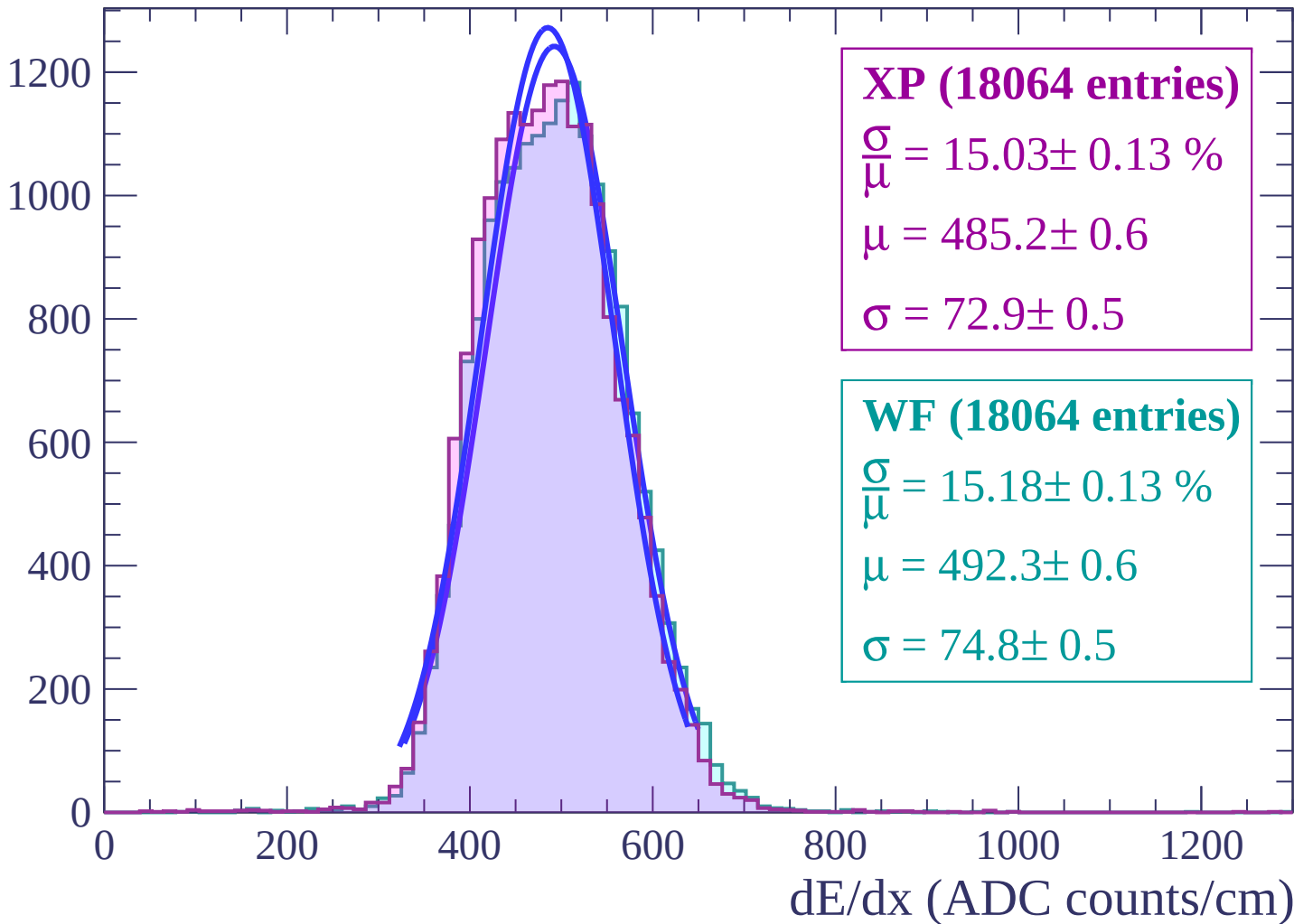


Energy loss | $-18 < \theta < -16$

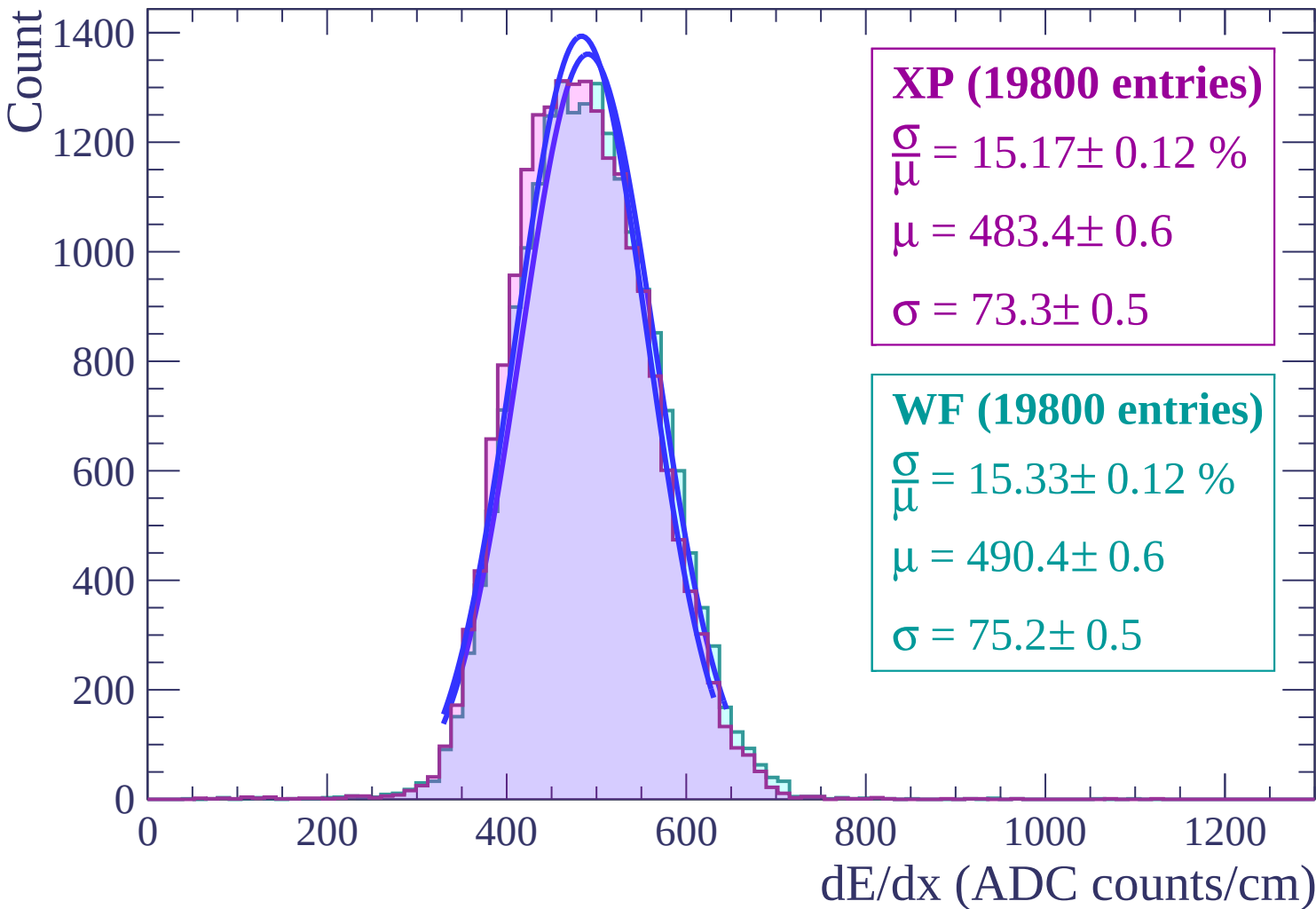


Energy loss | $-16 < \theta < -14$

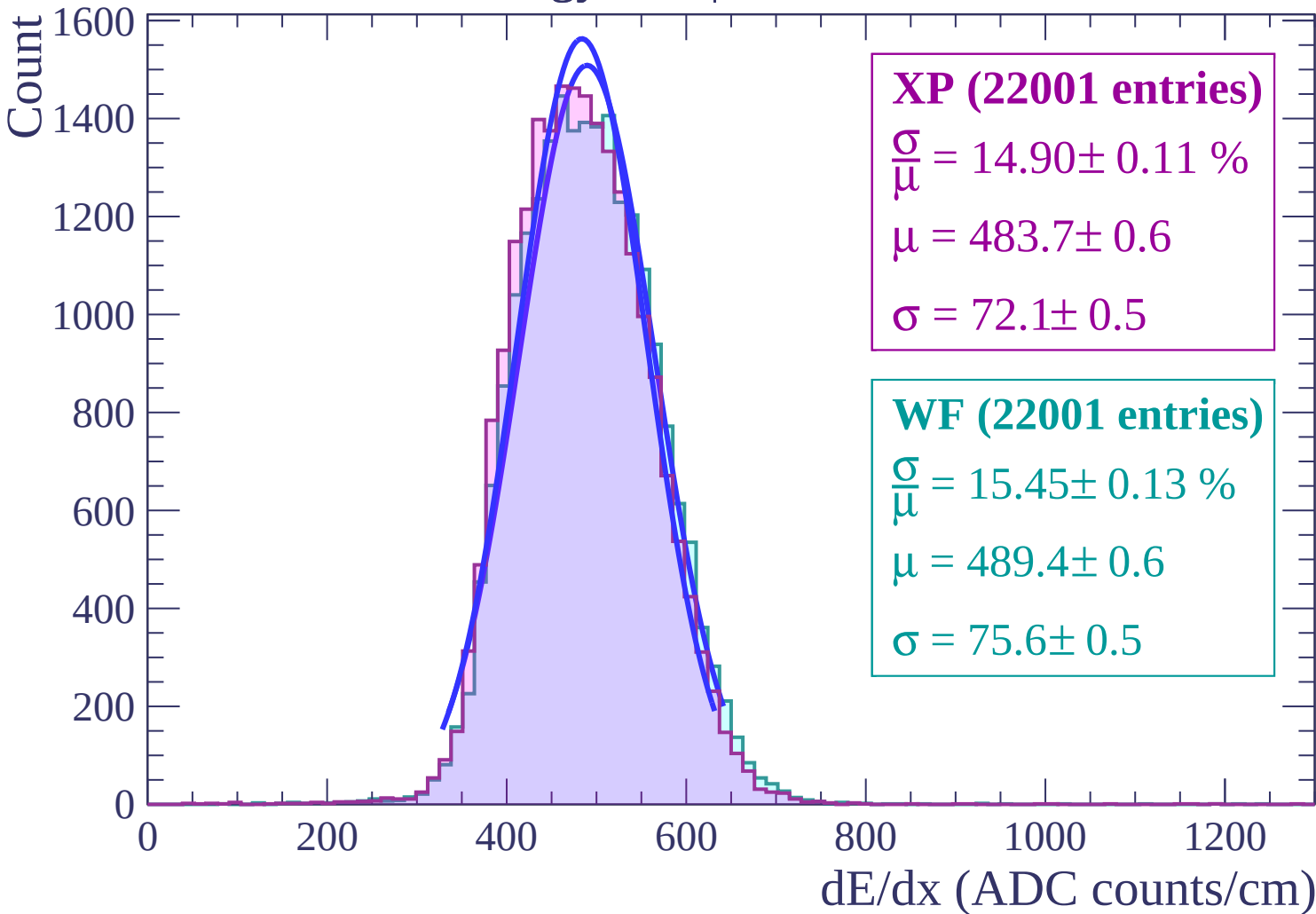
Count



Energy loss | $-14 < \theta < -12$

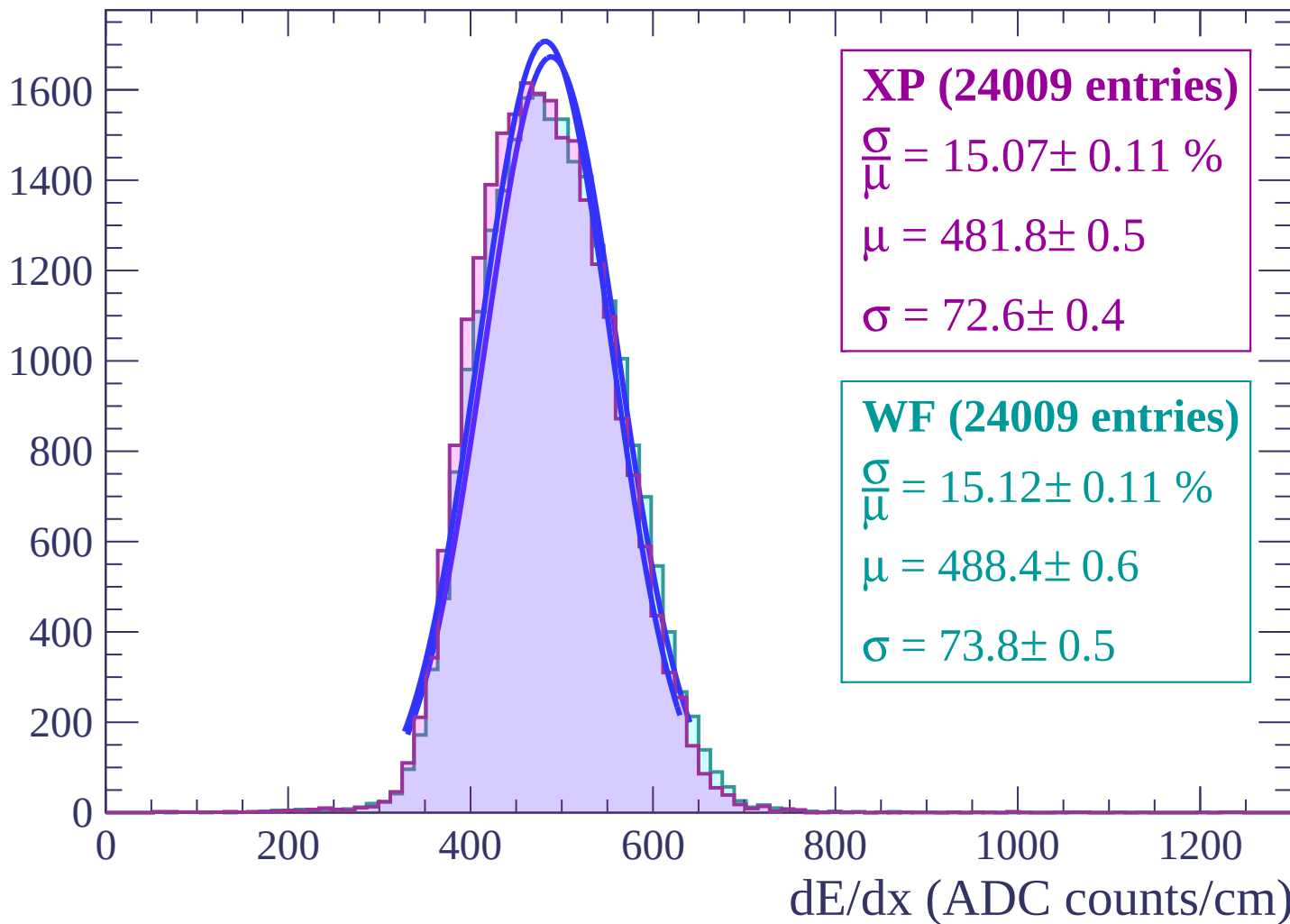


Energy loss | $-12 < \theta < -10$

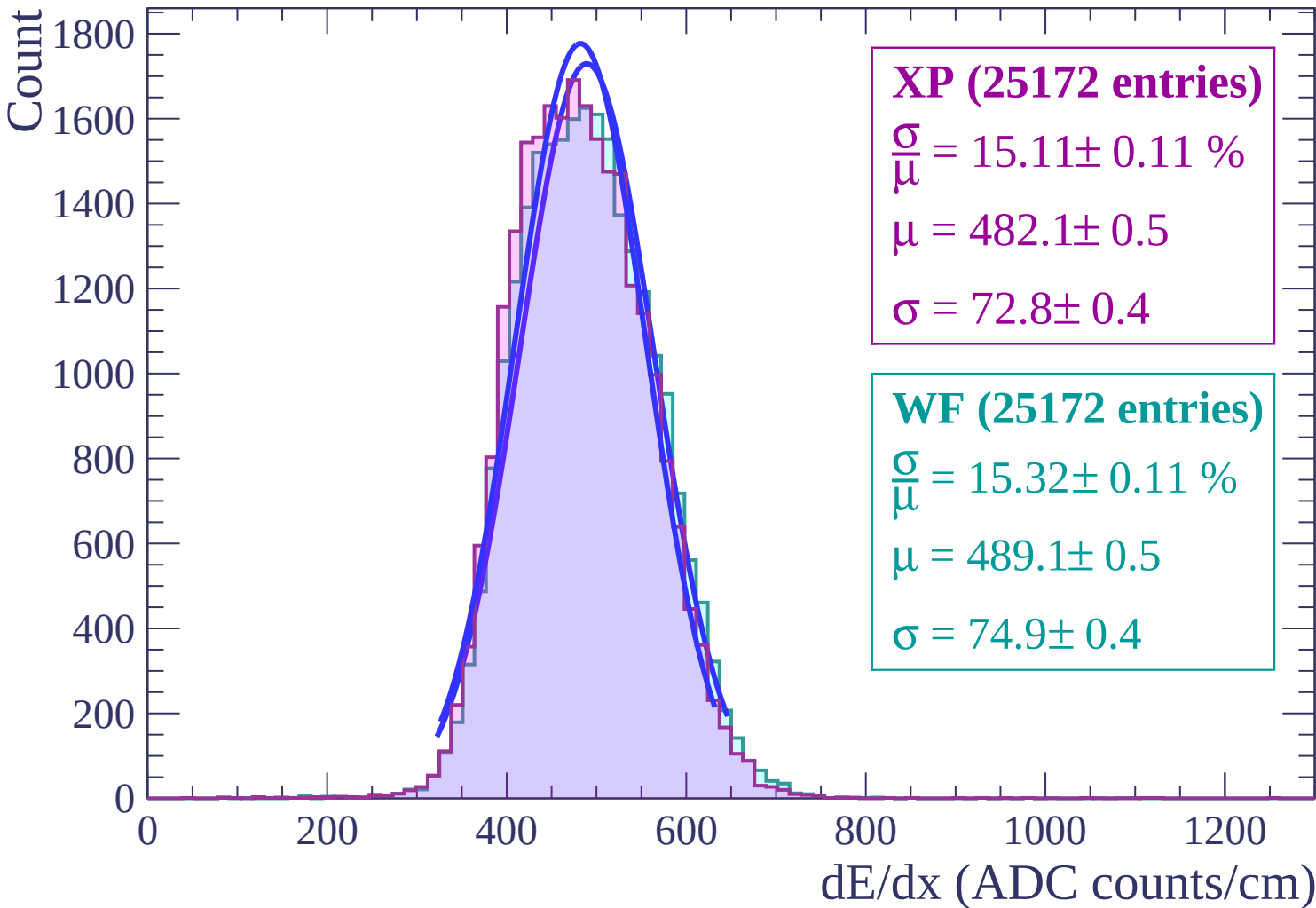


Energy loss | $-10 < \theta < -8$

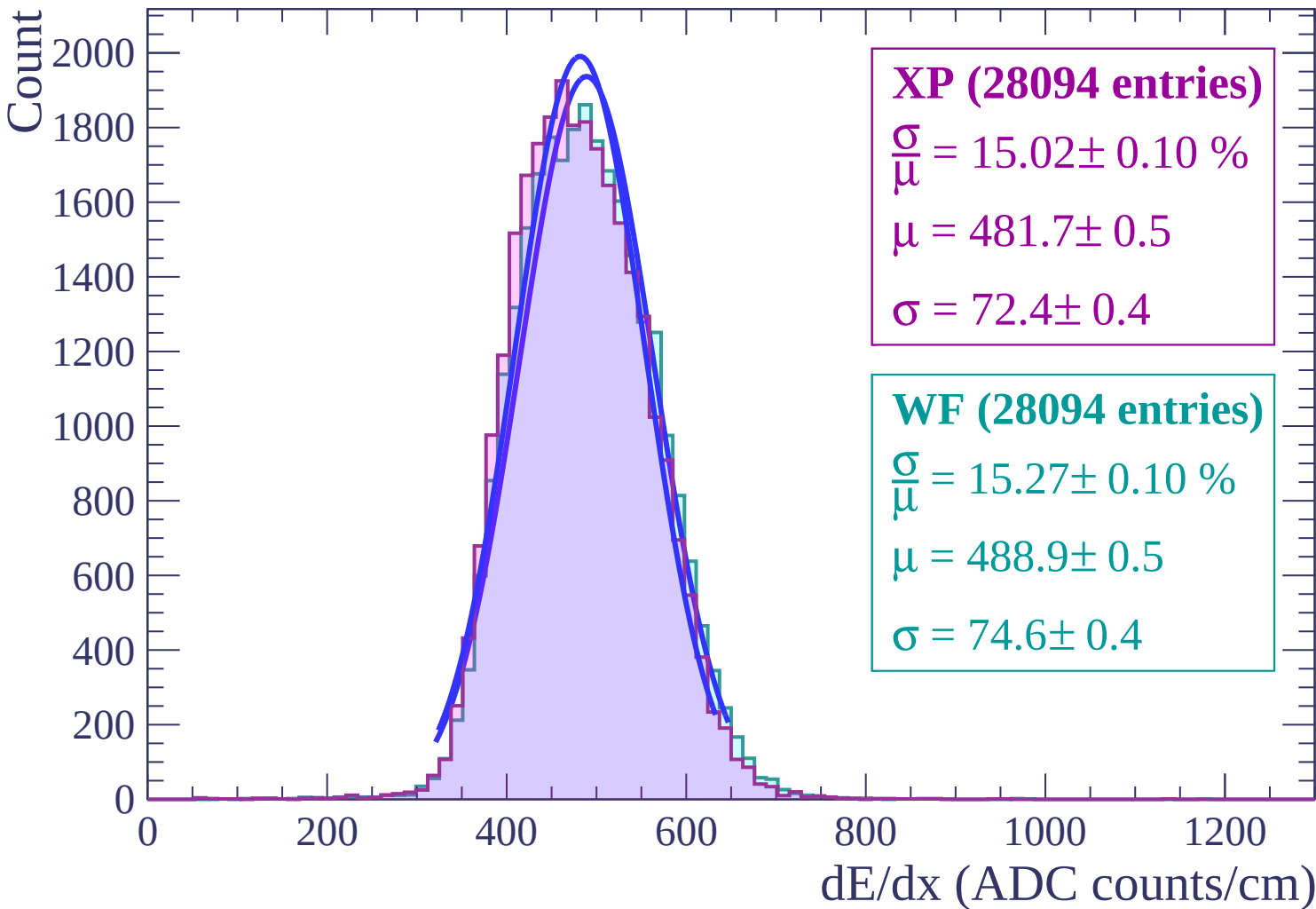
Count



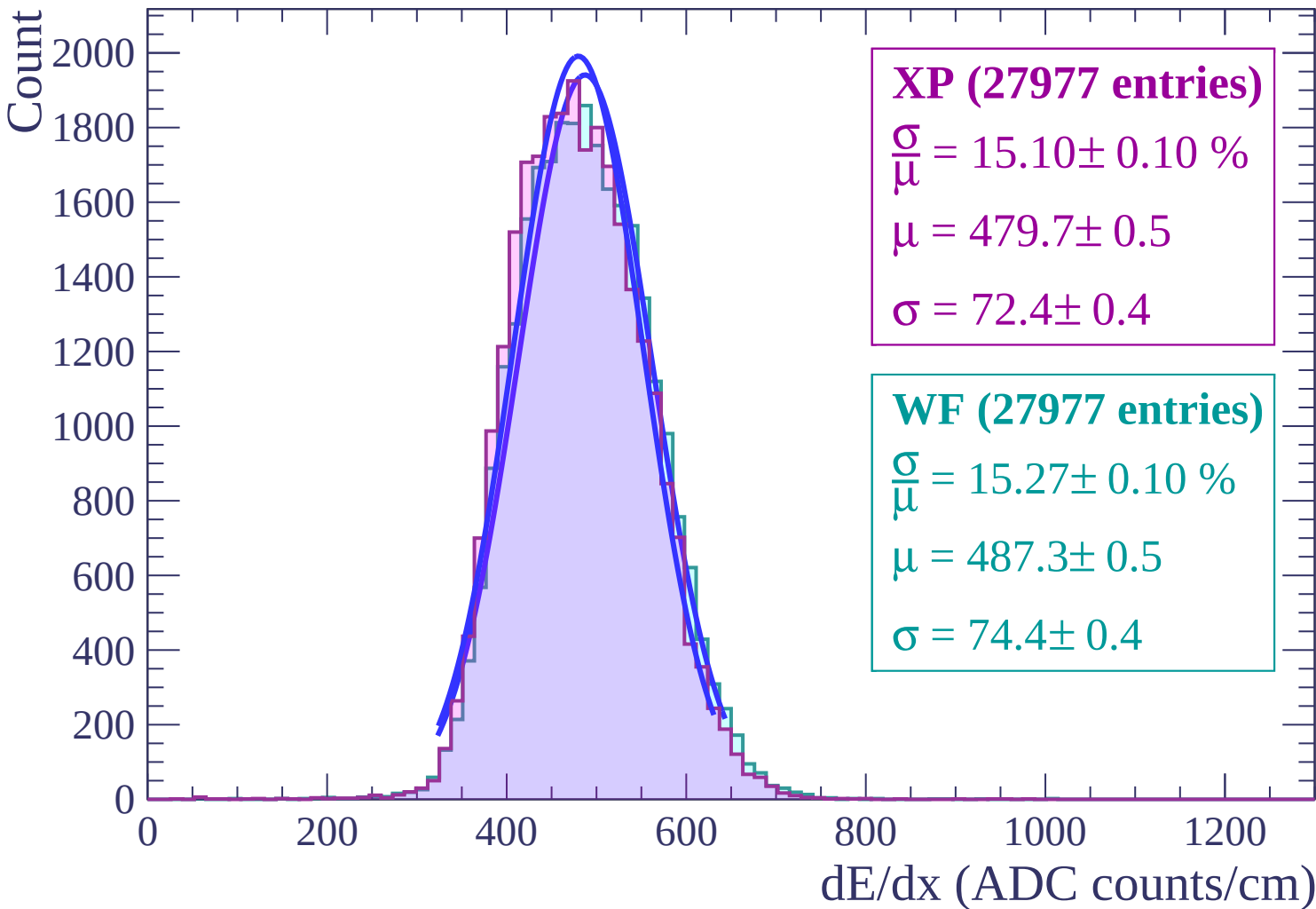
Energy loss | $-8 < \theta < -6$



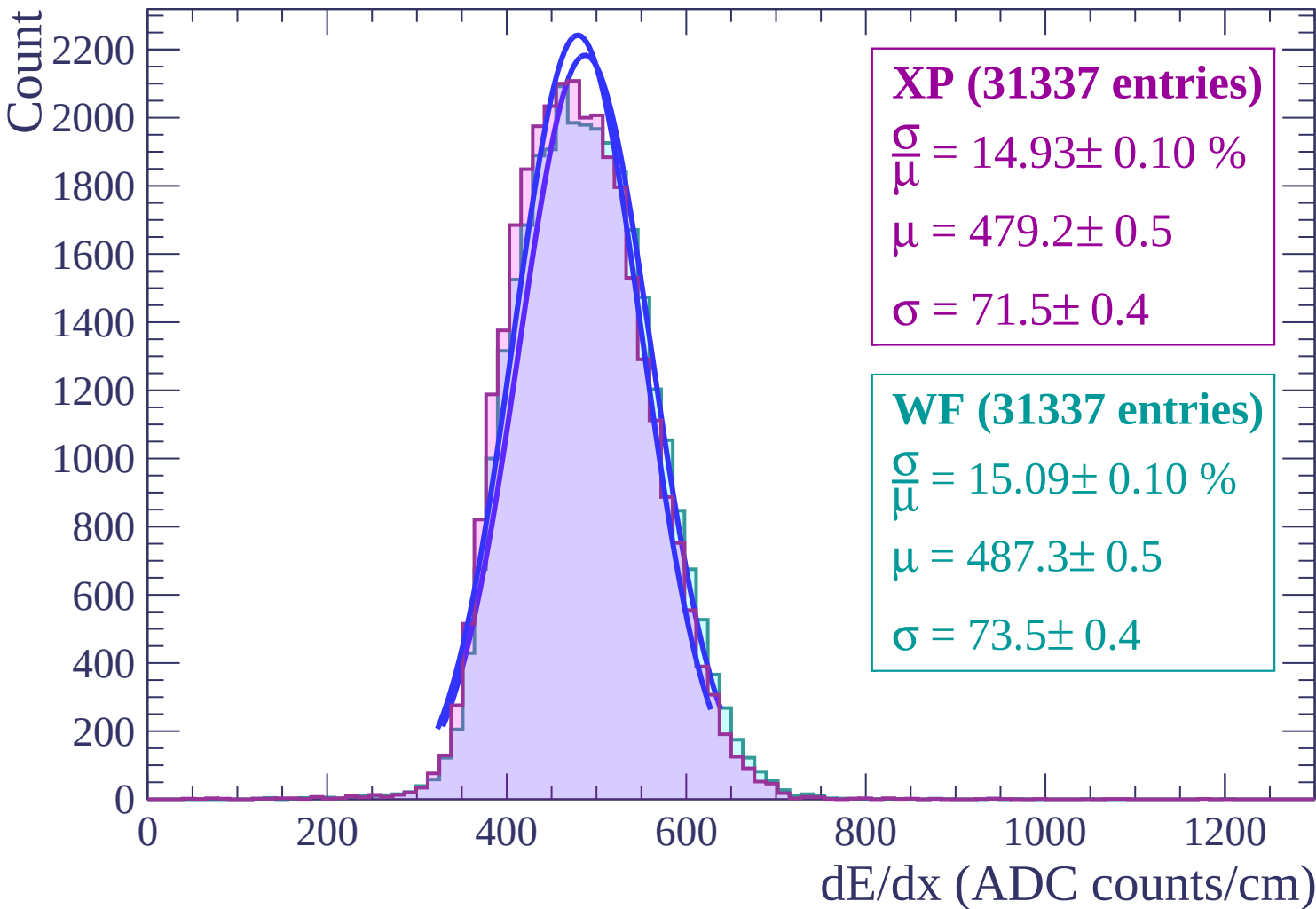
Energy loss | $-6 < \theta < -4$



Energy loss | $-4 < \theta < -2$



Energy loss $-2 < \theta < 0$



Energy loss | $0 < \theta < 2$

Count

3500

3000

2500

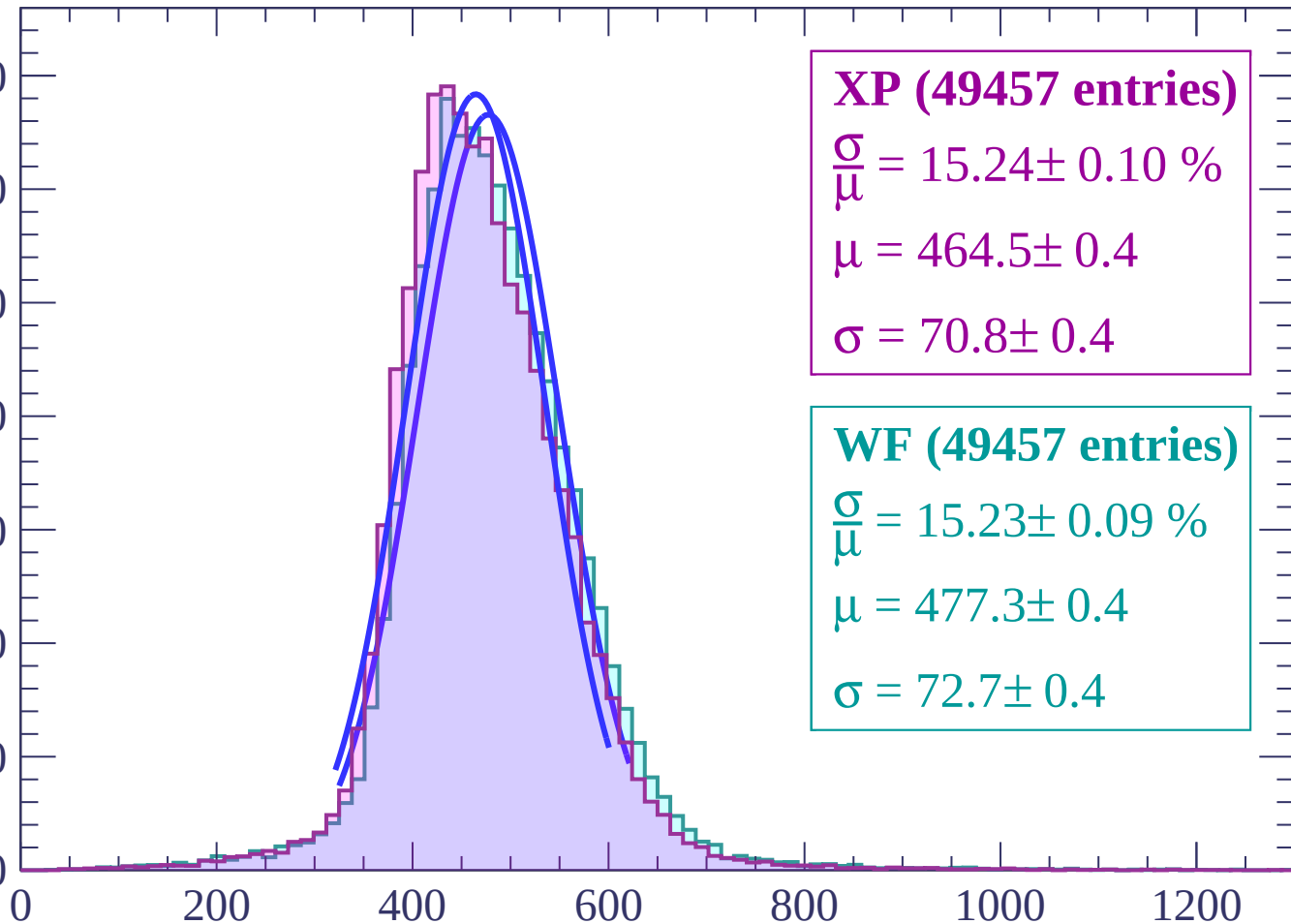
2000

1500

1000

500

0



XP (49457 entries)

$\frac{\sigma}{\mu} = 15.24 \pm 0.10 \%$

$\mu = 464.5 \pm 0.4$

$\sigma = 70.8 \pm 0.4$

WF (49457 entries)

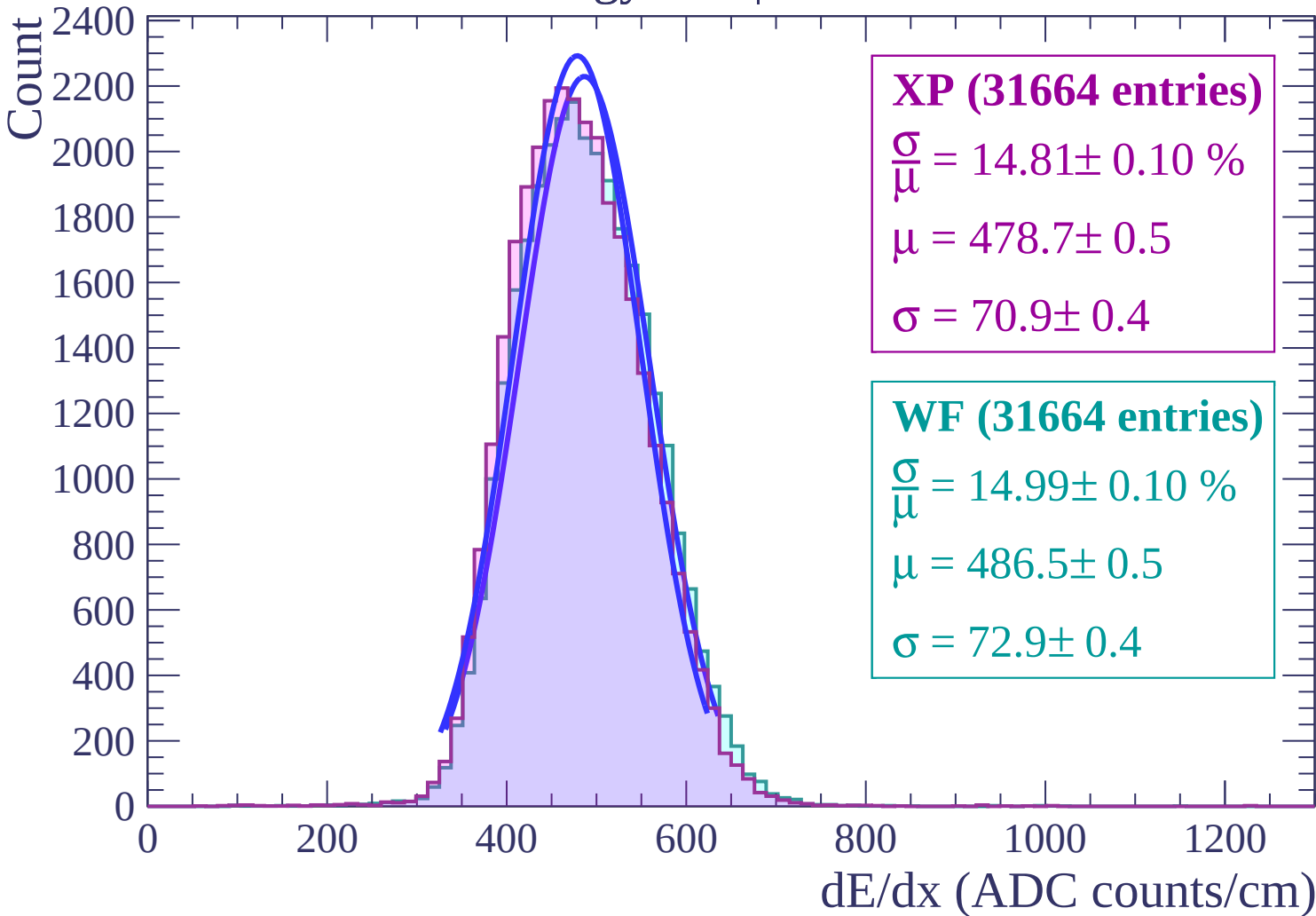
$\frac{\sigma}{\mu} = 15.23 \pm 0.09 \%$

$\mu = 477.3 \pm 0.4$

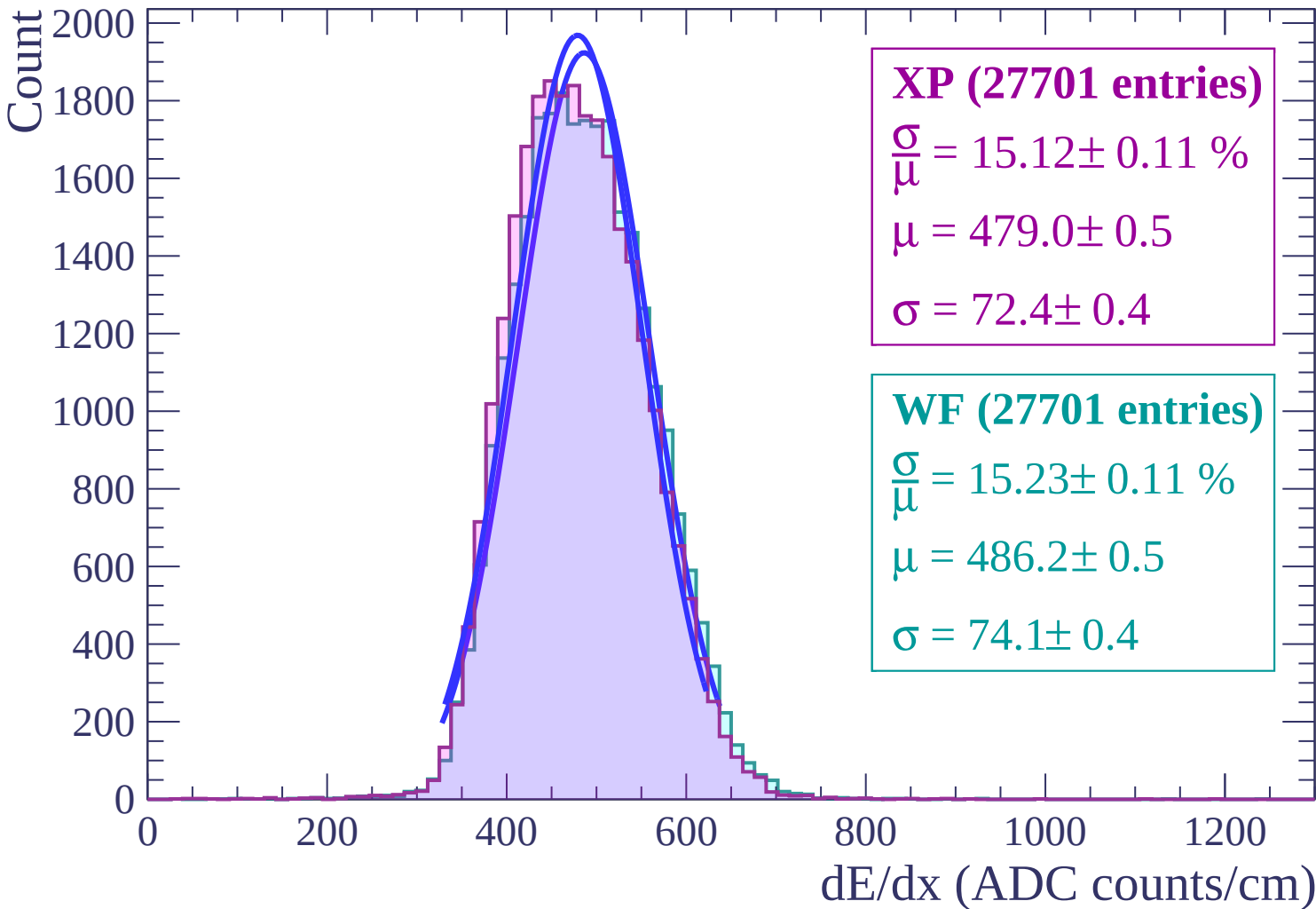
$\sigma = 72.7 \pm 0.4$

dE/dx (ADC counts/cm)

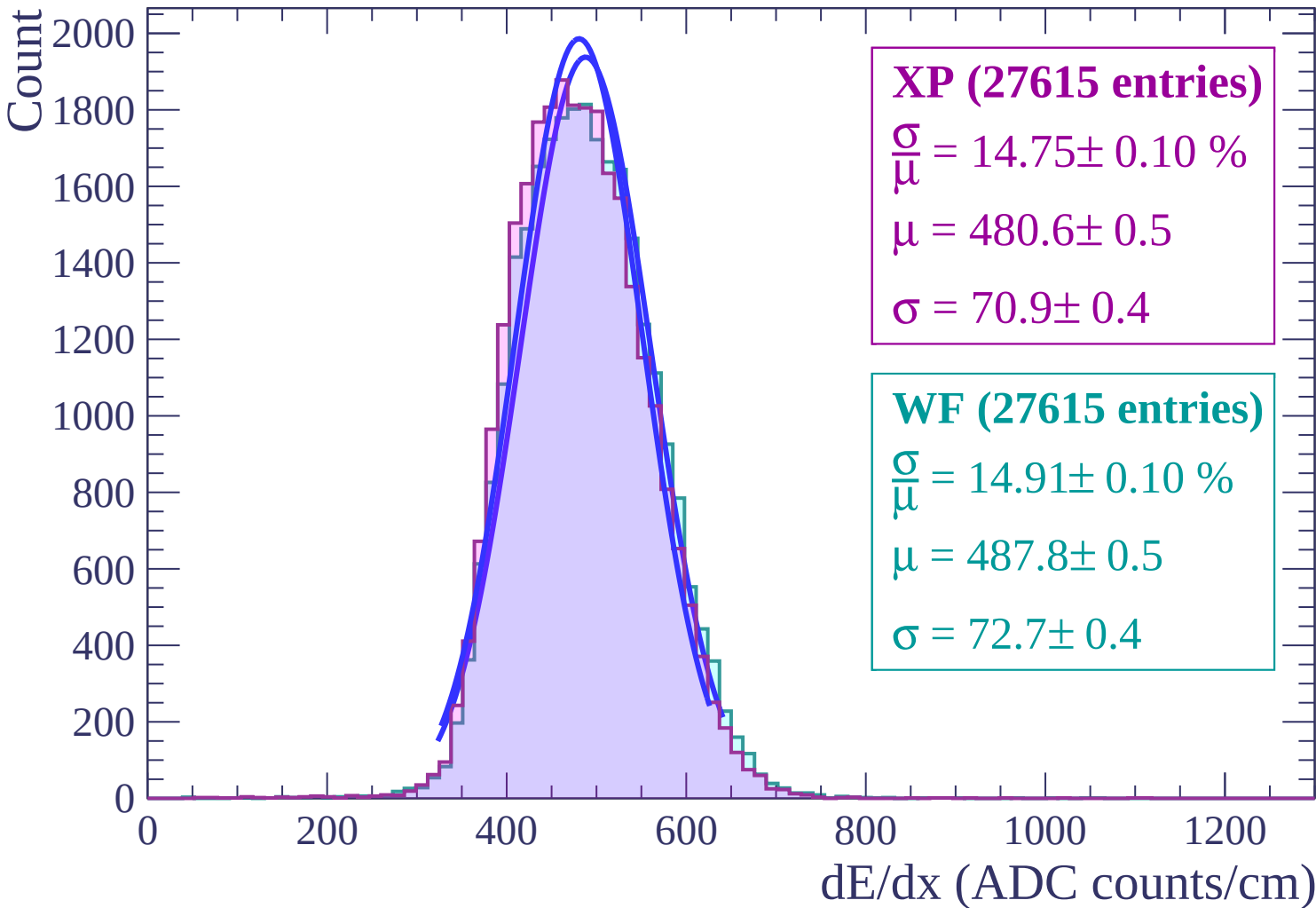
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$



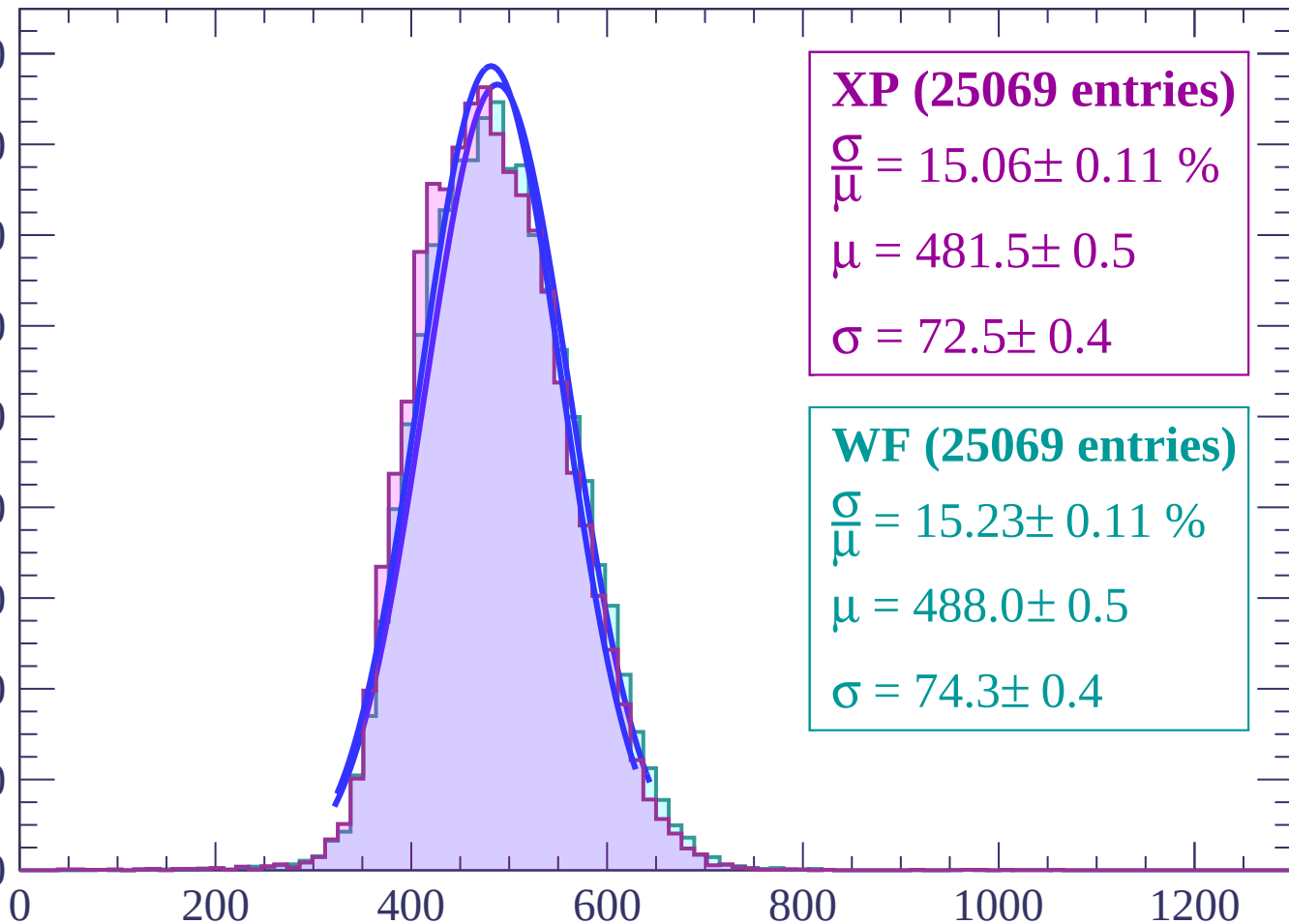
Energy loss | $6 < \theta < 8$



Energy loss | $8 < \theta < 10$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP (25069 entries)

$\frac{\sigma}{\mu} = 15.06 \pm 0.11 \%$

$\mu = 481.5 \pm 0.5$

$\sigma = 72.5 \pm 0.4$

WF (25069 entries)

$\frac{\sigma}{\mu} = 15.23 \pm 0.11 \%$

$\mu = 488.0 \pm 0.5$

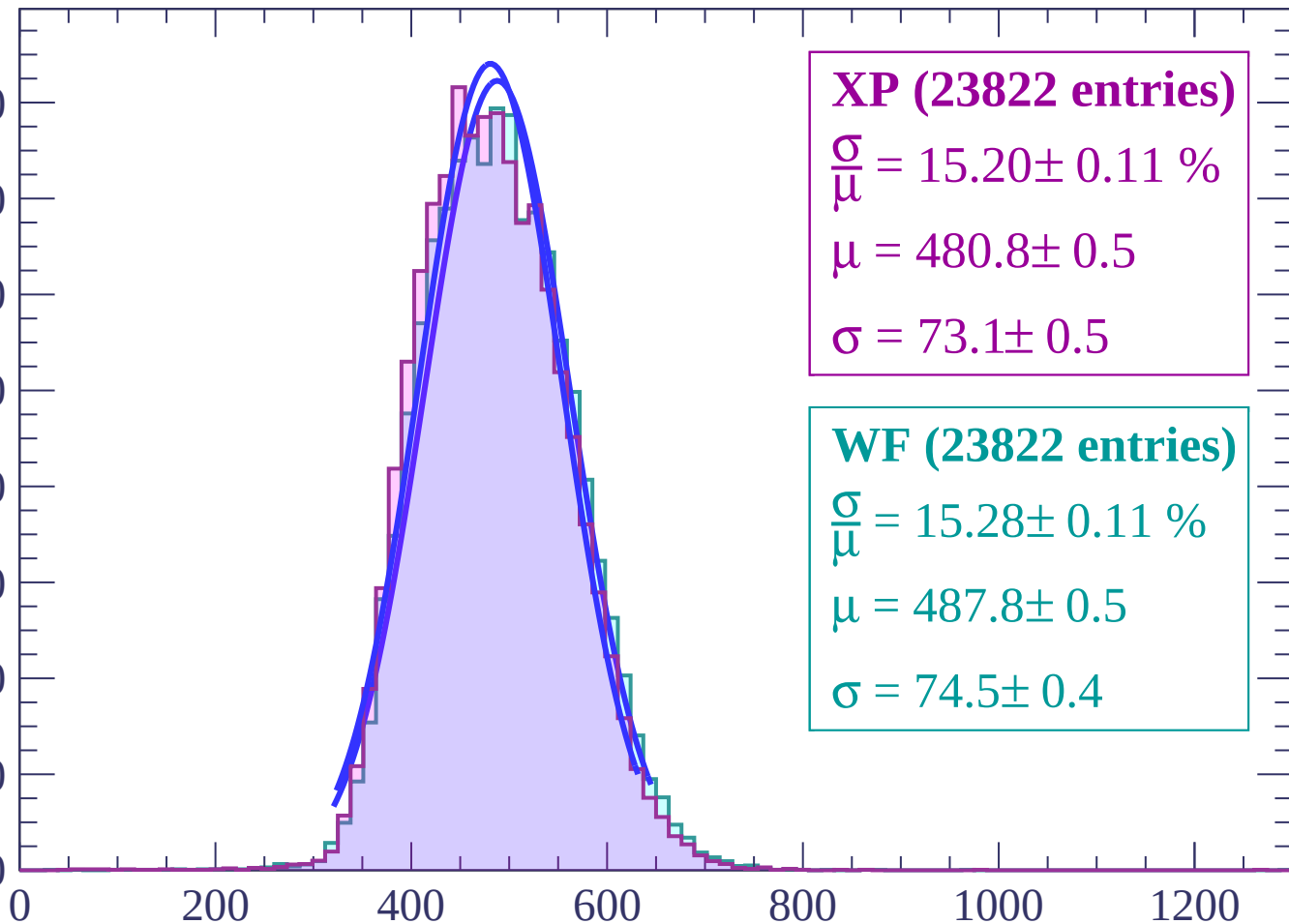
$\sigma = 74.3 \pm 0.4$

dE/dx (ADC counts/cm)

Energy loss | $10 < \theta < 12$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (23822 entries)

$\frac{\sigma}{\mu} = 15.20 \pm 0.11 \%$

$\mu = 480.8 \pm 0.5$

$\sigma = 73.1 \pm 0.5$

WF (23822 entries)

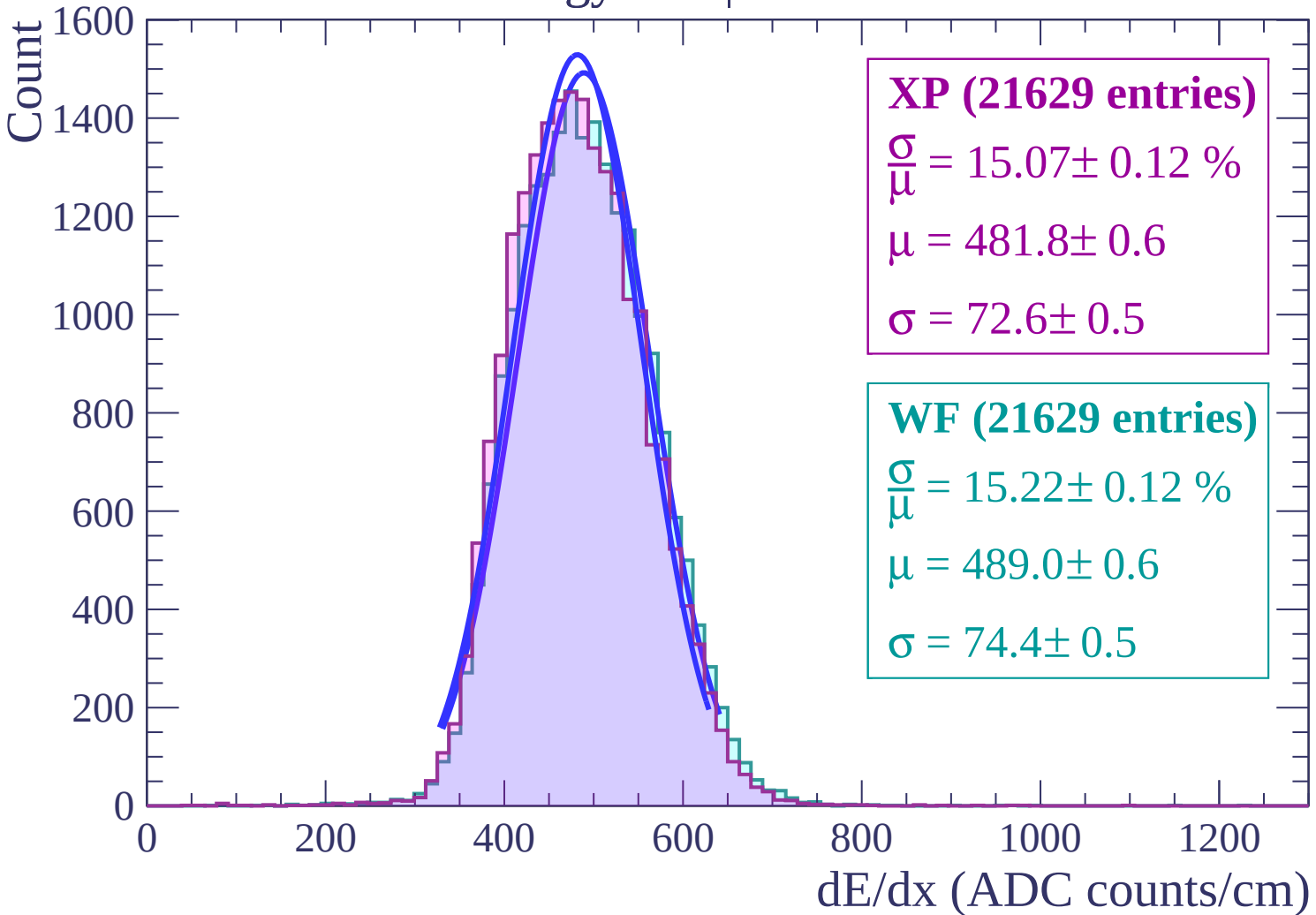
$\frac{\sigma}{\mu} = 15.28 \pm 0.11 \%$

$\mu = 487.8 \pm 0.5$

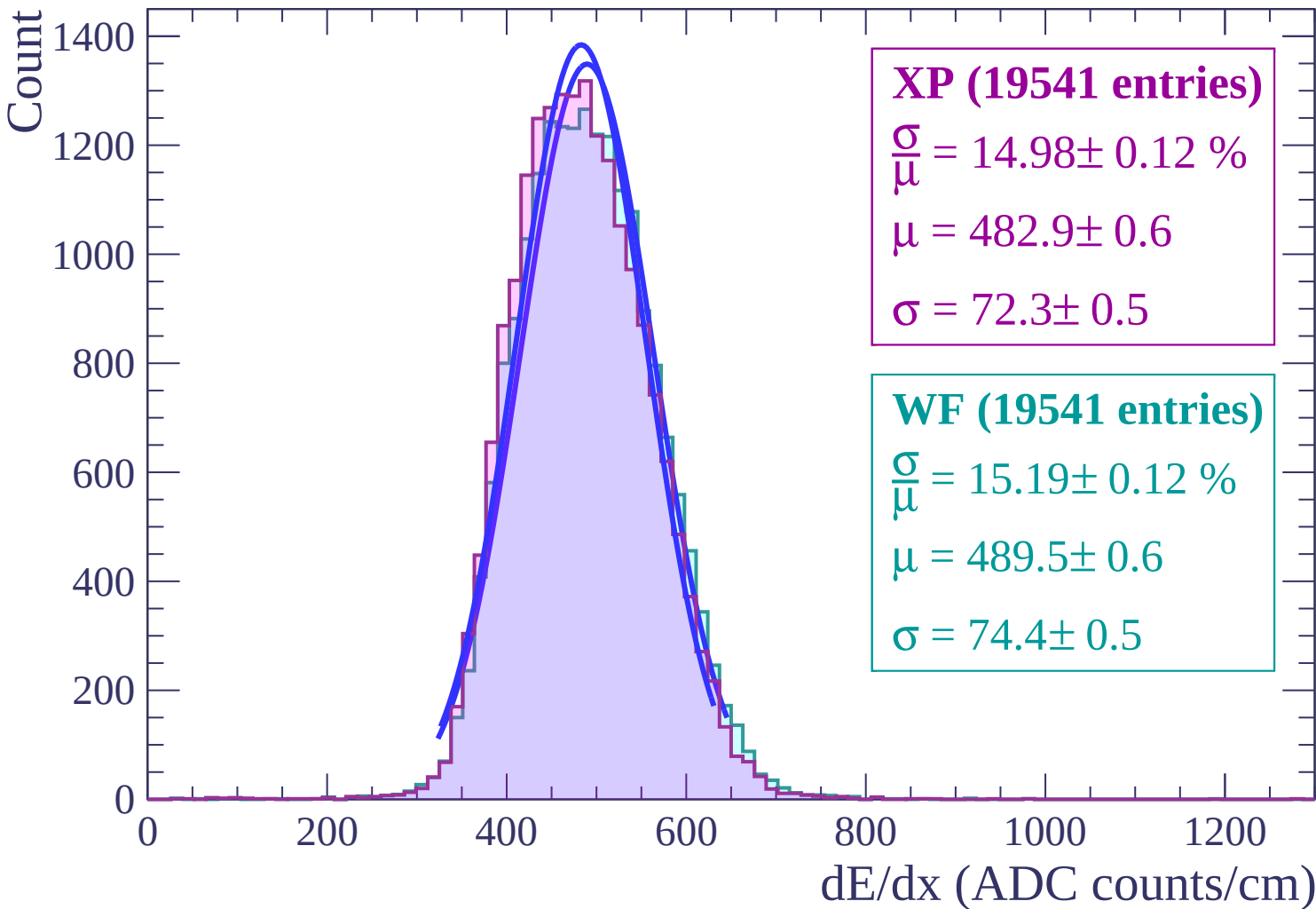
$\sigma = 74.5 \pm 0.4$

dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 14$



Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$

Count

1200

1000

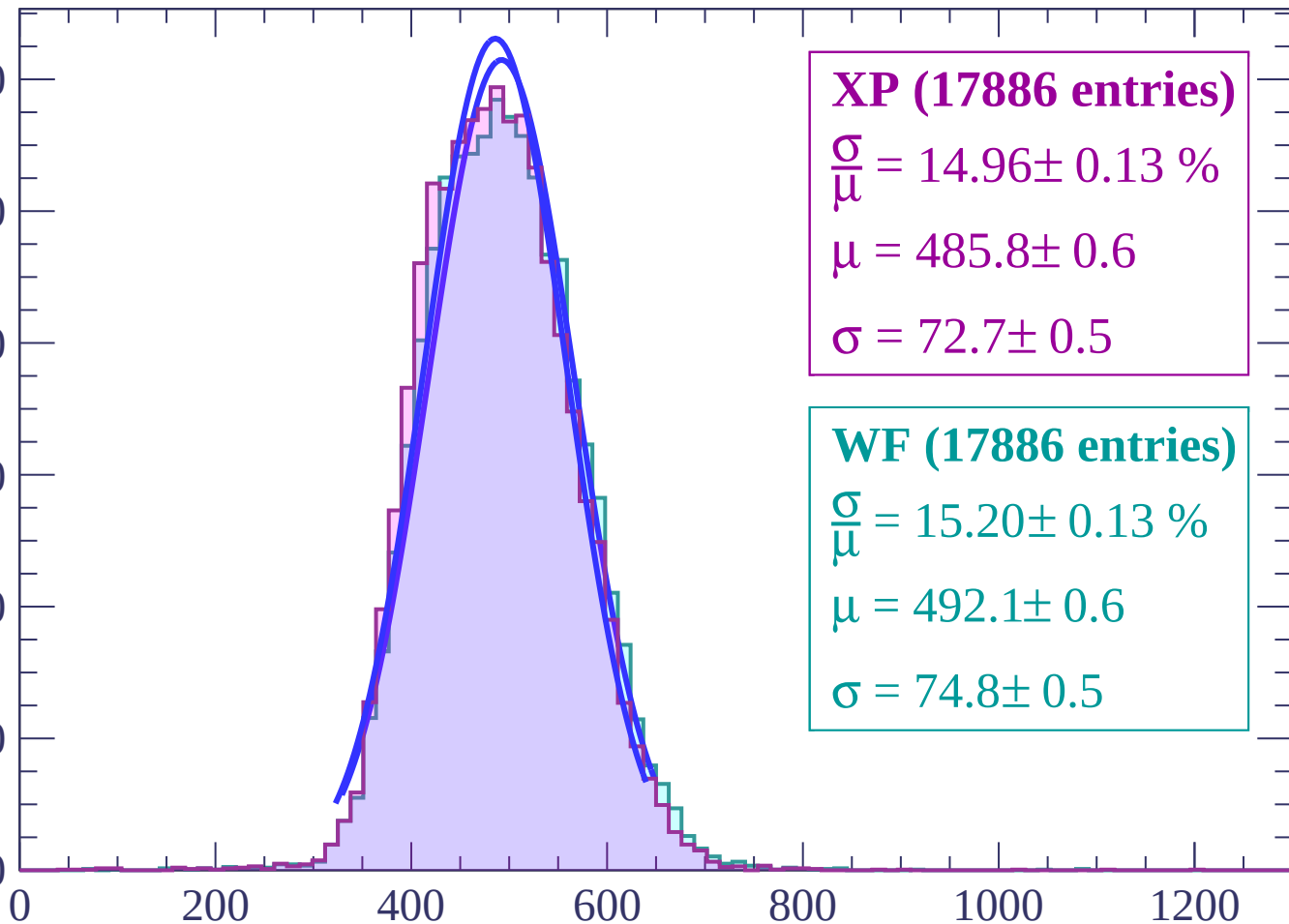
800

600

400

200

0



XP (17886 entries)

$\frac{\sigma}{\mu} = 14.96 \pm 0.13 \%$

$\mu = 485.8 \pm 0.6$

$\sigma = 72.7 \pm 0.5$

WF (17886 entries)

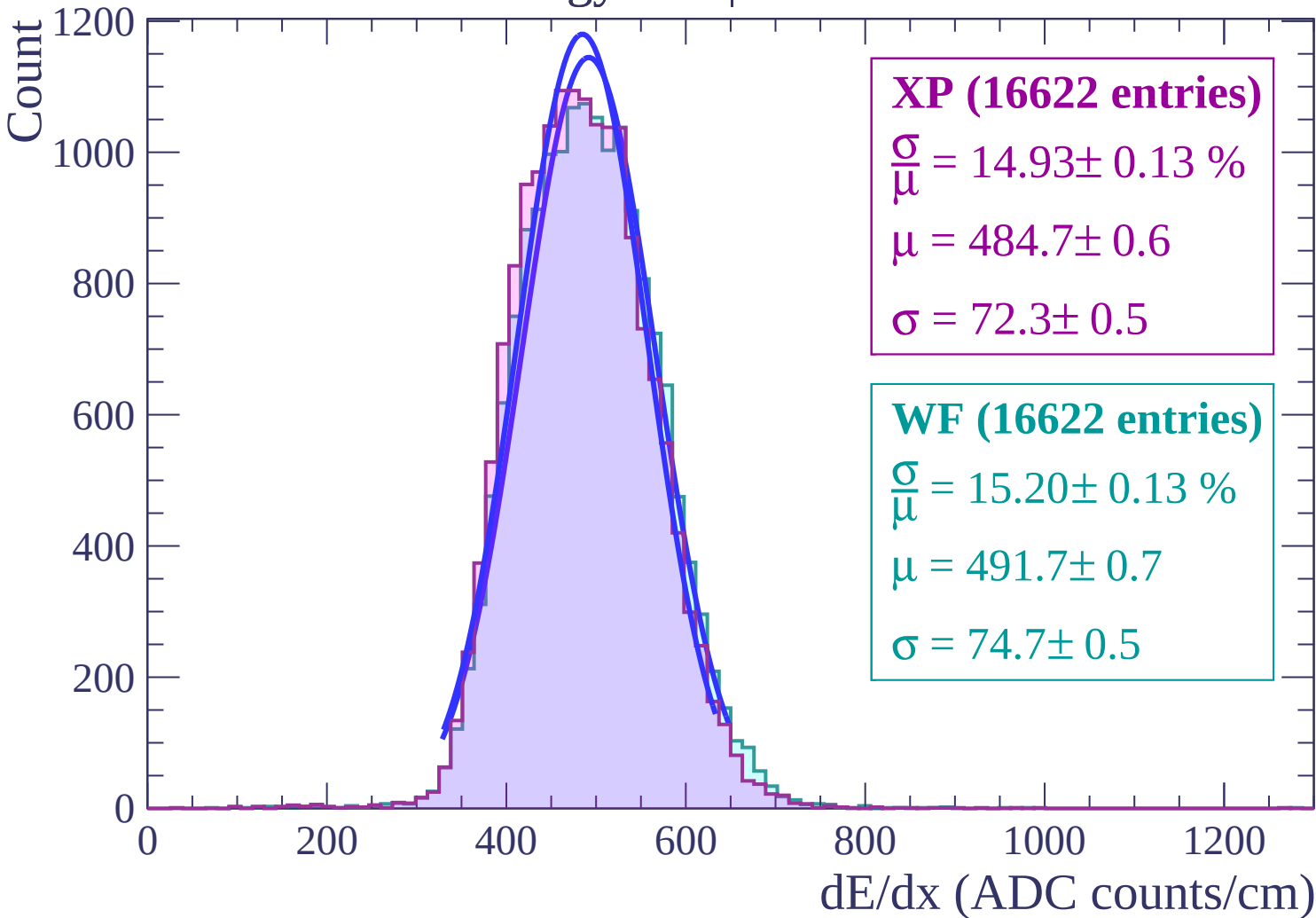
$\frac{\sigma}{\mu} = 15.20 \pm 0.13 \%$

$\mu = 492.1 \pm 0.6$

$\sigma = 74.8 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 20$



Energy loss | $20 < \theta < 22$

Count

1000

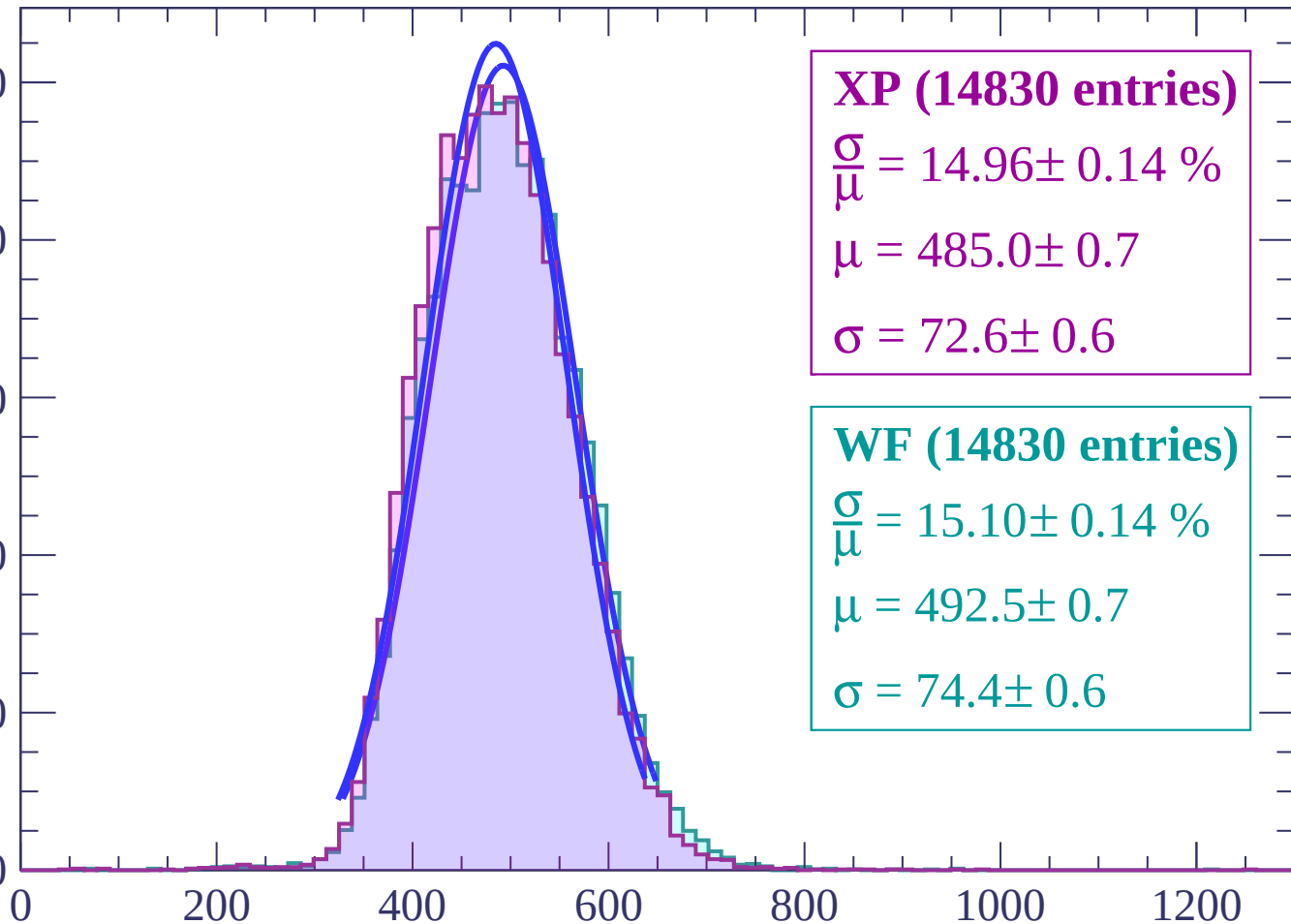
800

600

400

200

0



XP (14830 entries)

$\frac{\sigma}{\mu} = 14.96 \pm 0.14 \%$

$\mu = 485.0 \pm 0.7$

$\sigma = 72.6 \pm 0.6$

WF (14830 entries)

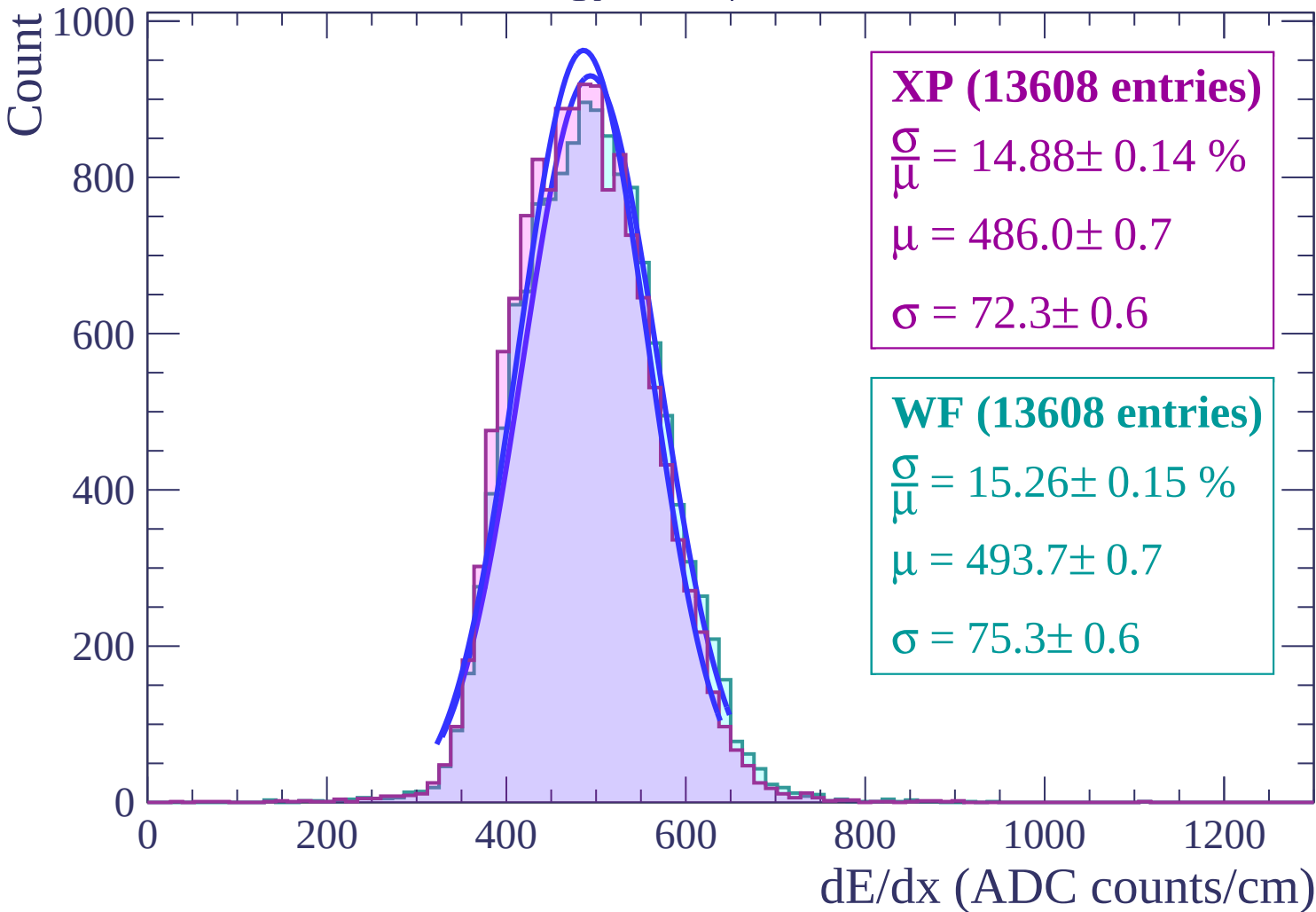
$\frac{\sigma}{\mu} = 15.10 \pm 0.14 \%$

$\mu = 492.5 \pm 0.7$

$\sigma = 74.4 \pm 0.6$

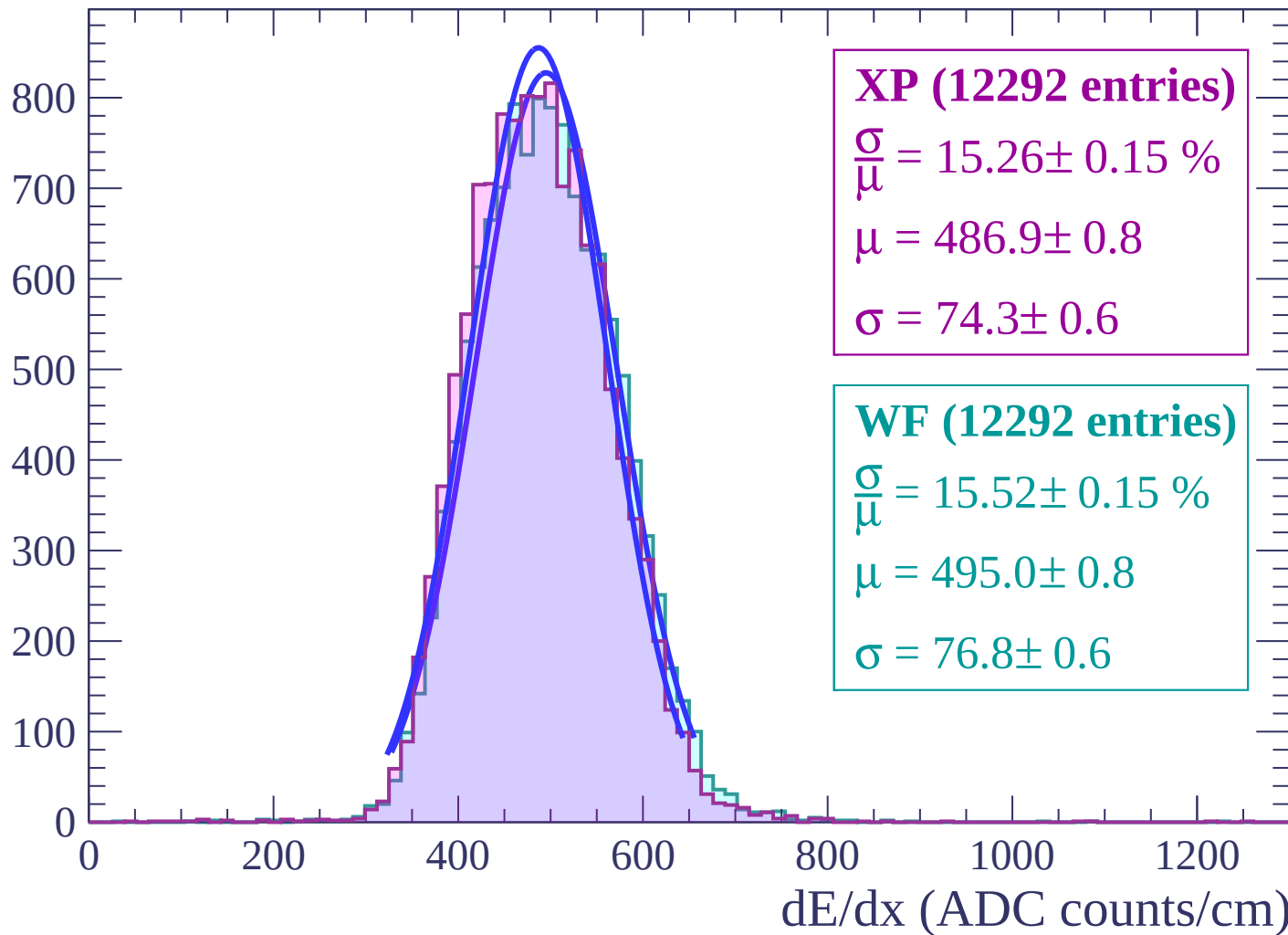
dE/dx (ADC counts/cm)

Energy loss | $22 < \theta < 24$



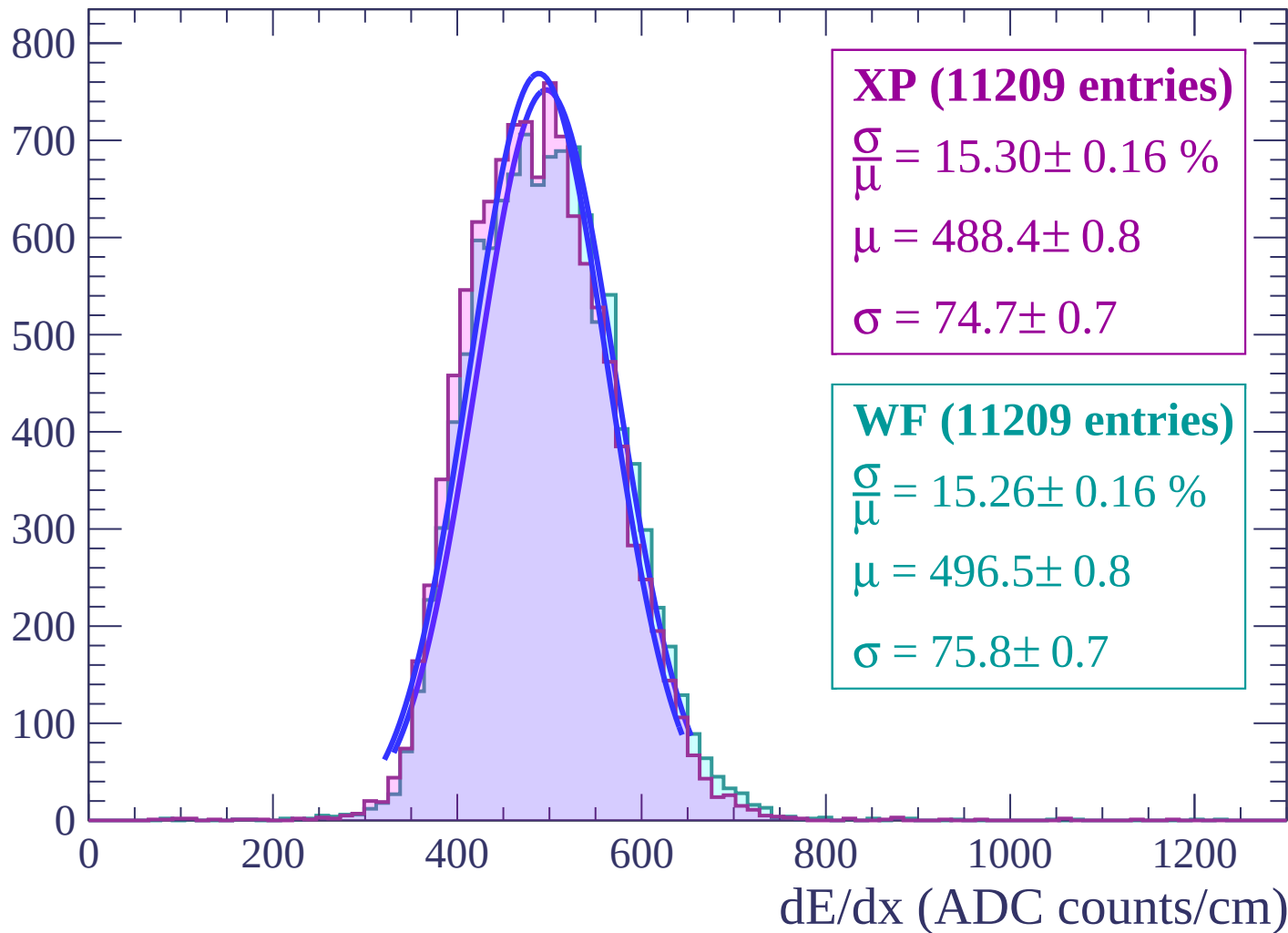
Energy loss | $24 < \theta < 26$

Count



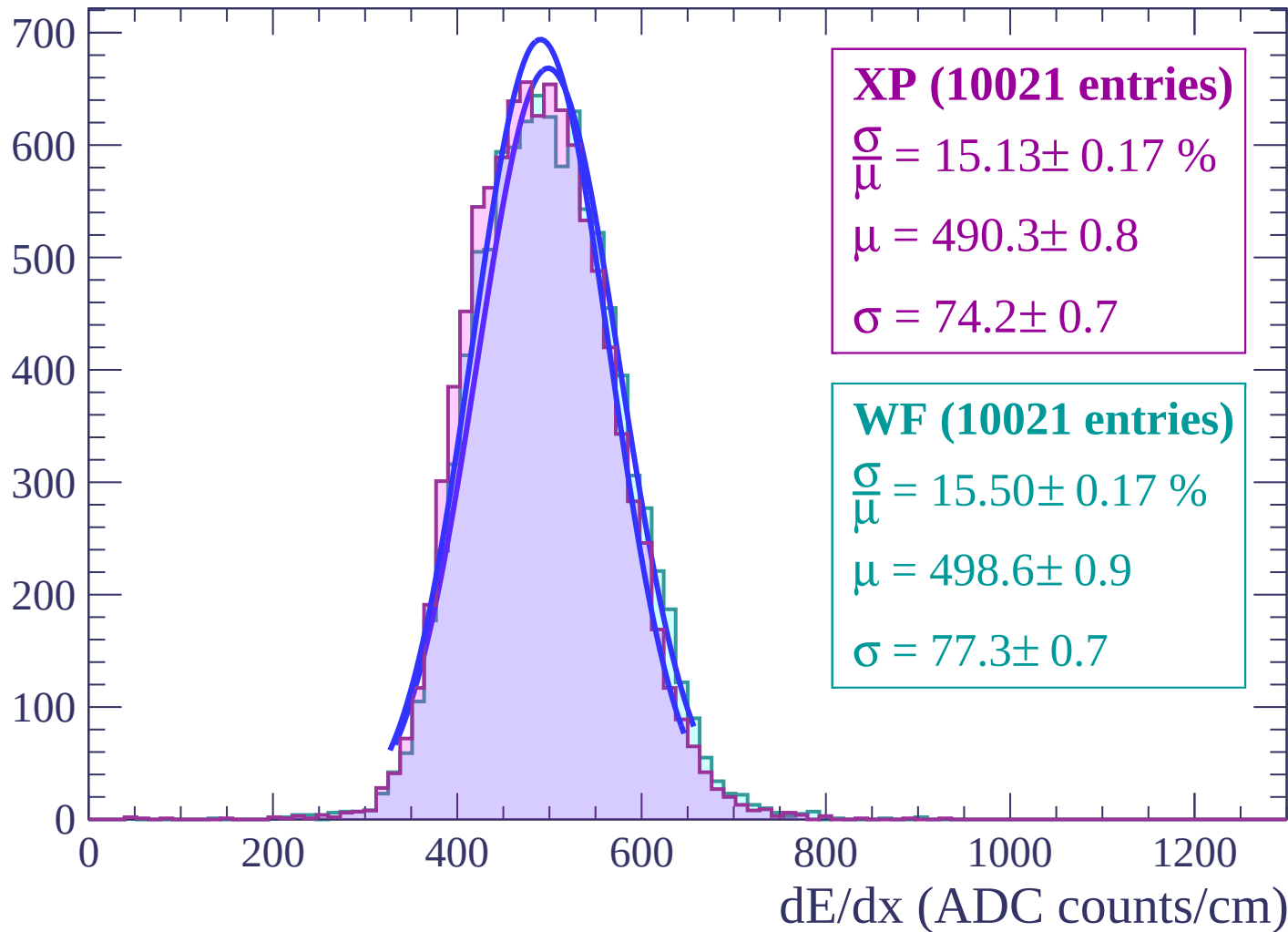
Energy loss | $26 < \theta < 28$

Count



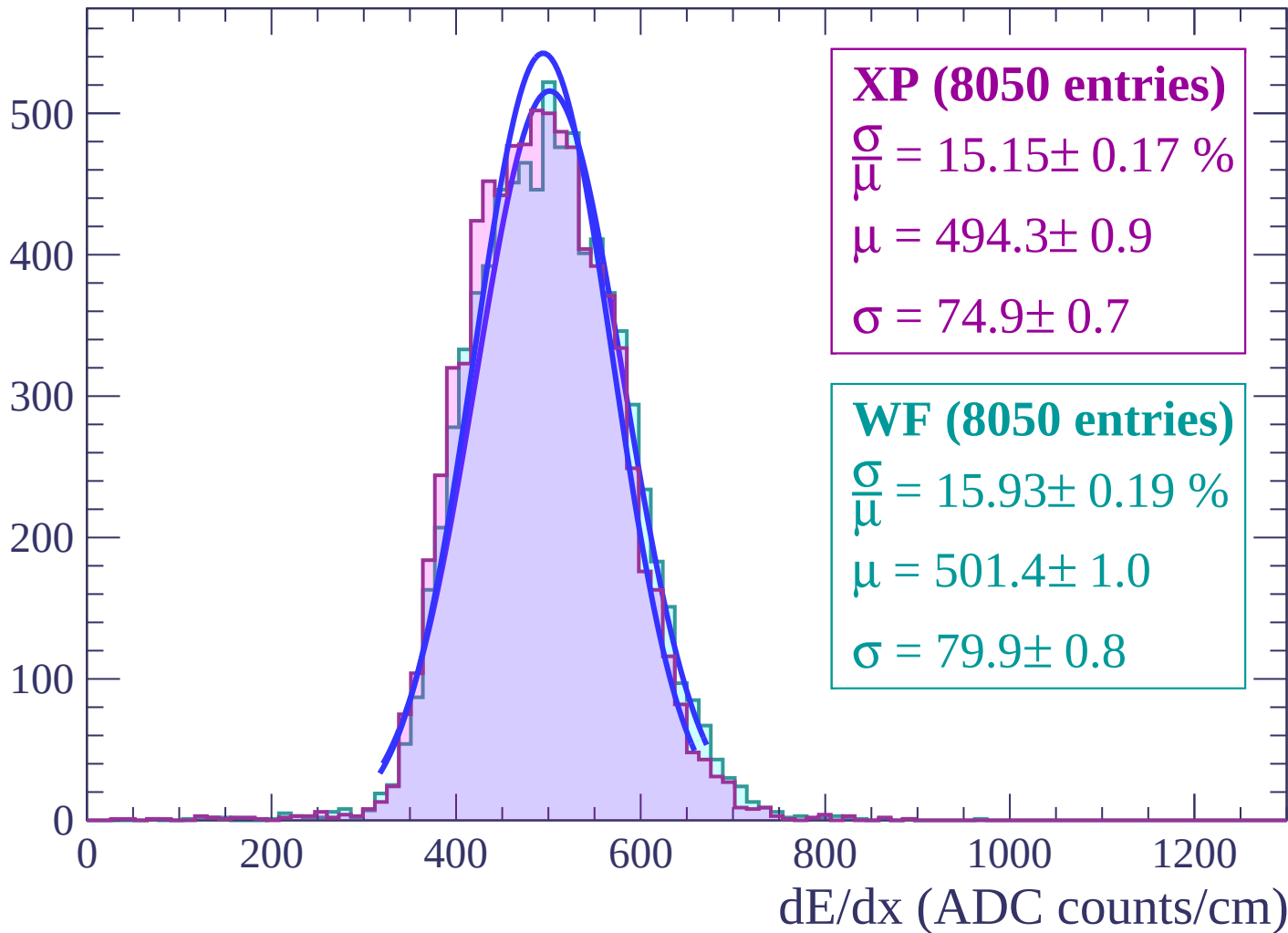
Energy loss | $28 < \theta < 30$

Count



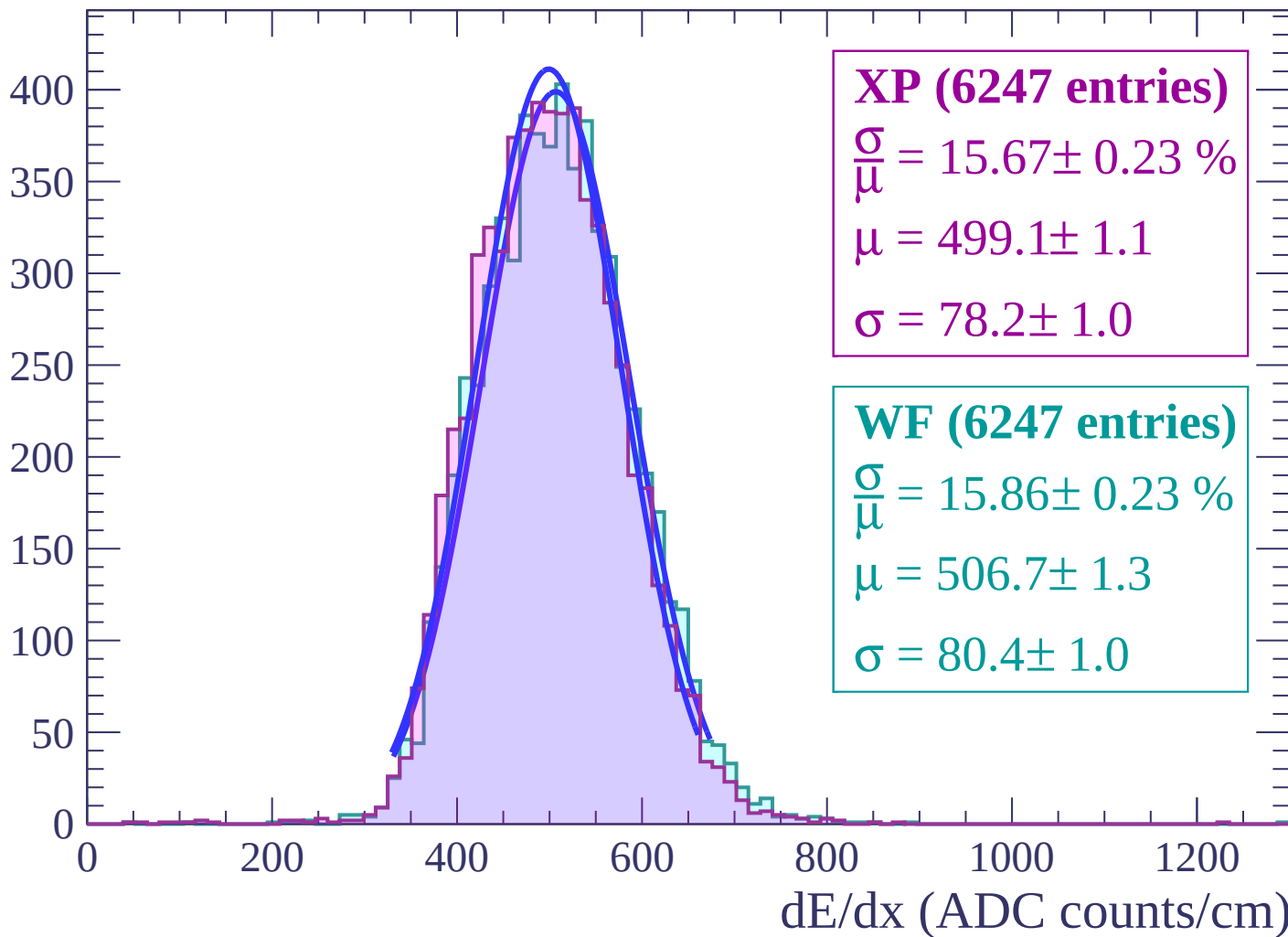
Energy loss | $30 < \theta < 32$

Count



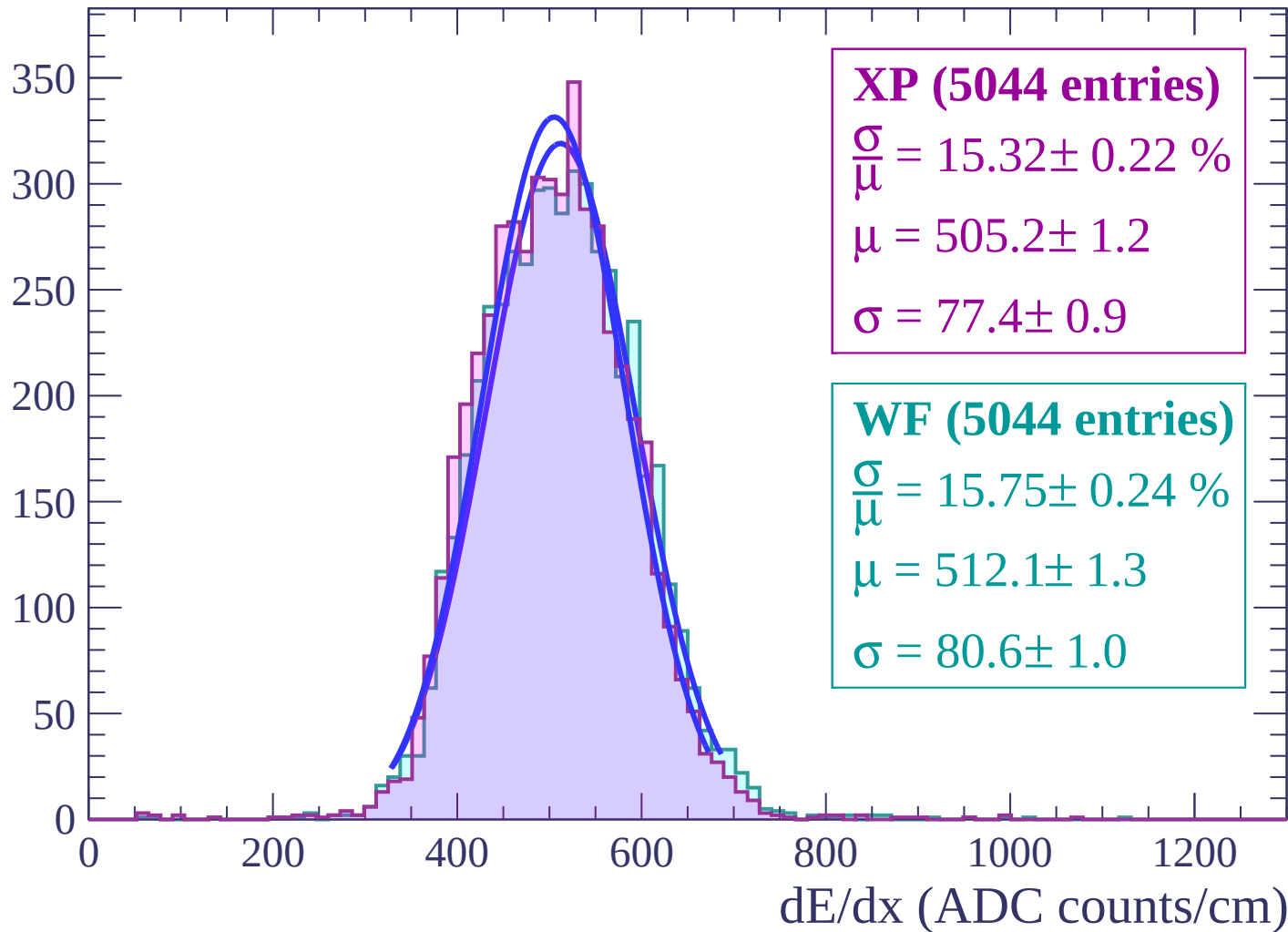
Energy loss | $32 < \theta < 34$

Count



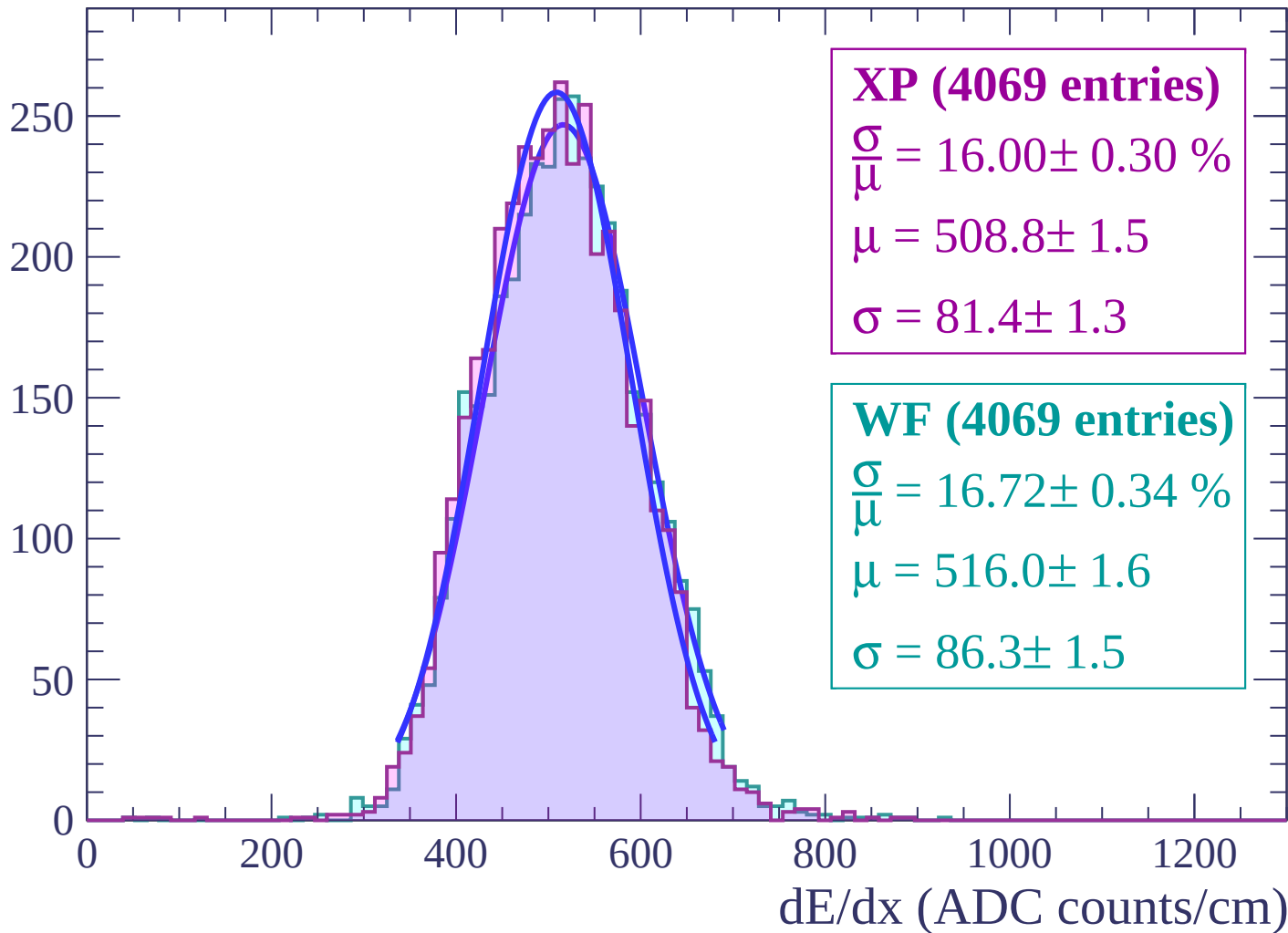
Energy loss | $34 < \theta < 36$

Count



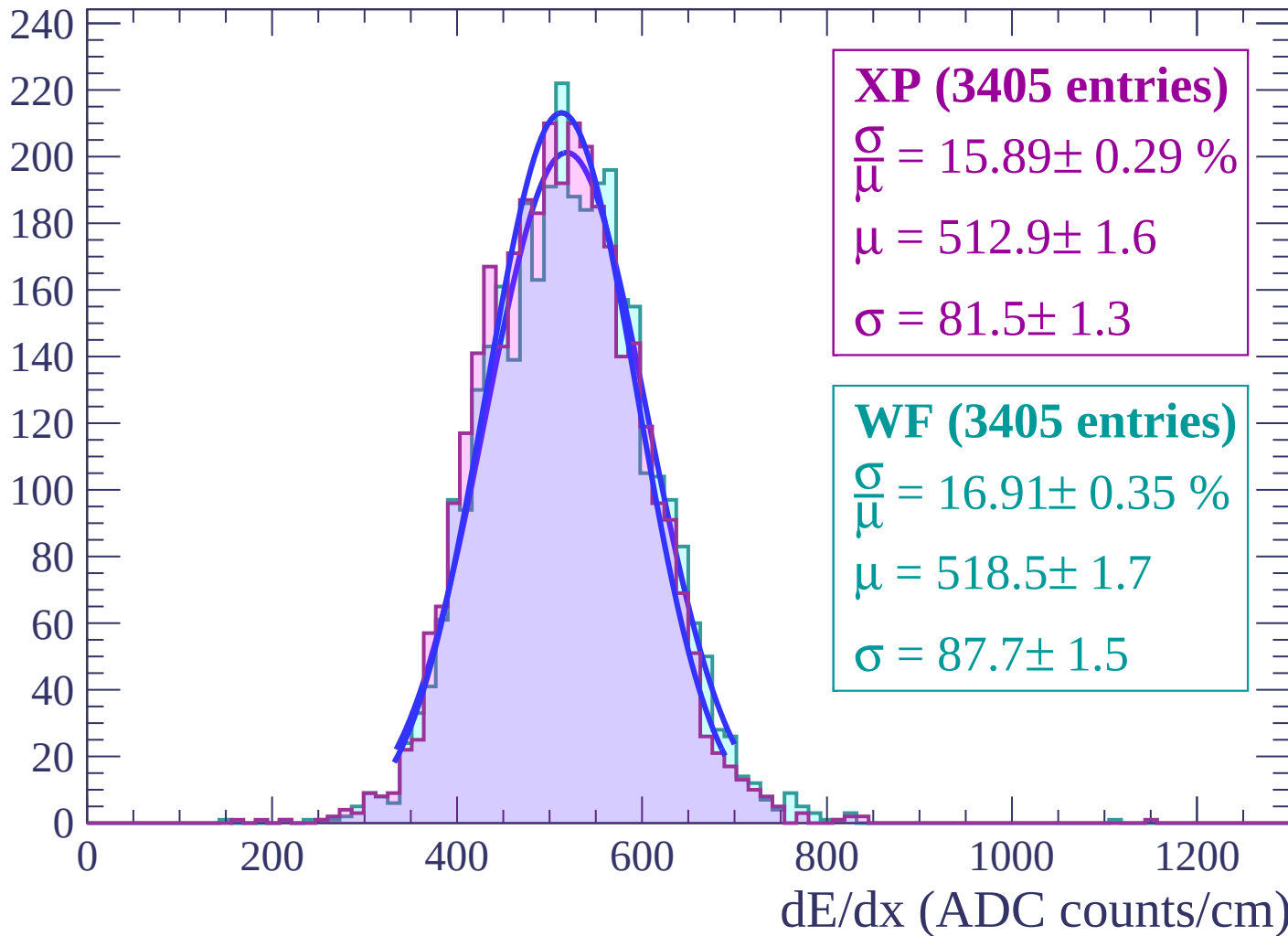
Energy loss | $36 < \theta < 38$

Count



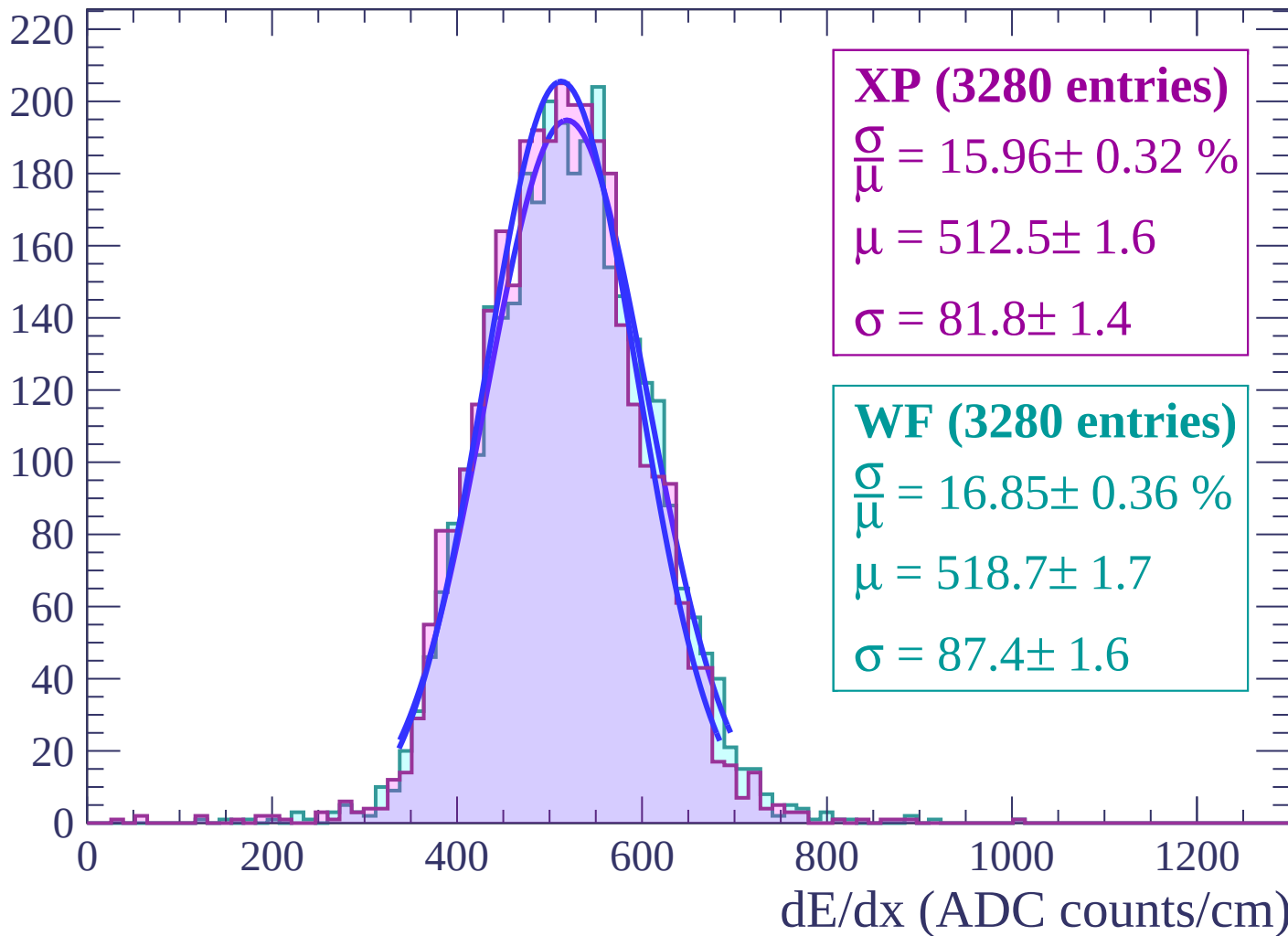
Energy loss | $38 < \theta < 40$

Count



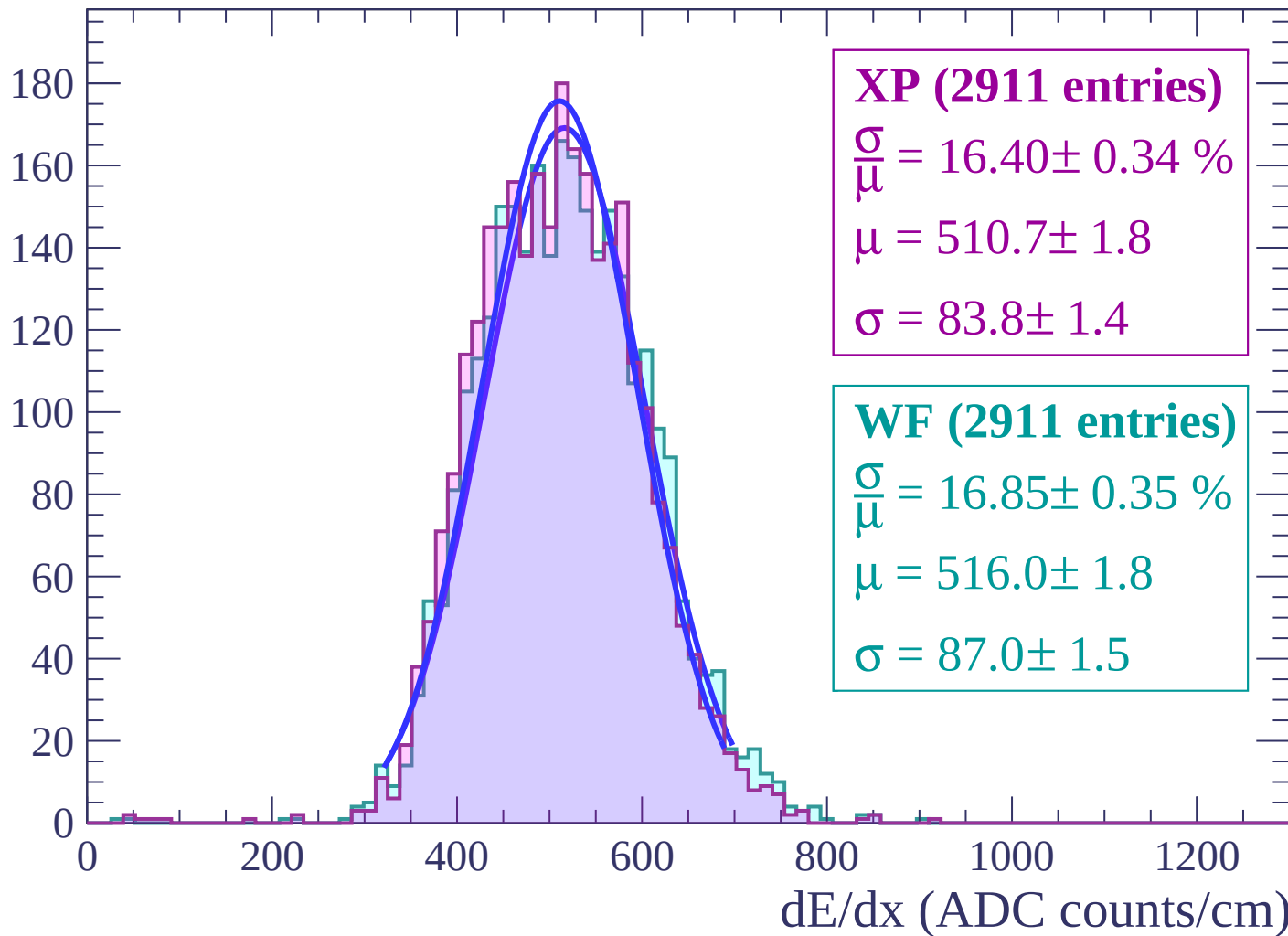
Energy loss | $40 < \theta < 42$

Count



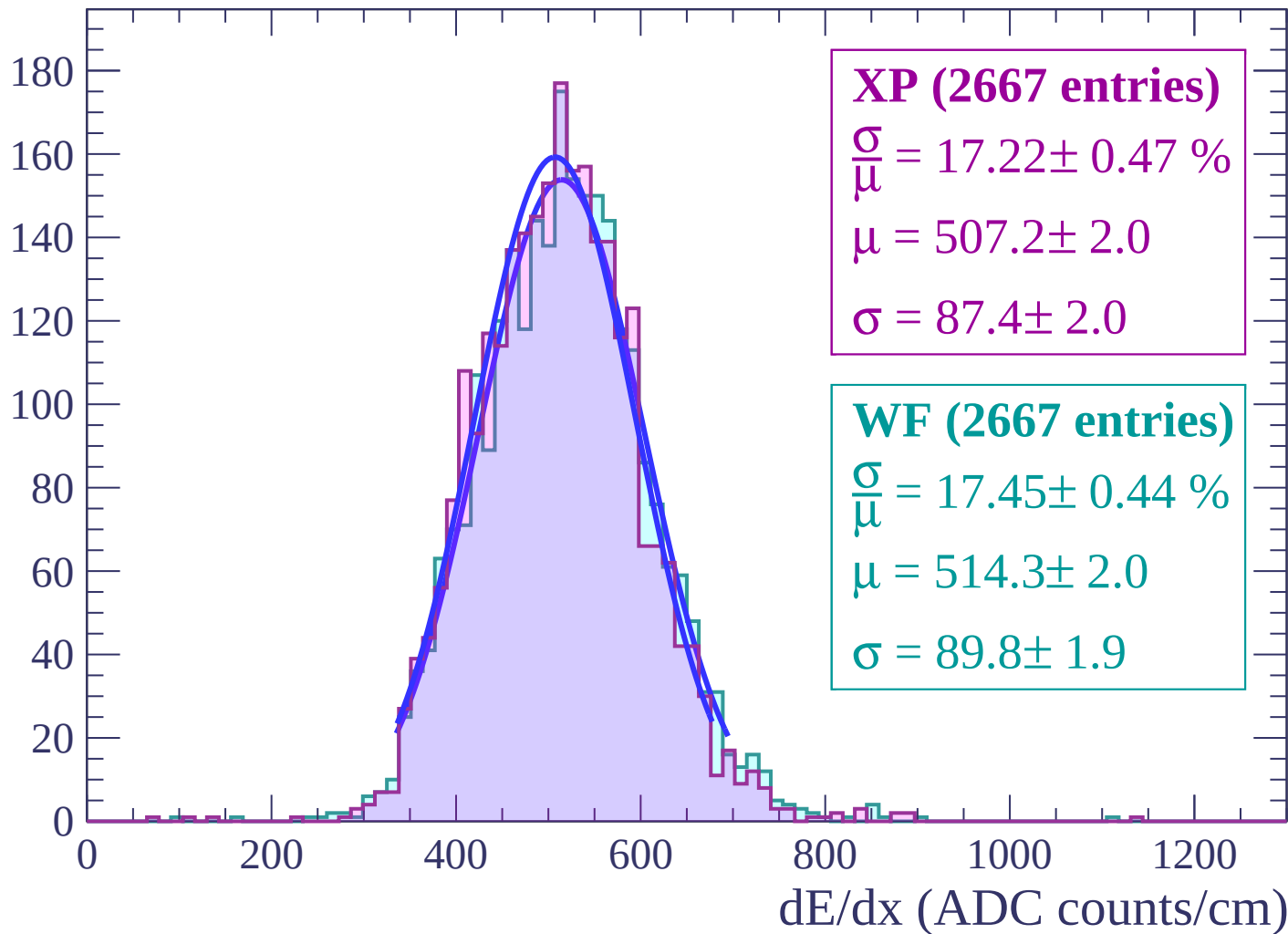
Energy loss | $42 < \theta < 44$

Count



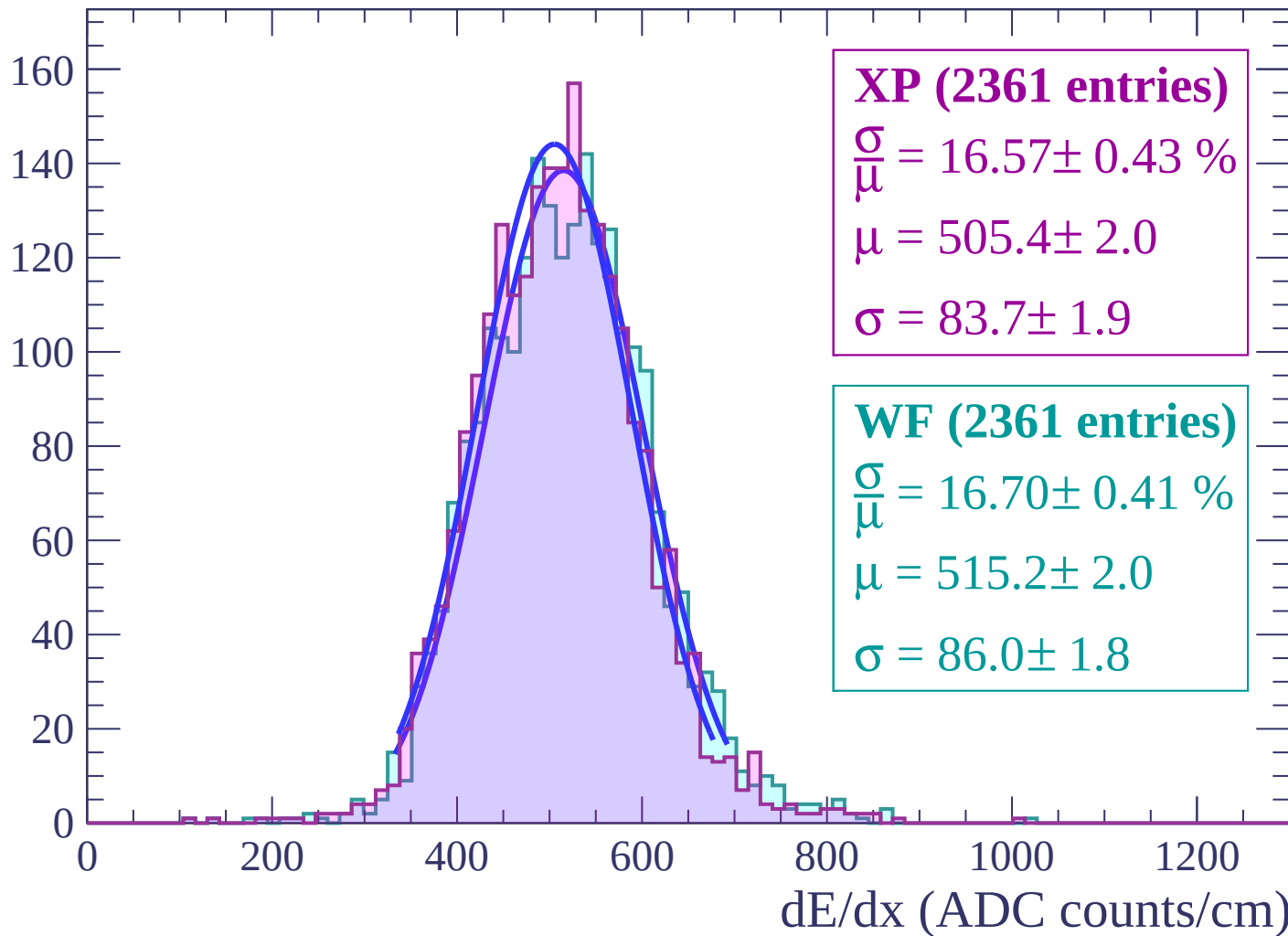
Energy loss | $44 < \theta < 46$

Count



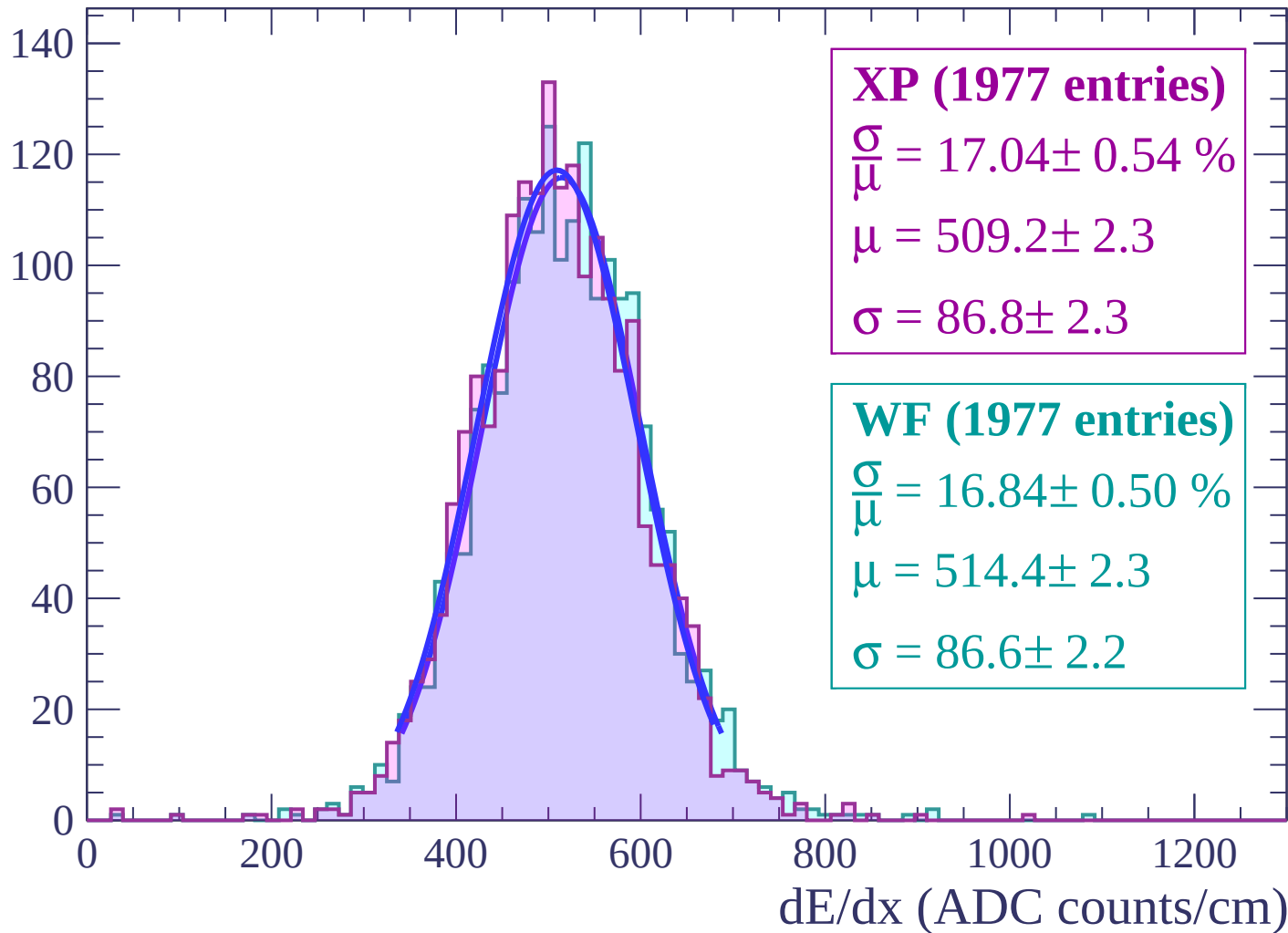
Energy loss | $46 < \theta < 48$

Count



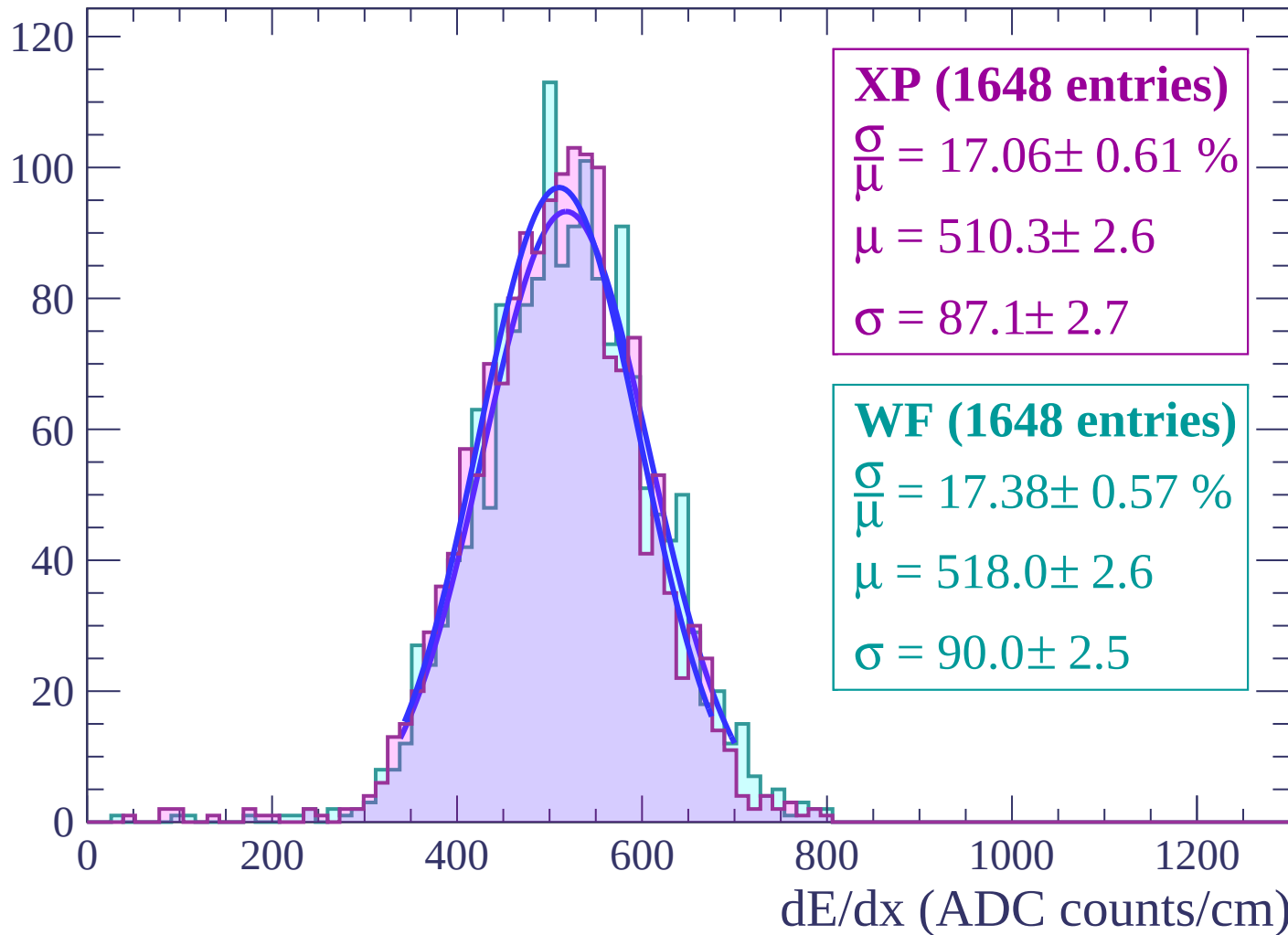
Energy loss | $48 < \theta < 50$

Count



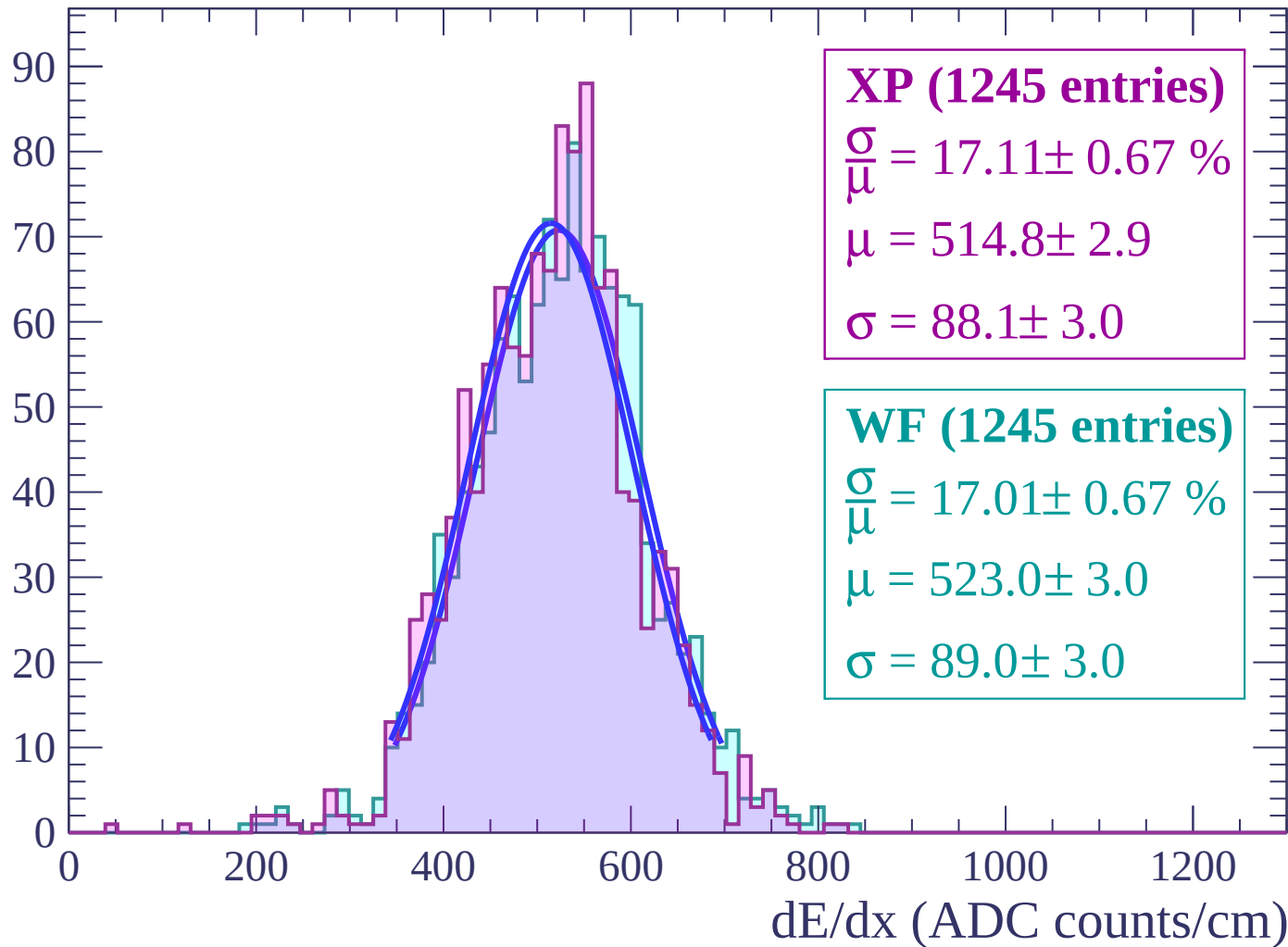
Energy loss | $50 < \theta < 52$

Count



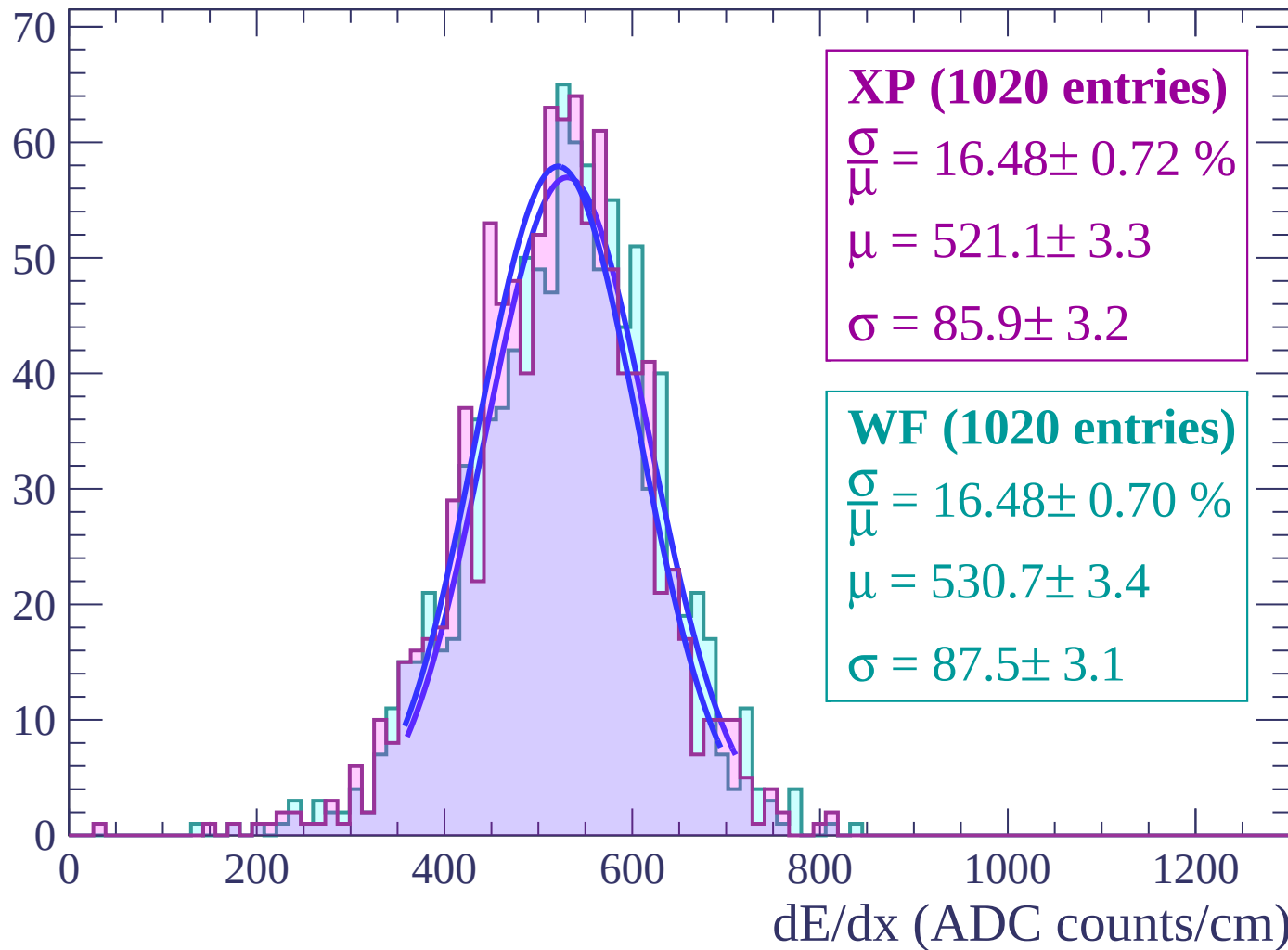
Energy loss | $52 < \theta < 54$

Count



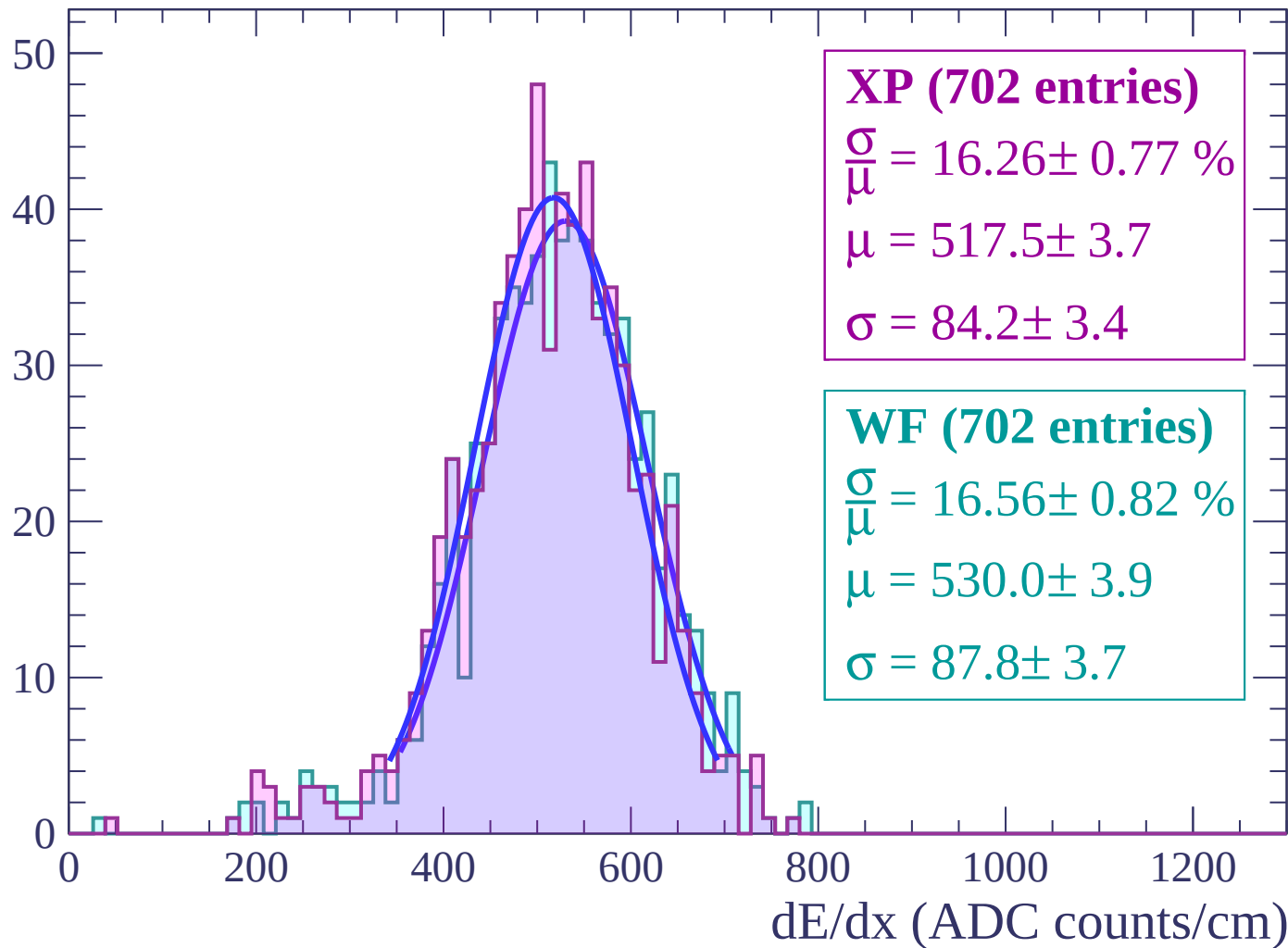
Energy loss | $54 < \theta < 56$

Count



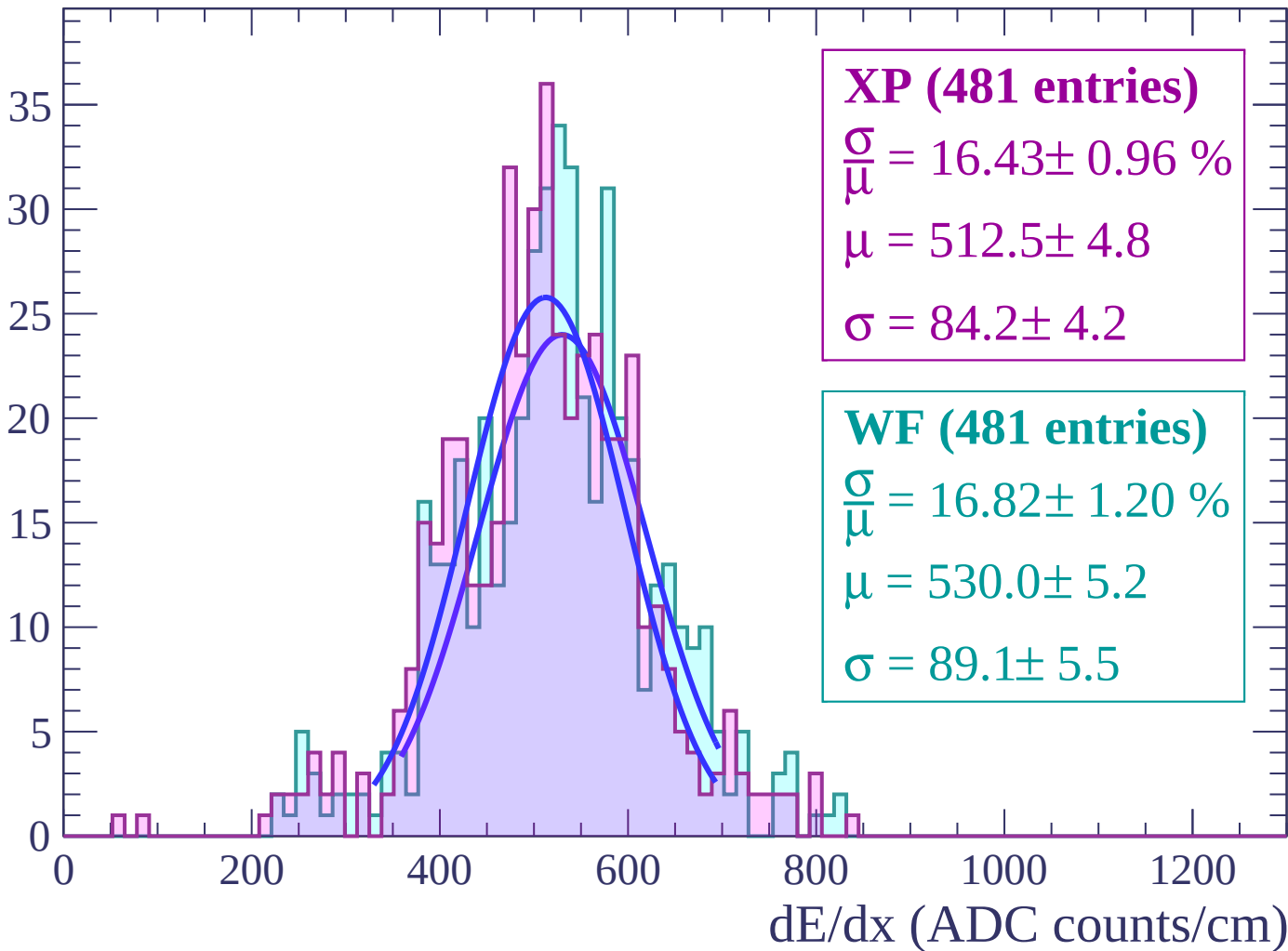
Energy loss | $56 < \theta < 58$

Count



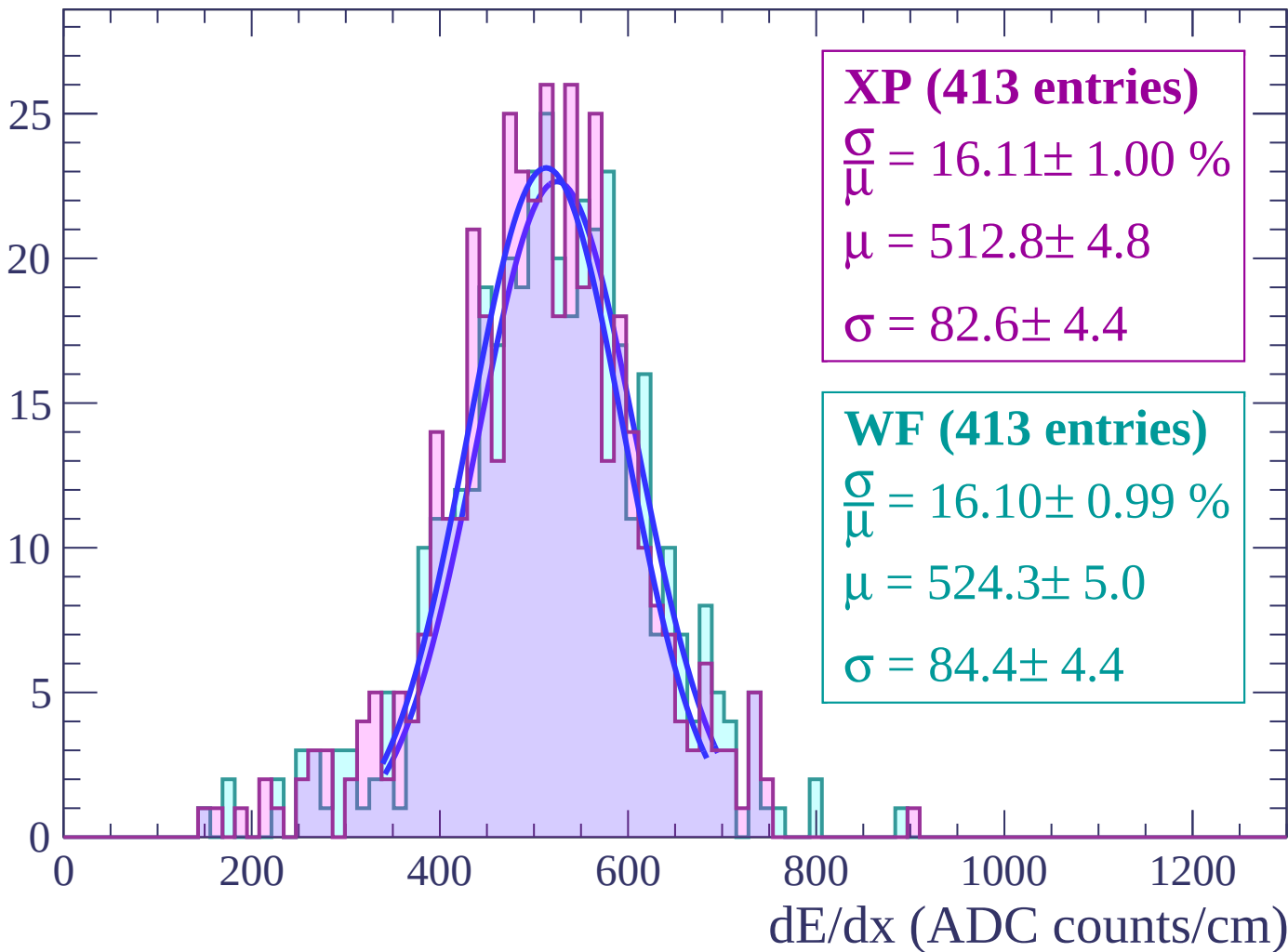
Energy loss | $58 < \theta < 60$

Count



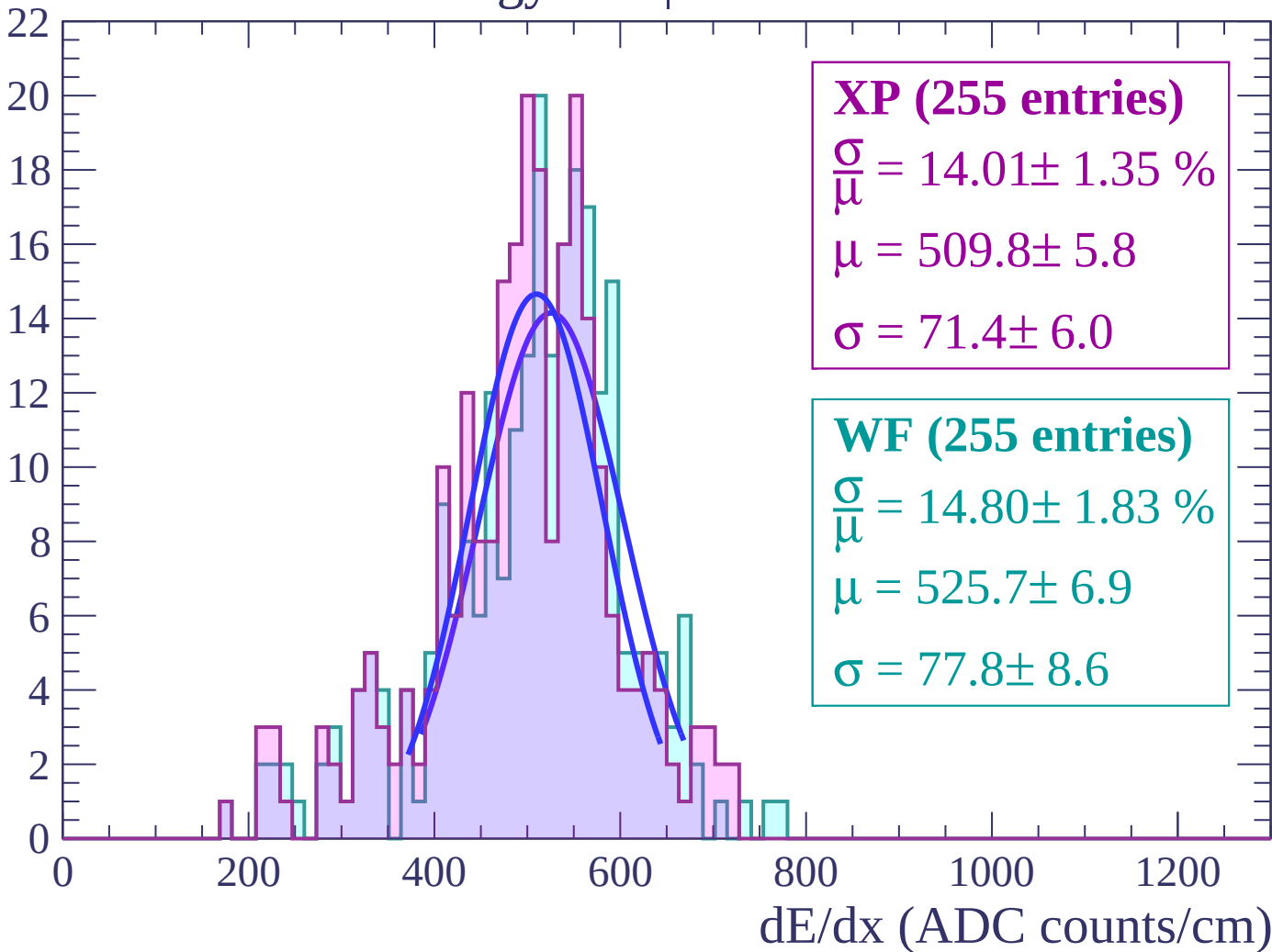
Energy loss | $60 < \theta < 62$

Count



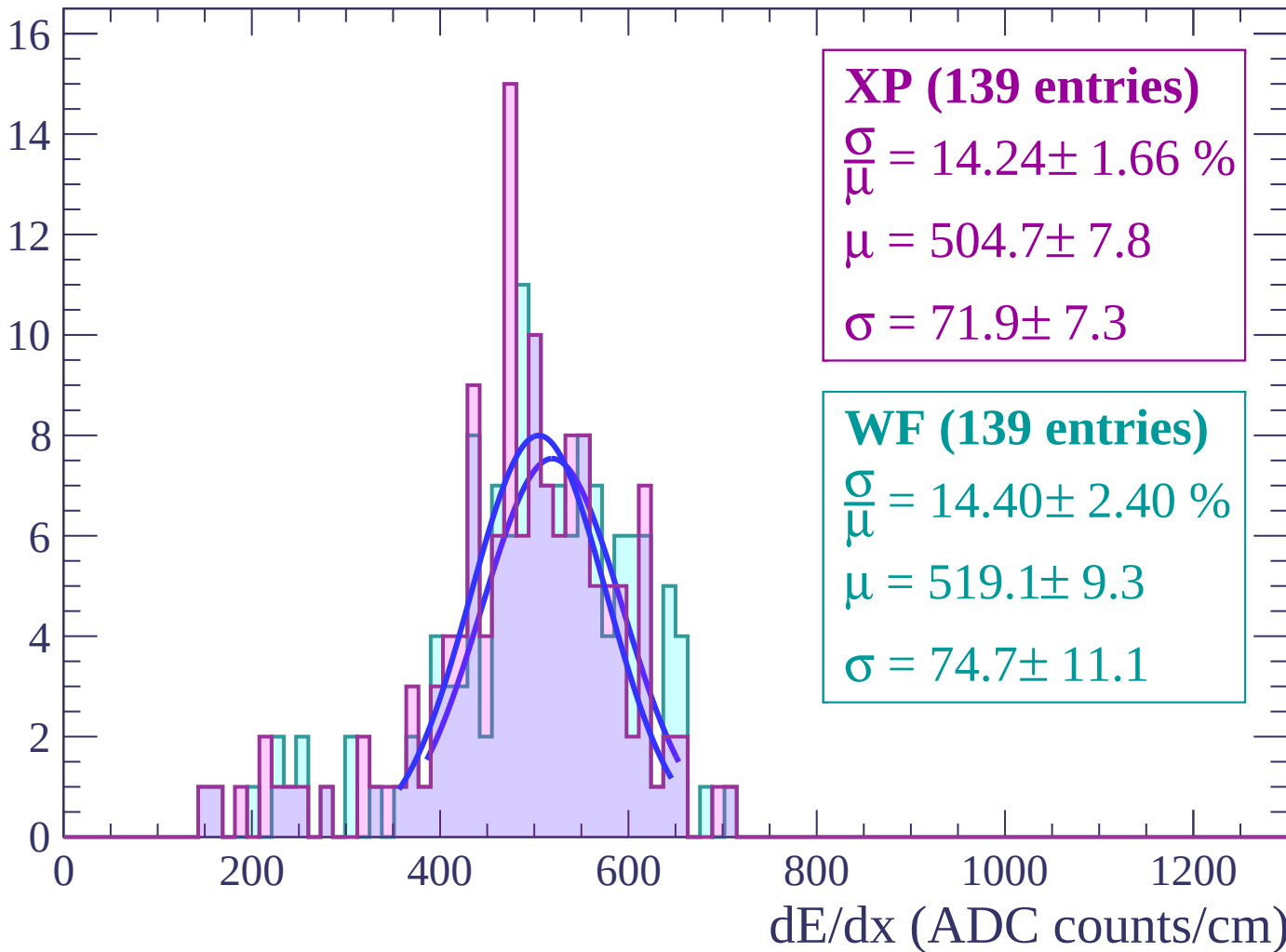
Energy loss | $62 < \theta < 64$

Count



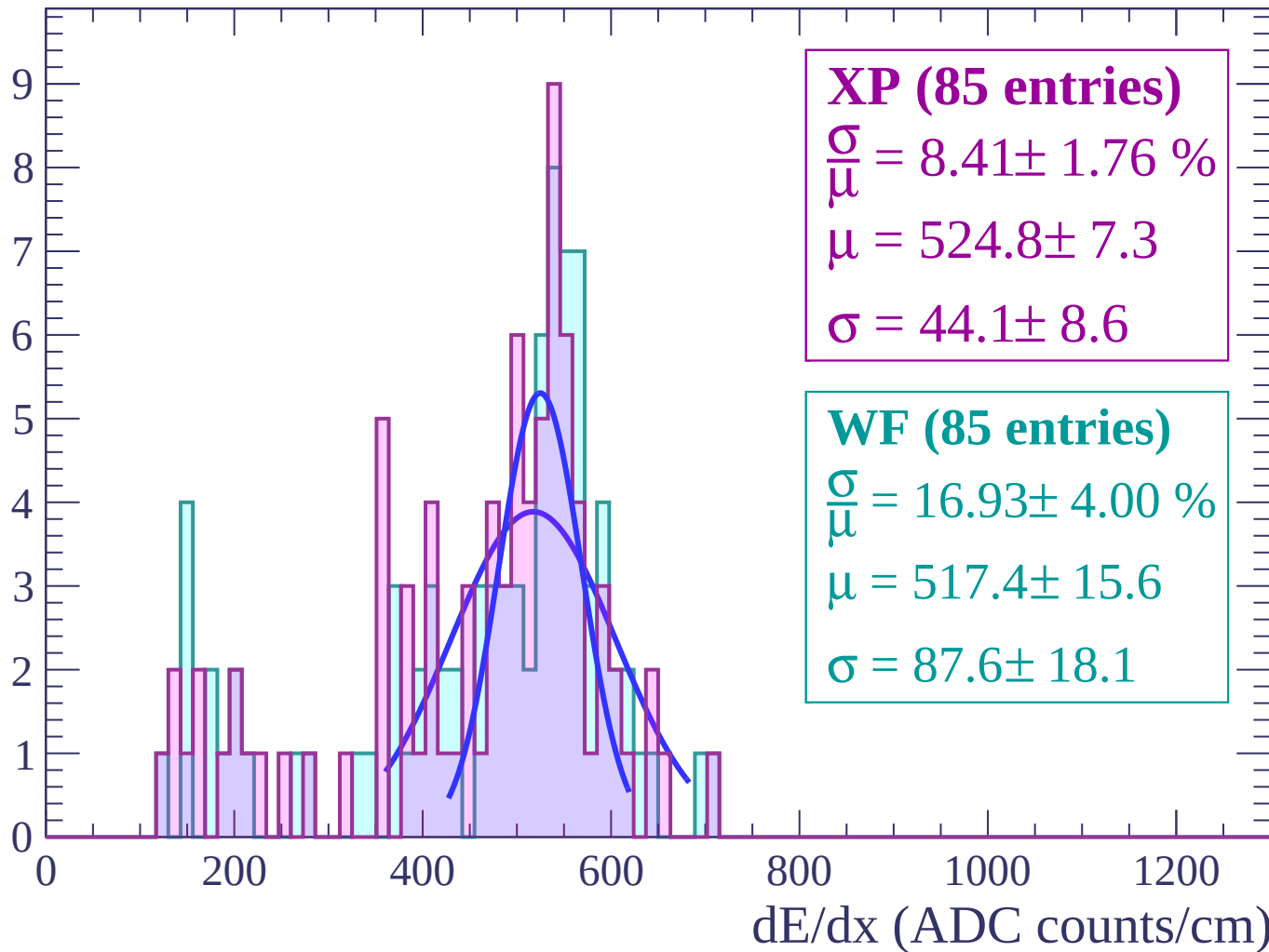
Energy loss | $64 < \theta < 66$

Count



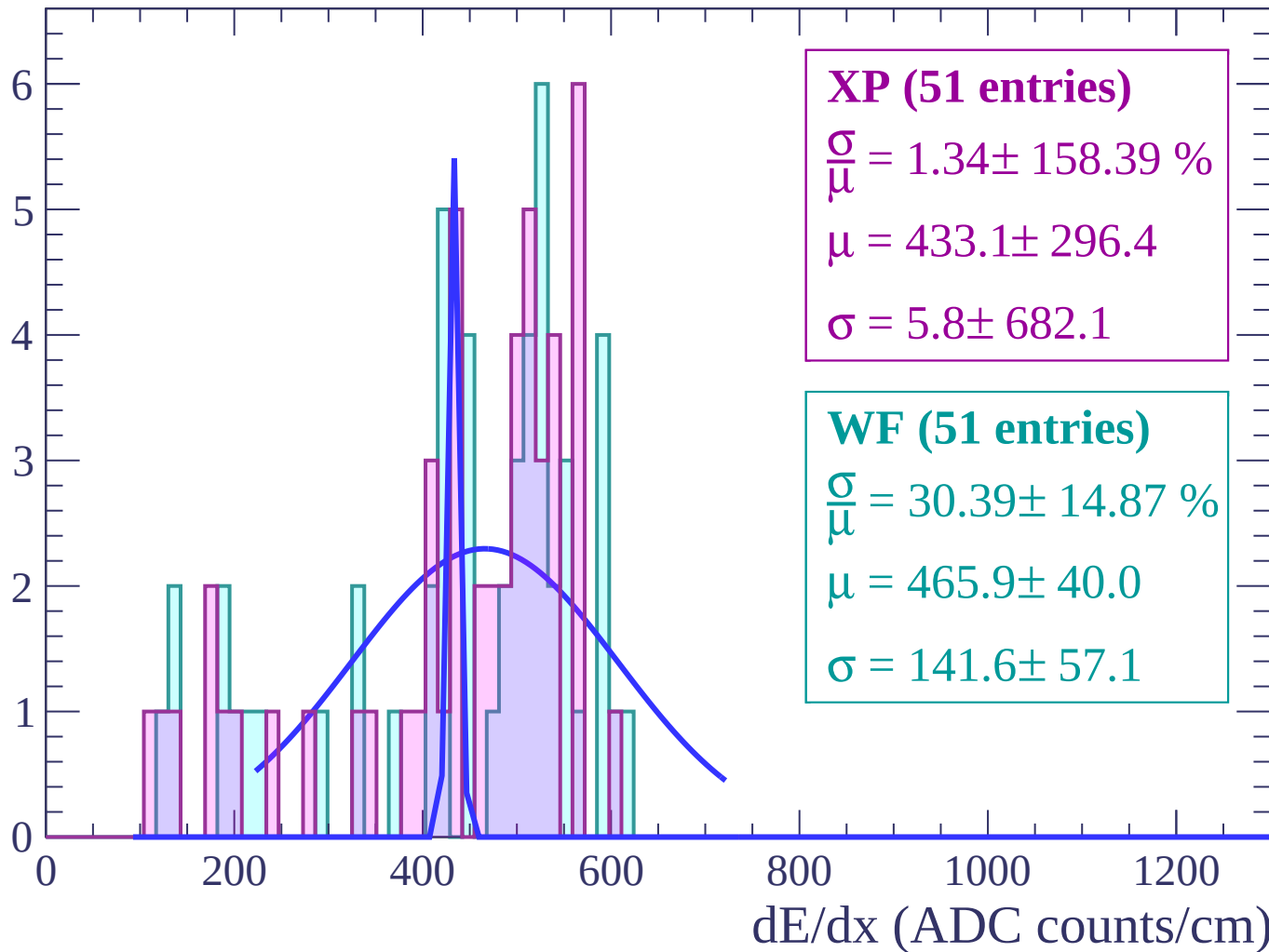
Energy loss | $66 < \theta < 68$

Count



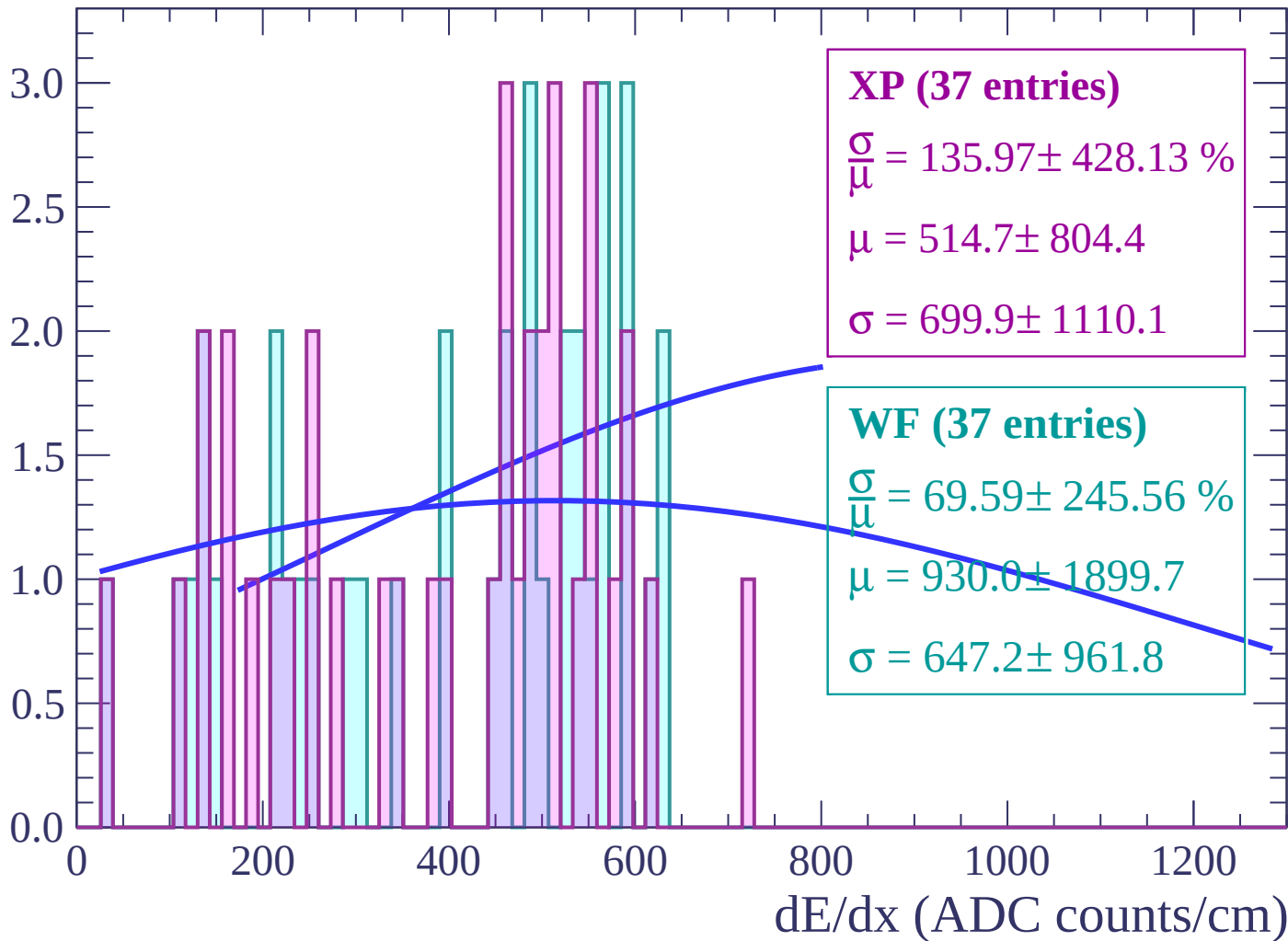
Energy loss | $68 < \theta < 70$

Count



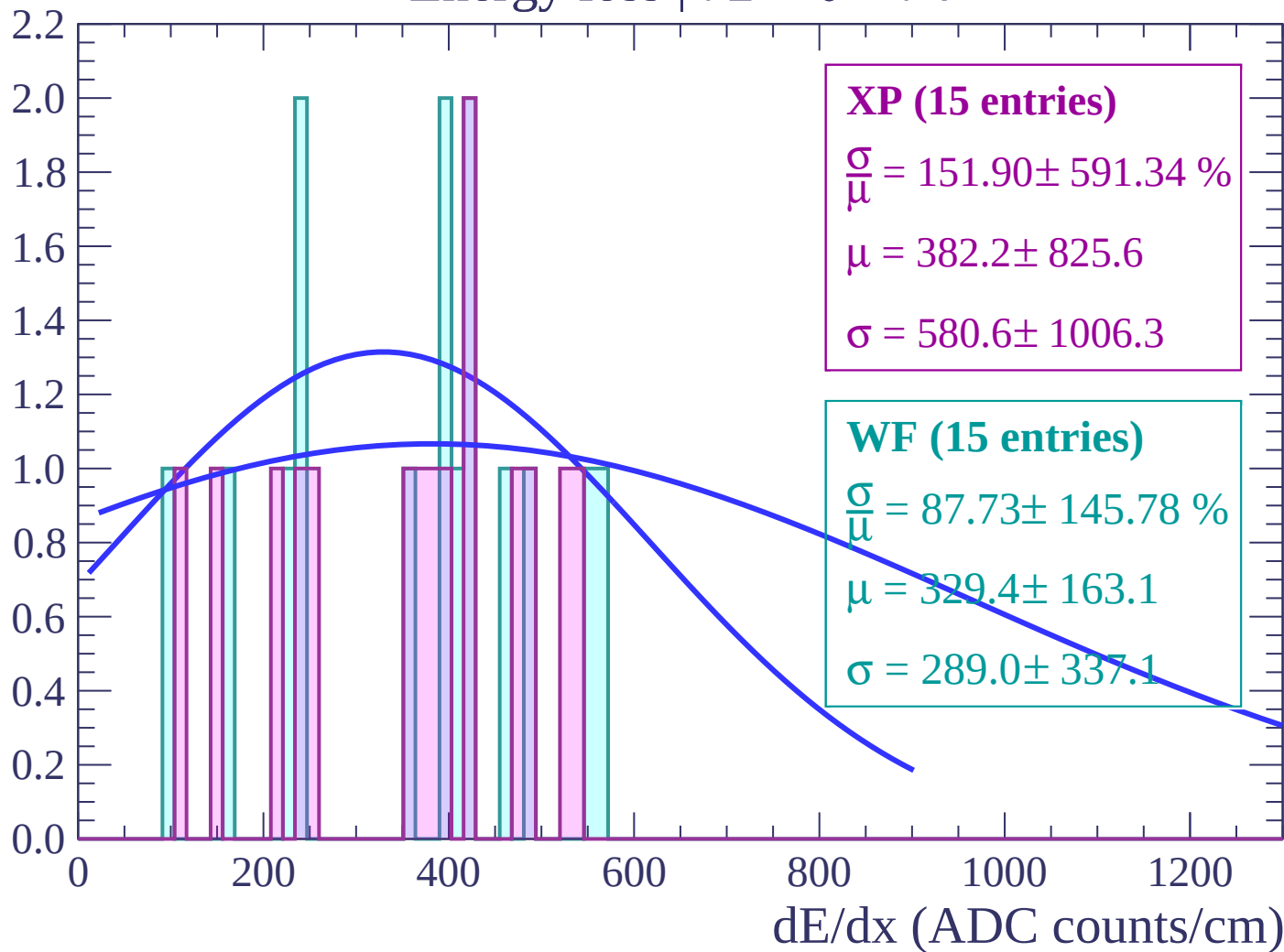
Energy loss | $70 < \theta < 72$

Count



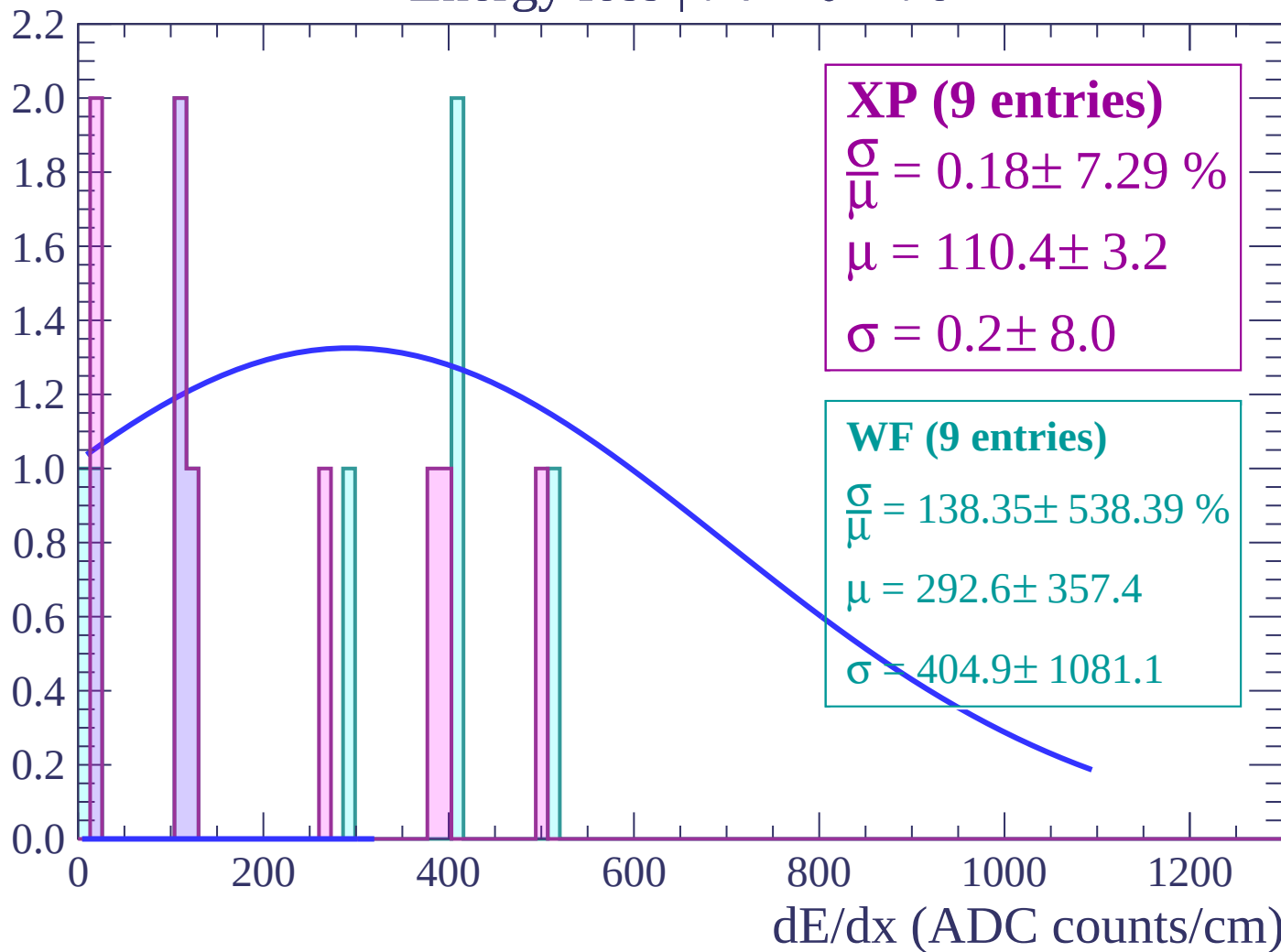
Energy loss | $72 < \theta < 74$

Count



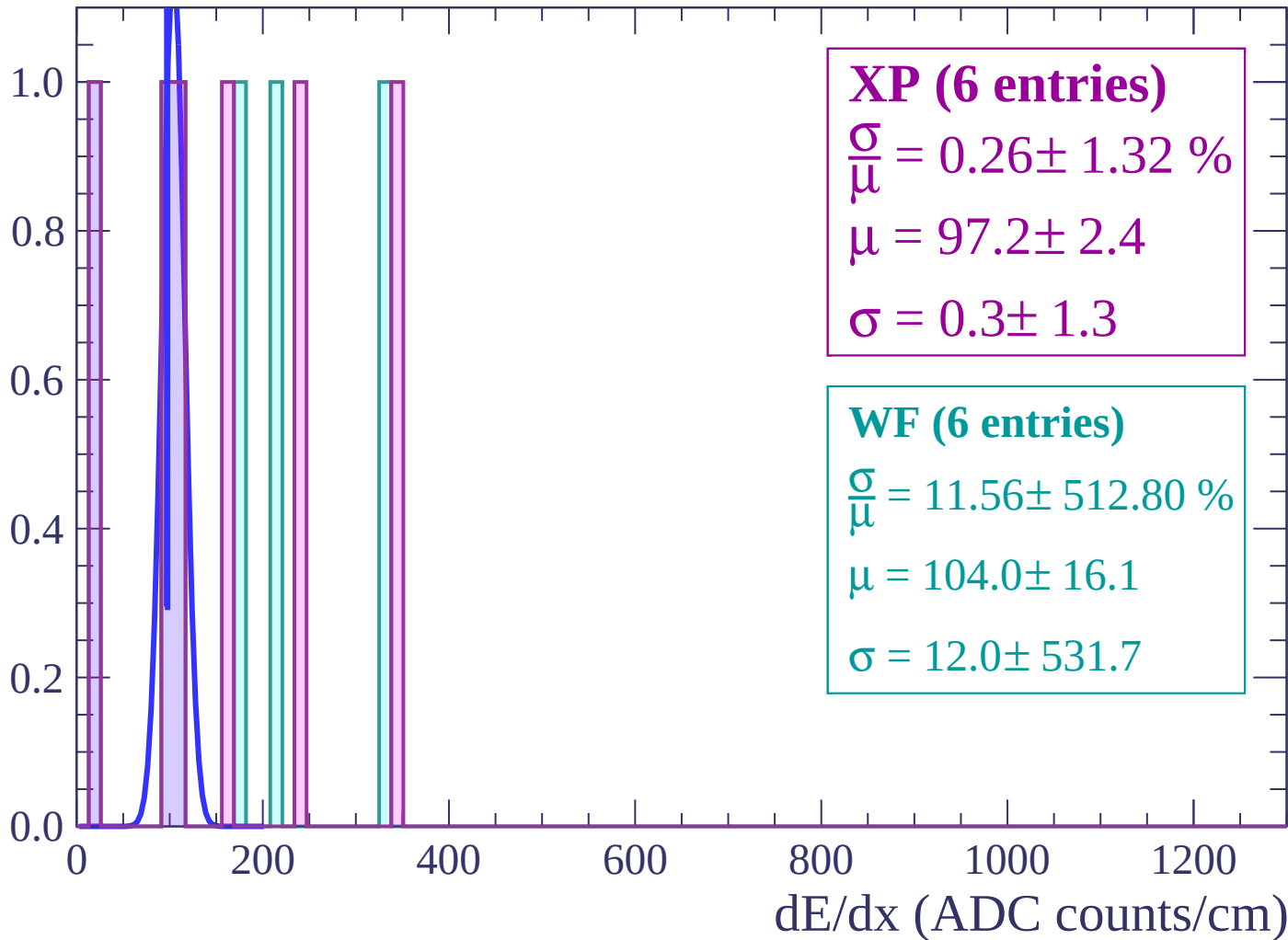
Energy loss | $74 < \theta < 76$

Count



Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

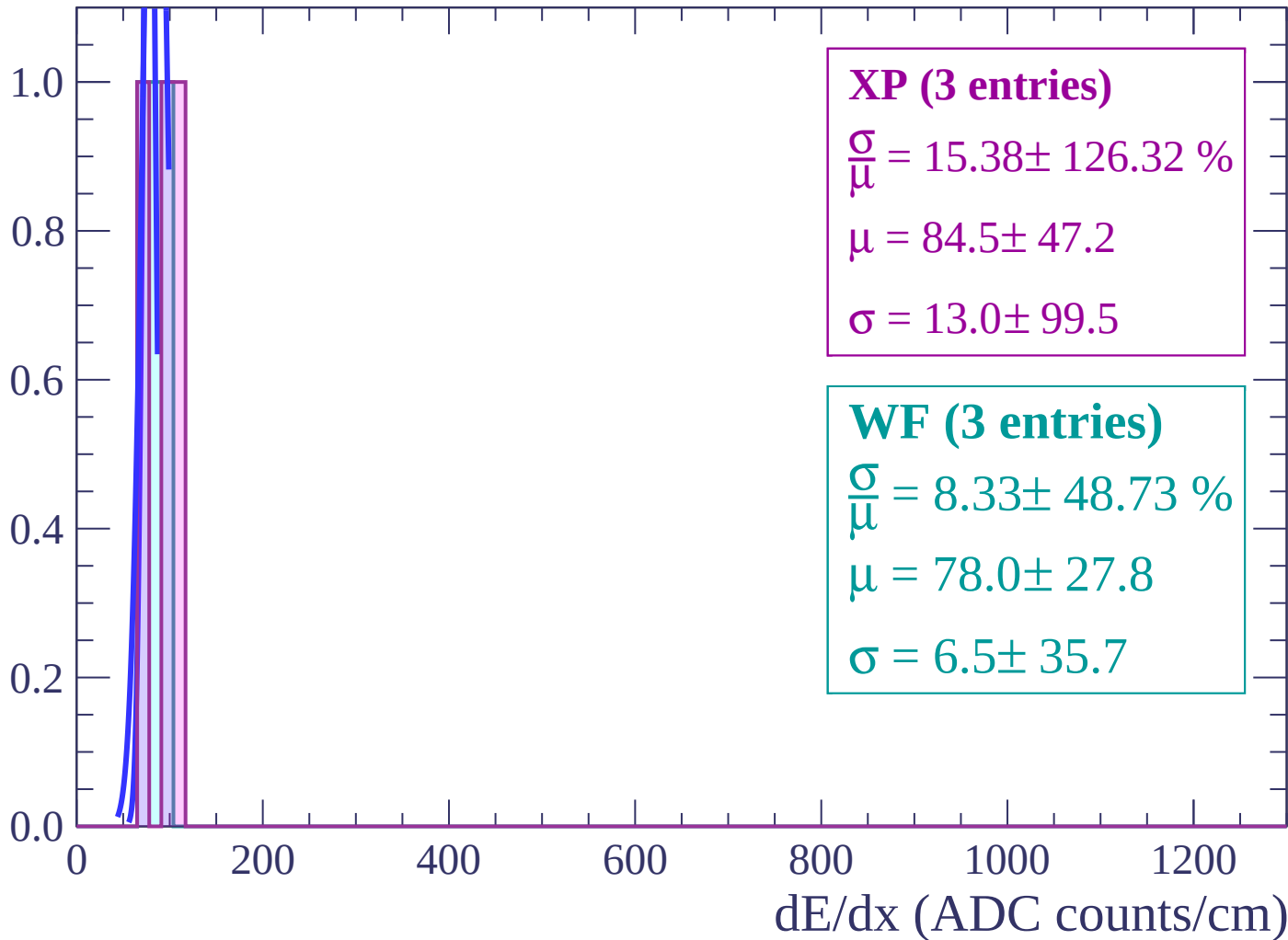
$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $80 < \theta < 82$

Count



Energy loss | $82 < \theta < 84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $86 < \theta < 88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

